FREE K.C.S.E REVISION SERIES 2012 to 2015

BIOLOGY

TOPICS HIGHLIGHTED

SECTION I & II QUESTIONS

- *This section deals with paper one and two questions. In paper one you will find simple questions requiring simple answers, single answer and/or a sentence.*
- *In paper two structured data based questions and essay questions.*

1. Classification I &II

1. Name **two** classes of phylum arthropoda with cephalothorax.

2. List any **three** distinguishing features of class mammalia.

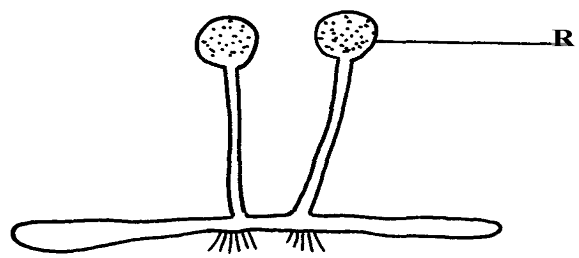
3. Give **two** characteristics that distinguish scientific names of organisms from the ordinary names

4. (a) In which kingdom do bacteria belong?

(b) Give any **two** benefits of bacteria to man

5. Name the phylum whose members possess notochord

6. The diagram below represents a bread mould:-



(a) Identify the kingdom to which the organism belongs:-

7. Give a reason why no moulting occurs during the adult stages of insects

8. Name the branch of Biology that deals with the study of animals

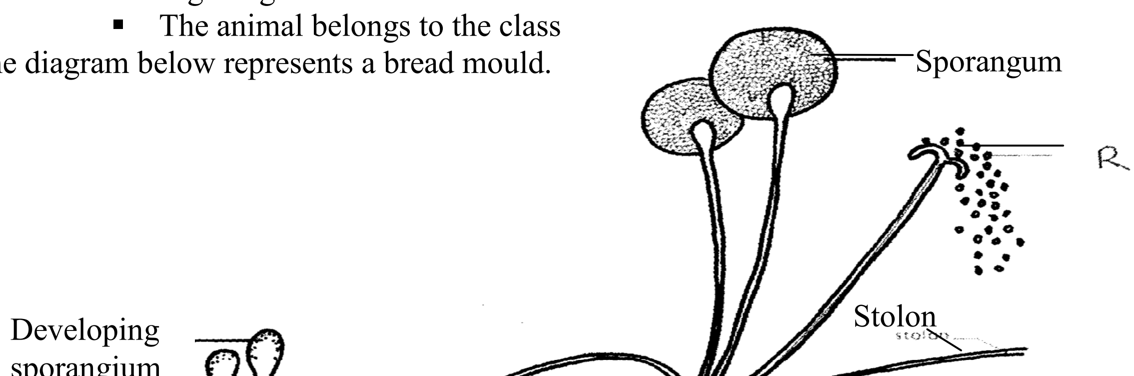
9. State **four** ways in which some Fungi are beneficial to human

10. During a class practical form four students came across a plant whose flower floral parts were in multiples of fours and fives. To which sub-division and class does the plant belong?

11. A student caught an animal which had the following characteristics:-

- Body divided into two parts
- Simple eyes
- Eight legs
- The animal belongs to the class

12. The diagram below represents a bread mould.



(a) (i) Name the Kingdom to which bread mould belongs.

(ii) Give **two** distinguishing characteristics of the Kingdom named in **(a)(i)** above.

(b) State the function of the part labelled **R**

13. (a) What is meant by the term taxonomy?

(b) The scientific name of a rat is Rattus norvegicus

(i) Write the name correctly

(ii) Identify the genus and species names

14. List **three** features that distinguish arthropods from other organisms

2. The cell – structure & functions of organelles

1. Name the organelles that perform each of the following functions:

a) Digestion and destruction of worn out organelles.

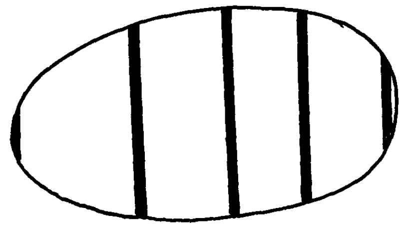
b) Osmoregulation

2. Explain why the following processes are important during the preparation of temporary

slides :- (a) Staining

(b) Use of a sharp cutting blade

3. In a class experiment to establish the size of an onion cell, a leaner observed the following on the microscope field of view.



If the student counted 20 cells across the diameter of this field of view, calculate the size of one cell in micrometers.

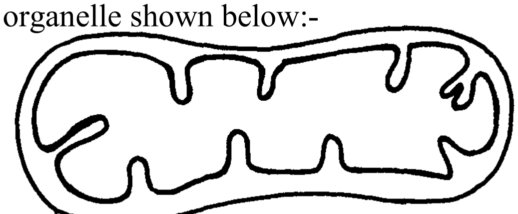
4. State the functions of the following cell organelles: (a) Nucleolus.
(b) Plasma membrane

5. What is the of nucleus of a cell made up of?

6. (a) In a laboratory exercise a student observing a drop of pond water under a microscope saw and drew a spirogyra. If the magnification of the eye-piece was x5 and that of the objective lens was x100, what was the magnification of the spirogyra?

(b) If the spirogyra has a length of 5cm at the above magnification, calculate the actual length in micrometers

7. (a) Identify the organelle shown below:-



(b) How is the organelle you have identified in **(a)** above suited to its function

8. Identify the structures of the cells that perform the following functions:-

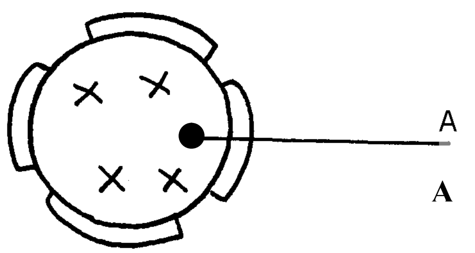
(a) Synthesize ribosomes

(c)Regulate exchange of substances in and out of the nucleus

9. (a) State the roles of enzyme catalase in living cells

(b) Which factor inactivates enzyme?

10. The figure below represents a certain cell organelle:-



(a) (i) Identify the cell organelle

(ii) What is the function of the part labelled **A**

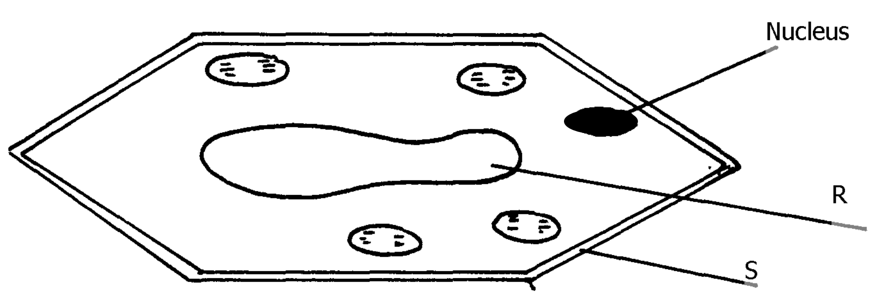
(b) Name the organelles that perform each of the following functions;

(i) Osmoregulation in amoeba

(ii) Carries out digestion and destruction of worn out cell organelles

11. State **three** properties of the cell membrane

12. The diagram below represents a plant cell



(a) Name a carbohydrate which forms part of the structure labelled **S**

(b) State **two** functions of the part labelled **R**

(c) Name **two** structures present in the diagram but absent in the animal cell

13. What do you understand by the following terms

a) Anatomy

b) Biochemistry

14. State the function of the following parts of a cell

a) Ribosome

b) Chloroplasts

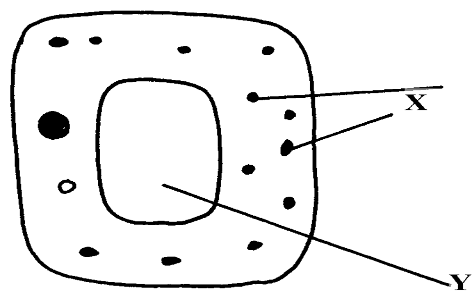
15. a) What is the formula for calculating linear magnification of a specimen when using a hand lens

16. State the function of the following cell structures:- a) Ribosome ;
b) Centrioles ;

17. What is the main structural component of:- a) Cell wall
b) Cell membrane

18. State **two** characteristics of the kingdom monera which are prokaryotes

19. The diagram below represents a cell



(a) Name parts labelled **X** and **Y**

b) Suggest why the structures labelled **X** would be more on one side than the other

20. During a practical class, form fours estimated the field of view to be 3.5mm. Using the low power objective, they observed spirogyra cells across the same field of view and counted 8cells.

Calculate the size of each cell and give your answer in micrometer

21. A student caught an animal which had the following characteristics:-

- Body divided into two parts
- Simple eyes
- Eight legs

a) To what class does the animal belong?

b) State **two** distinctive characteristics of members of the phylum from which the animals in this
question (15) belongs

22. Distinguish between the following terms :-

a) Magnification and resolution of a microscope

b) Mounting and staining of a specimen

23. Name the organelle that performs **each** of the following functions in a cell.

(a) Transport of packaged glycoproteins

(b) Destruction of worn out cell organelles

(c) Synthesis of proteins

24. Why are the following procedures done when preparing sections to be observed under a light
microscope?

(a) Making of thin sections

(b) Using a sharp blade to make the sections

(c) Staining

25. What are the functions of the following parts of a light microscope?

(a) Eye piece lens
(b) Condenser

(c) Diaphragm

26. Given that the diameter of the field of view of a light microscope is 2000um. Calculate the size of a cell in mm if 10 cells occupy the diameter of the field of view

27. State the importance of the following processes in microscopy:
(a) staining

(b) sectioning

28. A cell was found to have the following under a light microscope; cell membrane, irregular in shape, and small vacuoles. Identify the type of the cell above

29. State the functions of the following organelles;
(a) Lysosomes

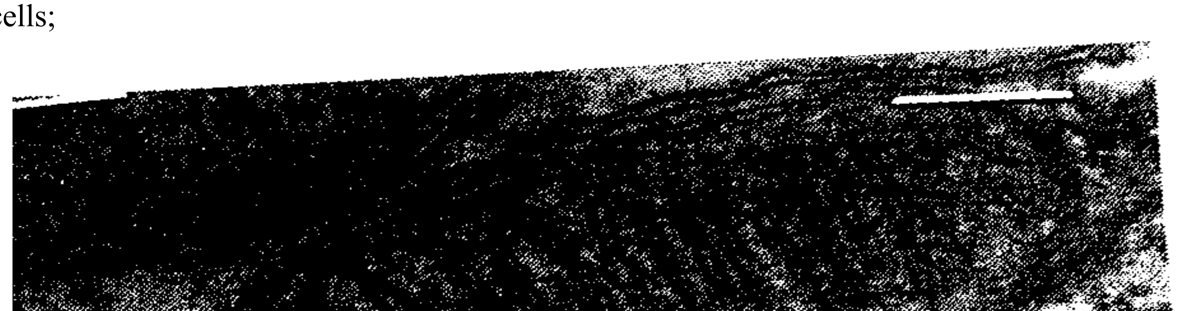
(b) Golgi apparatus

30. Name the class in phylum arthropoda which has the largest number of individuals

31. State the functions of each of the following parts in a microscope.
(a) The eye piece lens

(b) The objective lens

32. The figure below represents an electron micrograph of an organelle that is found in many cells;



(a) Identify the organelle

(b) State the function of the organelle

(c) What is the importance of infoldings in the inner membrane.

(d) Give **two** examples of tissues where you would expect many such organelles in animal body.

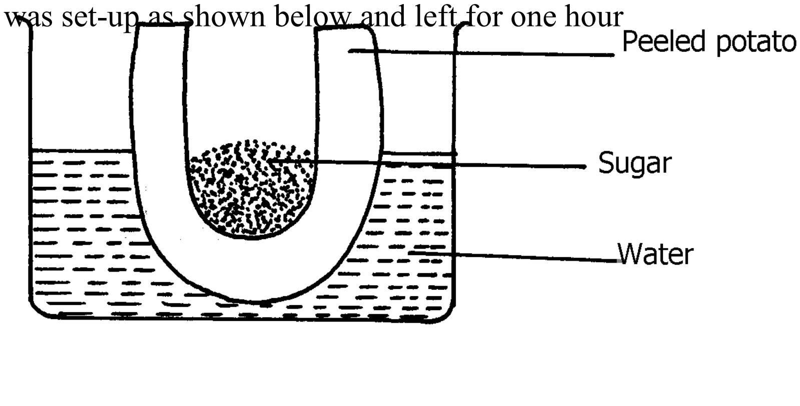
3. Cell Physiology – Osmosis, Diffusion and Active transport

1. Two equal strips **A** and **B** were from a potato whose cell was 30% of sugar. The strip **A** was placed in a solution of 10% sugar concentration while **B** was placed in 50% sugar concentration

   a) What change was expected in strip **A** and **B**

   b) Account for the change in strip **A**

2. An experiment was set-up as shown below and left for one hour



(a) State the expected result at the end of one hour

(b) Explain the observations made in this experiment

3. State what would happen in each of the following:-

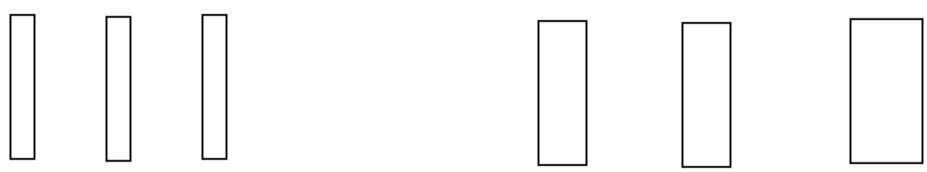
(a) A plant cell placed in: - (i) Strong salt solution
(ii) Distilled water

4. State **three** physiological processes that are involved in movement of substances a cross the cell

membrane

5. Potato cylinders were weighed and kept in distilled water evernight. They were then

reweighed.



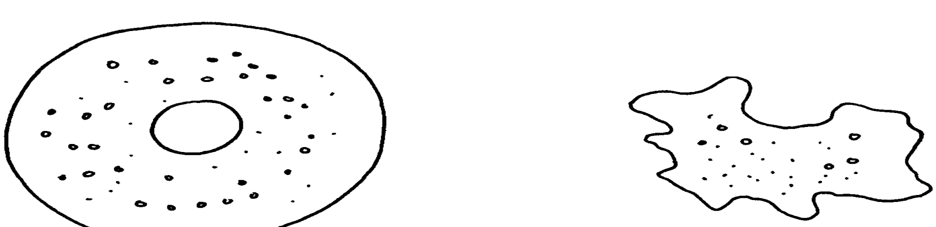
2.5 g 2.4g 2.7g 3.0 g 3.1 g 3.2g

At the beginning of the Experiment. At the end of the experiment

a) Calculate the average mass of a potato cylinders after reweighing. Show your working.

b) Explain why mass of the cylinders hand increased.

6. The diagrams below show a red blood cell that was subjected to a certain treatment.

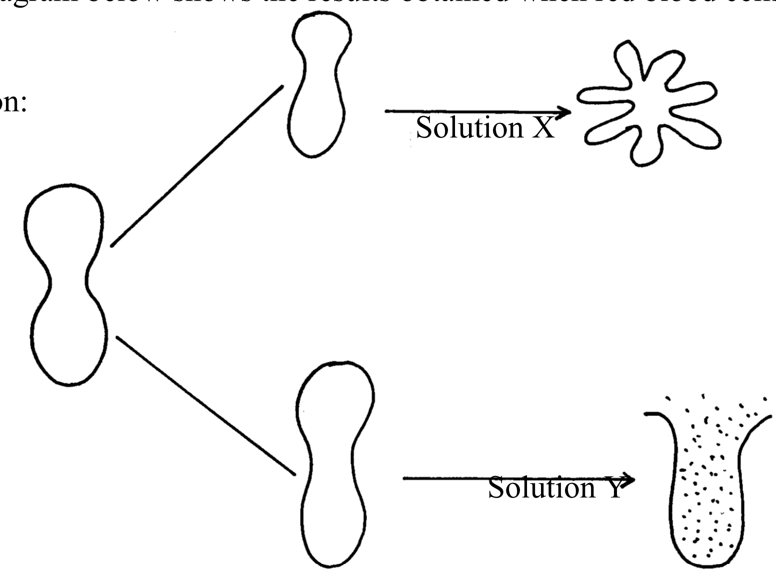


At the start of the experiment At the end of the experiment

a) Account for the shape of the cell at the end of the experiment.
b) Draw a diagram to illustrate how a plant cell would appear if subjected to the same treatment

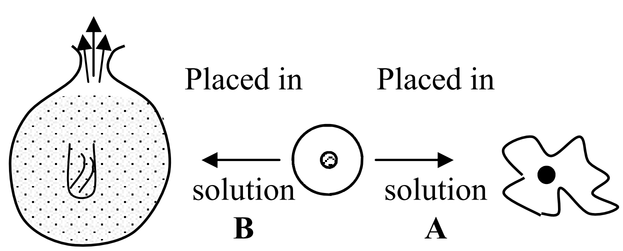
7. The diagram below shows the results obtained when red blood cells are placed in different

solution:



(a) What name is given to the process that occurs when the cell is placed in solution **Y**?
(b) Describe the process that would occur in a plant cell when placed in a similar solution as that of solution **X**

8. The figure below shows the results obtained when red blood cells are put in different solutions:-



(a) What is the name given to the process that occurs when the cell is put into solution **B**?
(b) Compare the results obtained when the cell is put in solution **B** to the results that would be

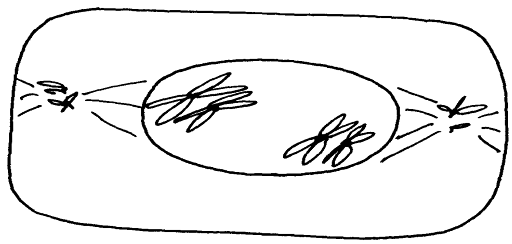
obtained if a plant cell was put in the same solution

9. Briefly state **two** adaptation for each of the following cells to their functions

(i) Spermatozoon

(ii) Palisade mesophlly cell

10. The diagram below represents a cell at a certain stage in meiotic cell division



a) Name the stage at which the cell drawn above represents

b) Give a distinguishing reason for your answer in **21(a)** above

c) State any **two** differences between mitosis and meiosis

11. What are **two** differences between tropisms and tactic movement

12. An experiment was carried out to investigate the effect of different concentrations of sodium

chloride on human red blood cells. Equal amounts of blood were added to equal volumes of the

salt solution but of different concentrations. The results are shown in the table below:

Set -up		**Number of red blood cells**	
	Sodium chloride concentration	**At start of experiment**	**At the end of the experiment**
A	0.9%	Normal	No change in number
B	0.3%	Normal	Fewer in number

(a) Account for the results in the set-up

(b) If the experiment was repeated using 1.4% sodium chloride solution, state the expected

results with reference to:

(i) the number of red blood cells

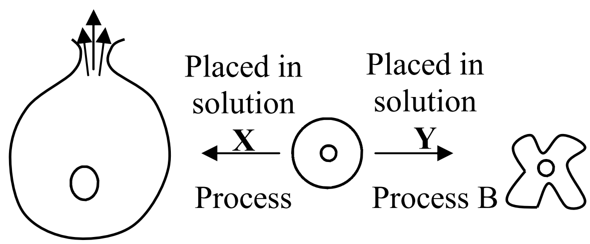
(ii) the appearance of red blood cells if viewed under the microscope

13. Name support tissues in plants characterized by the following

(i) Cells being turgid
(ii) Cells being thickened by cellulose
(iii) Cells being thickened by lignin

14. The diagram below illustrates the behaviour of red blood cells when placed into two different

solutions **X** and **Y**.



(a) Suggest the nature of solutions **X** and **Y**.

(b) Name the process **A** and **B**.

(c) What would happen to normal blood cell if it were placed in a solution isotonic.

15. Name **two** plant processes in which diffusion plays an important role

16. Two fresh potato cylinders of equal length were placed one in distilled water and the other in

concentrated sucrose solution:

(a) Account for the change in length of the cylinder in:

(i) Distilled water

(ii) Sucrose solution

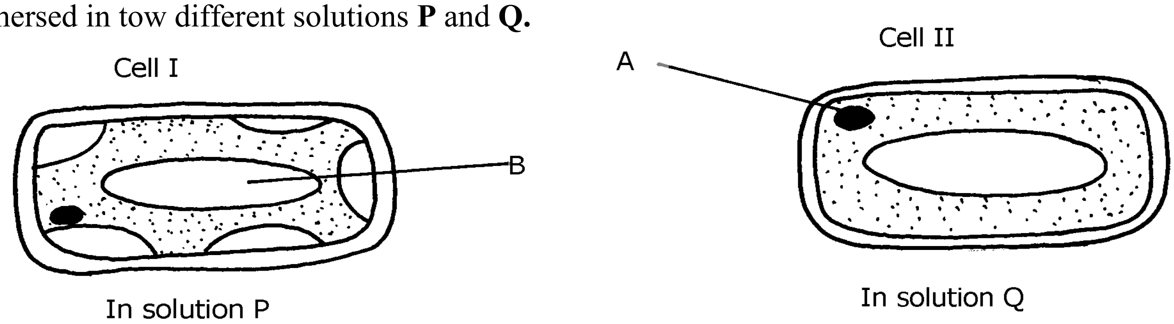
(b) (i) What would be the result in terms of length if a boiled potato was used?

(ii) Explain your answer in**(b)(i)** Above

(c) State **two** uses of the physiological process being demonstrated in the experiment

17. The two cells shown below are obtained from two different potato cylinders which were

immersed in tow different solutions **P** and **Q.**



a) i) Name the structure labelled **A**.

ii) State the function of structure B.

b) If eight of cell I were observed across the diameter of the filed of view of 0.5 mm.

Work out the actual diameters of each cell in micrometers.

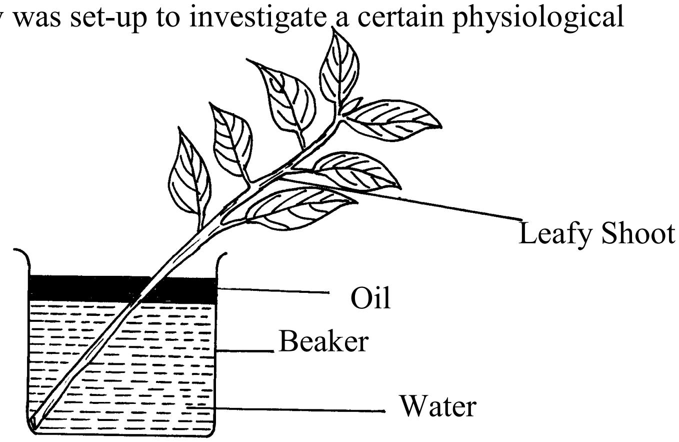
c) Suggest the identity of the solution Q.

d) Account for the change in cell I above.

e) State any **one** importance of the physiological process being demonstrated above in animals.

18. An experiment shown below was set-up to investigate a certain physiological process

in plants:-



(a) What process was being investigated?

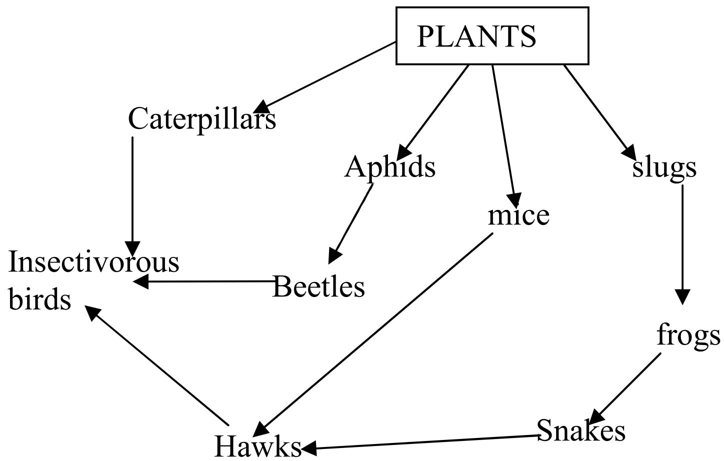
(b) Give the role of the oil layer in this experiment

(c) (i) What observation did the students make after leaving the set-up in bright sunlight for two hours?

(ii) Explain the observation in **(c)(i)** above

(d) What effect will the following have on the observation made?:-
(i) Fanning the shoot
(ii) Removing all the leaves from the shoot
(iii) Placing the set-up in the dark
(e) Suggest a suitable control for this experiment

19. Study the following food web and answer questions that follow:



(a) (i) Name the organisms that occupy the second trophic level

(ii) What is the other name for the second trophic level

(b) Write down **two** food chains from the food web that:

(i) End with hawks as tertiary consumer
(ii) End with hawks as quaternary consumer

(c) Giving reasons state;
(i) the organism with largest biomass

(ii) the organism with least biomass

4. Nutrition in (a) plants (b) animals

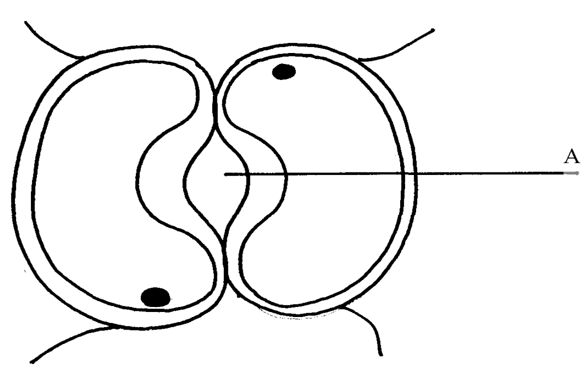
1. The chemical equation below represents a physiological process that takes place in living organisms:

$$C_6H_{12}O_6 + C_6H_{12}O_6 \xrightarrow{\textbf{R}} C_{12}H_{22}O_{11} + \textbf{Q}$$

(a) Name the process **R**

(b) Name the substance **Q**

2. The diagram below shows cells in plants:-



(a) Identify the cells shown above

(b) Explain how the cells are adapted to their function

(c) Explain how accumulation of carbon (IV) Oxide in the cells above would lead to the closure of structure **A**

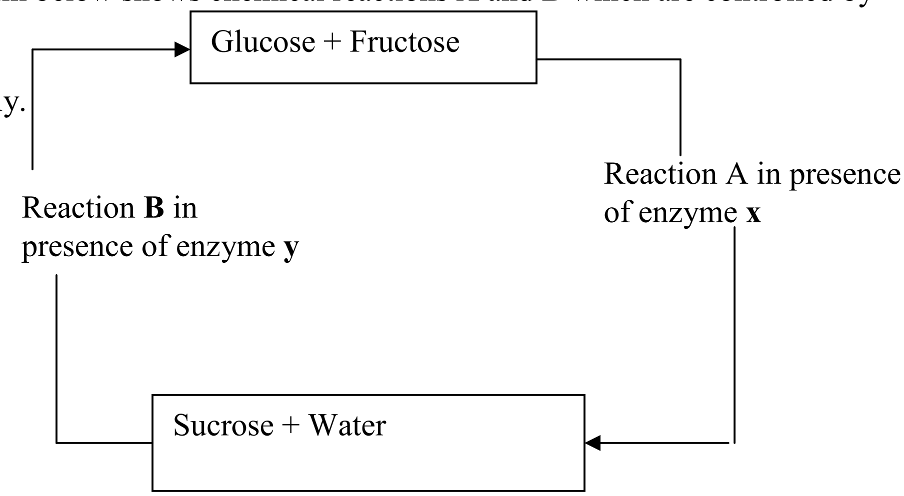
3. (a) A leaf of a potted plant kept in darkness for 48hours was smeared with Vaseline jelly then exposed to sunlight for 8hours. Explain why the test for starch in the leaf was negative

(b) Name **two** other processes that were interfered with in the plant

4. List **two** functional differences between plants and animals.

5. Explain how the guard cells are adapted to perform their function.

6. The diagram below shows chemical reactions **A** and **B** which are controlled by enzymes **x** and **y** respectively.



(a) Name: (i) Reaction **A**.

(ii) Enzyme **y**

7. What are the **two** functions of bile salts during the process of digestion?

8. State **three** adaptations of aquatic plants to photosynthesis

9. A biological washing detergent contains enzymes which remove stains like mucus and oils from clothes which are soaked in water with the detergent:-

(a) Name **two** groups of enzymes that are present in detergent

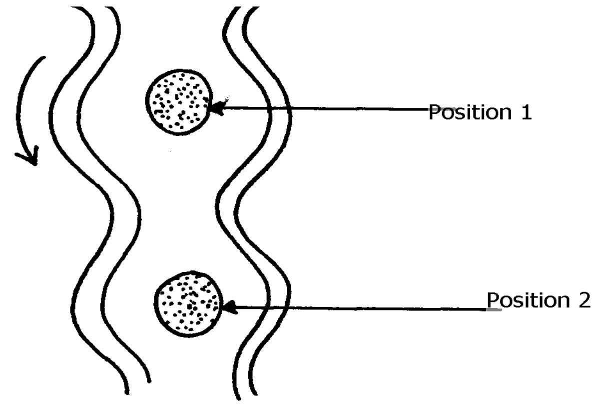
(b) Explain why stains would be removed faster with the detergent in water at 35^{o}C rather than at 15^{o}C

10. Name the diseases caused by deficiency of : (a) Iodine
(b) Vitamin C

11. Name **two** enzymes and **one** metal ion that are needed in the blood clotting process

12. The diagram below shows how food boles move along the human oesophogus and the

Intestine



(a) Identify the process illustrated in the diagram

(b) Briefly **state** how the movement of food boles from position 1 to position 2 is achieved

(c) Name **one** component of a persons diet that assists in the movement of food described in (b) above

13. State **two** adaptations of herbivores which enable them to digest cellulose

14. State **two** factors that affect the rate of osmosis

15. A certain organ **K** was surgically removed from a rat, later drastic increase in glucose level in the blood was reported but when substance **Q** was injected into the animal the whole process was reversed.

Identify: (i) Organ **K**

(ii) Substance **Q**

16. a) Name the component of a persons diet that is essential for peristalisis

b) Give **two** groups of food which are reabsorbed along the mammalion digestive system without undergoing digestion

17. State **three** roles of light in photosynthesis

18. State **two** ways in which the guard cells differ their adjacent epidermal cells

19. One of the components of bile is a chemical left over from destruction of red blood cells

i) Identify the chemical substance

ii) What is the role of bile in digestion

20. (a) What is peristalsis?
(b) Explain how the process above is brought about.

21. The following reaction may occur in a forward and backward direction.

Water + Carbon (IV) Oxide $\rightleftharpoons$ Glucose + Oxygen + Energy

(a) Name the organelle where the reaction occurs in:

(i) Forward direction
(ii) Backward direction

(b) Give **one difference** and **one similarity** for the two organelles named in **(a)** above

22. A solution of sugar cane was boiled with hydrochloric acid and sodium hydrogen carbonate was added to the solution, which was then boiled with benedicts solution. An orange precipitate was formed.

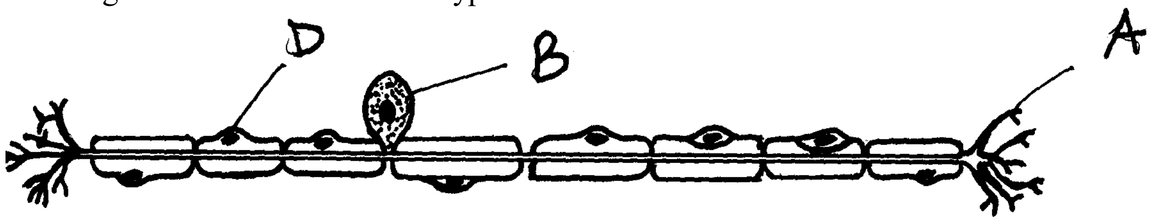
(a) Why was the solution boiled with hydrochloric acid and then sodium hydrogen carbonate added in it

(b) To which class of carbohydrates does sugar cane belong?

(c) State the form in which carbohydrates are:

(i) Transported in animals
(ii) Transported in plants

23. The diagram below is of a certain type of neurons

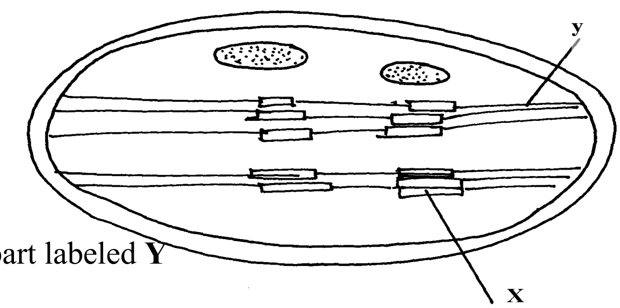


(a) Identify the type of neuron
(b) Give a reason for your answer in **(a)** above

(c) Give the functions of the parts labeled **A, B**, and **D**

24. a) The mitochondria organelle has cristae structure on the inner membrane. State the function of the cristae

b) The diagram below represents a cell organelle



i) Name the part labeled **Y**

ii) State the function of the part labeled **X**

25. a) State the role of emulsification in the digestion of fats in the alimentary canal

b) What is the function of hydrochloric acid in the alimentary canal

26. Briefly explain the effect of poisoning the roots hair on the uptake of nitrate by plants

27. Briefly explain the symbiotic relationship in the root nodule of a leguminous plant

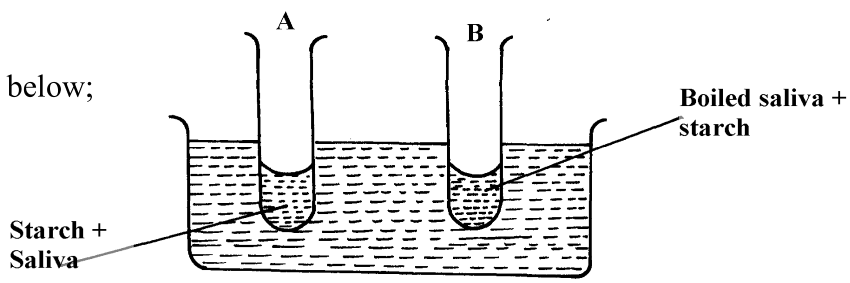
28. Explain how saliva is important in digestion

29. What is the fate of excess glucose in plants?

30. State **two** ways in which guard cells differ from other epidermal cells

31. Briefly explain the fate of the following products from the light stage of the process of
Photosynthesis: (a) Oxygen
(b) Hydrogen
(c) ATP

32. In an experiment to investigate on aspect of digestion, two test tubes A and B were set-up as
shown in the diagram below;



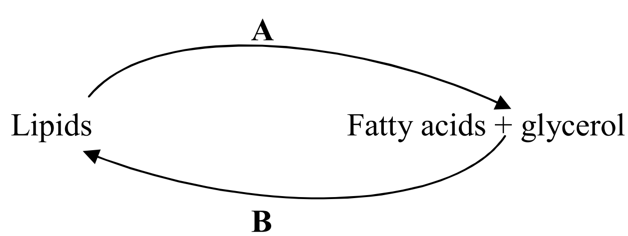
The test tubes were left in the bath for 30minutes.The content of each test tube was then tested for
starch using iodine solution:-

(a) What was the aim of the experiment?

(b) What results were expected in test-tube **A** and **B**

(c) Account of the results you have given in **(b)** above in test tube **A** and **B**

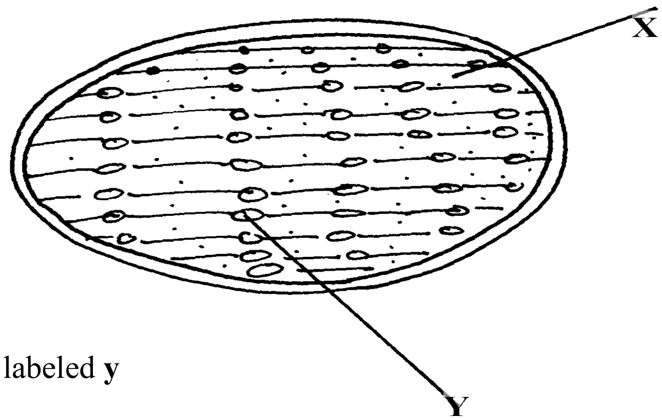
33. Below is a process that takes place along the mammalian digestive system:



(a) Name the processes represented by **A** and **B**

(b) Name part of the alimentary canal where the process **B** takes place

34. The diagram below represents a cell organelle



(i) Name the part labeled **y**

(ii) State the function of the part labeled **X**

(ii) State the function of the vitamin named in **(i)** above

36. (a) Name the disease caused by **schistosoma** parasites in man.

(b) How is **schistosome** adapted to its parasitic mode of life?

37. The table below shows **three** enzymes **A, B** and **C** and their respective optimum pH.

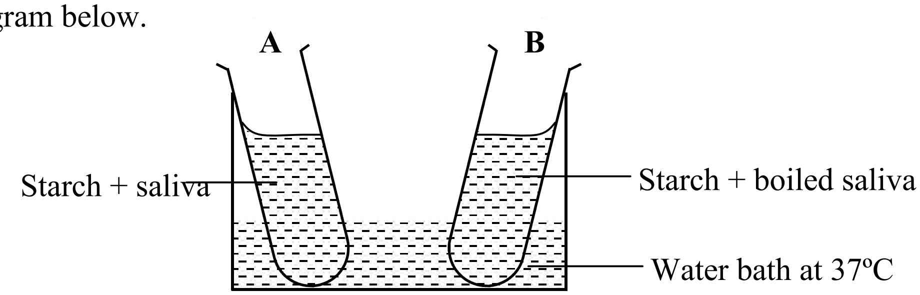
Enzyme	Optimum pH
A	6.8
B	2.0
C	8.0

(a) (i) Name the most likely region of the alimentary canal of a mammal where enzyme **B** would be found.

(ii) Give a reason for your answer in (a) (i) above

38. In an experiment to investigate an aspect of digestion, two tubes A and B were set up as shown

in the diagram below.



The test tubes were left in the water bath for 30 minutes. The content of each tube was then tested

for starch using iodine solution.

(a) What was the aim of the experiment?

(b) Explain the expected in the tube.

39. (a) Name the specific part of the chloroplast where the following processes occur.

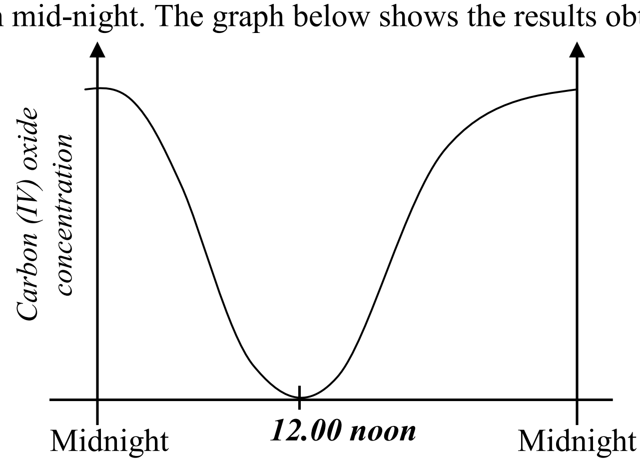
(i) Carbon IV oxide fixation

(ii) Photolysis

(b) State **one** way in which the dark reactions of photosynthesis depends on light reaction.

40. The concentration of carbon IV oxide in a tropical forest was measured during the course of 24

hour period from mid-night. The graph below shows the results obtained.



Account for the results obtained at: (i) Midnight.
(ii) At 12.00 noon.

41. State **three** ways by which the rate of enzyme controlled reactions can be increased.

42. Study the dental formula given below:

$$\mathbf{I\ \frac{0}{4};\ C\ \frac{0}{0};\ PM\ \frac{3}{3};\ M\ \frac{2}{3}}$$

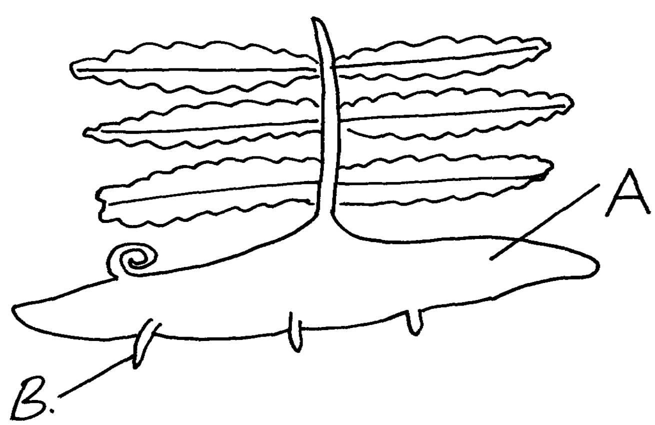
(a) Identify with reasons the mode of feeding of the animals whose dental formula is given above

(b) Calculate the total number of teeth in the mouth of the above animal

43. Explain why small mammals such as moles feed more frequently than larger ones such as elephants

44. State **three** ways by which plants compensate for lack of the ability to move from one place to another

45. Study the diagram below and answer the questions that follow



(a) Label the parts **A** and **B**

(b) State **one** observable difference between the structure above and the liverwort

46. What is glycolysis?

47. (a) State **two** difference between monosaccharide and polysaccharides

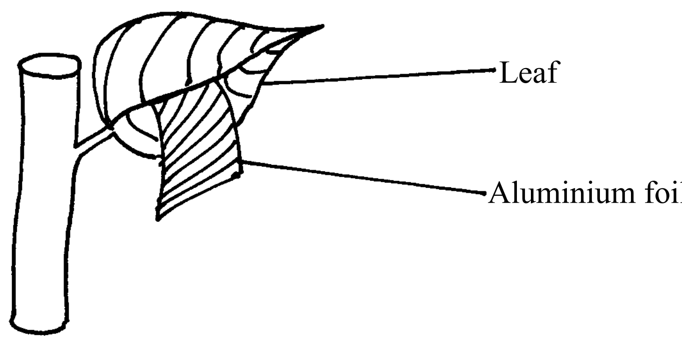
(b) Name the bond found in proteins

48. Name **two** products of light reaction used in the dark reaction

49. State **two** functions of the large intestine in humans.

50. The diagram below shows a leaf of a growing plant partly covered with aluminium foil.

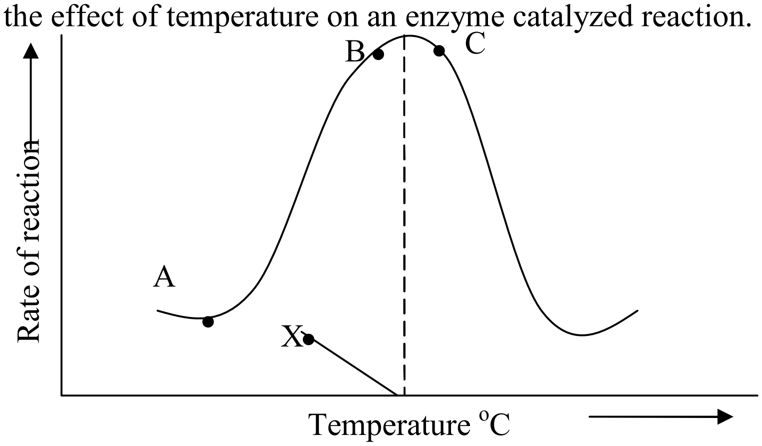
The plant was placed in the sun from morning to midday and then tested for starch.



(a) What was the aim of the experiment?

(b) State the observation made when the leaf was tested for starch

51. The figure shows the effect of temperature on an enzyme catalyzed reaction.

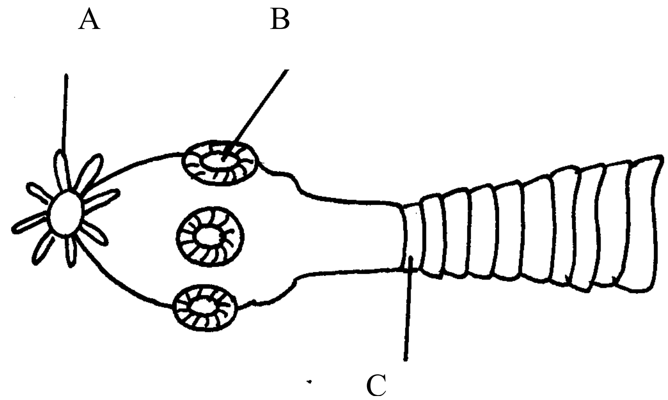


(a) Explain what happens between **A** and **B**

(b) What is **X**?

52. Name **two** mineral elements that are necessary in the synthesis of chlorophyll.

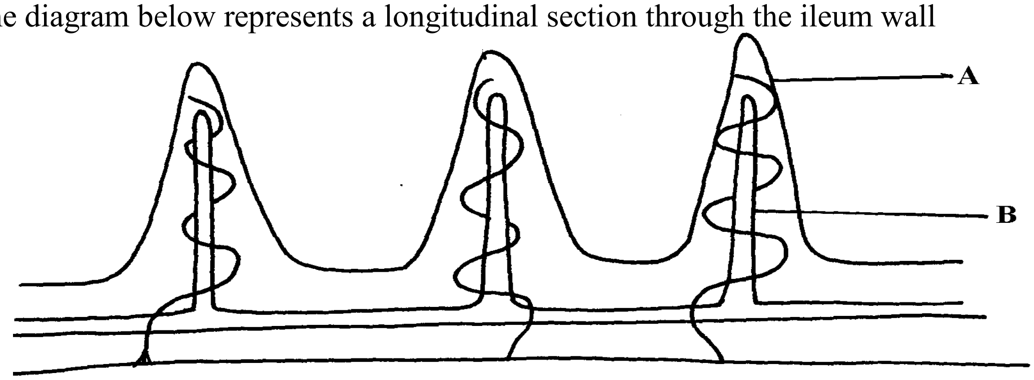
53. The figure below is a diagram of the anterior portion of the tapeworm. **<u>Taenia solium</u>.**



(a) Name the parts labeled **A, B**, and **C**

(b) What is the intermediate host of Taenia Solium?

54. The diagram below represents a longitudinal section through the ileum wall



a) Identify the structure labeled **A** and **B**

b) State **one** function of **A** and **B**

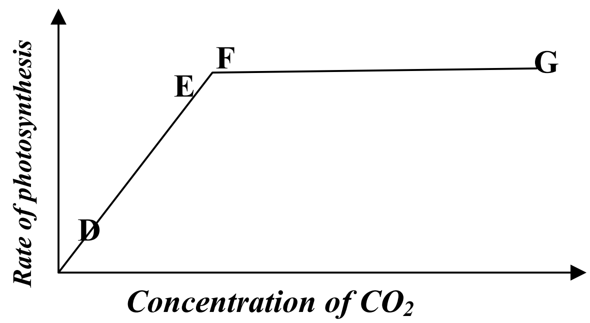
c) State **two** functions of the ileum

d) Explain the role of the liver in digestion

e) State the endocrine role of the pancreas in a mammal

55. The chart below shows the relationship between concentration of CO_2 around the plant and the

rate of photosynthesis



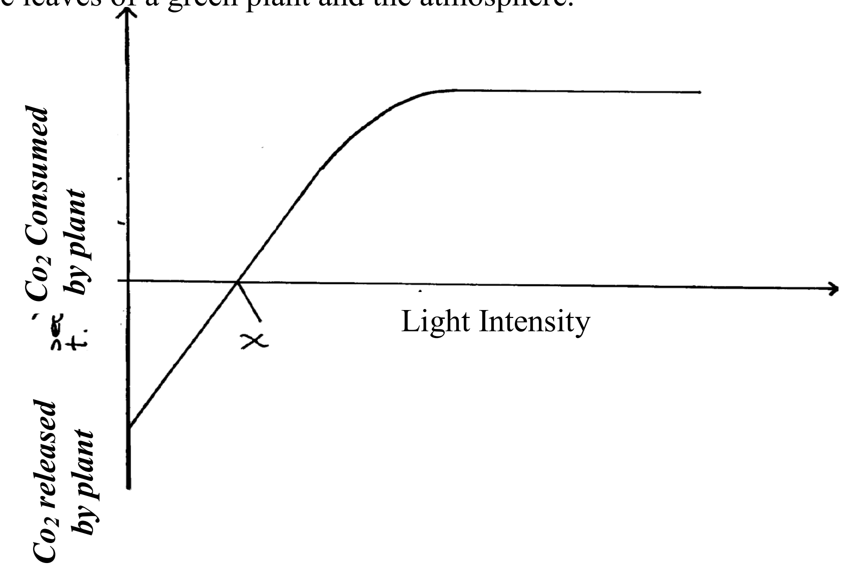
(a) Account for the rate of photosynthesis between **D-E**

(b) Account for the rate of photosynthesis between **F-G**

(c) Briefly describe the reactions during the light stage of photosynthesis

56. The diagram below shows the effect of varying light intensity on the exchange of carbon IV

oxide between the leaves of a green plant and the atmosphere.



a) What is the name given to the point marked x?

b) i) With reference to carbon IV oxide exchange state what happens at point **x**.

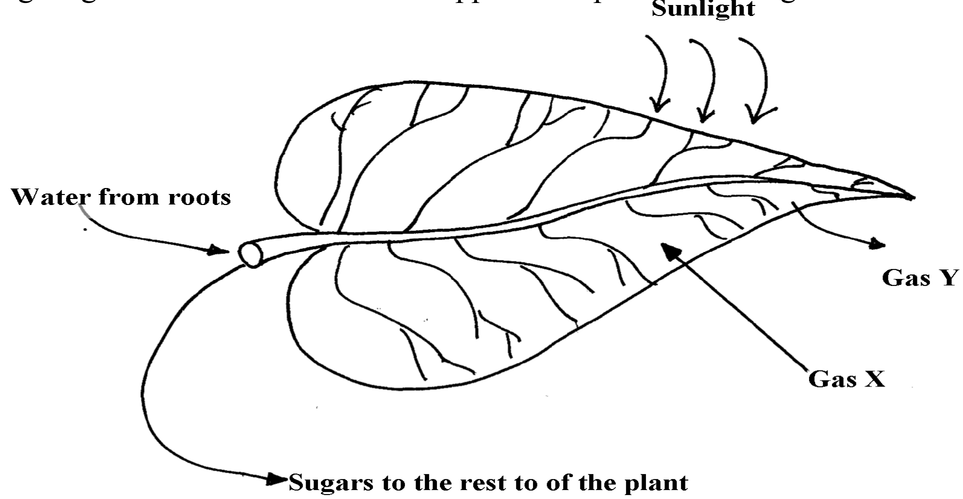
ii) Explain how the effect observed at point x occurs.

c) Explain why there is a net uptake of carbon IV oxide at light intensity above **x**.

d) What would happen to the plant if light intensity falling on it were maintained at **x** throughout?

e) What can you say about the exchange of oxygen between the plant and the surrounding air at intensities below **x**?

57. The following diagram of a leaf shows what happens in a pant leaf during photosynthesis:-



(a) Give **two** ways in which leaves are adapted to absorb light

(b) Name the gases labelled **X** and **Y**

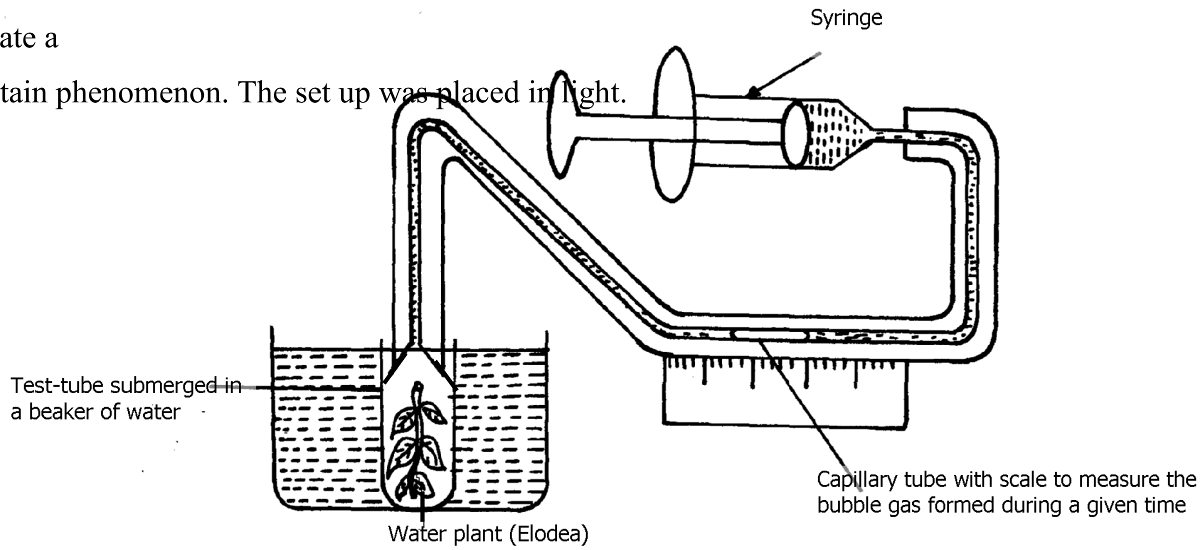
(c) Name the tissue that transports water into the leaf and sugars out of the leaf

(d) Explain why it's an advantage for the plant to store carbohydrates as starch rather than as sugars

58. (a) What is meant by digestion?

(b) Describe how mammalian small intestine is adapted to its function

59. Form one students from Inungo school arranged their apparatus as shown below, to investigate a certain phenomenon. The set up was placed in light.



(a) State the likely aim of the set up

(b) State the role of the syringe in the set-up above

. (c) (i) Name gas **X**
(ii) Write an equation to show how gas **X** was formed in the set-up

(d) State **three** factors that increase the rate of enzyme activity

(e) Give a reason why the test tube is immersed in a beaker of water

60. A student was culturing E. coli (a bacterium) in a Petri-dish. He placed the Petri-dish in an incubator at 30^{o}C.He removed it from the incubator the following day and found that five colonies of bacteria had grown. He decided to return it into the incubator to give it more time.

When he removed it fourteen days later, he could not observe any colony.

a) Why was there no colony on the fourteenth day?

b) Explain how bacteria cause spoilage of stored food in warm moist conditions.

c) Name other organisms which also cause food spoilage.

d) State their economic importance to nature.

61. The table below shows the results of an experiment carried out to determine the rate of photosynthesis at different light intensities and varying Carbon (IV) oxide concentrations. The rate was determined by counting the number of bubbles per minute. The temperature was kept constant

Light intensity in lux	**% carbon(IV)oxide concentration**						
	0.0%	0.3%	0.6%	0.9%	1.2%	1.5%	1.8%
1500	0	16	30	38	40	40	40
6000	0	52	80	96	100	98	100
10000	0	80	100	115	120	122	120

a) On a graph paper provided, draw a graph for each of the light intensities. All the three graphs should be plotted on the same axis (rate of photosynthesis on vertical axis and carbon (IV) oxide concentration on horizontal axis

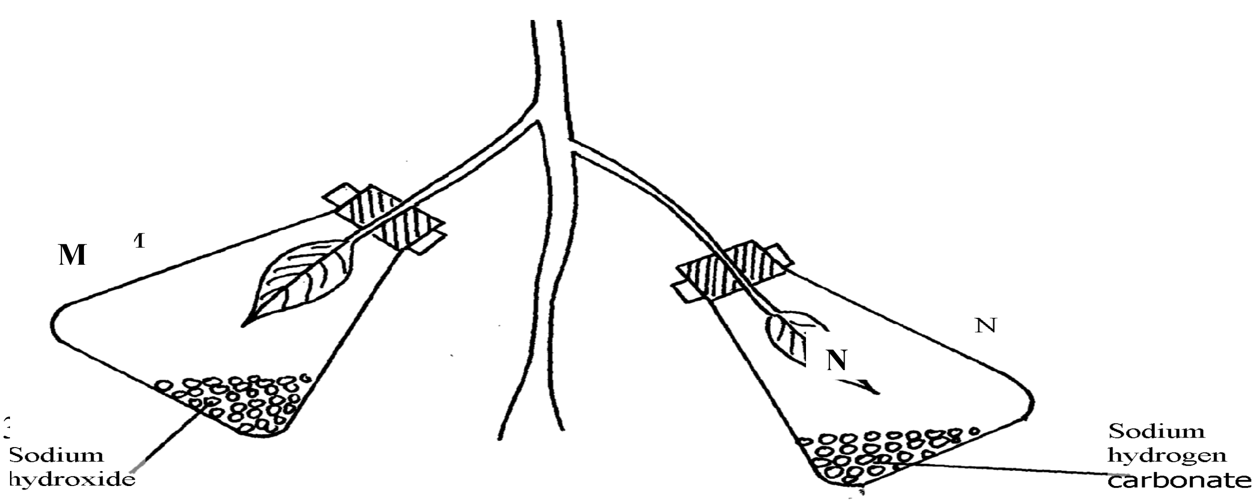
b) What is the effect of an increase in carbon (IV) oxide concentrations and light intensities

c) Briefly explain how aquatic green plants meet light intensities and carbon (IV) oxide requirement

d) Using the data provided in the table state **two** factors required by the green plants for food production

62. Explain how the mammalian intestines are adapted to perform their function.

63. A healthy plant was kept in the dark for 24hours following which two of its leaves were enclosed in glass flasks as shown below. The set up was the exposed to sunlight for a number of hours.



(a) Why was it necessary to keep the plant in the dark for 24 hours?

(b) Give the function of each of the following in the experiment
(i) Sodium hydroxide

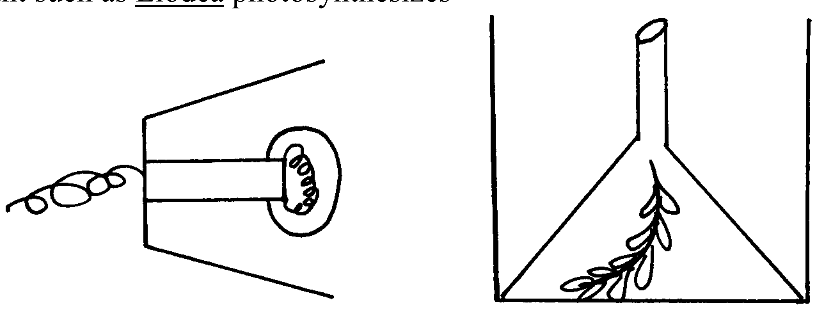
(ii) Sodium hydrogen corbonate
(c) Explain the expected results in leaf.
(i) **M** when tested for starch

(ii) **N** when tested for starch?

(d) Suggest a suitable control for this experiment

64. The diagram below shows an experiment that was carried out to measure how fast a were

plant such as Elodea photosynthesizes



The shoot was exposed to different light intensities and the rate of photosynthesis estimated by counting the number of bubbles of gas leaving the shoot in a given time. the results are given below;

Number of bubbles per minute	7	14	20	24	26	27	27	27
Light intensity (Arbitrary units)	1	2	3	4	5	6	7	8

a) Plot these data on apiece of graph paper provided

b) At what light intensity did the shoot produce ;
i) 18 bubbles per minute

ii) 25 bubbles per minute

c) Give **two** better ways of measuring the rate of photosynthesis than counting bubbles
d) What is the role of light intensity in photosynthesis

e) Account for the expected results of doing this experience at the following temperature;

i) 4ºC

ii) 34ºC

iii) 60ºC

f) Other than light intensity and temperature, name other factors that affect the rate of photosynthesis

65. In an experiment, a leaf from a plant which had been kept in the dark overnight was boiled in water for a minute. It was then boiled in alcohol and washed in warm water. Iodine solution was then added onto the leaf:

(a) Why was the loaf boiled in;- (i) Water

(ii) alcohol

(b) (i) What observation was made on the leaf after adding iodine solution

(ii) Give a reason for your answer in **(b)** above

(c) What was the aim of the experiment

(d) Why was it necessary to wash the leaf in warm water

(e) What is a variegated leaf?

(f) Write a word equation for the process of photosynthesis

5. Transport in (a) plants (b) animals

1. Explain why a fresh wound on the skin bleeds more on a hot sunny day than on a cold chilly day

2. State **three** adaptations of red blood cells to their functions.

3. How are sieve tube elements adapted to their function

4. Name the **polysaccharides** found in the following structures:- (a) **Exoskeleton**

(b) **Xylem vessels**

5. State **three** factors that maintain transpiration stream

6. (a) List **three** forces that facilitate the transport of water and mineral salts up the stem.

(b) Name the tissue that is removed when the bark of a dicotyledonous plant is ringed.

7. Study the dental formula of an organism below..

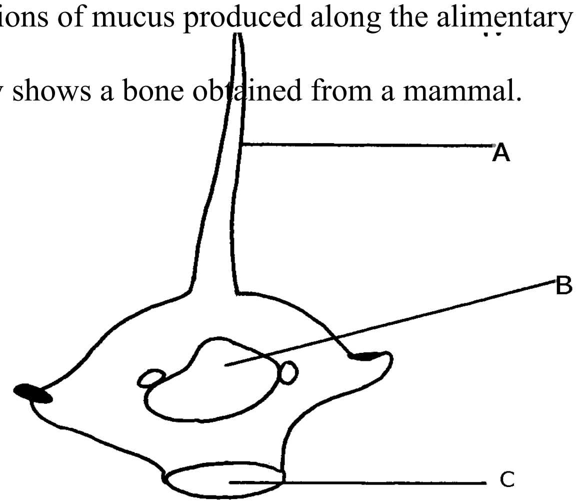
$\mathbf{I\ ^{3}/_{3},\quad C\ ^{1}/_{1},\quad Pm\ ^{3}/_{2},\quad M\ ^{1}/_{1}\ = x}$

(a) (i) What is the total number of teeth this organism possess?

(ii) What is the mode of feeding of the organism?

(b) State **two** functions of mucus produced along the alimentary canal.

8. The diagram below shows a bone obtained from a mammal.



(a) Name the part of the skeleton from which the bone has been taken.

(b) Label the parts **B** and **C**.

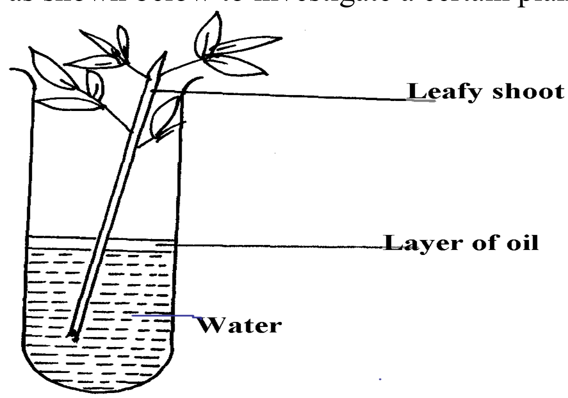
(c) State the functions of part **A**.

9. What is the destination of materials translocated in plants.

10. A person whose blood group is **AB** requires a blood transfusion, name the blood groups of the donors.

11. Explain why capillaries are: (i) Thin walled
(ii) Branched

12. An experiment was set-up as shown below to investigate a certain plant process:



(a) What process was being investigated above?

(b) What observation was made if;
(i) The experiment was left in strong wind for one hour?

(ii) All the leaves were removed from the plant?

13. How is aerenchyma tissue adapted to its function

14. (a) State **three** structural differences between arteries and veins in mammals

(b) Name a disease that causes thickening and hardening of arteries

15. Identify **two** forces that help in upward movement of water in plants

16. State **three** ways in which red blood cells are adapted to their functions

17. (a) Distinguish between tissue fluid and lymph
(b) Explain why deficiency of vitamin **K** leads to excessive bleeding even from small cuts

18. Name the type of circulatory system found in the phylum Arthropoda

19. Name the blood vessel that nourishes the heart

20. a) In which form is oxygen transported in the blood.

b) Why do plants not take in oxygen during the day although they need it for respiration

21. Name a disease of the blood characterized by excessive production of white blood cells

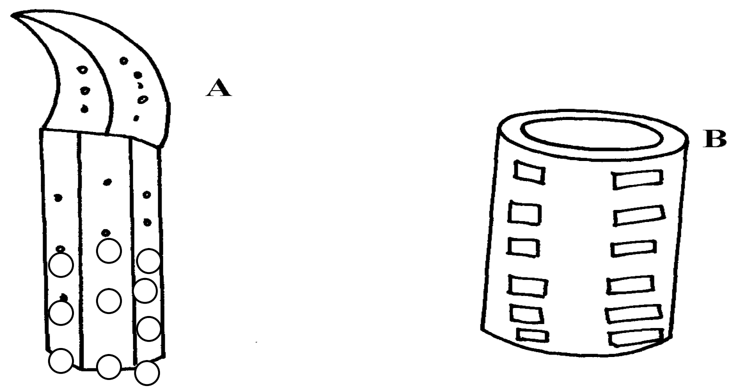
22. Laboratory analysis of a patient's urine revealed the following concentration of various substances:

Blood proteins	0.00%
Water	50%
Glucose	48%
Salts	0.8%
Urea	1.2%

a) From the analysis above, which disease is the patient suffering from

b) Name **two** symptoms of the disease in **3(a)** above

23. The diagrams below show two conducting elements of the xylem tissue



a) Identify each of them A and B

b) What makes the cellulose side walls of both A and B able to prevent collapsing?

24. Explain why the rate of transpiration is reduced when humidity is high

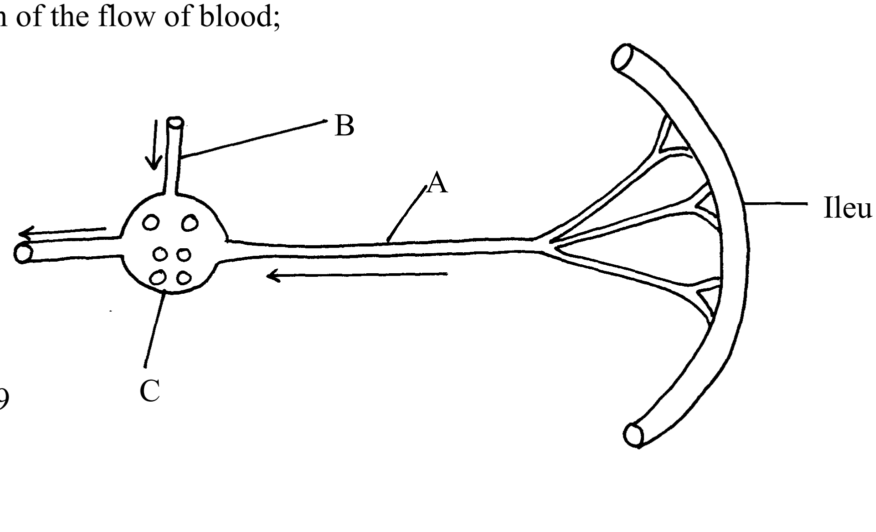
25. (a) State **two** functions of the xylem vessels,

(b) List **two** structural adaptations that make xylem vessels suitable to their function

26. (a) What is peristalsis?

(b) Explain how the process above is brought about.

27. The diagram below shows a part of a circulatory system. The arrows indicate the direction of the flow of blood;



(a) Identify the blood vessels labeled **A** and **B**

(b) Explain why it is important to transport food substances to organ **C** before being released for circulation to the rest of the body

28. Name **four** methods plants employ to remove excretory waste products

29. a) State the form in which oxygen is transported in the mammalian blood

b) Why is it dangerous to sleep in an enclosed room with a burning jiko

c) Why do plants not take in oxygen during the day although they need it for respiration

30. Name a disease of blood characterized by excessive – production of white blood cells

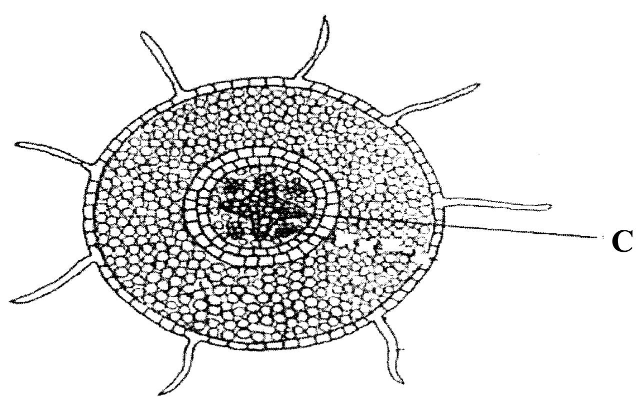
31. The table below is a representation of a chromosomal mutation

Before mutation	L	M	N	O	P	Q
After mutation	L	O	N	M	P	Q

(a) Name the type of chromosomal mutation represented above

(b) Name **one** mutagenic agent

32. The diagram shows a section through a plant organ.



(a) (i) Name the class of the plant from which the section was obtained belong.

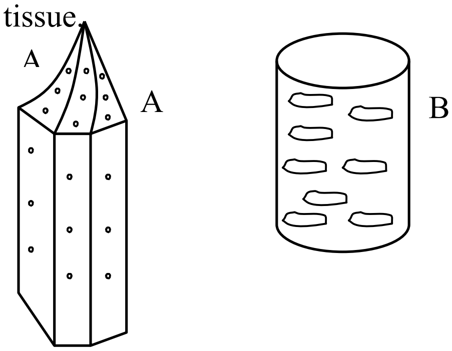
(ii) Give a reason for your answer in **(a)(i)** above

(b) How is the part labelled **C** adapted to its functions?

33. State **two** roles of transpiration to a plant

34. Uptake of water by plants is not affected by metabolic poisons. Explain.

35. The diagram below represents a plant tissue.



(a) Identify each of them.

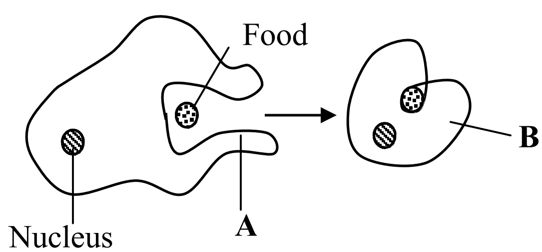
(b) What property makes **B** to be more efficient in function?

(c) What makes the walls of both **A** and **B** impermeable to water and solutes?

36. A woman gave birth to a child of blood group B+ (B positive). Name the two antigens that determined her child's blood group.

37. A transfusion of RH^+ blood was given to a patient with Rh^- blood. After one week a similar transfusion was given to the same patient. What was likely to be the effect of the second transfusion?

38. The diagrams below show stages in the process of feeding shown by amoeba.



(a) Name the part labeled **A**.

(b) Name the process illustrated in the diagram above.

(d) Name the type of cell in human beings that exhibit this process.

39. (a) Why are xylem vessels more efficient in the transport of water than tracheids?

(b) What is the significance of xylem vessels being dead?

40. Distinguish between guttation and transpiration

41. Other than transport, state one other function of xylem tissue in plants

42. State **two** functions of aerenchyma tissue in plants

43. (a) What is sickle-cell anaemia?

(b) Identify the part of the heart that initiates the heart beat

44. (a) Give a reason why the left ventricle muscles are thicker than the right ventricles muscles

(b) State the forms in which carbon (IV) oxide is transported in the blood

45. Explain how the following adaptation reduce transpiration in xerophytes

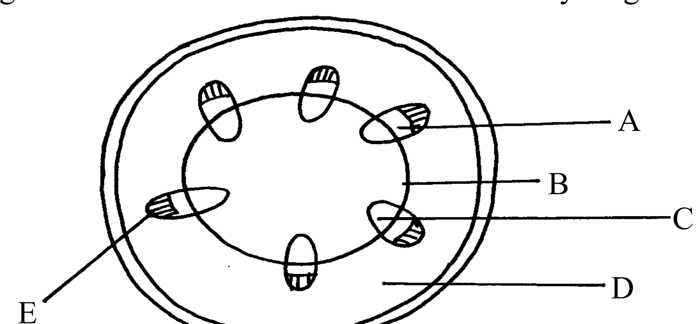
(a) Sunken stomata

(b) Thick waxy cuticle

46. Name the: (a) Material that strengthens xylem tissue

(b) Tissue that is removed when the bark of a dicotyledonous plant is ringed

47. The diagram below shows the traverse section of a young stem.



(a) What are the functions of the structures labeled **A, B** and **C**

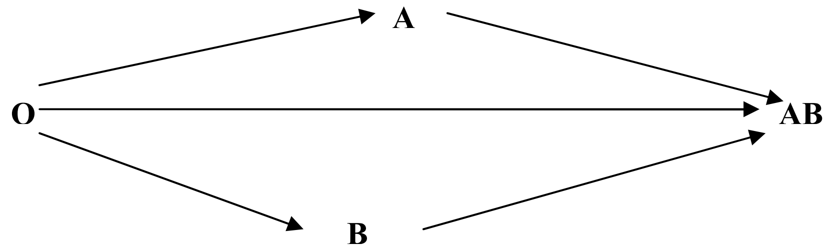
(b) What type of cells are fonjd in the parts labeleld **D**

(c) Name the tissue labeled **E**

48. Name the components of blood that do not enter the renal tubule in mammals

49 Outline the route taken by a molecule of glucose from the ileum up to the kidney.

50. The flow chart below shows a blood transfusion pathway

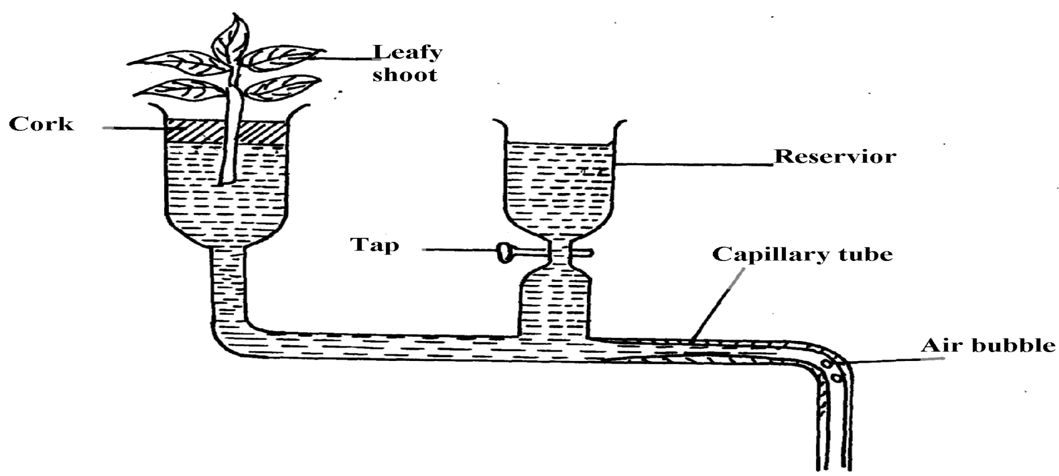


(a) What **three** conclusions can you draw from the flow chart?

(b) State **two** precautions that must be observed during blood transfusion

(c) Explain how blood clot is formed once a blood vessels is injured

51. The figure below represents a diagram of a photometer;



(a) What is the photometer used for ?

(b) State the precautions which should be taken when setting up a photometer

(c) Explain what you will expect if set up was placed under the following environmental conditions;

(i) Dark room
(ii) Leafy shoot enclosed in polythene bag
(iii) In a current of air created by a fan

52. The amount of blood flow through various parts of the body of a mammal was measured in

cm^3 per minute at rest and during different physical activities. Results are shown below.

	Blood flow in cm^3/min		
	At rest	**During light Exercise**	**During strenuous Exercise**
Heart muscles	200	300	1050
Gut	1300	1000	400
Skeletal muscles	1100	5050	23000
Kidneys	900	650	250
Skin	400	1300	600

a) Calculate the percentage change in blood flow through the skeletal muscles and gut when the mammal was exposed to strenuous exercise.

i) Skeletal muscles

ii) Gut

b) Account for the differences in amount of blood flow through the gut and skeletal muscles;

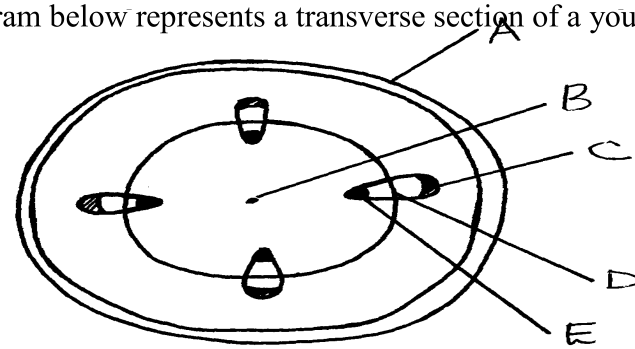
i) At rest

ii) During streneous exercise

c) Account for the result obtained for the skin during light exercise

d) Name **two** substances which are removed from the body by the kidney

53. The diagram below represents a transverse section of a young stem.



(a) Name the parts labeled **A, B** and **D**

(b) State the functions of the parts labeled **C** and **E**

(c) List **three** differences between the section above and the one that would be obtained from the root of the same plant

54. Describe the functions of the various components of the mammalian blood

6. Gaseous exchange in (a) plants (b animals

1. a) Name the site of gaseous exchange during breathing in mammals.

   b) State **three** characteristics of the site named in **(a)** above.

2. Why would carboxyhaemoglobin lead to death?

3. State **two** causes of coronary thrombosis

4. What adaptation do red blood cells have for transportation of carbon (IV) oxide?

5. (a) What is Respiration Quotient (RQ)?

   (b) (i) Calculate the RQ of the food substance shown by the equation below.

$$2C_{51}H_{98}O_6 + 145\ O_2 \longrightarrow 102CO_2 + 98H_2O + \text{Energy}$$

   (ii) Name the food substance being oxidized in b (i) above.

6. Outline **three** ways in which the gills of Tilapia fish are modified to perform their function.

7. Identify the surfaces of gaseous exchange in the following:-

   (i) Paramecium;
   (ii) Roots;
   (iii) Frog;

8. (a) Name **two** gaseous exchange

   (b) Explain how oxygen gets into the haemolymph of an insect

10. (a) Outline **two** physiological changes that occur in the body to lower the level of Carbon (IV)
    Oxide after vigorous physical exercise

    (b) Name the site of respiration in a cell

11. What is the importance of counter current flow in the exchange of gases in a fish

12. State **four** ways in which red blood cells (**RBC**) are adapted to the their function

13. (a) (i) Where in a cell does glycolysis take place?
    (ii) Name the product of the above process

(b) Briefly explain Kreb's cycle in a plant cell during anaerobic respiration

14. Describe the changes that occur to the rib cage and the diaphragm during inspiration

15. a) What is translocation

b) Name **two** forces that maintain transpiration stream

16. Most carbon (IV) oxide is transported form tissues to the lungs within the red blood cells and not in the blood plasma. Give **two** advantages of this mode of transport

17. Give a reason why halophytes have pneumatophores

18. Give **two** characteristics of respiratory surfaces in animals

19. Give a reason for each of the following on mammalian Red blood cells

(a) Absence of the nucleus

(b) Biconcave shape

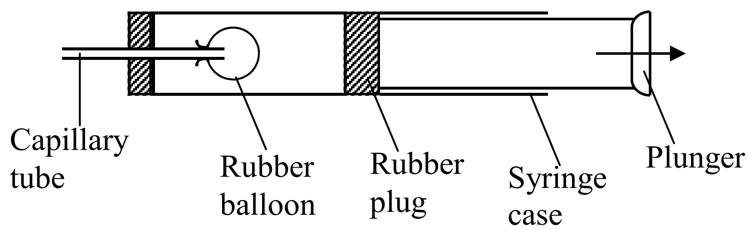
20. State **two** ways in which bodies of people living in high altitude areas respond to low oxygen concentration.

21. Explain what would happen to a mammalian Red blood cell 30 minutes alter being placed in distilled water.

22. (a) State **two** ways in which the surface area of the fish filaments is increased for efficient gaseous exchange.

(b) What is the importance of counter flow system in the filaments of a fish.

23. The apparatus below illustrates breathing in mammal.



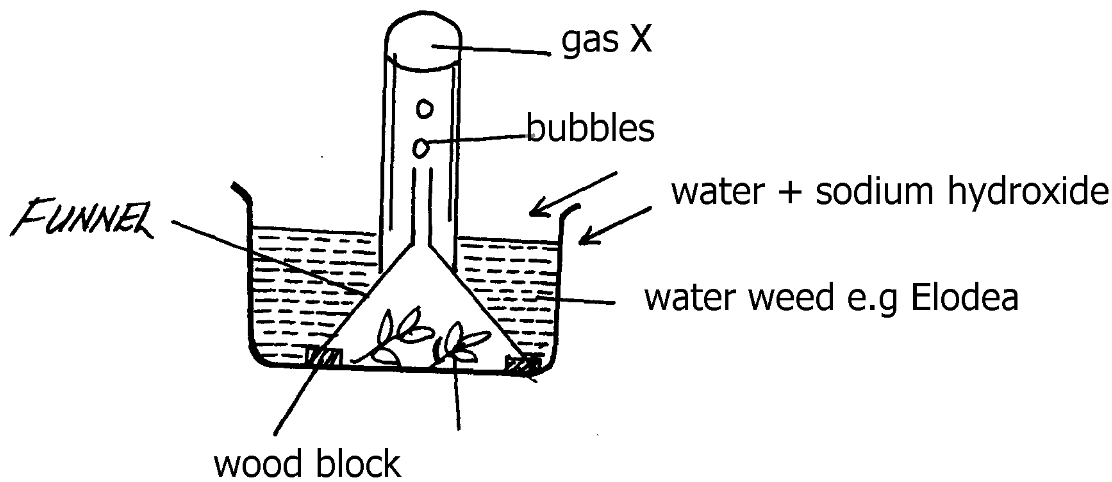
Describe what happens if the rubber plug is pulled in the direction shown by the arrow.

24. Describe the path taken by oxygen gas from atmosphere to the tissues of an insect.

25. Why should respiratory surfaces be: (i) Moist
(ii) Thin

26. The set up below represents an experiment to investigate the process of photosynthesis.

The set up placed in sunlight for six hours.



(a) Why was sodium hydrogen carbonate added to water in this experiment?

(b) Explain why the number of bubbles reduced by evening

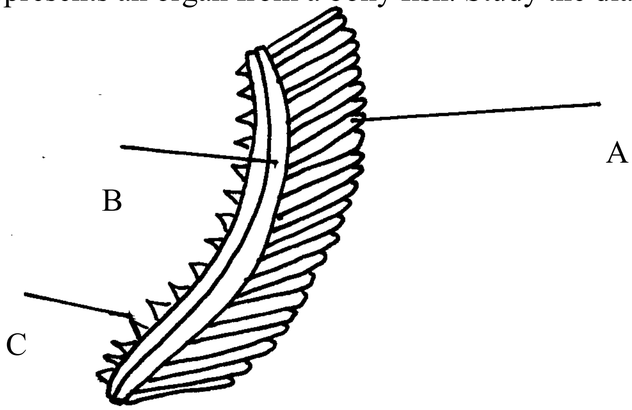
(c) Explain why the water was used in this experiment

(d) Explain why the water was used in this experiment

27. (a) State **two** adaptations of red blood cell to its functions

(b) Name **two** ways in which carbon (IV) Oxide is transported in mammalian blood

28. The diagram below represents an organ from a bony fish. Study the diagram and answer the
questions that follow:



(a) State the functions of each of the following:
(b) How is the structure labeled **C** adapted to its function?

29. State how the tracheal system in insects is adapted for gaseous exchange.

DAY	1	2	3	4	5	6	7	8	9	10	11	12
PULSES PER MINUTE	72	78	89	92	92	90	86	80	77	74	72	72

30. Differentiate between active immunity and passive immunity

31. Name **three** sites where gaseous exchange takes place in terrestrial plants.

32. An athlete training to take part in an international competition moved to a high attitude area where
he was to train for twelve days before the competition. He took his pulses per minute daily and
tabulated them as shown below:-

a) Other than pulse rate, name **one** other process which was affected by change of altitude
b) Account for the change in pulse rate from:- i) Day 1 to day 7

ii) Day 8 to day 12

c) Explain the advantage this athlete has over the one who trains in a lower altitude area

d) The equation below represents a reaction which takes place during rapid muscular movements in humans.

Glucose $\longrightarrow$ Lactic acid + 150KJ

i) State **two** effects of this reaction to an individual

ii) How is lactic acid finally eliminated from the muscle tissues of the human after the muscle

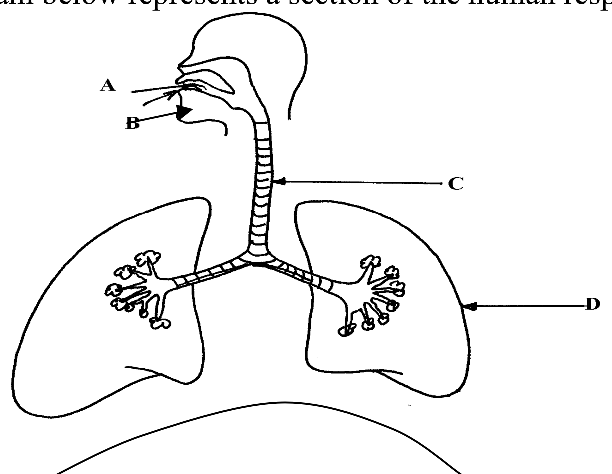
33. a) State any **two** structures used for gaseous exchange in plants.

b) Name any **two** sites where gaseous exchange takes place in a leaf of a terrestrial plant.

c) State any **two** types of leaves and their respective functions.

d) Briefly describe how stoma opens.

34. The diagram below represents a section of the human respiratory system:



(a) One can inhale through path **A**, or **B**. Giving reasons, state the more appropriate path.

(b) How is the part labbelled **C** adapted for its function?

(c) Explain the effect of regular tobacco smoking to the functioning on the organ labelled **D**

35. (a) How is the structure of mammalian gaseous exchange system adapted to its functions

(b) Describe the mechanism of opening and closing of the stomata using the photosynthetic theory

36. (a) Describe the mechanism of inhalation in man.

(b) Using photosynthetic theory explain the mechanism of opening of stomata.

37. In an experiment to investigate a certain processes in a given plant species, the rate of carbon (IV)

oxide consumed and released were measured over a period of time of the day. The results of the

investigation are shown in the table below:

Time of the day (hours)	6	8	10	12	14	16	18	20	22	24
Carbon (IV) oxide consumed in mm^3/min	10	43	69	91	91	50	18	0	0	0
Carbon (IV) oxide released in mm^3/min	38	22	10	3	3	6	31	48	48	48

b) Name the biochemical processes represented by;

(i) Carbon (IV) oxide consumption

(ii) carbon (IV)oxide release

(c) Account for the shape of the curve for carbon (IV) oxide consumption between;

(i) 6-16 hours

(ii) 20-24 hours

(d) Account for carbon (IV) oxide released between 12-16 hours

(e) (i) What is compensation point?

(ii) From the graph state the time of the day when the plant attains compensation point

(f) Explain how high temperature above optimum affects the rate of carbon (IV) oxide

consumption in the plant.

7. Respiration

1. (a) Distinguish between gaseous exchange and respiration

(b) Name the products of anaerobic respiration in plants

2. (a) State **two** phases of aerobic respiration

(b) With a reason, state the phase that yields more energy

3. A process that occurs implants is represented by the equation below:-

$C_6H_{16}O_6 \longrightarrow 2C_2H_5OH + 2CO_2 + \text{Energy}$
(Glucose) (Ethanol) (Carbon (IV) Oxide)

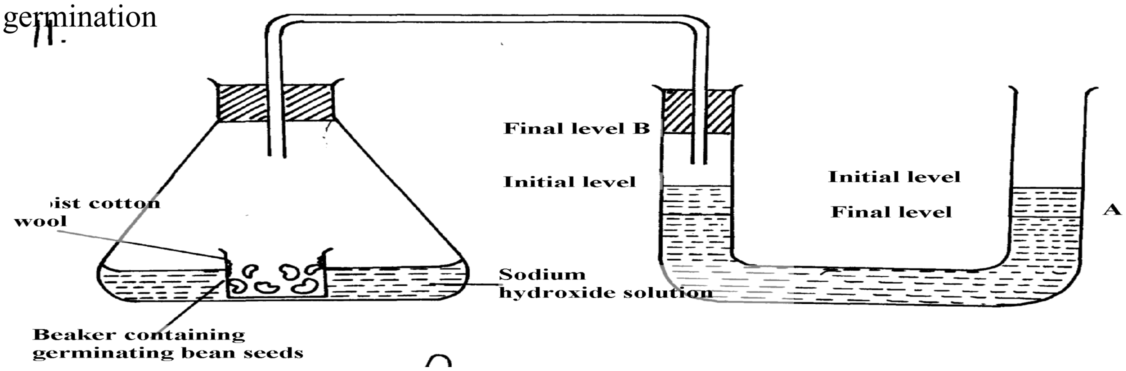
(a) Name the process
(b) State the economic importance of the process named in **(a)** above

4. Give a reason why it is difficult to calculate respiratory quotient (RQ) in plants

5. a) Explain what is meant by the term oxygen debt in human beings

b) What are the end products of anaerobic respiration in animals

6. The apparatus below was set up by a student to find out the changes in gases during germination



a) After 48 hours the level of water in the U-tube at **A** and **B** was as shown. Explain the observation

b) Calculate the respiration quotient (**RQ**) from the equation below:-

$2C_{51}H_{98}O_6 + 145O_2 \longrightarrow 102\,CO_2 + 98H_2O + \text{Energy}$

c) Identify the substrate being respired in the above equation

7. One molecule of lipid gives more energy than one molecule of glucose when respired aerobically but it is NOT always used as a respiratory substrate

a) Give **two** reasons for this
b) Name **two** disaccharides which are reducing sugar

8. (a) (i) Where in a cell does glycolysis take place?
(ii) Name the product of the above process
(b) Briefly explain Kreb's cycle in a plant cell during anaerobic respiration

9. How is the mammalian skin adapted to its protective function?

10. The oxidation state of a certain food is represented below by a chemical equation:-

$2\ C_3H_2O_2N + 6O_2 \longrightarrow (NH)_2CO_2 + 5CO_2 + 5H_2O$

a) Calculate the respiratory quotients (RQ) of the food substrate

b) Identify the food substrate

11. Whooping cough is a disease of the respiratory system name the causative agents and give **two** symptoms

12. How does the sunkness of stomata help in minimizing the rate of transpiration in plants

13. State **two** roles of adrenaline in man

14. Explain why a rat, though small eats more frequently than an elephant

15. Active yeast cells were added to dilute sugar solution in a container. The mixture was kept in a warm room. After a few hours bubbles of a gas were observed escaping from the mixture
(a) Write an equation to represent the chemical reaction above

(b) State **two** economic importance of this type of chemical reaction in industry?

16. (a) Give **two** reasons why fats are not the main respiratory substrates in the body of a mammal and yet they give a lot of energy when oxidized.

17. The equation below summarizes a metabolic process in plants.

Glucose $\longrightarrow$ Ethanol + carbon (IV) oxide+ Energy

State **two** industrial applications of the above equation.

18. (a) Differentiate between respiration and respiratory surface.

(b) Why is an effective respiratory system often associated with a circulatory system.

19. State **two** reasons why lipids are rarely used as a respiratory substrate compound to carbohydrates.

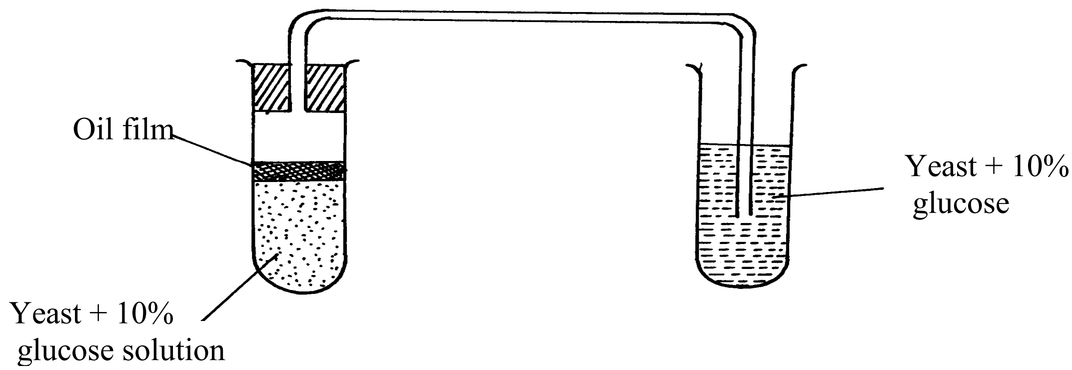
20. The equation below shows respiration for a certain food substrate. Study it and answer questions that follow:

$2C_{51}H_{98}O_6 + 145O_2 \longrightarrow 102CO_2 + 98H_2O$

(a) Calculate the respiratory Quotient, RQ

(b) Suggest with reasons the possible food substrate

21. The apparatus below was used to investigate anaerobic respiration:-



(a) How would you remove dissolved oxygen from the glucose before the experiment commencing?

(b) State what happens to the lime water as the experiment proceeds to the end

(c) Describe the reactions in the experiment

(d) Explain what would happen if the temperature of glucose solution and yeast was raised beyond 45°C?

8. Excretion and homeostasis

1. Explain the following:- i) Fresh water fish excrete ammonia

ii) Glucose is absent in urine yet present in glomerular filtrate

2. (a) State **two** functions of the kidney

(b) Name **two** substances that are not found in urine of a healthy person

(c) Name **two** diseases that affect the kidney

3. (a) State **two** structural modification of the kidneys of deserts animals like kangaroo rat.

(b) Describe how ingestion of very salty food may reduce the amount of water excreted in urine.

4. A student mixed a sample of urine from a person with Benedict's solution and heated, the colour

changed to orange.

(a) What was present in the urine sample?

(b) What did the student conclude on the health status of the person?

(c) Which organ in the person may not be functioning properly?

5. (a) If the human pancrease is not functional:-

(i) Name the hormone which will be deficient

(ii) Name the disease the human is likely to suffer from

(b) What is diuresis?

6. State **one** structural adaptation of nephron in the kidney of a desert mammal

7. Name the nitrogenous wastes excreted by the following organisms:-

Animal	**Nitrogenous Waste**
(i) Desert mole	
(ii) Marine fish	
(iii) Tilapia	

8. The table below shows description of sizes of glomeruli renal tubules of two animals which are

living in different environments

	Animal X	**Animal Y**
Glomeruli	Large and few	Small and many
Renal tubules	Short	Long

(a) Name the likely environment in which each animal lives: (i)Animal X

(ii) Animal Y

(b) What role does vasoconstriction play in thermoregulation?

9. The table below shows the approximate percentage concentration of various components in blood plasma entering the kidney, glomerular filtrate and urine of a healthy human being

Component	Plasma	Glomerular filtrate	Urine
Water	90	90	94
Glucose	0.1	0.10	0.00
Amino acids	0.05	0.05	0.00
Plasma proteins	8.0	0.00	0.00
Urea	0.03	0.03	2.00
Inorganic ions	0.72	0.72	1.50

(a) Name the process responsible for the formation of glomerular filtrate

(b) What process is responsible for the absence of glucose and amino acids in urine?

(c) Explain why there are no plasma proteins in the glomerular filtrate?

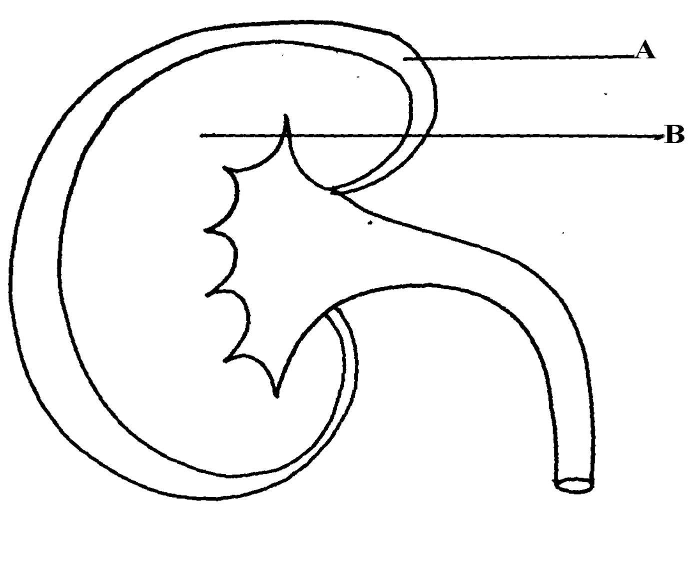
10. What is the importance of sebaceous glands in the human skin?

11. Explain why sweat accumulates on a person's skin in a hot humid environment

12. Distinguish between diabetes mellitus and diabetes inspidus

13. State **two** processes through which plants excrete their metabolic wastes.

14. The figure below shows a vertical section through a mammalian kidney.



(a) Label the parts A and B

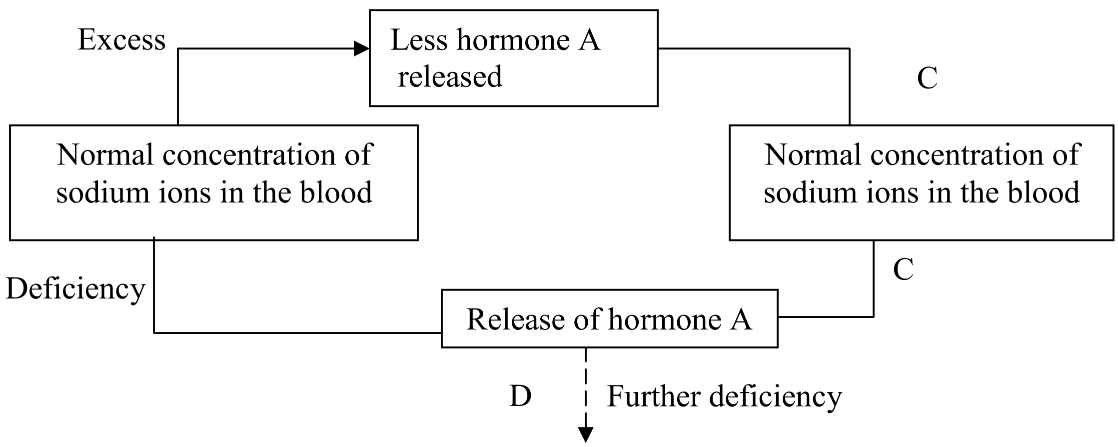
(b) Which part is the Bowman's capsule found?

15. (a) Explain the effects of the production of large amounts of Antidiuretic hormone in the human

body

(b) State **two** functions of the loop of Henle

16. Study the homeostatic scheme below:



(a) Identify the hormone labeled **A**
(b) Name the site of action of hormone **A**
(c) Identify the feedback labeled **D**

17. State **three** importance of Osmosis in plants

18. A patient was complaining of thirst most of the times. A sample of the patient's urine was found

not to contain a lot of sugar but was dilute:-

(a) Name the hormone the person's body was deficient of

(b) Which gland produces the above hormone

(c) Name the disease that the patient was most likely suffering from

19. State **two** features in the nephron that facilitate ultra filtration

20. The table below shows a description of size of glomeruli and renal tubules of two animals

which are adapted to living in different environment:-

Animal B	Animal A	
Glomeruli and many	large and few	small
Renal tubules	short	long

a) Name the likely environment in which animal **A** lives

b) Suggest the main nitrogenous waste produced by animal **B**

c) Name the organelle of osmoregulation in each of the following animal: i) Paramecium

ii) Insects

21. What role is played by the liver in excretion?

22. The equation below represents a metabolic process that occurs in the mammalian liver: Amino acids $\xrightarrow{\text{Enzymes}}$ organic compound + urea

a) Name the process

(b) What is the importance of the process to the mammals?

23. A person was found to pass out large volume of dilute urine frequently. Name the:-

(a) disease the person was suffering from?

(b) hormone that was deficient

24. Explain the effects of the following on the quantity and composition of urine

(a) Drinking large amount of clean water

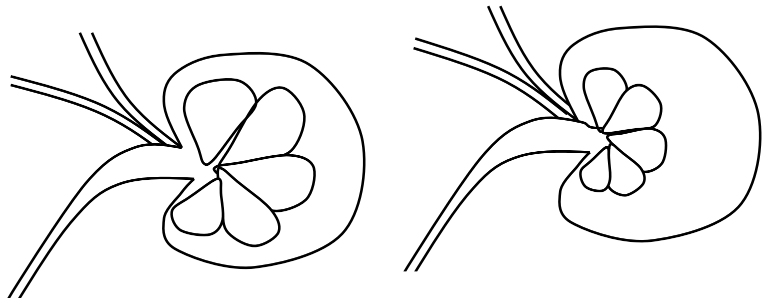
(b) Drinking very salty soup

(c) Removal of pancreas

25. (a) Distinguish between **excretion** and **egestion**

(b) State the importance of excretion in the bodies of living organisms.

26. The diagram below shows simplified structures of kidneys from two different animals.



Animal **N** Animal **M**

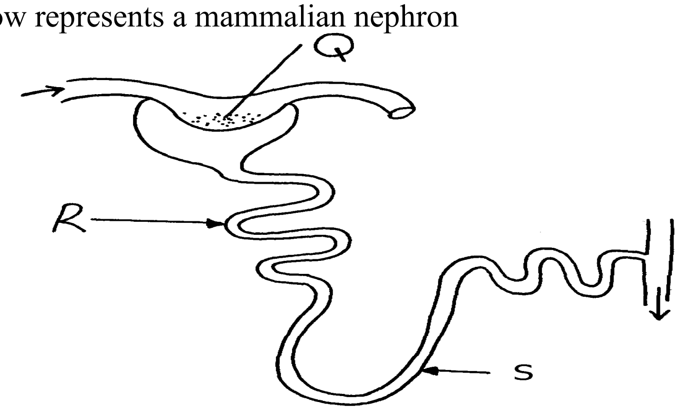
(a) Suggest possible habitat in which animal **N** is found.

(b) Give **two** reasons for your answer in (a) above.

27. (a) What is poikilotherm?

(b) State **two** classes of phylum chordata where all members are poikilothermic .

28. The diagram below represents a mammalian nephron



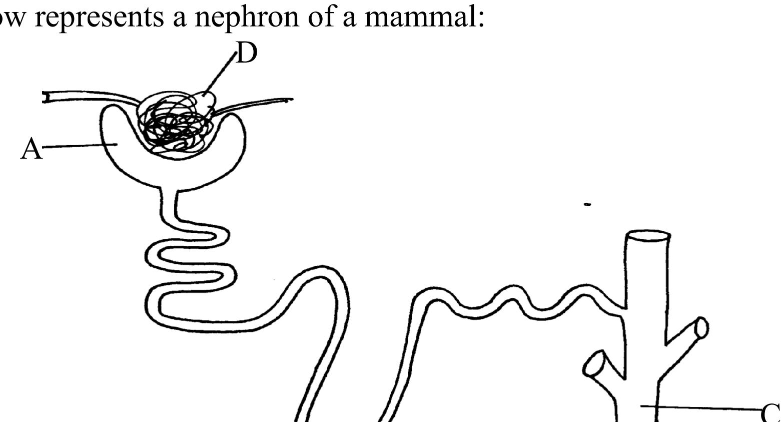
(i) Name the structure labelled **Q**

....

(ii) State **two** adaptations of part labeled **R**

29. Distinguish between internal environment and external environment as used in

30. The diagram below represents a nephron of a mammal:



(a) Name the parts labeled **A, B** and **D**

(b) Name a major substance in glomerular filtrate whose concentration remains the same
between **A** and **C**

31. Name the parts of the flower that are responsible for the production of gametes

32. The equation below represents a metabolic process that occurs in a certain organ in the m
mammalian body:-

Ammonia $\xrightarrow[\text{Carbon (IV) oxide}]{\text{enzymes}}$ Organic compound **Q** + water

a) Name the process represented in the equation.

b) Name the organ in which the process occurs.

c) Why is the process important to the mammal?

d) Identify the organic compound **Q.**

e) Explain the source of ammonia in the organ named in **(b)** above.
f) What happens to organic compound **Q**?

33. Kosgei and Onyancha collided during a football match and each got bruised. Kosgei's bruise stopped bleeding after ten minutes while Onyancha's bruise continued bleeding and he had
to be taken to hospital for treatment.

(a) Explain the process which brought about stoppage of Kosgei's bleeding

(b) Distinguish between blood clotting and haemagglutination.

(c) Name the disease, that Onyancha could be suffering from.

34. The table below shows the percentage of some substances in the glomerular filtrate and urine of a certain mammal:-

Substances	**Contents in glomerular filtrate**	**Contents in urine**
Water	90	90
Sodium ions	0.3	0.35
Chloride ions	0.37	0.60
Glucose	0.1	0.0
Urea	0.03	2.0
Proteins	0.0	0.0

(a) From the above table, account for ; (i) The absence of glucose in urine
(ii) The absence of protein in both glomerular filtrate and urine

(b) Explain the significance of the flow system in the nephron where the glomerular filtrate flows in opposite direction to that of blood in the surrounding capillaries

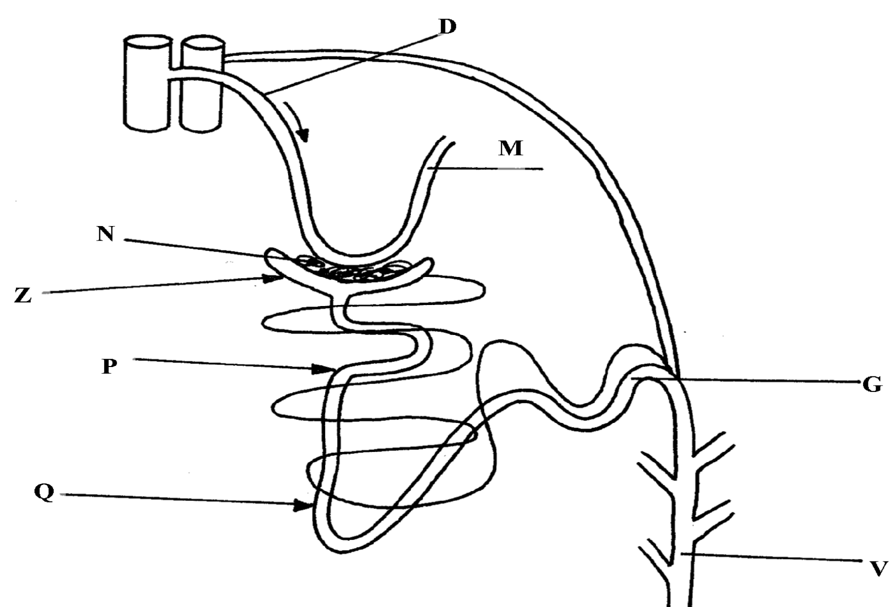
(c) Name the hormone that controls the percentage of water in urine and that which control the amount of salts

Percentage of water

Amount of salts

(d) List any **two** diseases /disorders of the kidney

35. Study the diagram below and answer the questions that follow



(a) Name the structure represented by the diagram
(b) (i) Name the parts labelled **D** and **M**

(ii) Name the hormones whose sites of action are Q and G
(c) Name **one** substance that is present in part **N** but absent in part **Z**
(d) The contents of part **V** were boiled with Benedicts' solution and an orange precipitate was formed. Account for the results

36. In an investigation, two persons **A** and **B** drunk the same amount of glucose solution. Their blood sugar levels were determined immediately and thereafter at intervals of one hour for the next six hours.

The results were as shown in the following table:-

Time (hrs)	**Blood glucose level (mg/100ml)**	
	Person A	**Person B**
0	90	120
1	220	360
2	160	370
3	100	380
4	90	240
5	90	200
6	90	160

(a) Draw a graph of blood sugar levels of persons **A** and **B** against time on the same axis
(b) Explain each of the following observations;-
(i) Blood sugar level increased in person **A** between 0 and 1 hour

(ii) The blood sugar level dropped in person **A** between 1 and 4 hours

(c) From the graph, what is the normal blood glucose sugar level for human beings

(d) Suggest a reason for the high sugar level in person **B**

(e) How can the high blood sugar level in person **B** controlled?

(f) What is the biological significance of maintaining a relatively constant sugar level in a human being

(g) Account for the decrease in the blood glucose level of person **B** after 4 hours

37. An experiment was carried out to determine the effect of drinking on excess amount of water on the flow of urine. A person drinks one litre of water and urine was collected at intervals of 15minutes.

The results were as shown below:

Time in minutes	0	15	30	45	60	75	90	105	120	135
Urine output ml/min	1.6	1.6	1.6	5.4	9.0	9.0	7.6	3.0	0.8	0.8

(a) Plot a suitable graph to represent urine output with time.

(b) Explain the rate of flow of urine between the following times;
(i) 15 and 60minutes.

(ii) 60 and 75minutes.

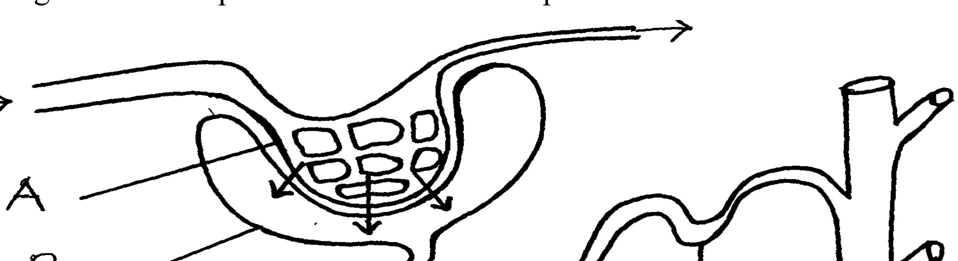
(iii) 75 and 135 minutes.

(c) Name **two** hormones responsible for regulation of relative amount of salts and water in man.

38. a) Explain how urea is formed in the human body

b) Describe the path taken by urea from the organ where it is formed until it is released from the human body

39. The diagram below represents a mammalian nephron.

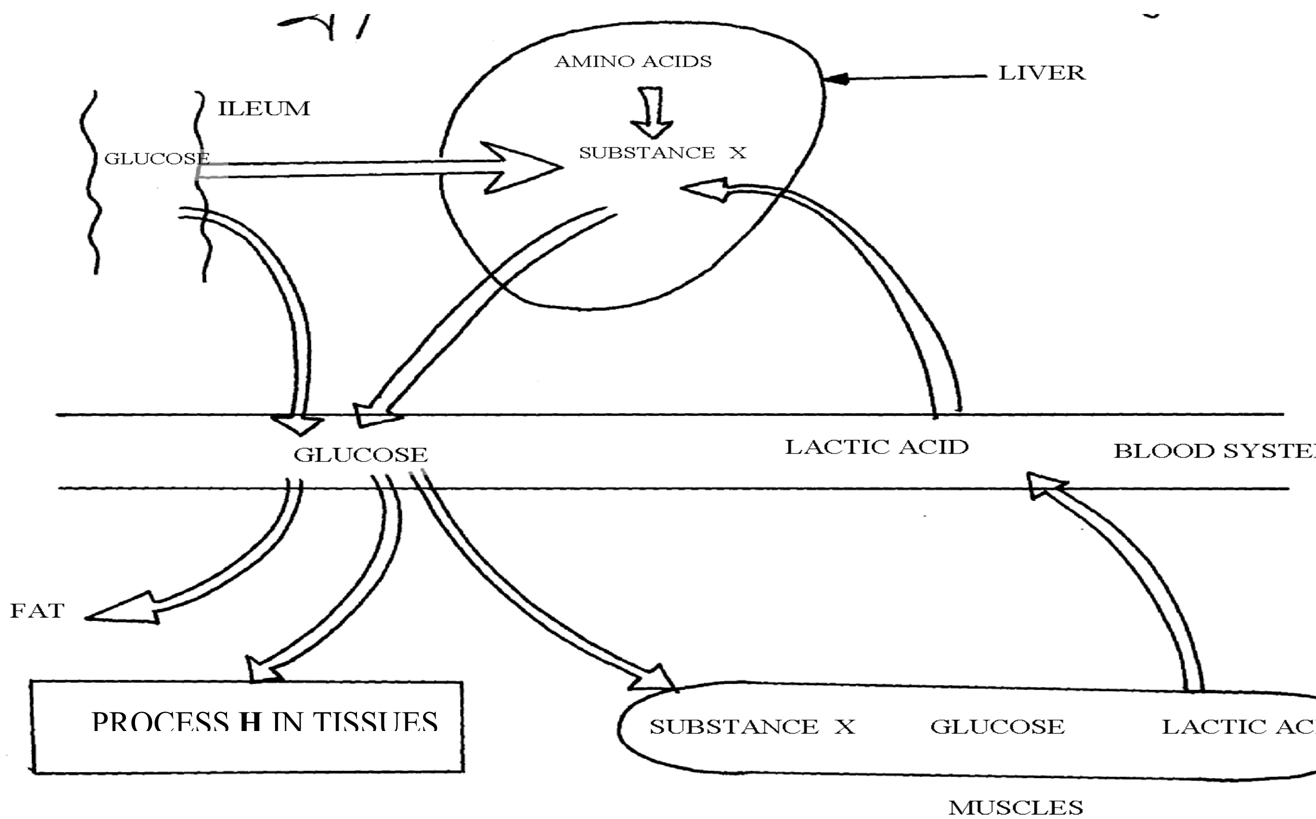


(a) Name the structures labeled **B,C** and **D**

(c) Name the process by which substances are reabsorbed from structure **C** into blood capillaries

(d) How is the pressure in structure **A** achieved?

40.

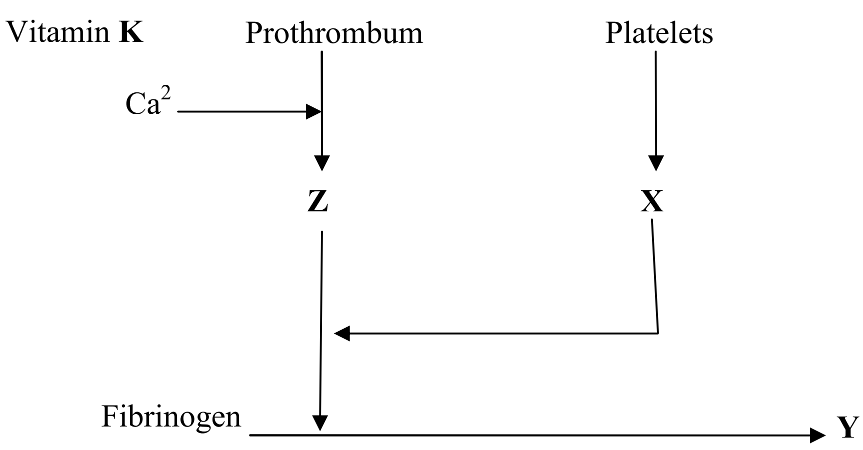


a) Identify **substance X**

(c) Give the end products of the process labelled **H**

(d) Give **three** other functions of the liver

41. The flow diagram below represents blood clotting process



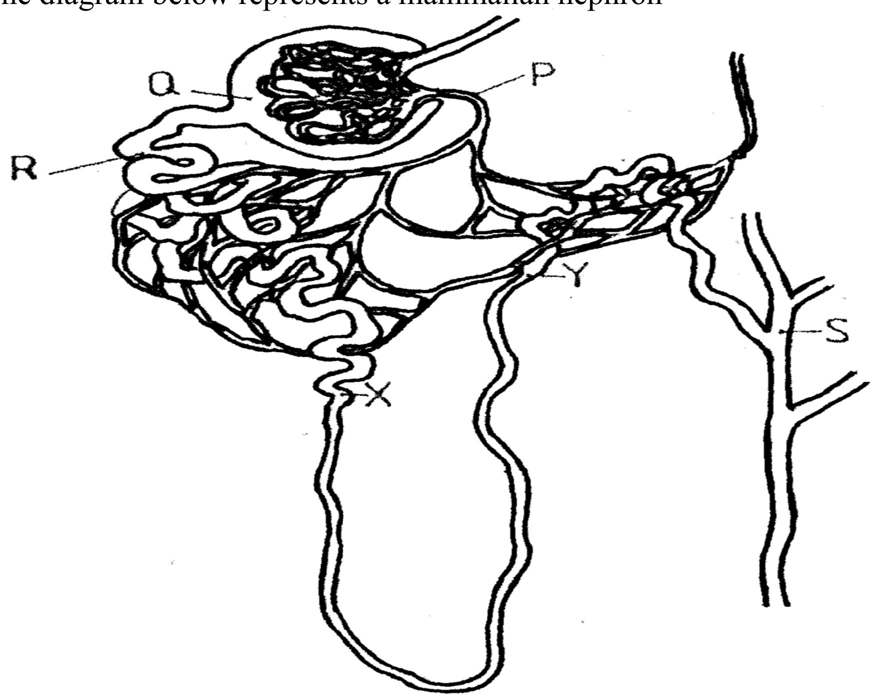
a) Name the proteins represented by the letters; V, Y, Z

b) State the importance of blood clotting

c) Why doesn't the physiological process above occur in undamaged blood vessels

42. How does an Endotherm respond to both heat gain and heat loss?

43. The diagram below represents a mammalian nephron



(a) Name the: (i) Structure labelled **P**

(b) State the structural modifications of the part label led **Q** for
(i) Desert mammals
(ii) Fresh water mammals

(c) (i) Name **one** substance present at point **R** but absent at point **S** in a healthy mammal.

(ii) The appearance of the substance you have named in **(c)(i)** above is a symptom of a certain disease. Name the disease

44. Describe how the mammalian skin regulates body temperature

9. Ecology

1. A student wanted to estimate the number of grasshoppers in 5km^2 grass field near the school compound.

Using a sweep net he captured 36 grasshoppers. He used a red felt pen to mark the thorax of each insect before releasing back into the field. Three days later he made another catch of grasshoppers. He collected 45 grasshoppers of which only 4 had been marked with red mark.

a) Name the above method used in the population estimation

b) Calculate the population of grasshoppers using the above data

2. What is the significance of the following in the ecosystem?

a) Decomposers

b) Predators

3. Birds feed on grasshoppers that feed on grass.

a) Draw a possible food chain from the above information

b) Explain why the biomass of organisms decreases at each preceding trophic level.

4. Define the following terms:-

(i) Autecology ;

(ii) Biomass;

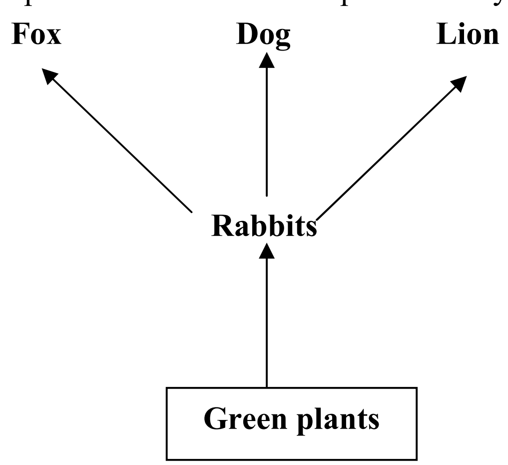
5. State **two** most important factors that favour exponential growth of a population of gazelle in a park

6. (a) Distinguish between habitat and niche.
(b) Explain why Biomass of producers is greater than that of primary consumers in a balanced ecosystem.

(c) State **two** advantages of a biological control method over the chemical control method of pests and parasites

7. Explain how oil as a pollutant may affect aquatic plants and animals?

8. The diagram below shows part of a food relationship in an ecosystem:-

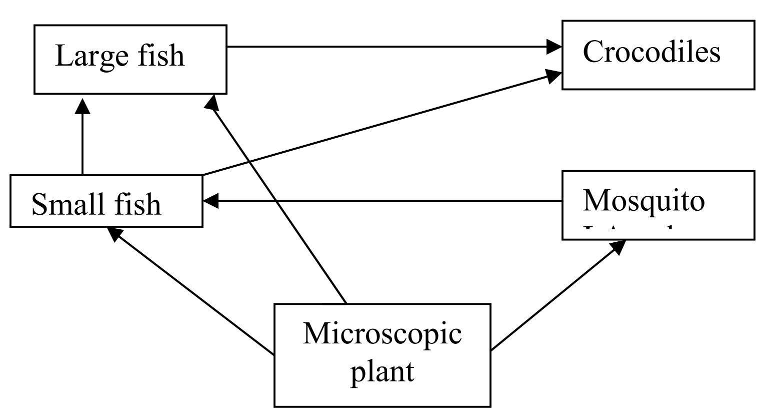


(a) Name the food relationship above

(b) How many trophic levels are shown in the diagram above?

(c) State main source of energy in the ecosystem

9. Use the food web below to answer the questions that follow:-



(a) Construct a food chain ending with crocodile as a quartenary consumer

(b) Name the organisms in the food web hat has only one predator

10. 50 black mice and 50 white mice were released in an area inhabited by a pair of owls. After four months, the mice in the area were recaptured and only 38 black mice and only a white mice remained.

(a) How would you explain these results?

(b) Name the theory of evolution that support the results in **(a)** above

11. In a certain school Form three class did an experiment to estimate the number of Tilapia in their fish pond. 725 tilapia were netted marked and released.

a) State the method used in this exercise
b) Calculate the tilapia population.

c) State **two** assumptions made by the students during the investigation

12. An investigation was carried out on a terrestrial ecosystem. The population sizes and species
biomass were determined and recorded as shown in the table

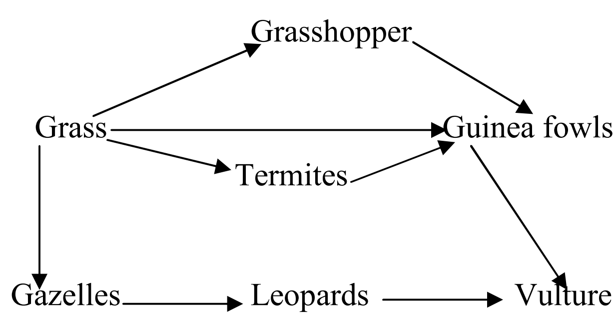
SPECIES	POPULATION SIZE	SPECIES BIOMASS
A	$1x10^3$	$1x10^3$
B	$1x10^3$	$1x10^{-1}$
C	$1x10^5$	1x10
D	1x10	$1x10^4$

a) If these organisms had a feeding relationship, construct a simple food chain involving all the organisms

b) Construct pyramid of numbers using the data provided above

c) State **one** disadvantage of using pyramid of number in expressing feeding relationships in ecological ecosystem

13. The figure represents a feeding relationship in an ecosystem



(a) Write down the food chain in which the Guinea Fowls are secondary consumers
(b) What would be the short term effects on the ecosystem if lions invaded the area
(c) Name the organism through which energy from the sun enters the food web

14. Outline **three** roles of active transport in human body
15. Distinguish between community and population

16. Describe how the belt transect can be used in estimating the population of a shrub in a
grass land

17. A fish farmer wanted to know the number of fish in a pond. He collected 10 fish from the pond
and labeled each, by a tag label on its fin and returned the ten fish to the pond to mix with

other fish . When he later collected 50 fish from the pond, he found only four of them had labels

a) Estimate the total number of fish in the pond (show your workings)

b) What **two** assumptions are being made in this methods of estimating population

18. What is the importance of saprophlic fungi and bacteria in an ecosystem

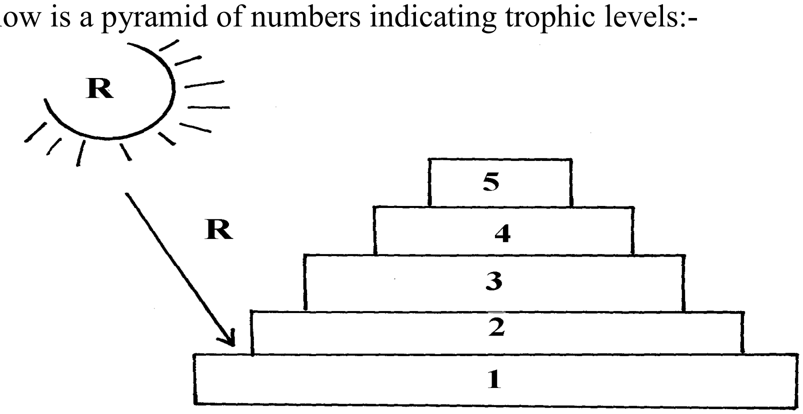
19. i) Name **one** main cause of global warming

ii) What are the effects of global warming

20. Explain how saliva is important in digestion

21. Give a reason why two species in an ecosystem cannot occupy the same niche

22. Below is a pyramid of numbers indicating trophic levels:-



(a) What do you understand by the term trophic level?

(b) Name the trophic level numbered 5 on the pyramid

(c) Name **Q**

(d)What is the significance of the arrow **R**

23. Two populations of the same species of birds were separated over a long period of time by an ocean. Both populations initially fed on insects only. Later, it was observed that one population fed entirely on fruits and seeds, although insects were available. Name this type of evolutionary change

24. To estimate the population size of crabs in a certain lagoon, traps were laid at random. 400 crabs were caught, marked and released back into the lagoon. Four days later, traps were laid again and 360 crabs were caught. Out of the 360crabs, 90 were found to have been marked

(i) Calculate the population size of the crabs in the lagoon

(ii) What is the name given to this method of estimating the population size

25. State the function of each of the following apparatus:

(a) Pooter …

(b) Sweep net …………

26. State the role of the following apparatus in the study of living things.

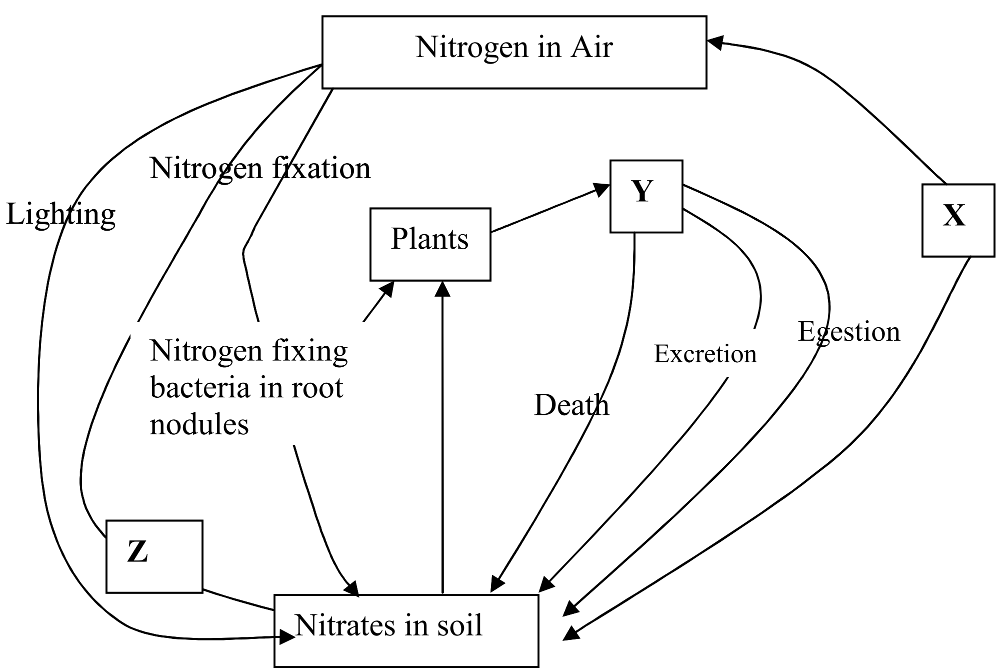
(a) Sweep nets.
(b) Pooter.
(c) Pit fall trap.

27. Name **three** density dependent factors in an ecosystem.

28. (a) What are the **two** main components of an ecosystem?

(b) Apart from mere observation of actual feeding suggest **two** methods that can be used to determine the type of food eaten by animals

29. The chart below represents a simplified nitrogen cycle.



What is represented by **X, Y** and **Z**?

30. In an ecological study, a locust population and that of crows was estimated in a grassland area
over a period of one year. The results were tabulated as shown below.

Months	J	F	M	A	M	J	A	S	O	N	D
No. of Adult locusts x 10^2	90	20	11	25	200	652	15	10	35	192	456
Number of crows	4	2	0	1	8	22	2	1	1	5	15
Amount of rainfall	20	0	55	350	520	350	10	25	190	256	350

a) Draw a graph of population of locusts and crows against time

b) i) State the relationship between rainfall and locust population

ii) Account for the relationship you have stated in **b (i)** above

c) What happens on the populations of locusts and crows in the months of January to March?
Give a reason.

d) If the area of study was one square kilometer, state **one** method used to estimate the
population of :- i) Locusts
ii) Crows

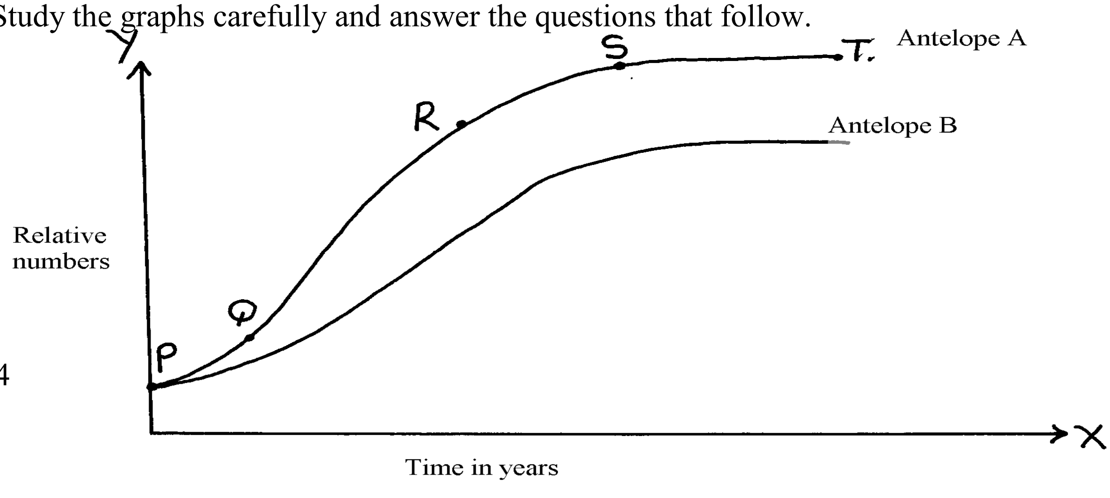
(e) (i) State the trophic levels of the (i) Locusts and (ii) crows

(ii) Construct a simple complete food chain involving these organisms

(f) If the locust were removed from the food chain, what would be its effect?

(g) Define **biomass**

31. Two species of antelopes were introduced into an ecosystem at the same time in equal numbers.
The graphs below show their relative numbers during the first eight years of their co-existence.
Study the graphs carefully and answer the questions that follow.



a) i) Which species of antelope has better survival adaptations.

ii) Give a reason for your answer above.

b) i) Name the type of curves shown.

ii) Name the phases labeled PQ, QR, RS, ST

c) Explain the shape of the curve for the species of Antelope **A** between

i) **Q** and **R**.

ii) **S** and **T.**

d) i) State the type of competition shown by the two species of antelopes.

ii) State any **two** symptoms of intraspecific competition in plants.

e) Suggest how the species B avoid competitive exclusion..

f) State any **three** adaptations that enable the antelopes to overcome predation.

g) State any **two** possible methods by which populations of the two antelopes' species were determined.

32. Explain **five** abiotic factors that affect the ecosystem

33. The data shown below was taken from Savannah grassland habitat. Examine it carefully and then answer the questions that follow:-

Organism	**Population**
Grasses	1000
Caterpillars	500
Squirrels	300
Frogs	200
Gazelles	300
Elephants	100
Snakes	50

Hunting dogs	40
Vultures	40
Lions	40
Hawks	10

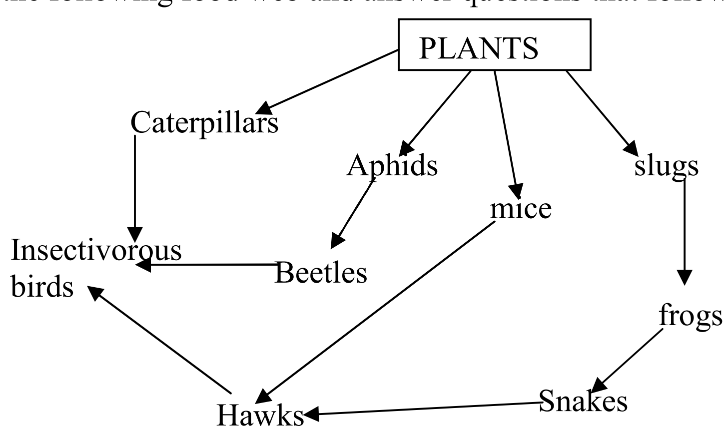
(a) Draw **three** food chains

(b) Draw a pyramid of numbers for a food chain with four trophic levels and indicate the trophic levels at which each member feeds

(c) State the effect of removing the hunting dogs

(d) Why is it advisable to feed 100kg of grain to man instead of using it to fatten steers then supply beef to human population?

34. Study the following food web and answer questions that follow:



(a) (i) Name the organisms that occupy the second trophic level

(ii) What is the other name for the second trophic level

(b) Write down **two** food chains from the food web that:

(i) End with hawks as tertiary consumer
(ii) End with hawks as quaternary consumer

(c) Giving reasons state; (i) the organism with largest biomass
(ii) the organism with least biomass

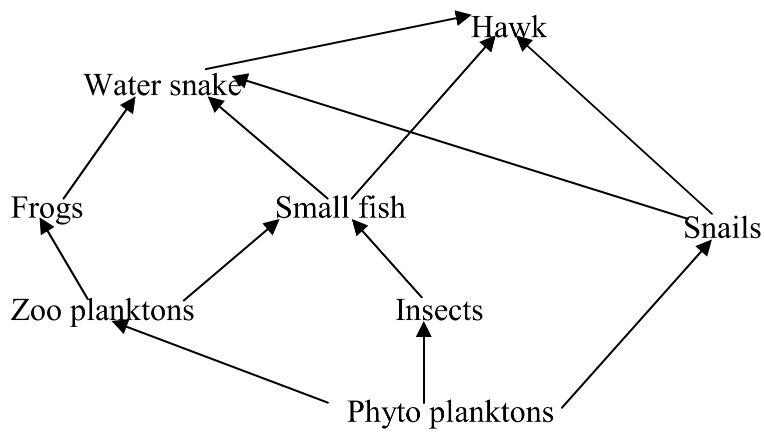
35. (a) Explain how food as a factor regulates the population of animals in an ecosystem

(b) Describe the flow of energy from the sun through the different trophic levels in an ecosystem

36. (a) Describe how a population of grasshoppers in a given area can be estimated

(b) Describe how the belt transect can be used in estimating the population of a shrub in a grassland

37. The flow chart below represents a feeding relationship in an ecosystem



(a) Name; (i) The producers in the ecosystem

(ii) Two organisms which are both secondary and tertiary consumers

(b) State **two** short term effects of immigration of insects in the ecosystem.

(c) Which organism has the least Biomass in the food web. Explain.

(d) State **three** disadvantages of using synthetic pesticides over Biological control.

(e) State the role of each of the following in an ecosystem;

(i) Saprophytes

(ii) Leguminous plants

(iii) Explain the role of producers in an ecosystem

(f) Name **one** method that would be used to estimate the population of small fish in the ecosystem

38. How are leaves of mesophytes adapted to their functions?

10. Reproduction in (a) plants (b) animals

1. a) Name the part of an ovule that develops into each of the following parts of a seed after fertilization
   i) Testa
   ii) Endosperm

   b) What is parthenocarpy?

2. State **three** roles of placenta during pregnancy.

3. Name **three main** methods through which HIV/AIDS is transmitted

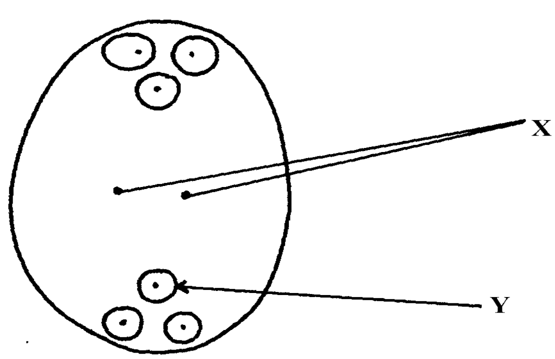
4. (a) Name the processes that lead to fruit formation without fertilization

   (b) Name the hormone that causes leaf, flowers and fruit abscission

   (c) What is the role of ecdysone hormone in insects

5. State the roles of oviduct in female reproductive system

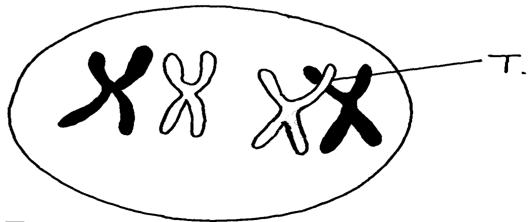
6. The diagram below represents a mature embryo sac. Study it carefully and answer the questions that follow:



(a) Identify structures **X** and **Y**

(b) Why is cross pollination more advantageous to a plant species than self pollination?

7. The diagram below shows a phenomenon which occurs during cell division.



(a) Name the part labeled **T**.

(b) (i) State the biological importance of the part labelled **T**.

(ii) Identify the type of cell division in which this phenomenon occurs.

8. (a) When are the **two** organisms considered to belong to the same species.

(b) Explain the term **alternation of generations.**

9. (a) Explain why Larmack's Theory of evolution is not accepted by biologists today.

(b) State the significance of mutation in evolution.

10. (a) Give **two** roles of the placenta.

(b) Explain why hormone testosterone still exerts its influence even when vas deferens have been cut.

11. Name **two** mechanisms that hinder self fertilization in flowering plants

12. State **three** ways in which plants compensate for lack of movement

13. (a) What do you understand by the term double fertilization?

(b) State **two** adaptations of animal dispersed fruits

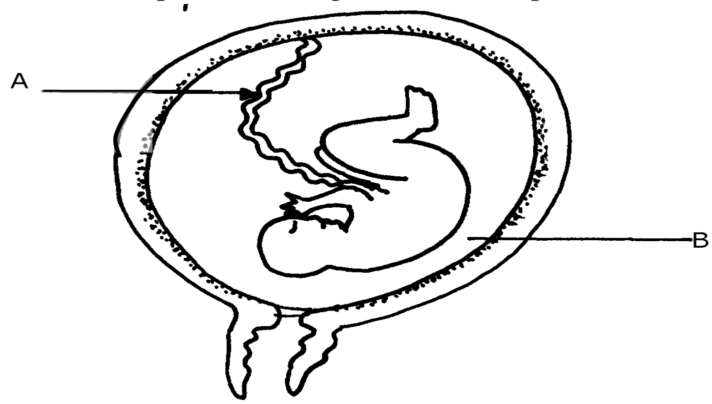
14. Name the hormone that;

(a) Stimulate the contraction of uterus during birth

(b) Stimulates the disintegration of the corpus inteum when fertilization fails to take place

15. State **three** ways in which flowers parent self pollination

16. The diagram below represents a stage in the development of human foetus



(a) State **one** function of each of the structures labelled **A** and **B**

(b) Apart from the size of the foetus what else from the diagram illustrates that birth was going to
occur in the near future

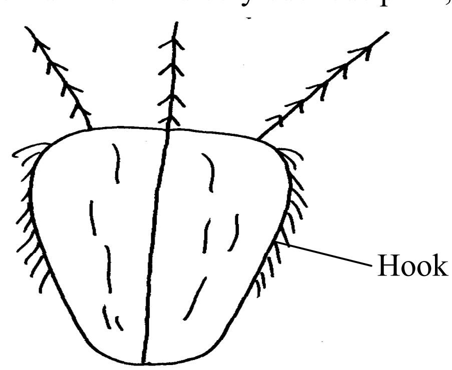
(c) Explain why a pregnant woman is supplied with doses of iron tablets regularly

17. Name the type of placentation where;
(i) Placenta appears as one ridge on the ovary wall

(ii) Placenta appears at the centre of the ovary with ovules on it and the dividing walls of carpels
disappear

18. The diagram below represents a mature fruit from a dicotyledonous plant, observe it and

answer questions that follow



a) To what group of fruits does the specimen belong?

b) Suggest the possible agent of dispersal of the fruit

19. Explain why menstrual periods stop immediately after conception?

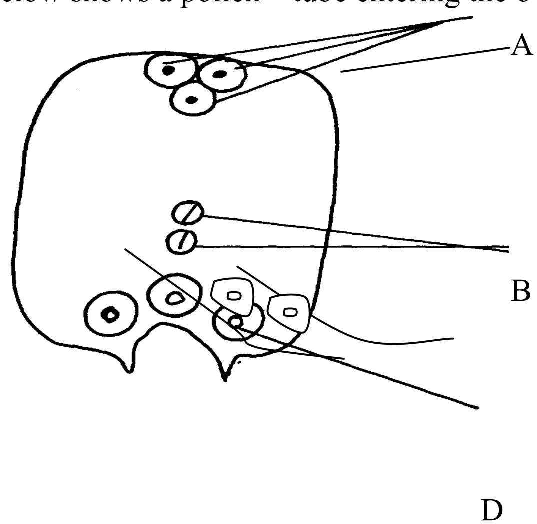
20. a) Why is sexual reproduction important in evolution of plants and animals

b) The calyx cells of a certain plant has 22 chromosomes. State the number of chromosomes present in the plant's

i) Endosperm

ii) Ovule cell

21. The diagram below shows a pollen – tube entering the ovule of a flowering plant



a) Name the parts labeled **A, B** and **D**

b) Name the kind of fertilization exhibited by the above flowering plant.

22. Donkey and zebra are closely related yet not of the same species. Explain

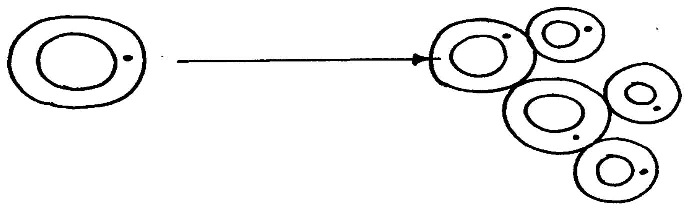
23. Name **two** factors in the environment which organisms respond to

24. What is meant by the terms:- a) i) Epigynous flower

ii) Staminate flower

b) Name the protective membranes surrounding the brain

25. The diagram below illustrate a process in a given species of organism



a) Name the organism that undergoes the process above

b) Identify the process shown to be taking place

26. State **two** ways by which HIV/AIDS is transmitted from mother to child

27. (a) State the role of centrioles during cell division

(b) (i) Explain the role of chlorophyll in photosynthesis
(ii) What is the **main** product of the dark stage of photosynthesis?

28. (a) At what stage of meiosis is the chiasmata formed?

(b) (i) What is the significance of the above part in living organisms?
(ii) State **two** importance of meiosis in living organisms?

29. (a) State **two** ways in which the male parts of a wind pollinated flower are adapted to their mode of pollination
(b) Differentiate between monoecious and dioecious plants

30. (a) What is seed dormancy?
(b) State **two** ways in which seed dormancy can be broken

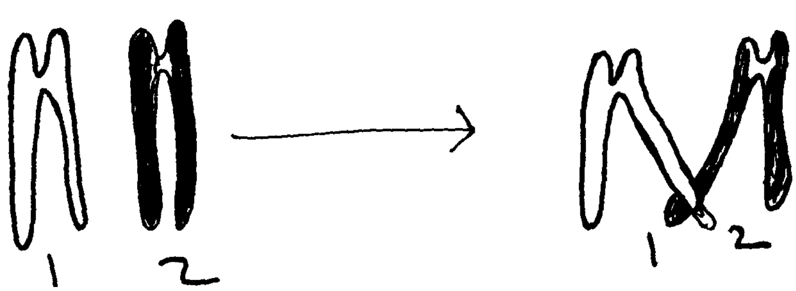
31. (a) Explain **two** importance of the adult stage in metamorphosis in insects
(b) What is the importance of the juvenile hormone in insects?

32. Describe the possible effects of discharging hot effluent from a factory into a slow flowing river

33. State **two** disadvantages of external fertilization in animals

34. State **three** roles of placenta in mammals

35. (a) The diagram below shows a stage during cell division



(i) Name the stage of cell division

(ii) Give a reason for your answer

(b) Name **two** structures in plants where male and female gametes are produced

36. State **two** advantages of metamorphosis to the life of insects

37. List **four** differences between Mitosis and Meiosis

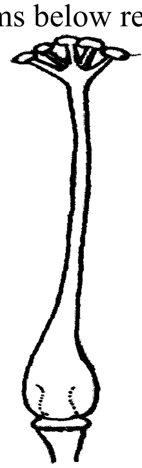
38. Give a reason why two species in an ecosystem cannot occupy the same niche

39. State the functions of the following hormones in the menstrual cycle :
(i) oestrogen

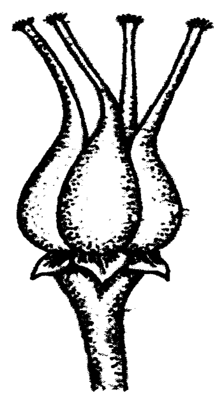
(ii) luteinizing hormone (L.H)

(iii) Follicle stimulating hormone (FSH)

40. The diagrams below represent two gynoecia **A** and **B** obtained from two different plants.



A



B

(a) What name is given to; Gynoecium **A?**
Gynoecium **B?**

(b) State the observable difference between the gnoecia **A** and **B**

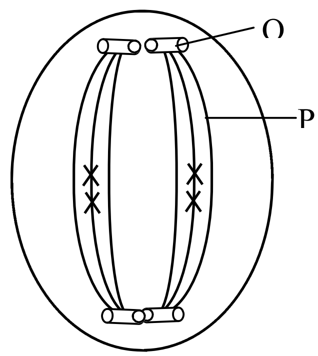
(c) State the role played by Heterostyly in plants.

41. State the difference between the sperm cell and the ovum.

42. (a) Name the parts of the flower in which pollen grains area formed.

(b) Name **two** nuclei found in pollen grains.

43. The diagram below represents a stage in cell division.



(a) Name the stage of all division shown in the diagram above.

(b) Give reasons for your answer.

44. Name the hormone that:

(a) Stimulate the contraction of uterus during birth.

(b) Stimulate the disintegration of corpus luteum when fertilization fails to take place.

45. State t**hree** ways in which seed dormancy benefits a plant

46. (i) State **two** major structural differences between fruit and a seed

(ii) Why is it advisable to use biological control of pests?

47. State the functions of the following parts in the male reproductive system

(a) Somniferous tubules
(b) Sertoli cells

48. (a) Name the parts of a flower responsible for gamete formation

(b) State **one** feature of pollen grains from a wind pollinated flower

49. Name the mechanisms that hinder self-fertilization in flowering plants

50. The eggs of birds are relatively much larger than those of mammals. Explain

51. Distinguish between the following terms:

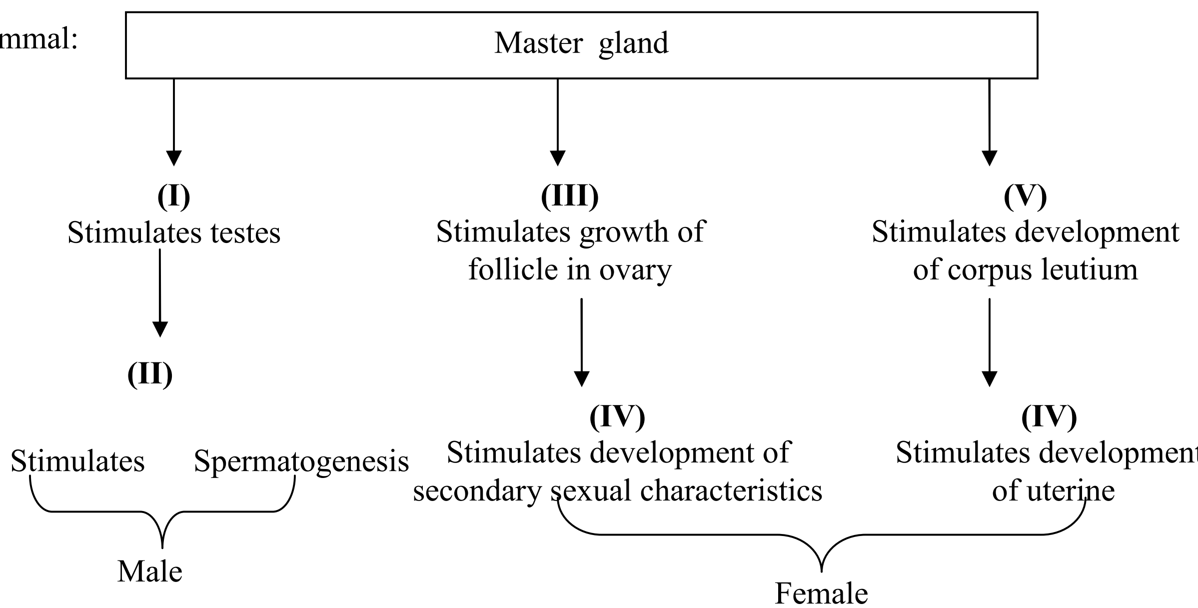
Pollination and **fertilization**

52. a) Describe the various mechanisms of fruit and seed dispersal.

b) Describe the varying events that follow a flower after fertilization.

53. Describe how fruits and seeds are suited to their mode of dispersal

54. The diagram below represents some hormones, their sources and functions in a mammal:



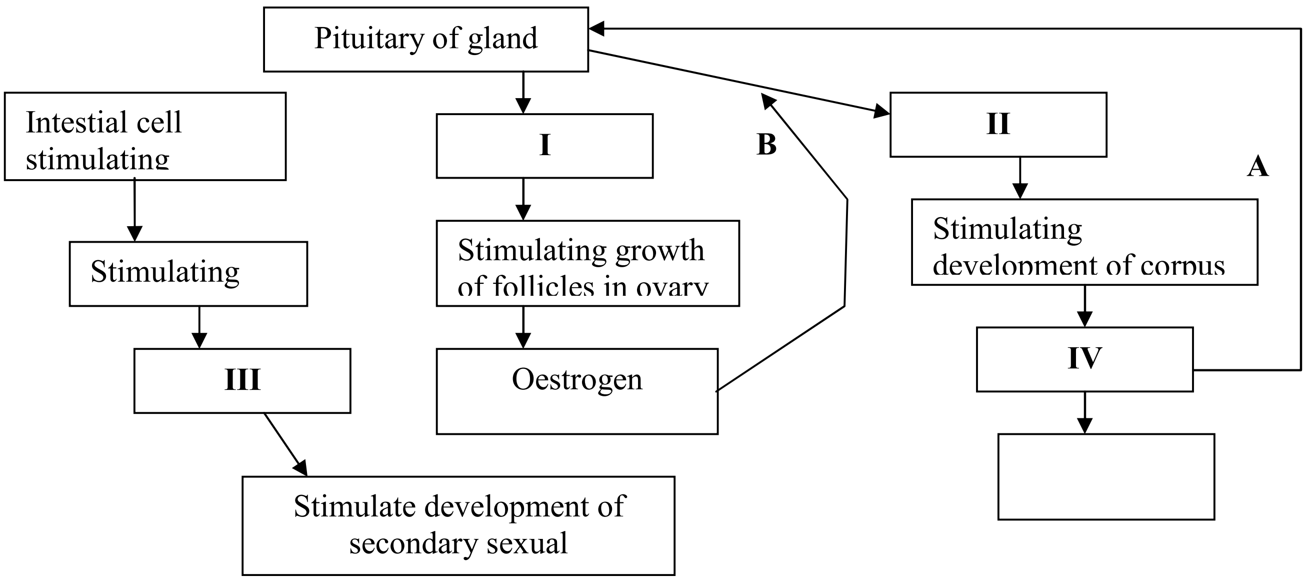
(a) Identify the master gland described above

(b) Name hormones **(ii), (iii), (v)** and (**iv)**

(c) Explain the consequences of deficiencies of hormone **(ii)** in man

(d) Other than stimulating the development of uterine wall, suggest one other function of hormone **(vi)**

55. The diagram below represents some hormones, their sources and functions in mammals.



a) Name the hormones **I, II** and **III**

b) Name hormones **IV** and state its function

c) Name the control labelled **A & B**

d) Name **one** secondary sexual characteristic common to both males and females

56. (a) State the role of spleen in human defense mechanism

(b) State **two** ways by which the HIV spread may be controlled through patients in hospitals

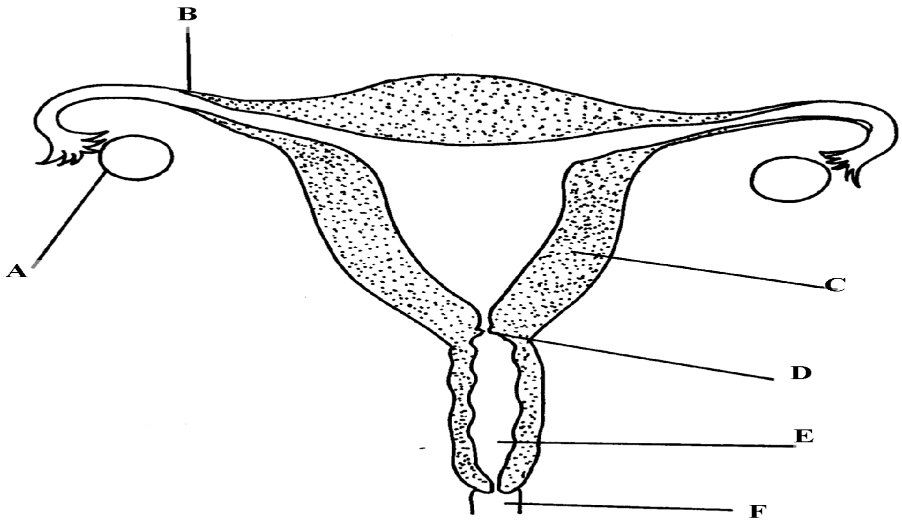
(c) What do you understand by the word Acquired Immunity Deficiency Syndrome (AIDS)

(d) Why is immunization against diseases encouraged by the government

(e) State how natural active acquired immunity is attained by an individual

57. Explain how seeds and fruits are adapted to the various methods of dispersal

58. The diagram below represents female reproductive system;



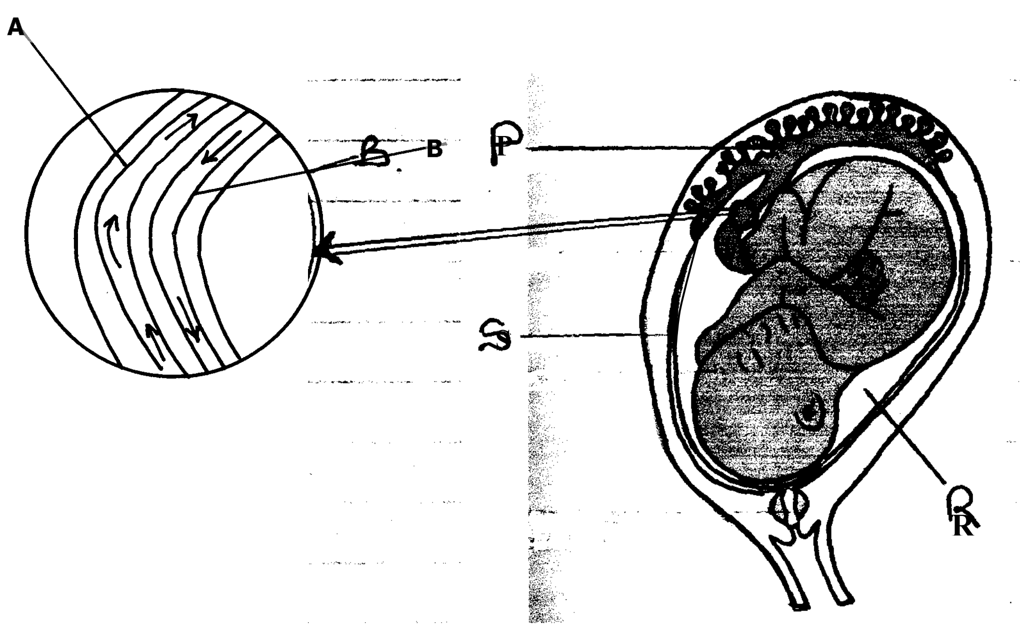
a) Name the part labeled; A, B, C and D

b) State **two** functions of structure **A**

c) How is part **C** adapted to its function?

d) Of what significance is part **E** to reproduction?

59. The diagram below represents a human foetus in a uterus



(a) Name the part labelled **S**

(b) (i) Name the blood vessels labelled **A** and **B**

(ii) State the difference in composition of blood found in vessels **A** and **B**

(c) Name **two** features that enable the structure labelled **P** carry out its function

(d) State the role of the part labelled **R**

60. An experiment was carried out to investigate the rate of growth of pollen tube against time.

The results are shown in the table below:

Time in minutes	Growth of pollen tube in millimeters
0	0
30	4.0
60	9.8
90	15.2
120	20.0
150	21.6
180	22.4

(a) (i) On the grid provided draw a graph of the pollen tube growth against time.

(b) (i) At what intervals was the growth of the pollen tube measured?

(ii) What was the length of pollen tube at; 130 minutes

(iii) At what time was the length of the pollen tube 18mm?

(iv) With reasons, describe the growth pattern of the pollen tube between:

- 0 to 120minutes
- Reason
- 120 to 180 minutes
- Reason

(v) State the importance of the growth of pollen tube to the plant

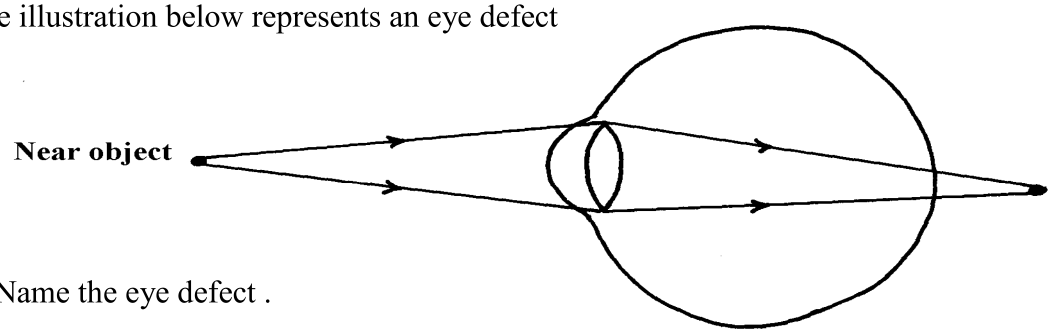
(c) State the changes that take place in a flower after fertilization

11. Growth and development in (a) plants (b) animals

1. a) Name the hormone which controls moulting in insects.

   b) State the importance of moultng in insects.

2. The illustration below represents an eye defect



   a) Name the eye defect .

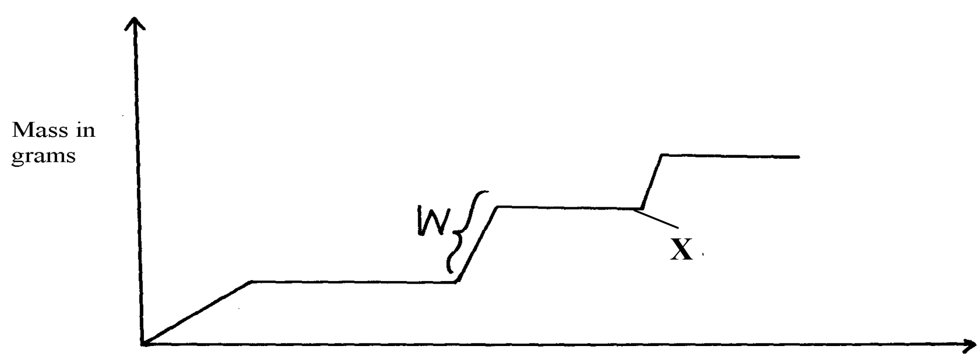
   b) Name the lenses that can be used to correct the defect.

3. (a) State **two** functions of the kidney

   (b) Name **two** substances that are not found in urine of a healthy person

   (c) Name **two** diseases that affect the kidney

4. The diagram below represents a growth pattern of arthropods.



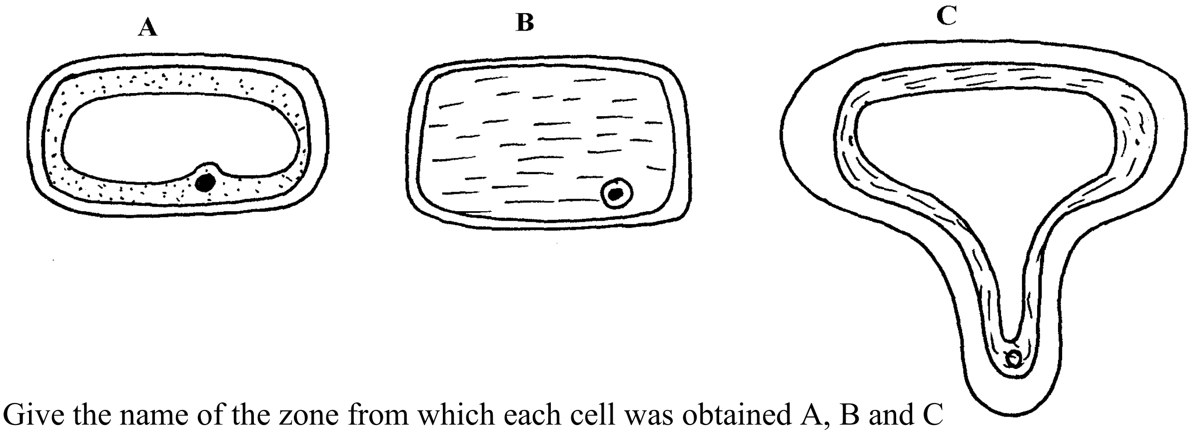
   (a) Name the type of growth pattern represented on the graph.

   (b)Identify the process represented by **X**.

   (c) Which hormone is responsible for process at **X** in 15 (b) above?

5. Distinguish between natural and acquired immunity.

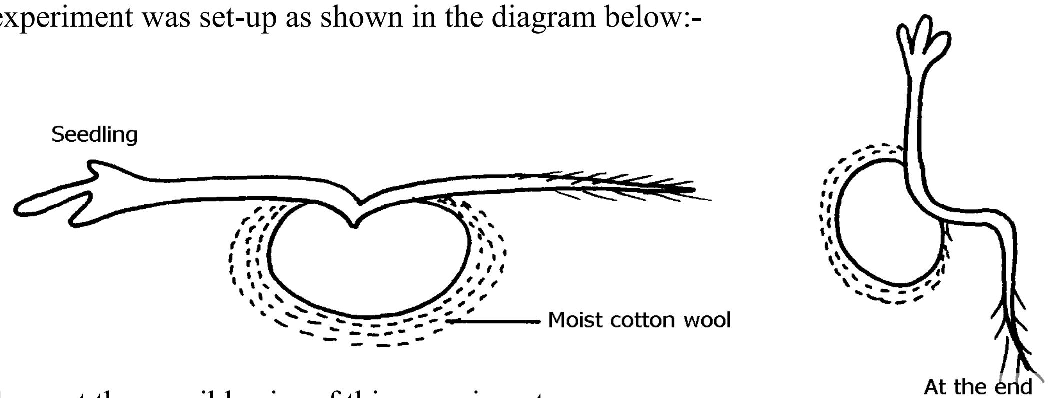
6. The cells shown below were obtained from different parts of a young root tip:



Give the name of the zone from which each cell was obtained A, B and C

7. Differentiate between continuous and discontinuous variations

8. An experiment was set-up as shown in the diagram below:-



(a) Suggest the possible aim of this experiment
(b) Account for the observation at the end of the experiment

9. State the location of **each** of the following plant meristematic tissues:-

(i) Vascular cambium

(ii) Intercalary meristem

10. Define the following terms: a) Growth

b) Development

11. State **two** advantages of metamorphosis in the life insects

12. State **one** disadvantage of exoskeleton in insects.

13. Distinguish between primary growth and secondary growth in a flowering plant

14. What is the role of the following to a germinating seed: (i) Oxygen

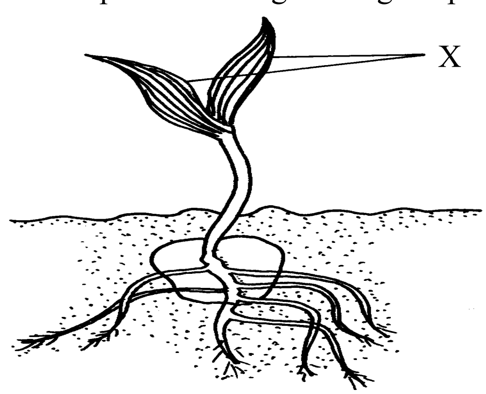
(ii) Cotyledons

15. Give **three** applications of plant growth hormones in agriculture

16. State **two** functions of calcium in the human body

17. State the biological importance of ecdysis in arthropods

18. The diagram below represents a stage during the process of germination.

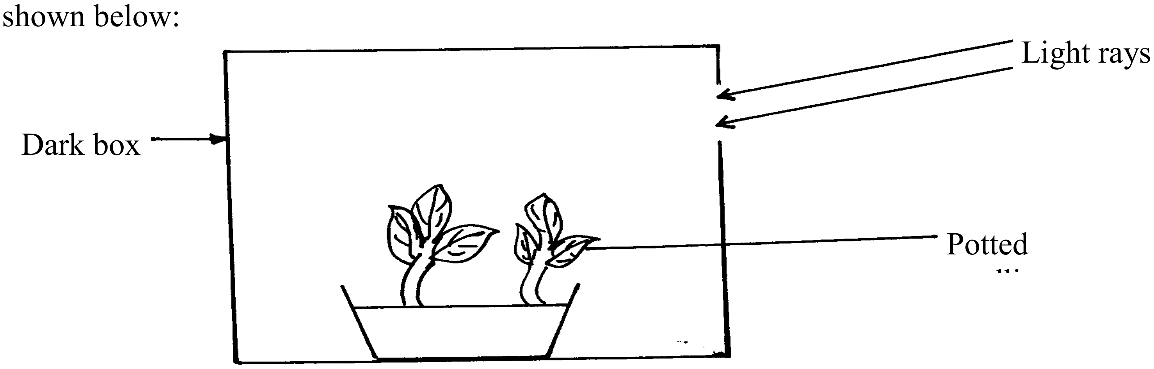


(a) (i) Name the type of germination illustrated in the diagram
(ii) Give a reason for your answer in **(a) (i)** above.

(b) Give **two** functions of the part labelled **X**

19. In an experiment young potted seedlings were placed in a dark box with unilateral light source

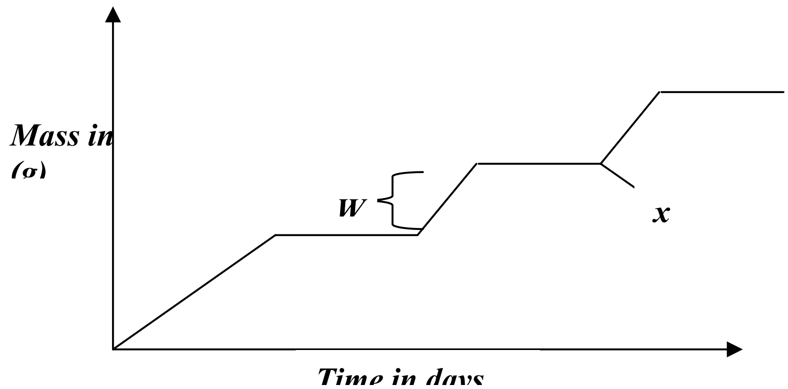
as shown below:



(a) What was the aim of the experiment?

(b) State the observations made on the seedlings after 3 days

20. The graph below represents the growth of animals in a certain phylum.



(a) Name the type of growth pattern shown on the graph.

(b) Identify the process represents by **x.**

(c) Name the hormone responsible for the process in B above.

21. (a) State the role of the vascular cambium in plant growth and development.

(b) Explain why monocotyledons plants do not undergo secondary thickening.

22. Explain how placenta is adapted to its functions

23. State the role of the following during germination:

(a) oxygen
(b) enzyme

24. Name the type of responses exhibited by:-

(a) (i) Marine crabs burrowing into the sand to avoid dilution of their body fluids

(ii) Chlamy domonas plant moving towards a region of high light intensity

(b) (i)What type of neuron is drawn above?

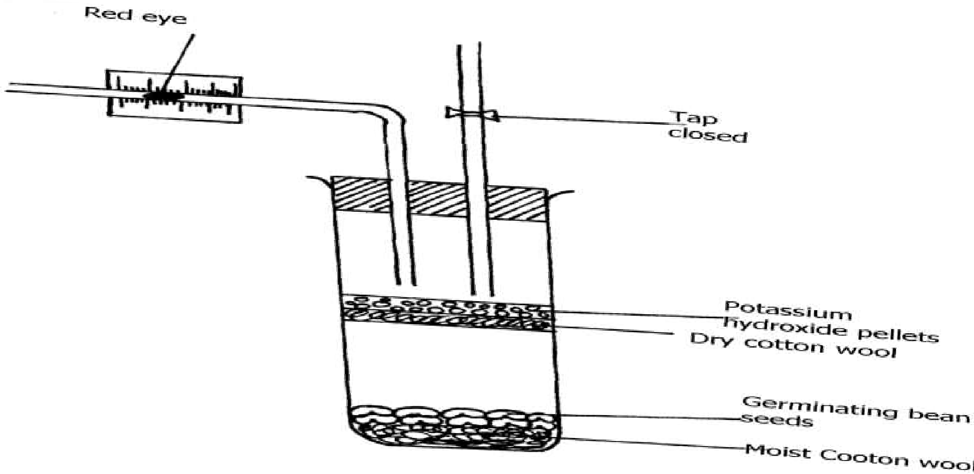
(ii) Using an arrow, show the direction of the nerve impulse

(iii) Name the part labelled **X**

(iv) State the function of part labelled **Y** .

(c) Give **two** differences between reflex action and conditioned reflex action

25. The experiment set – up below was designed to investigate an aspect of germination.



a) Why was potassium hydroxide pellets used in this experiment?

b) What was the role of moist cotton wool in this experiment?

c) i) By means of an arrow, indicate on the diagram the direction in which red dye would move during the experiment.

ii) Give reason for your answer in c(i) above.

d) Other than the factor investigated above, state any other **one** factor necessary for germination process.

26. The following data represents the development in dry mass of germinating seedlings within 18 weeks:

Time in weeks	0	1	2	4	6	10	13	15	16	18
Dry mass in grammes	0.1	2	3.2	10	18	32	44	45	44	38

(a) Using suitable scales plot a graph of dry mass against time

(b) Write reference to the graph, explain the changes in dry mass between:-
(i) Week 0 to 2
(ii) Week 5 to 13

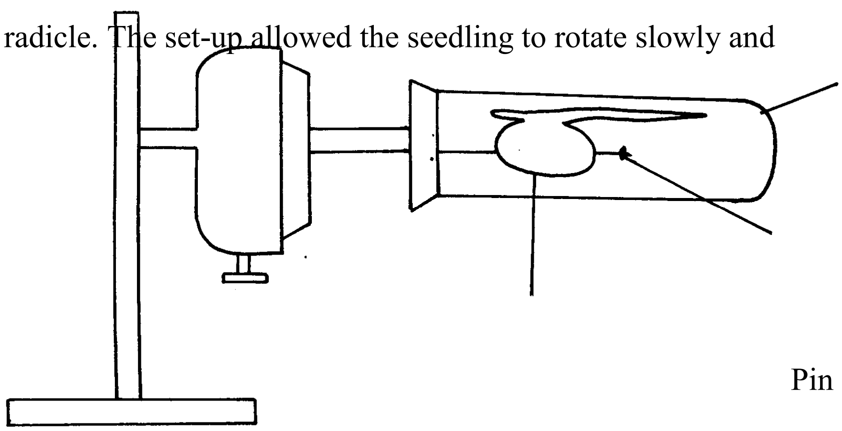
(iii) Week 16 - 18

(c) (i) What is the significance of time zero?

(ii) What difference would be expected from the above results if the experiment started with
the seeds? Give a reason for your answer

(d) (i) Describe how you carry out the experiment to obtain dry mass in the respective weeks
(ii) State **one** advantage of using dry mass instead of fresh weight in estimating growth of an
organism

27. The diagram below represents a set-up that was used to investigate the effect of rotation on the
growth of a bean radicle. The set-up allowed the seedling to rotate slowly and continuously for
seven days



(a) Name the piece of apparatus illustrated

(b) (i) State the observation made on the shape of the radicle after seven days

(ii) Explain the observation in **(b) (i)** above

(c) Suggest a suitable control for this experiment
(d) Give any **four** importance of tropism in plants

28. An experiment was carried out to determine the growth rates of variety of bamboo and a variety of maize in two adjacent plots. The average height and average dry weight of plants from the two

populations were determined over a period of twenty weeks. The data is as shown below:-

a) On the same axes, plot a graph to show the changes in average weight of the bamboo and maize plants over time

(b) (i) Which of the two plants had a higher productivity by the end of the experiment?

(ii) Give a reason for your answer in **(b)(i)** above

(c) Explain the following:

(i) Between weeks 14 and 18 the average height of maize plants remained constant while the average dry weight increased

(ii) Dry weight was used instead of fresh weight in this experiment

(iii) Describe how the average height and average dry weight of plants were determined in this experiment;

(d) Why was it appropriate in this experiment to use both weight and height?

(e) Give a reason why secondary thickening does not occur in bamboo and maize plants

29. (a) What is meant by the term **fertilization** ?

(b) (i) Name the type of cell division that produces gametes

(ii) Where does the type of cell division mentioned above occur in mammals?

(c) What happens to the wall of the uterus;

(i) before the release of an egg ?

(ii) if no fertilization occurs?

(b) How is the placenta adapted to its functions?

30. The relationship between seed fresh mass in the lupin *lupinus* and percentage seed germination, percentage seedling survival and seedling fresh mass is shown in the table;

Seed fresh Mass mg^{-1}	Percentage germination	Percentage of seedlings surviving 2 leaf stage	Mean seedling fresh mass 5 weeks after germination/mg
Below 16	41.9	84.6	24.3
17-25	90.2	96.8	44.2
26-35	95.6	98.8	60.7
36-45	97.5	100.0	86.4

Above 45	100.0	100.0	106.4

a) How was percentage germination in column two of the table calculated?

b) Why was seed fresh mass preferred to seed dry mass to take measurements of the seed weight
in the experiment

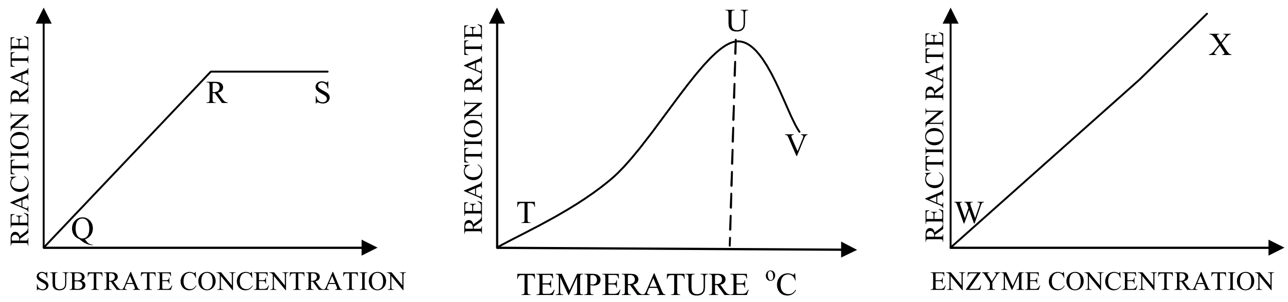
c) i) Explain why the measurements of mean seedling fresh mass (5) weeks after germinated
may not have been an accurate measurement of growth that had occurred

ii) How could more meaningful and accurate measurement been obtained in **c(i)** above?

d) With reference to the figures in the taste indicate the relationship between seed fresh mass
and percentage seed germination, percentage seedling survival and seedling fresh mass

e) Suggest an explanation why seedling produced from large seeds grow more rapidly than the
seedling produced from small seeds

31. The diagram below illustrate enzyme controlled reaction



a) State the relationship between rate of reaction and enzyme concentration

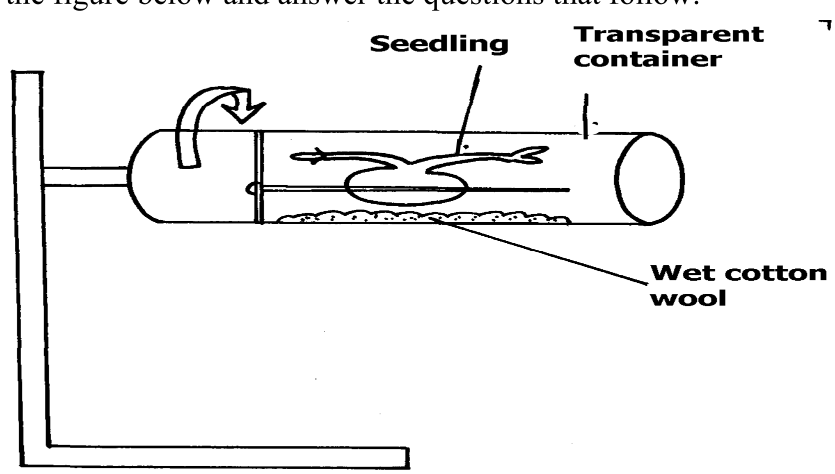
b) Account for the rate of reactions between; i) **Q** and **R**

ii) **R** and **S**

iii) **U** and **V**

c) Name **one** other factor that affects enzyme action, not illustrated above

32. Carefully study the figure below and answer the questions that follow:-



The seedling with straight radicle and plumule was attached to a machine horizontally as shown

above. The machine rotates making one revolution in 15minutes.

(a) Draw how the seedling would look like after one week

(b) Explain your drawing in **(a)** above

(c) Name the machine used in the experiment above

(d) What would happen if the seedling was put horizontally outside the machine

(e) Name the stimuli investigated and type(s) of response expected in the experiment

33. (a) Give the form in which each of the following substances ate transport in mammalian blood:

(i) Carbon (IV) oxide

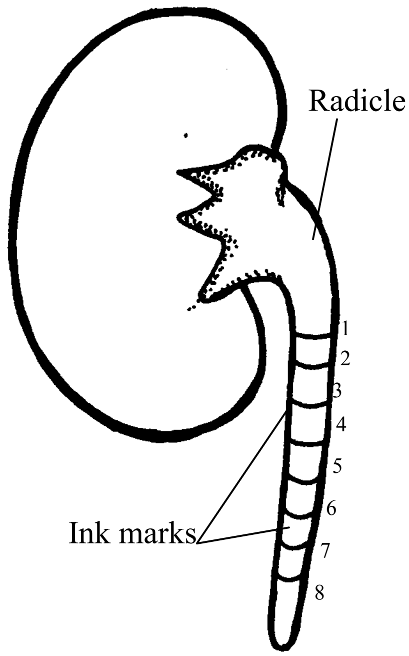
(ii) Oxygen

(b) Give **two** functions of pleural membrane

(c) Explain why formation of carboxyhaemoglobin in the blood of a mammal results in death

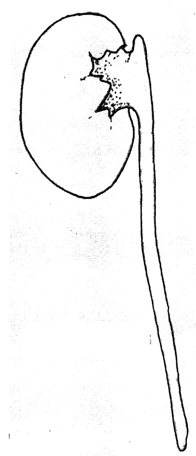
(d) Other than stomata, name **two** other gaseous exchange surfaces in plants

34. In an experiment the radicle of a seedling was marked equidistant using Indian ink as shown
in the diagram below:



(a) What was the aim of the experiment?

(b) On the diagram below mark on the radicle to show the appearance of the marks after 3days



(c) State **three** characteristics of cells found just behind the root cap of a radicle

(d) Give **two** factors inside a seed that causes seed dormancy

12. Genetics

1. A woman with blood group **A** gave birth to twins both having blood group **AB**.

Determine the genotype of:

a) Father
b) Mother

2. 50 black mice and 50 white mice were released into an area inhabited by a pair of owls. After four months, the mice in the area were recaptured and only 38 of the black mice and 9 of the white mice were remaining.

a) How would this observation be explained ?

b) Name the theory of evolution that supports the results in **(a)** above.

3. State **three** mechanisms that prevent self pollination in a flower that has both male and female Parts.

4. (a) Distinguish between complete and incomplete dominance

(b) State **two** sources of variation

5. Part of one strand of a DNA molecule was found to have the following base sequence.

G – T – C – A – G – T

(a) What is the sequence on m-RNA strand copied from this DNA portion?

(b) State **two** roles of DNA molecule.

6. State **three** ways by which plants compensate for lack of ability to move from one place to another.

7. A student mixed a sample of urine from a person with Benedict's solution and heated, the colour

changed to orange.
(a) What was present in the urine sample?

(b) What did the student conclude on the health status of the person?

(c) Which organ in the person may not be functioning properly?

8. Differentiate between continuous and discontinuous variations

9. Members of the same species of organism tend to differ due to variation. State **three** causes of variation in organisms

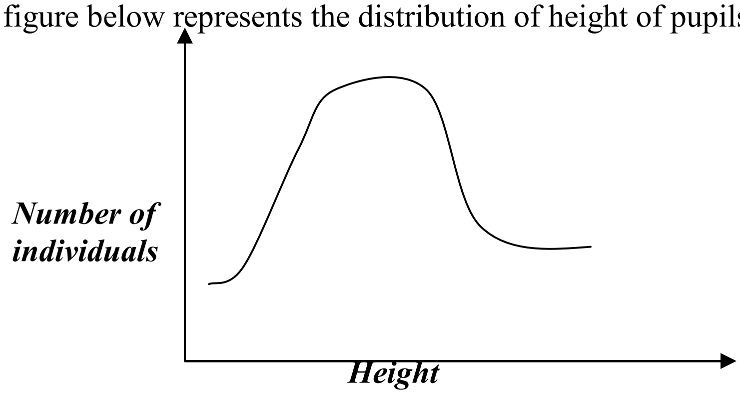
10. Identify the type of gene mutations represented by the following pairs of words:-

(i) Shirt instead of skirt
(ii) Hopping instead of shopping
(iii) Eat instead of tea

11. A DNA stand has the following base sequence: GCCTAGATCAC
What is the sequence of the : (i) Complementary DNA strand?

(ii) M-RNA strand coped form this DNA strand

12. The figure below represents the distribution of height of pupils in a school



(a) Name the type of variation represented by the curve

(b) Outline **two** possible causes of variation in height of individuals in man

13. a) Wekesa and Wanjiku who are siblings are both normal as their parents but have a hemophilic brother. Give the Genotype of their parents.

b) i)What are linked genes?

ii) What do you understand by the phase a test cross?

14. There are at least 205 known sex – linked recessive disorder

a) Name **any two** of them.

b) State a reason why sex – linked recessive why traits tend to effect the male child.

c) State why if a mother has the trait all her sons will have it

15. The table below is a representation of a chromatide with genes along its length. It undergoes mutation to appear as shown below:

Before mutation	L	M	N	O	P	Q
After mutation	L	O	N	M	P	Q

a) Name the type of chromosomal mutation represented

b) Name **one** mutagenic agent

16. The figure below is a structural diagram of a portion from a nucleic acid strand

a) Giving a reason, name the nucleic acid to which the portion belongs

b) Write down the sequence of bases of a complementary DNA strand

17. In an experiment, plants with red flowers was crossed with plants with white flowers.

All the plants in the $\mathbf{F_1}$ generation had pink flowers.

a) Give a reason for the appearance of pink flowers in the $\mathbf{F_1}$ generation

b) If plants in $\mathbf{F_1}$ were selfed, state the phenotypic ratio of the $\mathbf{F_2}$ generation

c) Explain; i) Why women should drink extra milk during pregnancy

ii) A pregnant women might want to urinate more often in late pregnancy

18. State the meaning of the following terms giving an example in each case:

(a) Sex-linked genes

(b) Multiple alleles

19. In a certain breeding experiment, a plant species with red flowers was selfed. It produced **119**

red flowered and **41 white** flowered offsprings.

(a) Using letter **R** to represent allele for the red flowers, state the genotype of the red flowered parent plant

(b) Determine the phenotypic ratio of red and white flowered plants. Show your working

20. Give an example of a sex-linked trait in human on:

(i) **Y** – Chromosome

(ii) **X** – Chromosome

21. Explain why growth of long hair on the pinnae of the ears in human occurs in males only

22. Explain why **prophase 1** of meiosis contributes towards genetic variation in living organisms.

23. A pure Red flowered plant was crossed with a pure white flowered plant. All the F_1 generation plants had pink flowers.

(a) Give an explanation for the absence of Red and white flowered plants in the F_1 generation. (b) If the F_1 generation pea plants were selfed, state the phenotypic ratio of the F_2 generation plants.

24. (a) Name a genetic disorder due to gene mutation that affects the malpighian layer of the skin in man.

(b) Give **two** functions of the fluid produced by sebaceous glands.

25. (a) Define the term "Gene mutation."

(b) Name the genetic disorders that result from gene mutation in human beings.

26. (i) What are mutations

(ii) Name **two** mutagens

27. A section of a DNA strand contains the following sequence CGGATAC

(a) Write the; (i) Complementary DNA strand

(ii) MRNA strand

(b) Name the site for protein synthesis in a cell

28. In a certain bird species, red flight feathers is controlled by gene **R** while white flight feathers is controlled by gene **r**. The heterozygous condition **Rr** results into pink flight feathers. The two genes are also sex linked and transmitted on x-chromosome.

a) By use of fusion lines, find the genotypes of across between a male with pink flight feathers and a female with white flight feathers

b) Which type of dominance is illustrated here?

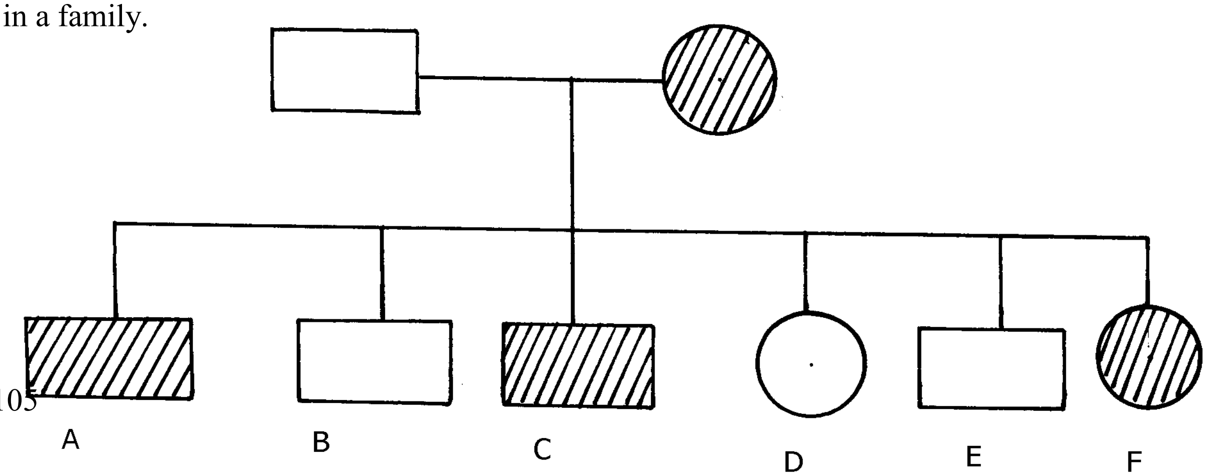
c) i)Identify the nucleic acid whose base sequence is shown below:

G-A-C-U-A-G-A-C-G

ii) Give a reason for your answer in **c (i)** above

iii) If the nucleic acid was involved in protein synthesis, how many amino acids would be present in the protein synthesized? Give a reason

29. Study the genetic chart below showing the inheritance of the gene responsible for haemophilia in a family.

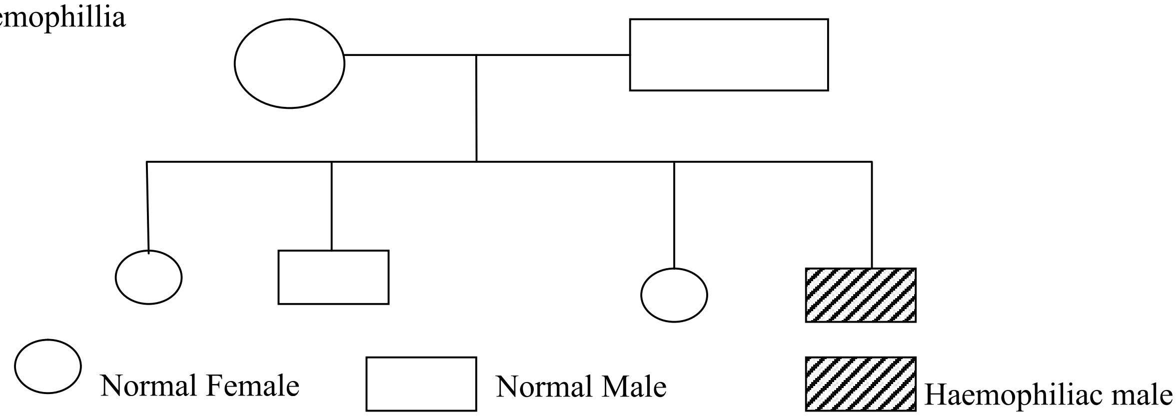


a) Write the genotype of individuals A, B, F

b) A member of this family labelled **F** marries a haemophiliac male. What will be the phenotypic
ratio of the offspring? Show your workings

c) Other than the condition stated above, state any other **two** common genetic disorders that
result from gene mutation.

30. Haemophilia is due to a recessive gene. The gene is sex-linked and located on **X** chromosome.
The chart below represents the offspring of parents who are phenotypically normal for
haemophillia



(i) What are the parental genotypes?

Explain your answer in **(i)** above

(ii) Work out the genotypes of the offspring

31. A cross between a red-flowered and a white flowered plant produced only pink – flowered

F_1 plants

(a) There was neither a red nor white –flowered F_1 plants. Explain

(b) The F_1 offspring were selfed to get F_2 generation. Using appropriate letter symbols, work out the genotypes of F_2 generation

(c) Give the genotypic and phenotypic ratios of F_2 generation

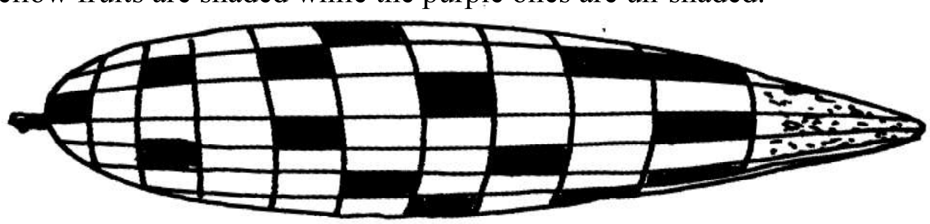
(d) Distinguish between dominant and recessive genes

32. A true-breeding purple maize variety was cross-pollinated with a true-breeding yellow

maize variety.

The offspring produced all purple fruits.

The plants grown from these F_1 grains were interbred among each other.
A typical cob of F_2 generation is shown below:
The yellow fruits are shaded while the purple ones are un-shaded.



(a) (i) In terms of flowers only, state why it is easier to work out genetic crossings using maize

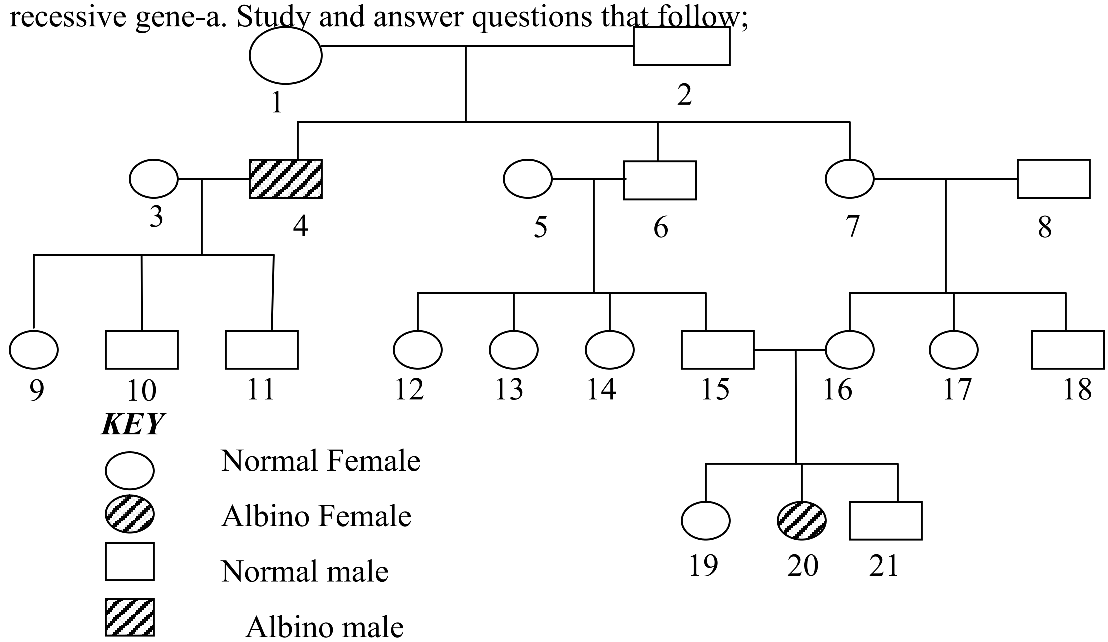
(ii) Count separately the yellow and purple grains and therefore find the rations of purple grains to yellow grains

(b) Using appropriate symbol, work out a genetic cross for F_2 generation

(c) From the above information, give the dominant gene

(d) State **two** practical applications of genetics in identity determination

33. The figure below is a pedigree chart showing incidence of albinism which is transmitted through a recessive gene-a. Study and answer questions that follow;



(a) Write down the genotype of persons 1 and 2. Give a reason for your answer

(b) Giving your reason state the most likely genotype of person 3

(c) The cross between person **15** and **16** represents mating between first cousins. Comment why it is not advisable for close relatives to marry

(d) Apart from albinism name **two** other effects of gene mutation

34. The table below shows results of test to determine blood groups of persons **Y** and **Z**.A tick (√)

Represents, agglutination while a cross (**x**) represents no agglutination;

Person	**Test with antibody (a)**	**Test with antibody (b)**	**Test with Rhesus antibody**	**Blood group**
Y- (male)	√	X	√	
X- (female)	X	√	X	

(a) Fill the blank space in table to show the blood group of the persons **Y** and **Z**
(b) In order to investigate the inheritance of Rhesus factor, work out a cross between a male with Rh^+ and female with Rh^- .Let **D** represent the presence of Rhesus factor and **d** to represent the absence of the Rhesus factor
(c) Determine the genotype of the cross in **(b)** above.

(d)Which of the children can donate blood to their mother?

35. Describe the behavioural adaptations of animals to temperature

36. In man blood group inheritance is controlled by multiple alleles in which allele **A** is co dominant to allele **B**. a woman laterozygous for blood group **A** married a man heterozygous for blood group **B**

a) State the genotype of both parents

b) Using a pun net square, show the genotypes of F_1 generation

c) State **one** application of knowledge of blood group inheritance in man

d) The nitrogenous bases in nucleic acids are Adenine (A), cytosine(C), Guanine (G), Thiamine (T) and uracil (U). Input of a molecule of DNA the sequence of bases is CTT.

Using the letters **A, C, G, T, U** where appropriate, write down the base sequence in;

i) Corresponding part of the complementary strand of DNA molecules

ii) Corresponding part in mRNA

iii) A change in the DNA molecules caused the base sequence in the triplets to change from CTT to CAT. State **one** factor which could have caused the change

37. In an investigation plants with red flowers were crossed with plants with white flowers. All the plants in the F_1 generation had pink flowers when the F_1 plants were crossed, he counted 480 plants in F_2 generation

(a) Using appropriate letter symbols, work out the cross between the F_1 plants to get the F_2 generation

(b) Give the phenotypic and genotypic ratios for the F_2 generation

Phenotypic ratio
Genotypic ratio

(c) How many plants in the F_2 generation had pink flowers? (show your work)

38. In an experiment, a black mouse was mated with a brown mouse. All the off springs in F_1 generation were black. The off springs grew and were allowed to mate with one another. The total number of F_2 generation offspring were 96.

(a) Using letter **B** to denote the gene for black colour. Work out the genotype of the F_1 generation. (Use a punnet square)

(b) State the following for the F_2 generation

(i) Genotypic ratio

(ii) Phenotypic ratio

(iii) The total number of brown mice

39. (a) Distinguish between Homologous structures and analogous structures. Give an example in each case.

Homologous structures
Example
Analogous structures
Example

(b) Explain why parasites develop resistance to certain drugs after a long time of exposure.

(c) (i) What is non— disjunction?

(ii) Give **one** example of a genetic disorder associated with non-disjunction
.

13. Evolution

1. a) Distinguish between homologous and analogous structures in evolution.
   b) Name **one** vestigial structure in mammals.

2. a) Give **two** examples of adaptive radiation in animals.

   b) State **two** disadvantages of using fossils as evidence of evolution

3. Distinguish between camouflage and mimicry.

4. State the role of light in photosynthesis

5. (a) Name the region of the **gut** where digestion of cellulose takes place.

   (b) State role of **cardiac sphincter** in the stomach.

6. (a) Give **two** limitations of fossil records as evidence of evolution

   (b) State any **two** similarities in structure between **Homo erectus** and **Homo Sapiens**

7. (a) (i) What is meant by vestigial structures?

   (ii) Give an example of a vestigial structure in human

8. Distinguish between the struggle for existence and survival for the fittest as used in the theory of natural selection

9. Give **two** factors that determine water reabsorption in the distal convulated tubule

10. Distinguish divergent and convergent evolution

11. (a) What are the advantages of natural selection

(b) All insects are believed to have arisen from a common ancestor. However, modern insects differ widely in a variety of ways such as in the adaptation of their mouthparts for different modes of feeding. What kind of evolution is this?

12. Explain why Lamacks theory of evolution is not accepted by Biologists today.

13. a) i) What is meant by vestigal structures

ii) Give an example of vestigal structure in human

b) Explain why certain drugs become ineffective in curing a disease after many years of use

14. (a) What is organic evolution?

(b) Briefly explain the term "***survival for the fittest***" as used in Darwin's theory of natural selection

15. Explain why insecticides become ineffective against insects if used for several years in succession

16. State **three** limitations of fossils records as an evidence of organic evolution

17. State **three** pieces of evidence that support the theory of organic evolution

18. What is meant by natural selection?

19. (a) Explain why Lamarcks theory of evolution is not accepted today

(b) State **two** limitations of fossils records as evidence of organic evolution

20. In a breeding experiment, plants with red flowers were crossed. They produced 123 plants with red flowers and 41 with white flowers:

(a) Identify the recessive trait

(b) Give a reason for your answer

(c) If white flowered plants were selfed, what would be the genotype of their offspring?
Show your working using appropriate symbols **(R, r**)

(d) What is a test cross?

21. a) What is organic evolution?

b) Describe the various evidences which support the theory of organic evolution.

22. (a) What is meant by the term natural selection

(b) Describe how natural selection brings about the adaptations of a species to its environment

(c) Distinguish between convergent and divergent evolution

(d) Discuss **four** evidences to show that evolution has taken place

23. Explain the various evidence for organic evolution

24. (a) What is organic evolution

(b) Explain why resistance to antibiotics is considered as an example of evolution

(c) List and explain various evidences of organic evolution

25. Pure breed red flowered plants were cross pollinated with pure breed white flowered plants.

The resulting F_1 offspring's had pink flowers.

(a) Using letter **R** to represent the gene for red colour and letter W to represent gene for white colour of flowers. Work out the genotype of the $\mathbf{F_1}$ generation

(b) If seeds from the $\mathbf{F_1}$ generation plants were planted and allowed to self pollinate. Work out the phenotypic ratio of the $\mathbf{F_2}$ generation

14. Irritability and sensitivity in (a) plants (b) animals

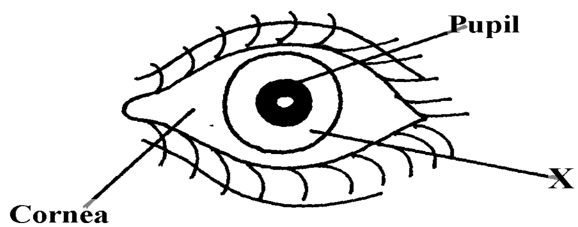
1. Give **two** functions of the exoskeleton in arthropods.

2. When shoots of young plants are exposed to unidirectional light they bend towards light;

   a) Name the type of response exhibited by the young shoots

   b) Explain the cause of the observation above

3. Study the drawing below and use it to answer the questions that follow :-



   a) Name the part labeled **X.**

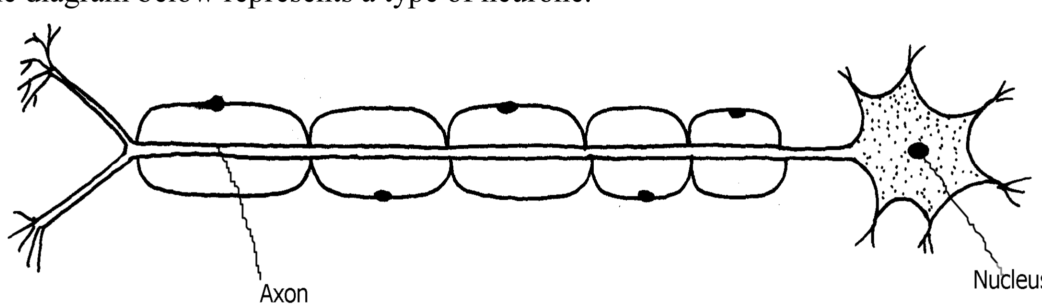
   b) Describe the changes that occur in the structure **X** in dim light.

   c) What is meant by the term **accommodation** with reference to the eye?

4. (a) State **two** differences between taxes and tropisms

   (b) Give **two** survival values of tactic movements to organisms

5. The diagram below represents a type of neurone.



(a) (i) identify the neuron above.

(ii) Give a reason for your answer in a (i) above.

(b) With an arrow, indicate on the diagram the direction of an impulse through the neurone.

(c) Name the chemical substance that brings about transmission of impulse across a synapse

6. A student was traveling from Nairobi to Mombasa. As the bus descended down hill he felt an

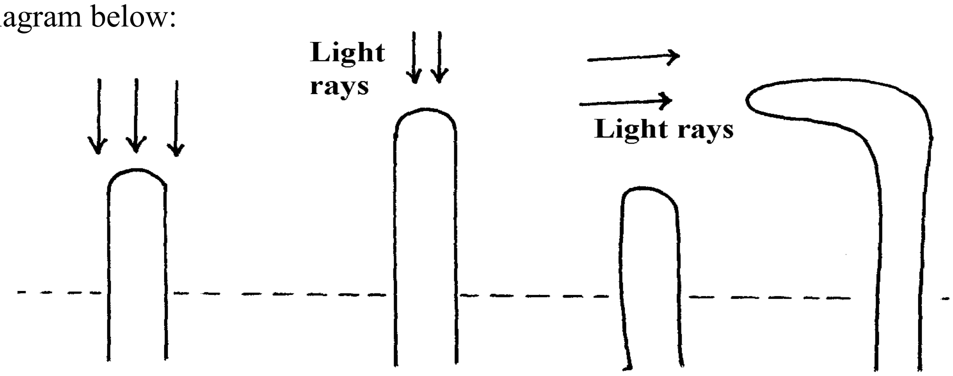
unpleasant sensation in the ear.

(a) How did the sensation come about?
*

(b) How can the unpleasant sensation be relieved?

7. An experiment was carried out to investigate a growth response in maize seedling as shown

in the diagram below:

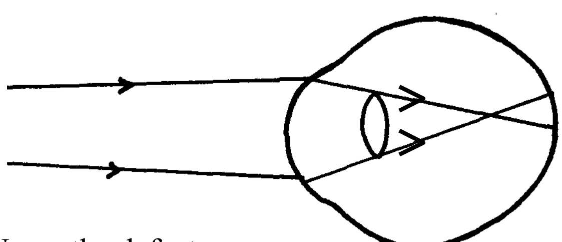


(a) State the type of response that is being investigated
....

(b) Explain the response exhibited by the shoot

8. State **three** genetic disorders caused by gene mutations

9. The diagram below shows the position of an image formed in a defective eye:-



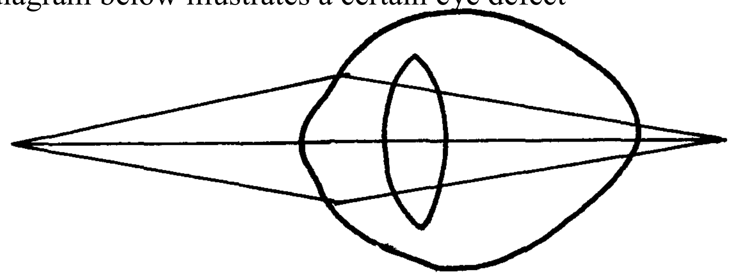
(a) Name the defect

(b)Explain how the defect named in **(a)** above can be corrected

10. (a) State **three** structural differences between arteries and veins in mammals
(b) Name a disease that causes thickening and hardening of arteries

11. (a) Name the part of the eye in which the light sensitive cells are located
(b) List the **two** types of sensory cells found in the part named in (a) above

12. The diagram below illustrates a certain eye defect



(a) State the defect
(b) On the diagram illustrate how the defect can be corrected

(c) State **one** advantage of having two eyes in human beings

13. Briefly explain the role of the following part of skin

a) Cornified layer

b) Malpighian layer

14. State the functions of the following structures of the mammalian ear

a) Eustachian tube
b) Essicles

15. a) Distinguish between conditioned and simple reflexes

b) State how the nerve cell structure is suited to its function of impulse transmission

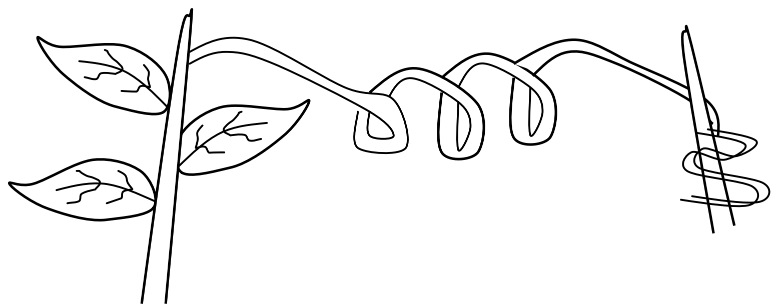
16. (a) Name the part of the mammalian eye that:
(i) Transmits impulses to the brain

(ii) Regulates the amount of light entering the eye
(b) State the changes that occur in the part of the eye named in **(a) (ii)** above when one moved from bright light to dim light conditions

16. Name the type of response exhibited by the following:

(a) A pollen tube growing towards the embryo sac
(b) Maggots moving from lit side of a box to the dark side

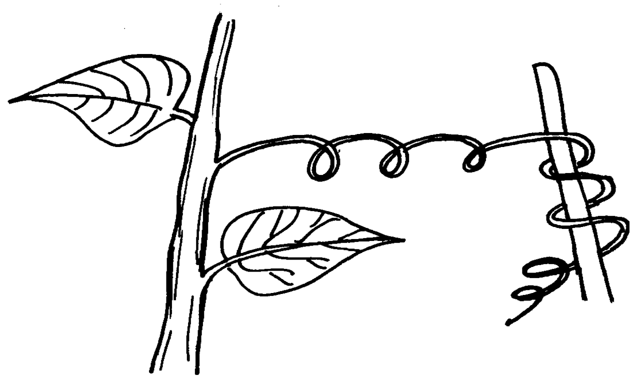
17. A response exhibited by a certain plant tendril is illustrated below:



(i) Name the type of response

(ii) Explain how the response named in **(i)** above occurs

18. A response exhibited by a certain plant tendril is illustrated below:-



Name the type of response

19. Removal of the apical bud from a shrub is a practice that results in the development of many lateral buds which later form branches

(a) Give reasons for the development of lateral branches after the removal of the apical bud
(b) Suggest **one** application of this practice?

20. In an accident a victim suffered brain injury. Consequently he had loss of memory which

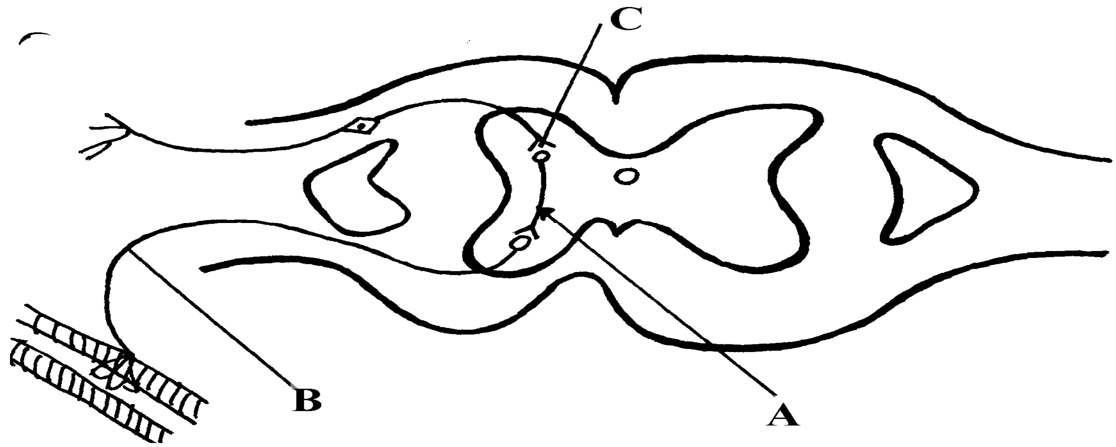
part of the brain was damaged?

21. A person was able to read a book clearly at arm's length but not at normal reading distance (a) State the eye defect the person suffered from

(b) Why was he unable to read the book clearly at normal distance?

(c) How can the defect be corrected?

22. The diagram below represents a simple reflex arc;



(a) Name the parts labeled **A** and **B**

(b) Explain how an impulse is transmitted across the gap labeled **C**

23. (a) State **two** functions of a mammalian ear
(b) How is the cochlea suited to its function

24. State **one** function of potassium ions in the human body.

25. State **two** functions of vitamin B_5 (pantothenic acid).

26. (a) What is the biological importance of tactic responses?

(b) A person had an accident and had problems with his vision, hearing and memory.
Identify the part of the brain that was affected

27. Identify the following responses shown by plants:- (a) Shoots grow towards light
(b) Roots grow towards gravity
(c) Tendril intertwine around an object

28. Name the type of responses exhibited by:-

(a) (i) Marine crabs burrowing into the sand to avoid dilution of their body fluids

(ii) Chlamy domonas plant moving towards a region of high light intensity

(b) (i)What type of neuron is drawn above?

(ii) Using an arrow, show the direction of the nerve impulse

(iii) Name the part labelled **X**
(iv) State the function of part labelled **Y** .

(c) Give **two** differences between reflex action and conditioned reflex action

29. In an experiment to investigate the effect of heat on germination of seeds, 12bags each

Containing 60 pea seeds were placed in a water bath maintained at 85^{o}C .
After every two minutes a bag was removed and seeds contained in it planted. The number that germinated was recorded. The procedure used for pea seeds was repeated for wattle seeds. The results were as shown in the table below:-

	Number of seeds that germinated	
Time (min)	**Pea seeds**	**Wattle seed**
0	60	0
2	60	0
4	48	0
6	42	2
8	34	28
10	10	36
12	2	40
14	0	44
16	0	46
18	0	48
20	0	49
22	0	47

(a) Using a suitable scale and same axes, draw graphs of number of seeds that germinated against time in hot water for each plant

(b) (i) At what time would number of seeds that germinated for each plant be same?
(ii) How many wattle seeds would have germinated if the 13th bag was available and was removed and seeds contained in it planted at 24minutes?

(c) Explain why the ability of pea seeds that germinated declined with time of exposure to heat
(d) Explain why the ability of the wattle seeds to germinate improved with time of exposure to heat

(e) Account for the shape of the graph for the wattle seeds which germinated between 20-24 minutes

(f) Some of the pea seeds were allowed to germinate and placed in a large air tight flask and left for four days:-

(i) Suggest the expected changes in the composition of gases in the flask on the fifth day

(ii) Give reasons for your answer **in (f)(i)** above

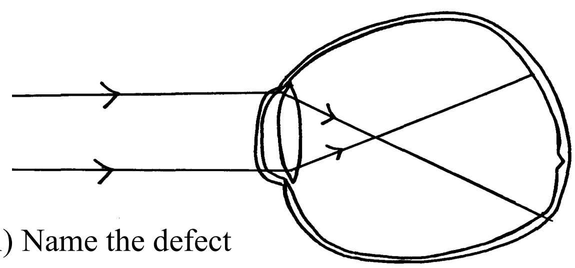
(g) Name **three** factors other than those investigated in **(a)** above which would affect dormancy

30. How is the mammalian skin adapted to its functions?

31. Explain how the mammalian skin is adapted to it's functions

32. Explain the structure and functions of the human eye.

33. The diagram below shows the position of an image in a defective eye.



(a) (i) Name the defect

(ii) State the causes of the defect

(b) Explain how the defect in **a(i)** above can be corrected.

(c) State the functions of cones

(d) How are nocturnal animals adapted to seeing?

5. Support and movement in (a) Plants (b) animals

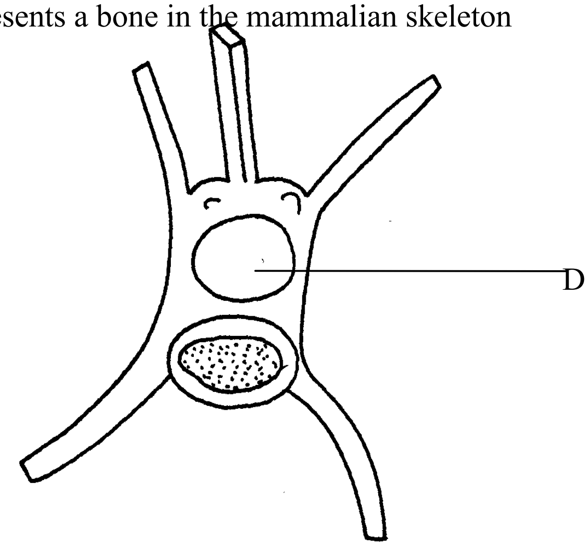
1. Explain how the following tissues are adapted to provide mechanical support in plants:-

a) Parenchyma

b) Collenchyma

c) Selerenchyma

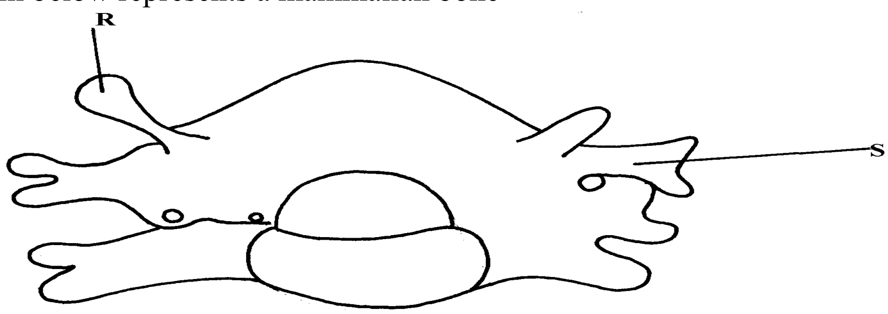
2. The diagram below represents a bone in the mammalian skeleton



a) Identify the bone with a reason

b) State the function of the part labeled **D**

3. The diagram below represents a mammalian bone



(a) Identify the bone shown above

(b) State the function of the parts labelled **R** and **S**

(c) State the region of the body in which the bone is found

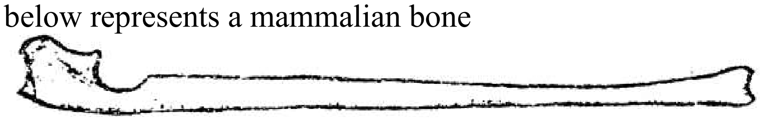
4. (i) Name **two** bones that form the ball and socket joint in the fore limb of a mammal

(ii) Name the fluid that is found in the above mentioned joint and its function

5. State **three** types of skeleton found in Kingdom animalia

6. State **three** differences between an animal's muscle cell and plant's palisade cell

7. The diagram below represents a mammalian bone



(a)Name the bone

(b) (i) Which bone articulates with the bone shown in the diagram at the notch

(ii) Name the type of joint formed when the bones in **b(i)** articulate

8. (a) Name the hard outer covering of the members of the phylum Arthropoda

(b) State **two** roles played by the structure named in **(a)** above

9. (a) State the role of lignin in the wall of the xylem vessel

(b) How does vascular bundles contribute to support in plants

10. (a) Distinguish between tendons and ligaments

b) State **one** way through which herbaceous plants achieve support

11. Name the ;

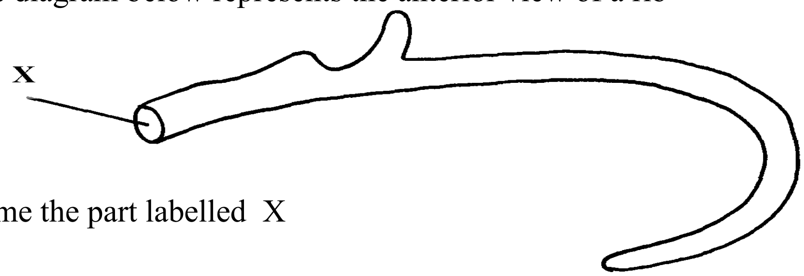
a) i) Material used to strengthen the xylem tissue

ii) Tissue that is removed when the bark of a dicotyledonous plant is ringed

b) State the areas of the plant where translocated materials are taken

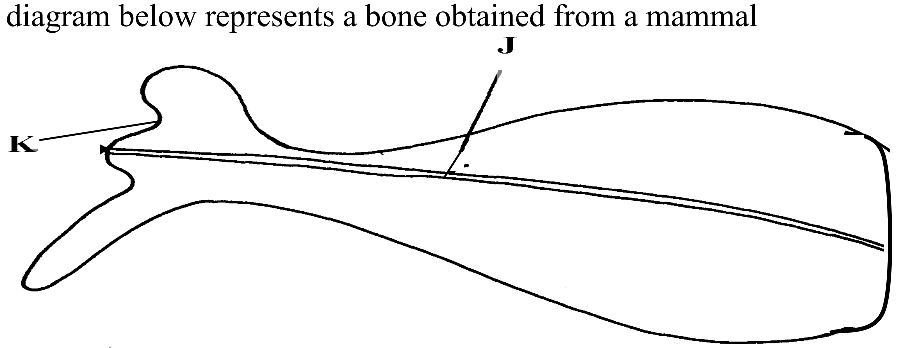
12. Give **three** importance of mammalian skeleton

13. The diagram below represents the anterior view of a rib



Name the part labelled X

14. The diagram below represents a bone obtained from a mammal

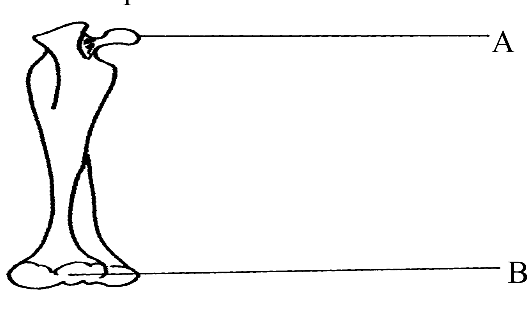


a) Name the bone

b) Name the:
i) Bones which articulate with the bone named in **(a)** above at the cavity labelled **K**
ii) Joint formed by the two bones at **K**

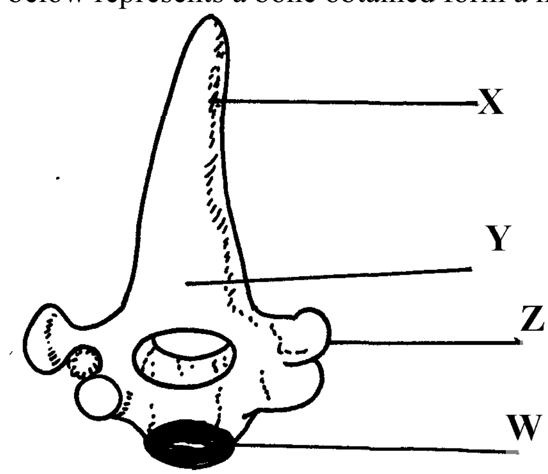
c) State functions of part labelled **J**

15. The diagram below represents a bone obtained from a mammalian skeleton:



(a) Identify the bone
(b) Name the:
(i) Bone it articulates with at point **A**
(ii) Type of joint that forms at point **B** in articulation with other bones

16. The diagram below represents a bone obtained form a mammal



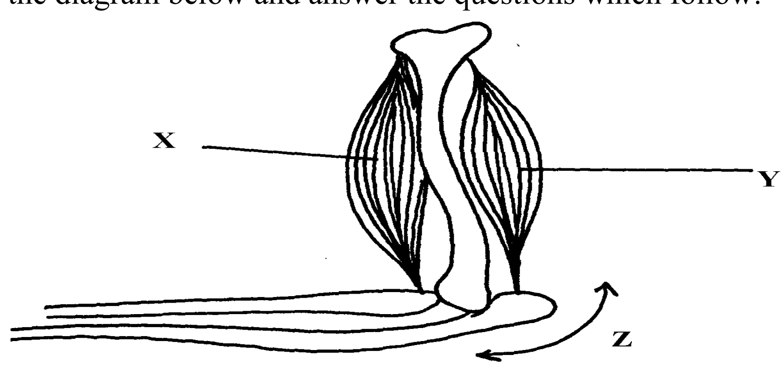
(a) Identify the bone
(b) Name the structures labeled **X** and **W**
(c) Name the bone that articulate with structure labeled **Z**

17. (a) Name the vertebra in a mammalian body that is characterised by presence of **odontoid**
process.

(b) State the function of the **odontoid process**

18. a) Name **three** supporting tissues in plants

b) Study the diagram below and answer the questions which follow:



i) Identify the muscle represented by **X** and **Y**

ii) Describe how muscles **x** and **y** cause straightening of the joint **z**

c) Name the joint **z**

19. (a) What is the importance of locomotion in animals?
(b) Explain how a bony fish is adapted for movement in its habitat

16. Human health

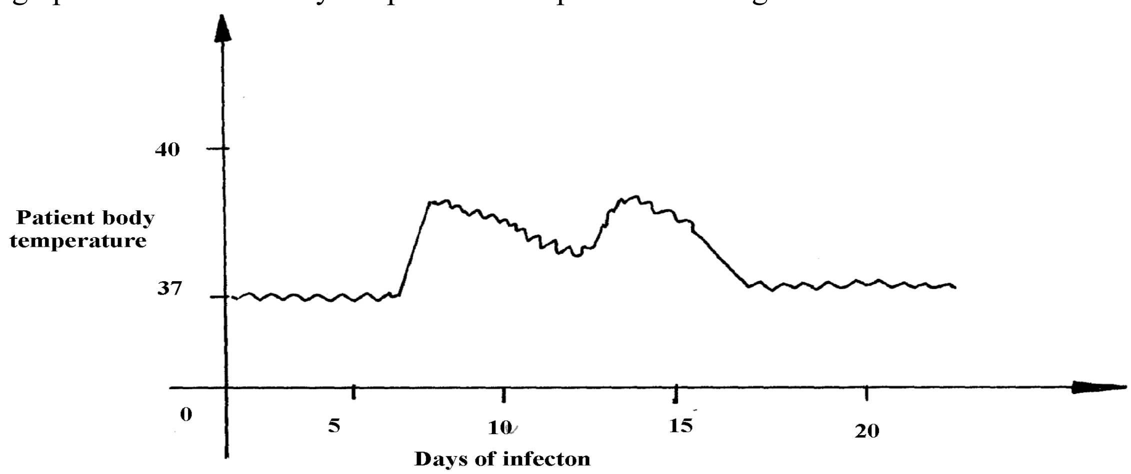
1. a) Name the causative agent of cholera.

b) Name the intermediate hosts in the life cycle of the following parasites;
i) *Ascaris lumbricoides.*

ii) *Schistosoma haematobium.*

c) How does the parasite *plasmodium vivax* gain entry into its host?

2. The graph below shows body temperature of a patient suffering from malaria



(a) What symptom of the disease is shown in the graph?

(b) Name the organism that causes malaria

(c) Suggest **one** method of controlling spread of malaria

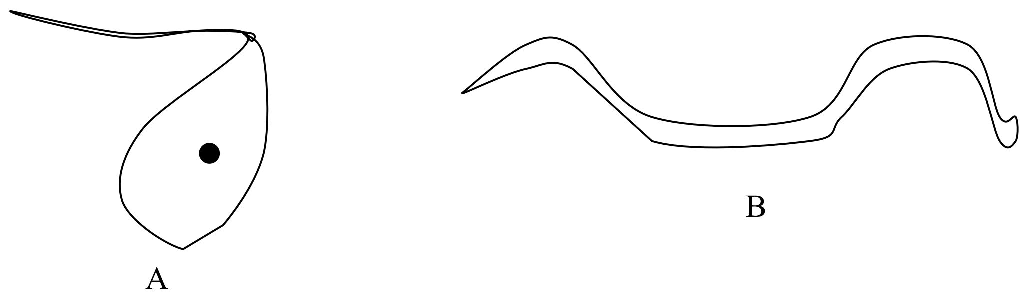
3. Name the causative agent of typhoid
4. Malaria is a common disease in Kenya:-
(a) What causes the disease?

(b) State **one** control measure of the disease

5. a) Name the causative agents of the following disease in humans:-

i) **Typhoid;**
ii) **Amoebic dysentery;**
b) Name the disease in human cased by plasmodium falciparum

6. Below are diagrams of disease causing micro-organisms. Use them to answer the questions that follow:



(a) State the kingdom to which they are found.

(b) Name the diseases caused by the organisms: A and B

(c) State **one** way in which the disease named for organism B can be prevented

7. Explain why it is important to go for voluntary counseling and testing (VCT) on HIV/AIDS

8. Name **one** human disease caused by each of the following parasites.

(a) Plasmodium falciparum…………………………………………………

(b) Entamoeba histolytica ………………………………………………….

SECTION I & II MARKING SCHEME

1. Classification I &II

1. arachnida; crustacean;

2. - Body is covered by fur or hair;

   - Have mammary glands (for milk production);
   - Have external earlobes;
   - Have highly developed brain;
   - Have muscular diaphragm that have sweat glands;
   - Have muscular diaphragm (that thoracic cavity from abdominal cavity); (first three)

3. – Two names i.e first genus and second species;

   - Genus names starts with capital letter while species starts with small letter;
   - Both names are written in italics, when printed or underlined when types or handwritten;

4. (a) Kingdom Monera;

   (b) Producing antibiotics; vaccines; hormones and in producing transigenic organisms in modern technology;

5. Chordata;

6. a) Fungi;
   b) Sporulation;

7. Prothoraic glands disintegrates hence no production of ecdysone / moulting hormone

8. Zoology;

9. - Mushrooms used as food;
   - penicilium are used to make antibiotic;
   - Yeast is used in brewing and bread baking;

10. Sub-division – Angiospermaphyta;

    Class – Dicotyledonae;

11. arachnida;

12. (a) (i) Fungi/mycophyta:
(ii) Non— green/ lacks chlorophyll;
- Body made up of hyphae/ mycelia;
(b) (Asexual) reproduction: OW WTE

13. (a) taxonomy is the classification of living organisms on their similarities and difference observed
(b) (i) Rottus norvegicus (1mk) (Genus name MUST begin with capital letter and be underlined separately)
(ii) Genus – Rattus;
Species – norvegicus;

14. - A segmented body;
- A hard exoskeleton;
- Jointed legs;

2. The cell – structure & functions of organelles

1. a) Lysosomes;

b) Contractile vacuoles;

2. (a) Make cells visible;
(b) Prevent distortion of cells;

3. Diameter of field of view

$= 4mm \times 1000mm = 4000\mu m;$

Size of each cell $= \frac{4000}{20}$

$= 200\mu m;$

4. a)Manufacture of ribosomes;
b) encloses cell contents; regulate movement of materials in and out of the cell;

5. Protein:

Nucleic acid (DNA – RNA);

6. (i) Mg = O.L.M x E.L.M;

= 100 x 5
= x500;
(ii) x 500 = 5 x 10,000 = 50000mμ
x 1 = ?
$= \frac{1 \times 50{,}000}{500}$;
= 100micrometer;

7. a) mitochondria;

b) -has cristae/inner membrane highly folded to increase surface area; for respiration.
-Has matrix medium for respiratory activities; (reject (b) if (a) is wrong.)
-Has matrix medium for respiratory activities; (reject (b) if (a) is wrong.)

8. a) nucleolus;

b) Centrioles;
c) nuclear membrane/pore;

9. a) catalyses the breakdown of toxic hydrogen peroxide; to harmless water and oxygen in active tissues;

b) Low temperature;

10. a) i) Nucleus.

ii) Formation of RNA / ribonucleic acid;
Formation of ribosomes;
b) i) Contractile vacuole;
ii) Lysosomes;

11. Sensitive to change in temp; sensitive to changes in PH; has both negative and positive charges;

12. a) Cellulose;

b) Store sugars, salt and food; carry out osmoregulation by inducing osmotic gradient that bring about water movement; maintain the shape of the cell;
c) Cell wall; and chloroplast;

13. Study of internal and external parts of the body of an organism; Study of the living organisms and their chemical composition;

14. a) Synthesis of proteins;

    b) Site for photosynthesis;

15. a) $\frac{\text{Length of drawing ;}}{\text{Length of object}}$

16. (a) Ribosomes:- Protein synthesis(1mk);
    (b) Centrioles – Spindle formation during cell division ;
    - Form cilia and flagella

17. (a) cellulose;

    (b) Lipoproteins/lipids and proteins;

18. - No organized nucleus;

    - Organelles not bound by membranes;
    - Lack mitochondria;

19. (a) X : Chloplasts;

    Y : Vacuole /sap vacuole;

    (b) More on the upper side to obtain optimum light intensity/ in bright light, they move away to avoid bleaching/ in dim light they move towards the source of light for maximum absorption of light;

20. Cell diameter = $\frac{\text{field of view in menometer}}{\text{Number of cells under the field of view}}$

    $\frac{3.5\text{x}1000}{8}$; $\frac{3500;}{8}$

    $= 437.6\mu m = 438\mu m;$

21. i) Arachnida

    ii) - Exoskeleton
    - Jointed appendages
    - Segmented body
    - Moulting;

22 a) Magnification – Ability of a microscope to enlarge tiny objects

Resolution – Ability of a microscope to separate between two tiny structures under magnification to appear distinct

b) Mounting – The placing of prepared slide on stage of a microscope;
Staining – Use of chemical stain on specimen for clear observation

23. (a) Golgi bodies/Golgi apparatus;
(b) Lysosome(s):
(c) Ribosomes;

24. (a) Make the sections transparent:
(b) To produce thin sections/ Not to distort the cells:
(c) To distinguish between different parts/organelles of the cells:

25. - Magnify the object further;

- Concentrates light onto the object;

- Controls amount of light illuminating the object;

26. Size of one cell = $\frac{\text{diameter of field view}}{\text{No. of cells arranged across the diameter}}$

$= \frac{2000\mu m;}{10 cells}$

$200\mu m = 0.2mm$

N/B = $1\mu m = 0.001mms$;

27. (a) To make the specimen /section more visible
(b) To allow light to pass through for easy viewing

28. Animal cell;

29. a) Stores hydrolytic enzymes for destruction of worn out organelles/ cells/ tissues/ digestion of bacteria/ pathogens; Acc digestion of food/ accept autolysis
b) Processing/ packaging synthesized and transporting of packaged cell materials; production of lysosomes/ secretions of packaged material;

30. Insecta; Reject insects/ exopoda

31. a) magnifying the image of the specimen;

b) Objective lens brings the image into FREE and magnifies it;

32. a) Mitochondria

b) early production/ respiration;
c) Increases surface area; for attachment of respiratory enzymes;
d) Nerve cells; skeletal muscles; cardiac muscles

3. Cell Physiology – Osmosis, Diffusion and Active transport

1. a) A- The strip increased in length/ size; B - Decreased in length/ size;
b)The sugar solution was hypotonic to the cell sap strip A; it gained water by <u>osmosis</u> hence increasing in length;

2. (a) The potato cup will be filled with solution;
(b) The solution in the potato cells is hypertonic to the water; hence water moves into the cell by osmosis; this makes the solution in the neighbouring cells to be hypertonic to the outer cells;
hence water moves from cell to cell until it eventually enters the potato cup;

3. (a) (i) Will lose water by osmosis and become plasmolysed;

4. Diffusion;

Osmosis ;
Active transport ;

5 a) 3.0 + 3.1 + 3.2 = 9.3 g;

Average = $\frac{9.3}{3}$ = 3.1g;

b) The cell sap had a higher concentration of solutes than distilled water, water therefore moves from the environment to the cell by osmosis ;

6. (a) red blood cells placed in a hypertonic solution and as a result lost water to the surrounding thorough osmosis hence shrunk/crenated ;
(b) Appearance of that cell if subjected to the same condition

7. a) Haemolysis
b) Plant cell will lose water the cell sap to the outside solution by osmosis; the cell becomes plamolysed/ flaccid; but it will retain its shape due to rigid cell wall;

8. a) Haemolysis ;

b) The plant cell will draw in water molecules by osmosis; it will swell and become turgid; but it will not burst because of the presence of cellulose cell wall;

9. i) Spermatozoon

- Tail – For swimming in vagina tract
- Numerous mitochondria – for provision of energy for swimming
- Streamlined – to reduce friction during movement
- Haploid nucleus – for fertilization of haploid ovum

Palisade mesophyll cell
- Numerous chloroplasts for photosynthesis
- Narrow and cylindrical – packed in small space
- Large sap vacuole for storage of manufactured food;

10. a) Prophase I Reject prophase alone

b) Homologous Chromosomes side by side or Bivalency

c_)

Mitosis	Meiosis
One phase	Two phases
Diploid daughter cells	Haploid daughter
No chiasmata formation	Chiasmata formation; Any two correct

Trophism	Tactic response
Growth is involved or brought about cell division	Locomotary
Slow	Fast

Set -up		Number of red blood cells	
	Sodium chloride concentration	At start of experiment	At the end of the experiment
A	0.9%	Normal	No change in number
B	0.3%	Normal	Fewer in number

11. A-no change in; number because 0.9% sodium chlorine solution is isotonic to RBC/blood;
B-fewer in number because 0.3% sodium chloride solution is hypotonic to RBC/blood therefore some water was drawn in to RDC by osmosis ;leading to haemolysis/boosting of

RBCs

b)i)number will not change;
ii)RBC will appear small in size/wrinkled/crenated/shriveled/shrink; 1mk
Rej. Flaccid/flabby/plasmolysed

12. (i) Paranchyma;
(ii) Collenchyma;
(iii) Xylem: and sclerenchyma

13. (a) X – hypotonic solution;
✓
Y – hypertonic solution;
✓
(b) A – haemolysis;
✓.
B – crenation /laking;
✓
(c) The cell will maintain/retain its normal shape.

14. Absorption of mineral salts by root hairs from the soil; Translocation of food from leaves to other parts of the plant; movement of salts from one cell to the next;

15. (a) (i) Increased in length, absorbed water through osmosis, (since cylinder cells were hypertonic/ at higher concentration) and become turgid.
(ii) Reduced in length, cylinder host water to the hypertonic sucrose solution/become flaccid.
(b) (i) No change in length
(ii) Cells are dead and cannot carry out osmosis.
(c) - opening and closing of stomata
- Support in plants
- Movement of water from cell to cell
- Feeding in insectivorous plants
- Absorption of water by root hairs
- Absorption of water in the intestines
- Reabsorption of water in kidney nephron.

16. (a) (i) Nucleus

(ii) Maintain the shape of he cell providing support to herbaceous plants; stores sugar and salts; (mark first one)
(b) $\frac{0.5 \times 100}{8}$; 62.5µm;
(c) Hypotonic solution;
Accept -highly concentrated salt/sugar solution

(d) The potato cell sap were lowly concentrated than the surrounding solution; hence lost water molecules by osmosis through the semi permeable membrane to become plasmolysed;
(e) Re-absorption of water from the kidney tubules/ hence important in osmoregulation;

17. a) transpiration

b) prevent evaporation of water from the surface
c) the level of water dropper
d) i) faster drop in water level ii) no change in water level;
iii) slower/very slow drop in water level;
e) another set up using a leafless twig;

4. Nutrition in (a) plants (b) animals

1. a) Condensation;
b) water;

2. (a) Guard cells;
(b) Cells walls are thicker on the inner side then the outer side; which enables them to pull inwards when the cells are turgid; contains chloroplasts that are able to phosynthesize and produce sugars which enable them to absorb water; (any two points)
(c) Accumulation of carbon (IV) oxide in the leaf forms a weak carbonic acid; lowering the pH which favours conversion of sugar to starch; causing the guard wells to lose turgidity; and close;

3. (a) Stomata on the epidermic were blocked; thus no carbon (IV) oxide entered the heat; therefore photosynthesis did not take place;

(b) Respiration; Excretion/ transpitation;

4. (a)

Etiolated plant	Normal plant
- Yellow leaves/stems - small leaves - long inter-nodes and thin stems - weak stem/feeble stem	- green leaves/stems; - large leaves; - short internodes and thick stem; - strong /firm stem;

(b) Enables plants to grow faster towards light for photosynthesis;

5.

PLANTS	ANIMALS
- Make their own food through the process of photosynthesis	- Depend on plants and other animals for food;
- They do not move from one place to another	- They move from one place to another;
- Respond slowly to stimuli	- Respond faster /quickly to stimuli;

6. They have thick inner membrane and thin outer membrane to allow them to bulge outwards when turgid to open stomata; Have numerous chloroplasts, to carryout photosynthesis, forming sugars to control opening and closing of stomata;

7. Reaction A – condensation;

Enzyme Y – Sucrose;

8. - To emulsify fats;

- To provide an alkaline condition for enzyme activities;
- To provide an alkaline condition for enzyme activities;

9. Have stomata on upper surface;

- Large leaf surface to increase surface are for absorption of light;
- Presence of aerenchyma tissues, allows them to float on water hence accessing sunlight;

10. (a) – Protease;

- Lipase

(b) At 35^{o}C optimum temperature for enzyme to act; at 15^{o}C enzymes in active since temperature is low;

11. a) Goiter;

b) Scurvy;

12. Enzymes – Thrombin; Thromboplastin/ Thrombokinase;

Metal ion – Calcium ions;

13. a) Peristalsis;

b) Circular and longitudinal muscles on the wall of oesophagus and intestines contract alternately;

c) Roughage;

14. Long gut / many chambers to provide large surface area for digestion; bacteria in rumen has enzyme cellulase which digest cellulose (to glucose/ sugars).

15. Concentrated of the solutions separated by a semi-permeable membrane; existence of concentration gradient; temperature of the solution;

16. i) Pancreas; ii) Insulin;

17. a) Roughage;
b) Water, vitamins, mineral salts;

18. Photolysis – Splitting water into H^+ and oxygen gas;
- Synthesis of ATP to be used during dark stage;
- Synthesis of chlorophyll necessary for photosynthesis;

19.

Guard cells	**Other epidermal cells**
- Have chloroplasts/photosynthesize - Have thick inner walls/thin outer walls Bean shaped	- No chloroplasts/do not photosynthesize - Walls uniformly thickened block shaped (any correct pair)

20. (i) Biliverdin ; Bilirubin ;
(ii) Emulsify fats;

21. a) Involuntary movement of food along the alimentary canal
b) Rhythmic contraction and relaxation of the circular and longitudinal muscles along the gut;

22 . a) i) Chloroplast;

ii) Mitochondrion;

b) Similarity — Both have a double membrane;

Difference Chloroplast Mitochondrion;
- Grana Cristae;

- Stroma Matrix;

23. a) HCl — to hydrolyse complex sugar to simple sugar

    $NaHCO_3$ — To neutralize the HC1
    b) Disaccharides;
    c) i) Glucose;
    ii) Sucrose;
24. a) Sensory neuron;

    b) Cell body is off the axon;
    c) A — Conduct coming signals / Receives impulses;
    B — Receives impulses rough dendrites / coordinates the nerve cell;
    D — produce myclin sheath that protects and insulates the axon;

25. a) Increases surface area for attachment of respiratory enzymes;

    b) i) Intergrana;
    ii) Accept site 4 photolysis; contains chlorophyll pigment absorbs light;

26. a) Increases surface area of fats for purpose of digestion;

    b) Accept any two correct
    - Destroys any ingested pathogens;
    - Provides acidic media for protein digesting enzymes (pepsin);
    - Converts/ activates pepsinogen inactive form to pepsin;

27. Poison acts as competitive inhibitor for active site of respiratory enzymes; energy production
    for active transport of nitrates is impaired;

28. Rhizobium bacteria benefits by getting Shelter & carbohydrates;
    - Leguminous plant obtains nitrates fixed by the bacteria;

29. - Enzymes amylase digests starch to maltose

    - Mucus lubricates food

30. They are converted to starch; then stored in organs and tissues;

31. -Guard cells have chloroplast;
    -They are bean shaped;

32. Oxygen-releases to the atmosphere or used by plants for respiration;
    Hydrogen-enter dark stage, where it combines with CO_2 to form simple sugar;

ATP- provide energy during the combination of hydrogen atoms with CO_2in dark stage;

33. a) to investigate the effect of boiled saliva on starch/to show the effect boiled/denature enzyme amylase has on starch;
b) A-brown colour/colour of iodine persists;
B- blue black/blue/dark colouration;
c) A-starch has been digested/starch has been broken down/amylase hydrolyses starch hence no colour changes;
B-enzymes/amylase denatured hence no starch digested;

34. a)A-condensation;
B-hydrolysis;
b)Duodenum; (any correct Rj .wrong spelling)
-ileum;

35 i)stroma
ii)side of light reaction of photosynthesis /site of water photosynthesis and adenosine triphosphare production (ATP)

36. (i) (Vitamin D/calciferol;
(ii) Prevents rickets/Osteomalacia;

37. a) Schistosomiasis/ Bilharzia:
b) -Has suckers for attachment to the host:
- Has secondary host/snail to increase its chances of survival:/increase chances of transfer to several hosts;
- Its larvae/Eggs produces lytic enzyme to soften the hosts tissues hence allow prevention into the host:
- Larva covered with cysts to remain dormant for a long time;
- Goes through various forms of lifecycle/miracidia. cercariae and redia to make it difficult to eradicate/increase chance of survival/transmission;
- Adult produces chemical substances to cover the body to protect it against hosts defence mechanism;
- Separate sexes to ensure dispersed eggs are fertilized before shed into blood vessels.

38. (a) (i) Stomach
(ii) Presence of hydrochloric acid to provide acid conditions

39 (a) To investigate the effect of heat on salivary amylase.
(b) A – The brown colour of iodine was retained because the starch was digested by enzyme

amylase in the saliva;
✓ 1.
B – The colour changed to blue black/black; because amylase in the saliva was denatured by heat;

40. (a) (i) stroma;
✓
(ii) Granum;
✓
(b) – Provide energy – ATP;
- Provide H^+ - ves H_2 GAS /atoms;

41. Midnight – There was no photosynthesis at night; and carbon IV oxide was not used hence the high concentration;
Noon - Carbon IV oxide was used in photosynthesis and therefore CO_2 concentration dropped.

42. - By increasing the enzyme /substrate concentration;
- By increasing the temperature below the optimum upto the optimum temperature;
- Providing suitable /favourable /optimum pH.

43. (a) - Mode of feeding is herbivorous. Reject Herbivore
- Absence of upper incisors but have hony pad
(b) 30

44. Small mammals have large surface area to volume ratio; hence lose heat quickly to environment; to replace the heat , lost, their metabolism is high making them to feed more frequently

45. - Plants are able to synthetize their own food
- Plants are able to use pollination rather rhan moving to seek mating partners
- Use seed and fruits dispersal to colonize new habitats (3x1=3mks)

46. a) A- Rhizome
B- Adventitious roots
(b) The liverwort body form is thalloid while the fern has 3body parts, roots, stem and leaves

47. The break down of glucose into pyruvic acid

48. (a)

Monosaccharide	Polysaccharides
- Are soluble in water	- Are insoluble in water

- Form sweet tasting solution	- Do not have a sweet taste
- Reduce Copper(II) ions in benedicts solution to Copper (I) ions when heated together	- They do not reduce
- Are crystalizable	- Are not crystallizable

(b) Peptide

49. H^+/H atom; rej H2/Hydrogen gas
- ATP/energy

50. Absorption of water; accept absorption of salts/ calcium/ iron; secretion of mucus;

51. a) To show that light is necessary for photosynthesis;

b) Only the uncovered areas turned blue- black with iodine; the part covered with aluminum foil did not receive light and thus could not carry out photosynthesis;

52. a) As the temperature increases, the rate of the reaction also increases; this happens because an increase in temperature increases molecular movement, thus increasing the chances of collision between the enzyme and substrate molecules;

b) X – is the optimum temperature/ It is the temperature at which the reaction proceeds

53. Nitrogen;
Magnesium;
Iron;

54. a) A- Hook;
B – Sucker;
C – Youngest proglottid;

b) Intermediate host – pig;

55. a) A – Villus
B- Lacteal
b) A __ Increases surface area for maximum digestion and absorption;
B – Absorption of fatty acid and glycerol;
c) - Final digestion of undigested foods;
- Absorption of soluble end products of digestion;
d) Produces bile juice which contains bile salts that emulcify fat;

e) Produces insulin and glucagons hormones;
Reject if only one hormone is mentioned

56. (a) Rapid increase (in water of photosynthesis) due to increase in concentration of CO_2
(b) Constant rate/no increase rate and no decrease, other factors /light/temperature water
become limiting/inadequate.
(c) chlorophyll traps energy.
Light energy react water into hydrogen ions and oxygen/photolysis.
Hydrogen is picked by hydrogen.
Acceptor/NAD/NADP (and becomes reduce, * ACCEPT NADPH,NADPH
ATP adenosine triphosysbate formed.

57. (a) Compensation point
(b) (i) There is no net uptake or release of Carbon (VI) oxide by the plant;
(ii) The rate of respiration and photosynthesis in the plants are equal; therefore all the
Carbon (VI) Oxide released during respiration is used in photosynthesis;
(c) At light intensity beyond/above X, the rate of photosynthesis is higher than the rate of
respiration; and this requires a net uptake of Carbon (IV) Oxide (to sustain the increasing
rate of photosynthesis);
(d) Growth would cease because all the products of photosynthesis would be utilized in
respiration;
(e) The plant will take up oxygen from the surrounding air since the rate of respiration is
higher than the rate of photosynthesis;

58. (a) Broad and flat to absorb maximum light
Have chloroplast with chlorophyll to trap light.
Transparent cuticle to allow light to pass through
(b) X – Carbon (IV) Oxide
Y – Oxygen
(c) Xylem – Transports water
Phloem – Sugars out of the leaf
(d) Starch is insoluble in water, hence osmotically inactive; This reduces effect on absorption of
water.

59. a) breakdown of complex food, substance; into simple diffusible substances;
b)intestines relatively long/coiled /folded ;this allows food enough time for absorption.
Intestines long /have villi; to increase the surface area for absorption and digestion ;

The walls have glands which secrete enzymes for digestion;(examples of correct enzymes
e.g. Maltose, sucrose lactose etc).some glands /goblet cells also produce mucus; which protects
The intestinal wall from autodigestion/being digested; and reduce friction;
Intestines have opening of ducts which allows bile pancreatic juice into the lumen;
The intestines have circular and longitudinal muscle, whose contraction and relaxation/peristalsis;
Leads to mixing of food with enzymes/juice; facilitating rapid digestion and help push food along the gut; the intestines are well supplied with blood vessels to supply oxygen/ remove digested food from an efficient absorption and transporting system to move the food away from the small intestines;
Have lacteal vessels for transport of fat/lipid; have thin epithelial lining; to facilitating fast absorption /diffusion;
Note. Allow increases in surface are for absorption only once

60. (a) To investigate the rate of photosynthesis;
(b) It is used to draw the bubbles of gas through the apparatus;
(c) (i) Oxygen gas;

(ii) $6\ CO_2 + 6H_2O \xrightarrow[\text{Chlorophyll}]{\text{Light energy}} C_6H_{12}O_6 + 6O_2$;

Acc. Either word or chemical equation
If chemical, must be balanced, symbols capital.
or Carbon (IV) Oxide + water $\xrightarrow[\text{Chlorophyll}]{\text{Light}}$ Glucose + Oxygen;

(d) - Optimum
- Optimum PH
- Absence of inhibitors.
- Presence of co-factors or co-enzymes.
- Low substrate concentration.

(e) - To minimize temperature changes.

61. a) A- Rhizome
B- Adventitious roots
(b) The liverwort body form is thalloid while the fern has 3body parts, roots, stem and leaves

62. The break down of glucose into pyruvic acid

63. (a)

Monosaccharide	Polysaccharides
- Are soluble in water	- Are insoluble in water
- Form sweet tasting solution	- Do not have a sweet taste
- Reduce Copper(II) ions in benedicts solution to	- They do not reduce

Copper (I) ions when heated together	
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Magnesium;
Iron;

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become limiting/inadequate.
(c) chlorophyll traps energy.
Light energy react water into hydrogen ions and oxygen/photolysis.
Hydrogen is picked by hydrogen.
Acceptor/NAD/NADP (and becomes reduce, * Accept NADPH, NADPH
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Carbon (VI) Oxide released during respiration is used in photosynthesis;
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respiration; and this requires a net uptake of Carbon (IV) Oxide (to sustain the increasing
rate of photosynthesis);
(d) Growth would cease because all the products of photosynthesis would be utilized in
respiration;
(e) The plant will take up oxygen from the surrounding air since the rate of respiration is
higher than the rate of photosynthesis;

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Have chloroplast with chlorophyll to trap light.
Transparent cuticle to allow light to pass through
(b) X – Carbon (IV) Oxide
Y – Oxygen
(c) Xylem – Transports water
Phloem – Sugars out of the leaf
(d) Starch is insoluble in water, hence osmotically inactive; This reduces effect on absorption of
water.

74. a) breakdown of complex food, substance; into simple diffusible substances;
b)intestines relatively long/coiled /folded ;this allows food enough time for absorption.
Intestines long /have villi; to increase the surface area for absorption and digestion ;
The walls have glands which secrete enzymes for digestion;(examples of correct enzymes
e.g. Maltose, sucrose lactose etc).some glands /goblet cells also produce mucus; which protects
The intestinal wall from autodigestion/being digested; and reduce friction;

Intestines have opening of ducts which allows bile pancreatic juice into the lumen;
The intestines have circular and longitudinal muscle, whose contraction and relaxation/peristalsis;
Leads to mixing of food with enzymes/juice; facilitating rapid digestion and help push food along the gut; the intestines are well supplied with blood vessels to supply oxygen/ remove digested food from an efficient absorption and transporting system to move the food away from the small intestines;
Have lacteal vessels for transport of fat/lipid; have thin epithelial lining; to facilitating fast absorption /diffusion;
Note. Allow increases in surface are for absorption only once

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(b) It is used to draw the bubbles of gas through the apparatus;
(c) (i) Oxygen gas;

(ii) $6\ CO_2 + 6H_2O \xrightarrow[\text{Chlorophyll}]{\text{Light energy}} C_6H_{12}O_6 + 6O_2$;

Acc. Either word or chemical equation
If chemical, must be balanced, symbols capital.

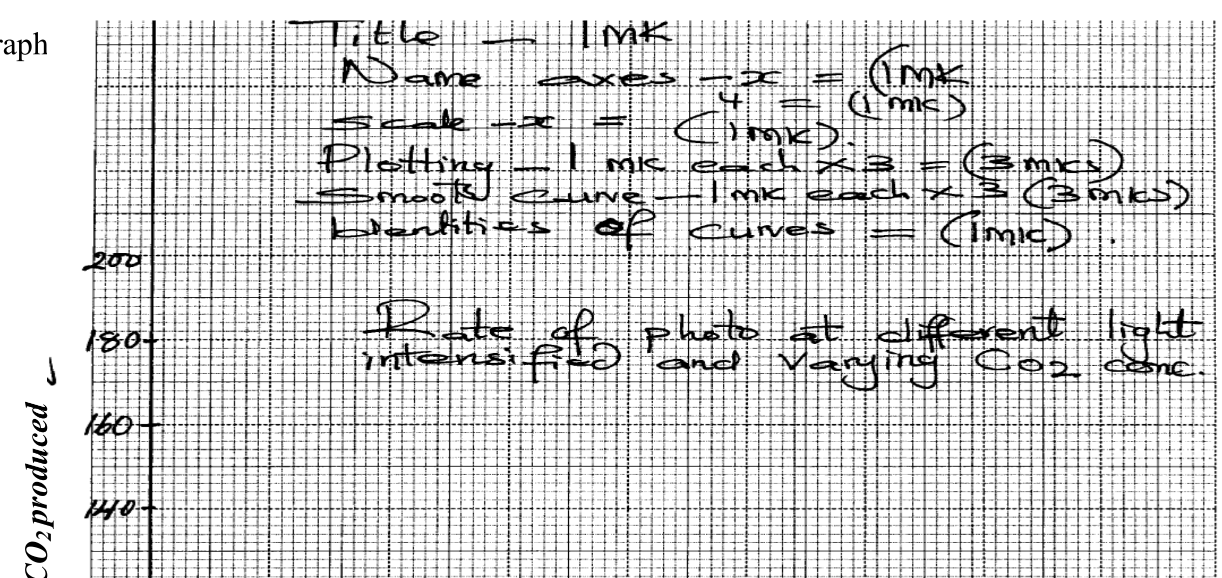
or Carbon (IV) Oxide + water $\xrightarrow[\text{Chlorophyll}]{\text{Light}}$ Glucose + Oxygen;

(e) - Optimum
- Optimum PH
- Absence of inhibitors.
- Presence of co-factors or co-enzymes.

(e) - To minimize temperature changes.

76. (a) The bacteria ad exhausted the available food materials and they died;
(b) They multiply very fast as they feed on the substances; release toxic waste on food then die
there causing food spoilage
(c) fungi;
(d) – Speed up recycling of matter in the ecosystem;
- Breaks down /decompose dead complex organic matter

77. (a) Graph



title -1mk
labelled axes) –
plotting –)
curve – (free hand) (Continuous not dotted - Rej. curve if joined with a ruler

(b) As carbon dioxide concentration increase rate of photosynthesis also increases up to a limit beyond which there is no increase even of CO_2 concentration is increased.

- Increase in light intensity increased also rate of photosynthesis up to a limit

(c) - Have a darker colour/ light sensitive chlorophyll which to maximumly absorb any light rays penetrating water

- They either float next to water surface to be exposed /closer to light or floats on water surface.
- Have thin or no cuticle to allow easier diffusion of dissolved CO2

(d) – Carbon (IV)- Oxide concentration;
- Light intensity ;

78.

- Is relatively long/ cooled/ folded to allow food (enough) time/ increase surface area for absorption of digested food and for digestion
- Lumen has projection called villi; villi has projections called microvilli; to increase surface area for absorption
- Walls have glands which secret enzymes for digestion; e.g. maltase/ sucrose/ lactase/ enterokinase/ peptidases

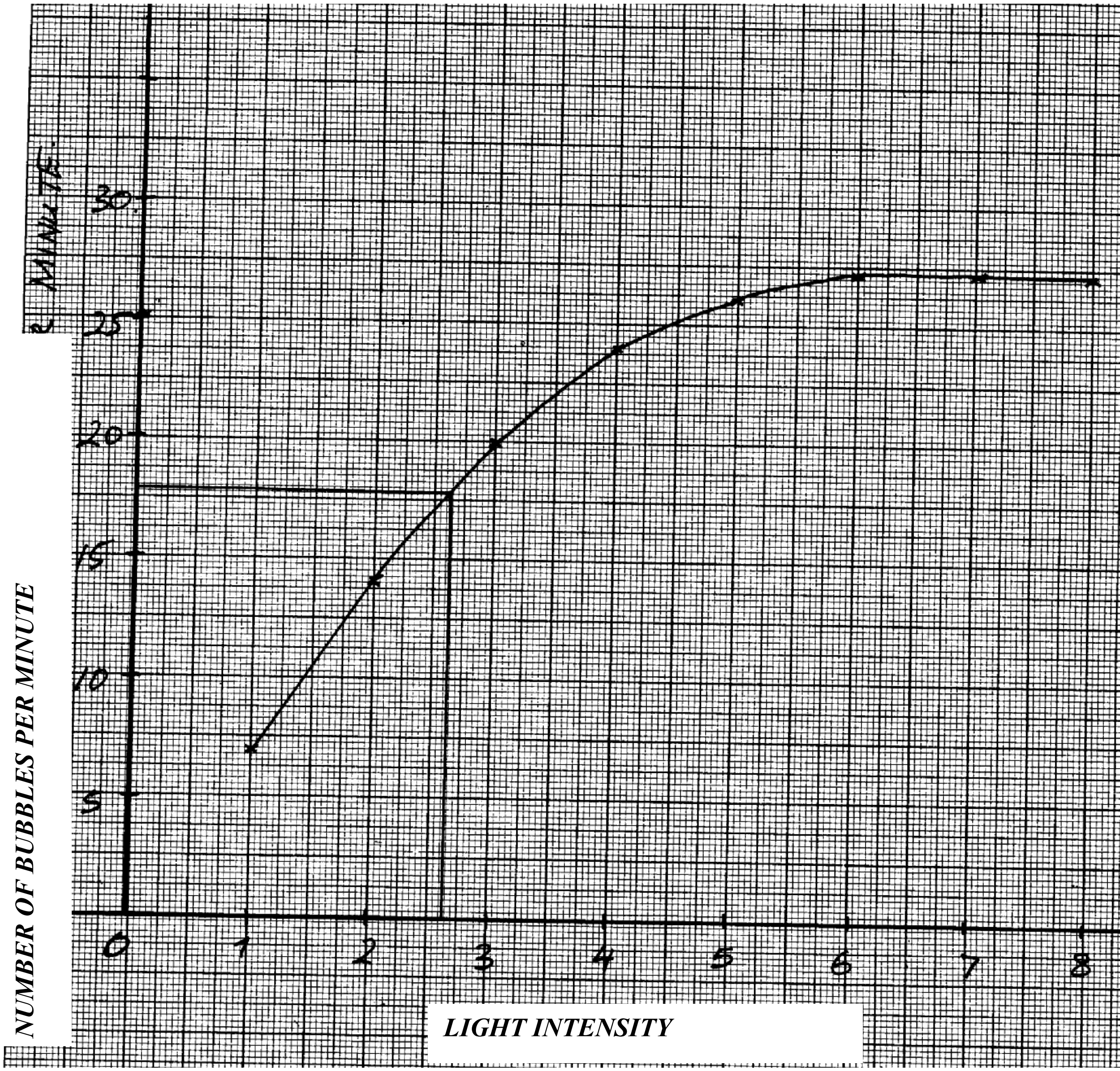
- Some glands/ goblet cells produce mucus; which protects the intestinal wall from being digested and also reduce friction
- Have openings of ducts which allow bile/ pancreatic juice into the lumen
- The intestines have circular and longitudinal muscles; whose contraction and relaxation/ peristalsis leads to mixing of food with enzymes/ juices; facilitating rapid digestion; and helps push food along the gut
- Intestines are well supplied with blood vessels/ highly vascullarized; to supply oxygen/ remove digested food
- Lacteal vessels; transport fats/ lipids
- They have thin epithelia; to facilitate fast/ rapid absorption/ diffusion

79. (a) To destarch the plant leaves;

(b) (i) To absorb carbon (iv) oxide in the flask;
(ii) To enrich the air in the flask with carbon(iv) oxide;
(c) (i) leaf M – Sodium Hydroxide absorbed Carbon (IV) oxide in the flask;
- No photosynthesis occurred and so the leaf retained the brown colour of Iodine;
(ii) Leaf N – Sodium hydrogen carbonate enriched the flask with carbon (IV) oxide;
- Photosynthesis occurred and starch formed reacted with iodine to give the leaf the characteristic blue-black colour;
(d) Conical flask covered with aluminium foil and no sodium hydroxide or sodium hydrogen carbonate;

80. a)Graph

b) i) 2.5 – 2.7; i.e. 2.6 + 0.1
ii) 4.5 + 0.1
c) - Volume of CO_2 consumed/ volume of O_2 liberated
- Change in dry mass (due to photosynthesis);
d) - Photolysis of water
ATP synthesis
e) i) Rate of photosynthesis very low
Enzymes inactivated
ii) Rapid rate of photosynthesis
Optimum temperature for enzyme reaction
iii) Very low rate of photosynthesis
Enzymes denatured
f) Chlorophyll concentration (in leaves)
CO_2 concentration
Water availability



79. a) i) to kill cells/expose starch graduals/stop biology processes;

ii) to decolourise the leaf/to dissolve chlorophyll;

b)i) leaf retained brown colour of iodine;

ii) starch was absent (since no photosynthesis had taken place);

c) to investigate the necessity of light in the process of photosynthesis;

d) to soften the leaf and wash off alcohol;

e) leaf with some parts/patches looking chlorophyll;

f) water + carbon(iv)oxide $\xrightarrow[\text{chlorophyll}]{\text{light}}$ Glucose +oxygen; (carbohydrate)

5. Transport in (a) plants (b) animals

1. On a hot sunny day blood vessels are dilated hence more blood is lost; on a cool chilly day the
blood vessels have constricted hence less blood flows on the surface of the skin;

2. - Biconcave disc shaped to increase surface area for gaseous exchange;

- Have no nucleus to increase room for the package of red blood cells;
- Numerous in number to increase surface area for the transportation of oxygen
- Have haemoglobin which has a high affinity of oxygen;

. - Cytoplasmic filaments/strands along which food streams;
- Companion cells have mitochondria that provide energy for translocation;
- Sieve plates with sieve pores thorugh which cytoplasmic filaments pass.
- Photoplasmic material pushed on the sides to create lumen space for translocation;

4. (a) Chitin;
(b) Lignin:

- Root pressure;
- Cohesion – adhesion forces

5. Transpiration pull;

6. (a) – transpiration pull;

- Cohesion and adhesion;
- Capillarity;
- Root pressure;

(b) Phloem;

7. (a) (i) (15 x 2) =30;
(ii) Carnivorous; reject carnivore
(b) – To lubricate the food;
- To protect the alimentary canal wall from digestion by protein digesting enzyme /protoelytic enzyme;
- Make the food adhere together during swallowing;

8. (a) Thoracic vertebrae;
(b) B – Neural canal;
C – Centrium;

(c) For attachment of back muscles;

9. - Growing regions (e.g meristems); storage organs for storage (e.g stems, roots, fruits)
- secretory organs (e.g. flower nectarines);

10. A, AB, B, O; for all blood groups

11. (i) Efficient diffusion of substances e.g. food, gases and waste products;

(ii) Efficient transport of food/gases/waste products to and from cells;

12. (a) Transpiration;

(b) (i) The level of water in the boiling tube reduced significantly;
(ii) The level of water did not reduce;

13. Aerenchyma tissues have large and numerous air spaces; hence facilitation buoyancy;

14.

Arteries	Veins
- Thick muscular - No valves (except pulmonary artery and aorta at the base - Narrow (small) lumen	- Thin muscular walls - valves present; - Wide lumen (large) lumen;

a)

b) Arteriosclerosis; reject Artheroma

15. Transpiration pull; Capillarity;
Cohesion and adhesive forces; Root pressure;

16. -numerous to increase surface area
-Biconcave to increase surface area for packaging hemoglobin alter shape to fit narrow
lumens of capillaries;
-No nucleus to increase surface area for oxygen leading;
-Have hemoglobin which has high affinity for oxygen;

17. a) Tissue fluid is a fluid / liquid found surrounding cells/ between cells formed as a result of

ultra filtration from blood while lymph is inter cellular fluid which nutrients and oxygen have
been taken and is rich in waste materials
(mark as a whole)
b) Vitamin K is needed for formation of prothrombin which is activated to thrombin which
helps in clotting of blood.

18. Open circulatory system

19. Coronary Artery;

20. a) Oxyhaemoglobin;

b) Use oxygen released from photosynthesis process;

21. Leukemia (acc. blood cancer)

22. (a) Diabetes mellitus
(b) - Symptoms of diabetes mellitus
- Passing urine frequently;
- Constantly feeling thirsty;
- Dehydration;
- Loss of weight;
- Poor resistant to infection;

23. (a) A – Tracheid; B – Vessel;
(b) - Side walls are impregnated wit lignin/deposited with lignin /walls are lignified/pressure of
lignin nucleus not enclosed by a membrane ;

24. There is high concentration of water vapour around the leaf/less space for water vapour form the leaf to occupy low saturation deficit /low diffusion gradient / the diffusion between the
concentration of water vapour in the atmosphere and the air spaces is greatly reduced.

25. a) Transports water and dissolved mineral salts; provides mechanical support due to
lignification of cells
b) Narrow lumen of vessels and tracheids – enhances capillarity forces;
Presence of pits on lignified walls follows for lateral movement of water;

They are hollow to allow uninterrupted/ continuous flow of water from roots to leaves;

26. a) Involuntary movement of food along the alimentary canal
b) Rhythmic contraction and relaxation of the circular and longitudinal muscles along the gut;

27. a) A – Hepatic portal vein; B- Hepatic artery;
b) Excess glucose must be converted to glycogen; for storage
c) Burning charcoal produces carbon (II) Oxide which combines with haemoglobin to
28. a) Oxyhaemoglobin

form carboxyhaemoglobin that is stable/ does not dissociate; reducing efficiency of
haemoglobin in carrying oxygen leading to death; Ref death alone
leaf fall;
exudation;
gutation;
transpiration

29. - Sebum – from sabaeceous glands – antiseptic ;

- Confined layer of dead cells- impenotratable by bacteria/ fungi/ viruses
- Sweat – saline and kills bacteria and viruses
30. Leukamia/ blood cancer;

31. a) inversion;
b) mustard gas/ gamma rays/ x-rays/ beta rays/colchicines;

32. (a)(i) Dicotyledonae;
(ii) Star shaped xylem/phloem between the arms o the xylem;
(b) Lignified walls to prevent it from collapsing/keep it hollow open throughout:
- Hollow/Lack cross walls for continuous flow of water and mineral salts any 1
- Narrow Lumen to enhance capillarity;

33. - Creates transpiration pull:
- Absorbs latent heat of vaporization hence cools leaves of the plant: (2marks)

34. Water absorption does not involve active transport that requires energy from respiration
facilitated enzymes ; hence no metabolic inhibition involved;

35. (a) A – Tracheids ; B – Xylem vessel;
(b) B is hollow at the middle therefore the substance flowing through it gets to their destination

faster as compared to that of A;
✓
(c) – Lignification ;
✓

36. - Antigen B;
- Rhesus antigen / Rhesus factor /rhesus protein;

37. After the first transfusion the patient would produce rhesus antibodies; second transfusion
rhesus antigen would react with rhesus antibodies; causing agglutination;

38. (a) Pseudopodium;
✓
(b) Phagocytosis;
✓
(c) White blood cells.

39. (a) Xylem vessels are hollow (lack cross walls) , hence more efficient in transporting water
than tracheids which have trapped ends with perforation;
(b) Xylem vessels are dead due to heavy lignification on their walls hence provision of support
to the plant as well preventing collapse ;

40. Transpiration is the loss of water vapour, while gutation is loss or exudation of liquid water through hydathodes

41. Support

42. – Storage of air
- For buoyancy

43. (a) A blood disorder where red blood cells appear sickle shaped
(b) Sinoatrio node/pace maker

44. a) To generate high pressure to pump blood; to all parts of the body/ to furthest distance;
b) Hydrogen carbonate (HCO_3)
Carbonic acid;

45. a) Sunken stomata form pits; in which water vapour accumulates reducing rate of transpiration
b) Water proof to reduce the rate of transpiration;

46. a) Lignin;

b) Phloem;

47. a) A – Transport of organic food substances from sites of manufacture;
B – Formation of new xylem and phloem tissues;
C – Transport of water and mineral salts from the roots to the leaves;

b) Parenchyma cell;

c) Sclerenchyma;

48. - Blood cells;
- Plasma proteins;

49. Ileum —***Absorbed***→ Blood stream → Hepatic portal vein → Liver → Hepartic vein → Inferior vena cava → Right auricle —***Systole***→ Right ventricle —***Systole***→ Pulmonary artery → Lungs → Pulmonary Vein → Left auricle —***Systole***→ Left ventricle —***Systole***→ Aorta → Renal artery → Kidney

Acc. Descriptive form   Rej. – From appoint of a mistake

50. (a) - Blood group O is a universal donor (as it donates to all other blood groups); *UGU*
- Blood group AB a universal receipient (as they receive blood from all other groups.
- Blood group A can receive blood from group O and A only.
- Blood group B can receive blood from O and B only.
- Blood group O does not receive blood from other blood groups except O.
- Compatibility of blood group
- Absence of pathogens in blood.
- The Rhesus factor matches.

(c)- When blood vessel is injured, exposed platelets rupture to release thromboplastin (enzymes); which converts prothrombin to thrombin; in presence of Ca^{2+} thrombin activates conversion of fibrinogen into fibrin; which forms mesh work of fibre in the cut surface;

51. a) Measure rate of transpiration;
b) -Assemble apparatus under water;
- Apply vasectomy between cork shoot contacts;
- Open the reservoir tap;
c) i) The air bubble will not move
ii) Water droplets will be seen in he polythene;
iii) Air bubble will move faster

52. (a) (i) $\frac{(23000 - 100)}{1100} \times 100 = 1,990\%$;

(ii) $\frac{(1300 - 400)}{1300} \times 100 = -6.92\%$;

(b) (i) At rest, the gut is more active than skeletal muscles as this is the time when digestion is taking place; more blood goes to gut to transport the absorbed food;
(ii) During strenuous exercise, skeletal muscles are more active; and a lot of blood is diverted to help it contract and relax while very little blood flow through the gut which becomes less active;
(c) During light exercise, the skin becomes more active; thus give the highest blood flow compared to other times to release excess heat, sweat and wastes.
(d) – Excess water;
- urea, ammonia, uric acid; (OWTTE)

53. a) A- Epidermis
B- Pith
b) C- Transports manufactured food/ products of photosynthesis/ translocates food
E- Transports water and mineral salts
c)

Section above	Section from root
Xylem/ phloem form around cambium	Xylem star shaped and centrally placed
Pith at the centre	No pitch
Roof hairs absent	Roof hair present
Epidermis has cuticle	Epidermis has no cuticle

54. Geographical distributed of organism;

Theory supposes that at sometime the present day continents found a large single land mass; animals migrated freely all over the land mass; the land broke up into parts which drifted from one another forming the present day continents; this drive isolated animals from common ancestry; leading to the formation of new differed species distinct; from those found in other climatically similar but separate regions.

Comparative embryology;

Embryos of different groups have been found to have similar morphological feature during their early stages of development. This similarly suggest a common ancestry

Comparative anatomy;

When comparing the firm and structure of different organism; some groups shows basic structural similarities; which suggest a common ancestry as observed in homologous and analogous structures

Homologous structures are those that have common embryonic…. But are modified to perform different functions e.g. vertebrate fore limbs

Analogy structure those that have different embryonic origin bad have evolved to perform similar functions due to exploitation of similar environment e.g. bad and insect wing)

Cell biology;

Cells of all higher organism show basic similarities in their structure and functions; cell membrane and cell organelles such as ribosomes; biological chemicals in common e.g. ALP &DNA. This strongly indicate that all cell types have a common ancestral origin

-blood pigments among also show the same ancestral origin

Comparative serology;

Analysis of blood proteins and the antigens to reveal phylogenetic relationship. Those species that are more phylogenetical reacted contain more similar blood proteins

An immunological reaction between human beings and chimpanzees produces a lot of precipitate showing a close phylogenetic relationship

-red blood cells; carry oxygen; to all parts of the body/from lungs /to tissues; transport CO_2; to lungs /from tissues;

-platelets/thrombocytes; produce in enzymes/thrumbokinace /thrumboplastin; necessary for blood clotting;

-leucozytes/W.B.C; produces antibodies for defense against disease; they also engulf foreign bodies/pathogens;

-plasma; transport nutrients; hormones; distribute heat; carbon(iv)oxide; nitrogenous waste/urea; mineral ions; fibrinogen; plasma bathes the tissues allowing for exchange of materials

Acc. Plasma proteins for fibrinogen (20)

6. Gaseous exchange in (a) plants (b animals

1. a) Alveoli

b) Thin walled/ epithelium highly vascularised, has large surfaces area; moist;

2. It does not easily dissociate; thereby reducing the capacity of haemoglobin to transport *KKE*
oxygen;

3. Smoking; Diet with lots of fats and carbohydrates/cholesterol fat; (2mks)
KKE

4. - Carbonic anhydrase; facilitates formation of /ionization of carbonic acid;
- It has haemoglobin which readily combines with bicarbonate.

5. (a) The ratio of the amount/volume of Carbon (IV) Oxide produced to that of oxygen consumed
during respiration;

(b) (i) $RQ = \frac{\text{Vol. of } CO_2 \text{ produced}}{\text{Vol. of oxygen consumed}}$

$= \frac{102}{145}$;

$= 0.703$;

(ii) Fats;

6. Gill filaments are thin/one cell thick to facilitate faster/rapid diffusion of respiratory
gases; the surfaces of the gill filaments are moist to facilitate dissolution of respiratory gases ; The gill filaments are numerous to prove a large surface area for gaseous exchange; the gills have numerous rakers that filter food/solid particles that may damage the gill filaments; The gill has a gill bar which is long and curved to provide a large surface area for attachment of gill filaments; the gill is highly vascularised to ensure efficient transport of respiratory gases;

7. (i) Cell membrane;

(ii) Lenticels;
(iii) Skin, lungs and mouth cavity;

8. a) (Moist) skin/ buccal cavity; lungs; mark the first two
b) – (oxygen) dissolves in the water film; in the tracheoles; and diffuses in to the haemolymph (along the concentration gradient)

9. a) Increased rate of breathing; increased rate of heart beat;
b) Mitochondrion;

10. - Importance of counter current flow in fish : - It maintains a steep concentration gradient

across the respiratory surface; thus ensuring there is a maximum exchange of O_2 from water to the blood;

11. Four adaptations of red blood cells

- -lack of nucleus to create large surface are for dense packing of haemoglobin required for oxygen transportation;
- Have biconcave shape to provide large surface are for oxygen transportation;
- have thin membrane to facilitate rapid diffusion of respiratory gases;
- have numerous /many haemoglobin densely packed to increase the rate of oxygen transportation;
- are pleumorphic /can change shape easily thus can squeeze through narrow capillaries;

12. a) i) Cytoplasm
ii) Pyruvic acid
b) Pyruvic acid is broken down; into ethanol and CO_2

13. - Ribcage moves upwards and outwards;

- Diaphragm muscles contracts; hence;
- Diaphragm flattens;

14. a) Process of movement of food substances from site of manufacture to other storage organs
b) - Capillarity
- Root pressure
- Transpiration pull

15. -PH of blood, plasma is not altered homeostasis maintained;

-Within RBC there is an enzyme (carbonic anhydrate) which helps in fast loading / dissociation / combination and offloading/dissociation of CO_2; (award 1st two2mks)

16. Have lenticels: for gaseous exchange:
*

17. - Moist;

- Thin epithelium; Mark 1st two

- Highly vascularised:
- Large surface area;

18. (a) Create more room/space for packing of more haemoglobin:
(b) To provide a large surface area for diffusion of a lot of respiratory gases:

19. -Increase in Red blood cell count/Total number of red blood cells;

- Increase in haemoglobin content of RBC

20. Distilled water is hypotonic to RBC (OWTTE); hence water is absorbed by osmosis; the RBC
bursts haemolysis (due to absence of cell wall)

21. (a) Numerous/ many;
(b) Long:
(b) Blood in the gill filaments flow in the opposite direction to water over the gill filaments:
to create a deep diffusion gradient; for rapid ?faster diffusion of respiratory gases:(2marks)

22. When the rubber plug is pulled there is an increase in volume and decrease in pressure in the
syringe; Therefore due to this the atmospheric pressure exceeds the pressure in the syringe case
causing air to flow in the balloon; leading to the increase in size of the balloon;

23. - Air containing oxygen from the atmosphere gets to trachea; through spiracles; on to the tracheoles from where it diffuses; to the tissue;

24. (i) Moist to dissolve respiratory gases prior to diffusion;
(ii) Thin to reduce the distance through which diffusion has to take place/to facilitate
rapid diffusion;

25. (a) Adds carbon dioxide to the water
(b) At evening the light intensity has reduced hence reduction in the rate of hotosynthesis.
(c) Water plants are able to extract dissolved carbon(IV)oxide in water (1x1=1mk)

26. (a) - Red blood cells are biconcave in shape increase surface area to pack more haemoglobin
- They are numerous for efficient transport of oxygen
- Red blood cells lack nucleus, creating large surface area to more haemoglobin
(b) – In form of hydrogen carbonate by plasma, carboamino haemoglobin or carbonic acid in plasma

27. a) A – Ciil rackers act as a screen preventing entry of food and other particles that might damage the delicate gill lamella;
B – Gill bar for attachment of gill rakers and gill filament
C – Gill filaments – the surface on which gaseous exchange take place

b) Filaments are supplied wit a dense network of blood capillaries for the efficient transport of gases;

28. .- Ventilated through spiracles on either side of the insects body;
- Trachea branches too numerous tracheoles increasing the surface area for gaseous exchange;
- Tracheoles are moist to allow gases to diffuse in solution form;
- Tracheoles membrane is very thin to provide a short distance for diffusion
- Trachea has circular rings of chitin to prevent collapsing. This keeps the air passages always open;
- Spiracles have valves to enhance movement of gases into the trachea, and also to prevent drying of the trachea;

29. - Active immunity is immunity that is produced when an animal's body reacts to an antigen by producing antibodies;
- Passive immunity is immunity that is produced when antibodies are transferred from one individual to another;

30. - Lenticels;
- Cuticles
- Mesophyll cells/ spongy mesophyll/ palisade mesophyll/ stomata/ substomatal chambers;

31. a) Ventilation
b) i) lower concentration of oxygen in high altitude areas; raises the demand of oxygen by body cells
ii) Number of red blood cells has increased; hence enough oxygen is reaching all body cells adequately
c) Has a higher capacity of transporting enough oxygen to body cells; due to higher number of red blood cells; in the body (has lower oxygen demand)
d) i) Muscle crumps; Muscle fatigue
ii) It is completely oxidized by oxygen to form water, carbon IV oxide and energy;

32. a) Red Blood Cells

- Lacks nucleus to provide greater space for packing more haemoglobin; oxyhaemoglobin;

- Thinner membrane for faster diffusion of gases through a shorter distance
- Biconacave to increase the surface area for maximum transport of gases
- Shorter life cycle for increasing more efficiency in gass transport;
- Numerous to increase the surface area for maximum transport of gases;

White blood cells - Have a lobbed nucleus to carry out engulfing and digestion process of
pathogens more effectively

Platelets - Has thromboplastic enzyme which catalyses the activation of prothrombin to thrombin during blood clotting process;

FibrinogenIt is highly sensitive to thrombin whose presence changes it into insoluble fibrin;

Plasma - Has water with a high specific heat capacity which enables it to maintain the
temperature of the body within a narrow range

- Water also dissolves and act as a medium of transport of dissolved substances;

33. (a) Stomata; cuticle; lenticels; any two

(b) Spongy mesophyll layer; Palisade mesophyll; sub-stomatal air spaces/chambers;

(c) Foliage leaf – photosynthesis;
scale leaf v - protection;
floral leaf – attraction of agent of pollination/photosynthesis;
cotyledon leaf – storage of food / photosynthesis

(d) Guard cells photosynthesize food, accumulate monosaccharide and become osmotically
active; they absorb water from neighbouring epidermal cells and stoma opens as they
expand/swell;

34. (a) Path A (Nose) has mucous lining which trap foreign particles in air; has sensitive
cells to smell in nose limit inhalation of poisonous gases; air is warmed in the nose before
reading the lungs; hair in the nose filter solid particles in the air;

(b) Has a lumen/tubular for air passage; has mucous membrane to trap foreign particles and filter
dust; Has cartilage to prevent collapsing / to keep it open; Has elastic muscles to allow
compression and flexibility;

(c) Soot/smoke particles block the passage (bronchi/alveoli) of the gases; may cause cancer
/stimulate the epithelium membrane/lining to secrete a lot of mucus which may block the

passage;

35. (a) Adaptations of the air ways (trachea and bronchi)

- The walls of the trachea and bronchi are lined by rings of cartilage; which prevent them from collapsing and keep them open for air passage;
- The inner passage of air ways is lined with mucous membrane; which contain ciliated cells; whose movements to and from the pharynx cause a sweeping action that collects mucus containing dust towards the pharynx hence preventing their entry into the air ways;
- The mucous membrane contains mucus secreting cells; which produce mucus that trap dust and pathogenic particles which would find their ways into the air ways;
- The mucous membrane has a rich supply of blood; which helps to keep the incoming air warm and moist for easy diffusion into the lungs;
- The epiglottis and other structures on top of the trachea prevent food, drinks and other soil particles from gong into the trachea during swallowing;

<u>Adaptations of the lungs</u>

- It has numerous alveoli; that provide a large surface area for efficient gaseous exchange;
- Epithelial lining between alveoli wall and the blood capillaries is thin; to provide a shorter diffusion distance for easy gaseous exchange;
- The lung is spongy and has numerous air sacs; that accommodate large volume of gases (oxygen);
- It is highly supplied with blood capillaries that transports oxygen and carbon (IV) oxide to and from the body tissues respectfully;
- Its epithelial lining is covered by a thin layer of moisture; to dissolve oxygen for easy diffusion into the blood stream;
- The lung is connected to tree – like system of tubes (the trachea, bronchi and bronchioles); that supply oxygen and removes carbon (IV) oxide from the lung;
- The whole lung is covered with the pleural membrane which is gas-tight thus changes in pressure within the lungs can occur without external interference;

N/B- Mark as a whole)

(b) <u>Opening</u>

-In the guard cells there are chloroplasts; which carry out photosynthesis in the presence of light; (in the day)

-During photosynthesis glucose is produced in the guard cells; this increases osmotic pressure; compared to the neighbouring epidermal cells; water then moves into the guard cells by osmosis; and increases their turgidity;

-The inner walls of guard cells are thicker than the outer walls; so outer walls stretch more than the inner walls causing guard cells to bulge outwards causing the stomata to open;

Closing

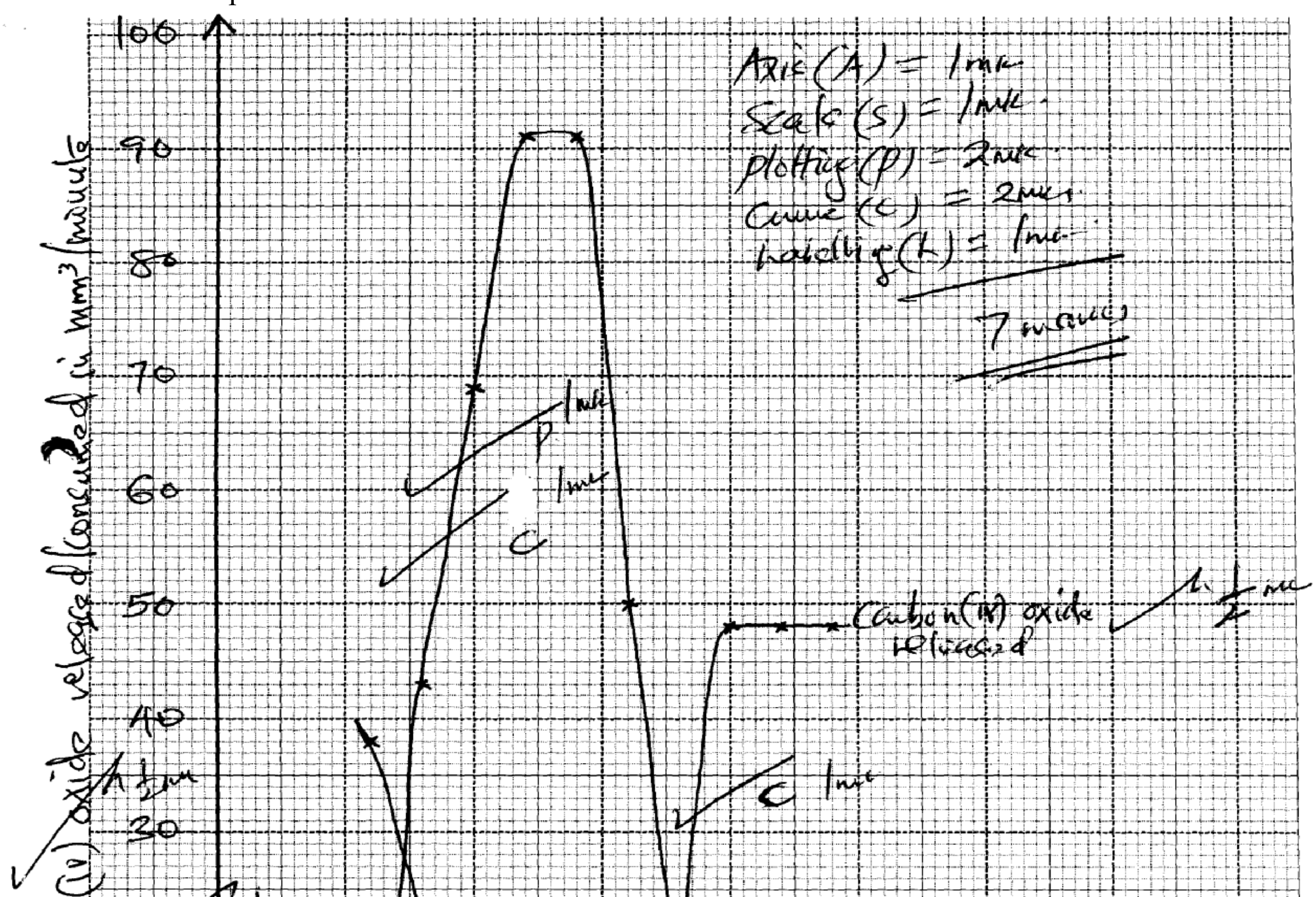
-During the night when there is no light; no photosynthesis takes place in the guard cells; Glucose in the guard cells is converted into starch; this lowers the osmotic pressure of the guard cells than the neighbouring cells;

-Water is then drawn away from the guard cells by osmosis; into the neighbouring cells, making them to be flaccid;

- Thinner outer wall shrink and the curvature of the thicker inner wall reduces; then the stomata closes;

36. a) External intercostals muscles contract; internal intercostals muscle relax; Rib cage move outwards; and upwards; Diaphragm muscles contract; diaphragm flatten; Volume in thoracic cavity increases; pressure reduces;
Atmospheric air enters the lungs; lungs inflate;

b) Guard cells have chloroplast; which photosynthesis in the presence of light, to form sugar; the osmotic pressure of guard cell increases; water move from neighboring cells into guard cells; by osmosis. Guard cells become turgid; inner walls of guard cells being thicker than outer walls. Causes the outer wall to stretch more resulting in guard cells budging outwards, stoma opens



37. (a)
b) i) Photosynthesis;
ii) Respiration
c) i) Rapid increase in amount of carbon (iv) Oxide consumed; As time increase amount of light increases', thus increasing rate of photosynthesis
ii) No carbon (iv) Oxide consumed, No light hence no photosynthesis
d) Low amount of carbon (iv) oxide released, carbon (iv) Oxide consumed for photosynthesis; respiration rate very low
e) i) Point when rate of photosynthesis equals rate of respiration
ii) At 18 hrs
f) It denatures enzymes/ stops photosynthesis; hence consumption of carbon (iv) Oxide

7. Gaseous exchange in (a) plants (b animals

1. (a) Gaseous exchange is the movement of gases across a respiratory surface; while respiration is the biochemical breakdown of food molecules to produce energy (and carbon IV oxide);
(b) Ethanol/Alcohol;
Carbon (IV) oxide; and energy; (any two)

2. (a) Glycosis;

Krebs cycle;

(b) Krebs cycle; became oxygen is used to oxidize acid to water, Carbon (IV) Oxide and energy;

3. a) anaerobic respiration/fermentation;

b) -baking of bread
-brewing industry

4. Carbon (IV) oxide produced in respiration is utilized in photosynthesis; oxygen produced in photosynthesis is used in respiration;

5. a) Amount of oxygen required to get rid of lactic acid that accumulates in the body tissues when oxygen available is lower than the demand

b) Energy/A.T.P/ Lactic acid

6. (a) Germinating seeds respired using oxygen in the conical flask and produced CO_2, which was absorbed by the sodium hydroxide solution. A partial vacuum was created in the conical flask. The atmospheric pressure being higher pushes the water down to A and upto B.

(b) RQ – $\frac{\text{Vol of } CO_2 \text{ produced}}{\text{Vol. of } O_2 \text{ used}} = \frac{102}{145} = 0.70$;

(c) Lipids;

7. (a)
- Complete oxidation of lipids require a lot of oxygen;
- Lipids are insoluble in water hence difficult to transport in the body
- Complete oxidation of lipids take a longer time

(b) Maltose
Lactose

8. a) i) Cytoplasm
ii) Pyruvic acid

b) Pyruvic acid is broken down; into ethanol and CO_2

9. a) RQ = $\frac{CO_2 \text{ produced}}{O_2 \text{ consumed}}$

$= \frac{5}{6}$; $= 0.83$;

b) Protein;

10. Bacteria, bacteria/ Symptoms

- Prolonged coughing and vomiting
- Convulsions and coma
- Conjuctival haemorrhage
- Severe bronchopneumonia

Causative agents

Symptoms

11. - Lowers saturation deficit by trapping H_2O moisture;

    - Protects direct sunlight to the stomatal pore;

12. They form depressions such that when wind blows it does not carry away water molecules.

13. - Increase rate of respiration

    - Speeds up the heart beat rate

14. A rat has a large surface area to volume ratio thus loses a lot of energy on form of heat therefore eats a lot to replace the lost energy;

15. a)Glucose $\longrightarrow$ water + carbon(iv) oxide + energy/210kj

    Or

    $C_6H_{12}O_6 \longrightarrow H_2O + CO_2$ + ATP (energy) (mark as a whole) 1mk

16. Insoluble hence not easily transported to respiratory sites;

    - They require more oxygen to be oxidized;

17. - Making of beer/Brewing/Ethanol/alcohol;
    - Baking industry/Raising of the dough:

18. (a) Respiration – Chemical breakdown of food to release energy. ✓

    Respiratory surface – Surface across which respiratory gases exchange.

    (b) Circulatory system transports the respiratory gases to and from tissues; hence maintains steep concentration gradient around the respiratory surface;

19. - Not every soluble/not readily soluble therefore not easily transported to the site of respiration;
    - A lot of oxygen is required to oxidize one gram of fat/liquid than one gram of glucose;

20. a) RQ = $\frac{\text{Volume of } CO_2 \text{ given out}}{} = \frac{102}{} = 0.70$;

Volume of O_2 used; 145

b) Fats/ oil/ lipid;

Reason: RQ for lipids/ fats/ oils is always less than 0.8; more oxygen is used than carbon IV produced;

21. (a) Boiling
(b) becomes milky/cloody /precipitate.
(c) Yeast produces enzyme amylase which catalyze breakdown of glucose anerosiccally into
energy (heat)
CO_2 and Ethanol
CO_2 makes lime water to become cloudy

(d) High temperature donators enzymes, reduces/stops respiration/stops the reaction.

8. Excretion and homeostasis

1. i) Ammonia is highly soluble in water and requires a lot of water for excretion hence assists in the removal of excess ammonia;

ii) All the glucose is reabsorbed at the proximal convoluted tubule;

2. (a) – Excretion;

- Osmo-regulation;

(b) – Glucose
- Amino acids;

(c) – Nephritis;
- kidney stones /Gall stones;
- Hepatitis A and B;

3. (a) Extra long loop of henle; Have fewer and smaller glomeruli;

(b) Salty food increased the salt concentration in blood; Blood becomes hypertonic to kidney tubules; more water is reabsorbed from kidney tubules; hypertonic urine is thus produced;

4. (a) Glucose;

(b) The person was a sufferer of diabetes mellitus;
(c) Pancrease;

5. a) i)insulin;

ii) Diabetes mellitus;

b) Diuresis is a condition which is characterised by production of large volumes of dilute urine;

6. i) urea;

ii) Triethylamine;

iii) Ammonia;

7. a) i) Fresh water; reject water

ii) Desert/ Arid areas; reject land

b) Reduces blood flow to the skin as more blood is stored in the spleen, reducing heat loss through the skin;

8. a) Ultra filtration;

b) Selective reabsorption;

c) Proteins have large molecular weights hence not ultrafiltrated;

9. Produces sebum to keep hair and epidermis supple and water proof; and protect skin against bacteria (through antiseptic substances);

10. a) Sweat produced does not evaporate due to high humidity;

b) Body does not cool hence more sweat is produced leading to accumulation;

11. Diabetes mellitus

- Caused by failure of the pancreas to secrete enough insulin;
- High glucose concentration in the blood than normal;

Diabetes insipidus

- Inability of the pituitary gland to secret anti-duretic hormone;
- High concentration of solutes in blood ;

12. Two processes through which plants excrete metabolic wastes:-

-Gaseous exchange;

-Transpiration;

-Shading leaves;

-Production of resins and gums;

-Storage of wastes in seeds/bark/fruits;

13. Has got long loop of henle in order to maximize water reabsorption thus conserving it;

14 .i) urea;
ii) Triethylamine;
iii) Ammonia;

15. (a) i) Fresh water; reject water
ii) Desert/ Arid areas; reject land
b) Reduces blood flow to the skin as more blood is stored in the spleen, reducing heat loss through the skin;

16. a) Ultrafiltration;
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19.

Diabetes mellitus	**Diabetus insipidus**
-Caused by failure of the pancreas to secrete enough insulin -High glucose concentration in the blood than normal - Caused by failure of the pancreas to secrete enough insulin - High glucose concentration in the blood than normal	-Inability of the pituitary gland to secret anti-duretic hormone -High concentration of solutes in blood Inability of the pituitary gland to secret anti-duretic hormone High concentration of solutes in blood

20. Two processes through which plants excrete metabolic wastes:-
-Gaseous exchange
-Transpiration
-Shading leaves
-Production of resins and gums
-Storage of wastes in seeds/bark/fruits

21. a) A – medulla; B – Cortex;
b) Cortex;

22. a) Enhances more reabsorption of water; leading production little but conc urine;

b) Reabsorption of water; Na^+/CL^- ions;

23. a) Aldosterone;
b) Loop of Henle;
c) Positive feed back;

24. - Reabsorption of unuseful substances in the kidney;
- Absorption of digested food from the ileum;
- Removal of metabolic waste products from kidney;

25. a) A D H / Vasopression;

b) Pituitary gland;
c) Diabetes Inspidus;

26. a) - Afferent vessels are wider than effereal vessels;

- Presence of pores on capillary and Glomerula membrane;
- Highly coiled narrow capillaries to reduce speed of flow of blood and increase ;
pressure

27. a) Arid/ semi arid areas
b) Ammonia
c)i) Contractile vacuole
Malpigian tubules

28. - Deamination

- Detoxification
- Breakdown of haemoglobin

29. a)Deamination;

b)-Removal of excess amino acids;
-Availing of energy in the body;
-Formation of glycogen /fats for storage; (award any one)

30. a)diabetes insipidus;
b)antidiusetic hormones (ADH);

31. a) large quantities of dilute urine;
b) Small quantities of concentrated urine ;(renal failure if habitual)
c) Production of urine containing glucose/sugar;

32. (a) Excretion — Separation and elimination of waste products of metabolism from
bodies of living organisms:

Egestion; Removal of undigested materials from food vacuoles/alimentary canals of animals:

(b) Removes waste products metabolism to create/pro'. idea suitable internal environment for best working of cells

33. (a) N – desert/arid/semi arid;

(b) Small sized glomeruli; to reduce ultra filtration longer loop of henle; to increase reabsorption of water – conservation of water.

N.B – Reject 12(b) if 12 (a) is wrong.

34. (a) – Organisms whose body temperature varies with the environmental temperature; ✓

(b) – Reptilia - rej. Reptile;
- amphibia - rej. Amphibians;

35. Glomerulus;

Adaptations of part R

- Coiled to increase the surface area for re-absorption of some glomerular filtrate
- Presence of numerous Mitochondria to promote active transport of glucose, amino acids
- covered by dense network of blood capillary for absorption of useful glomerular filtrate

36. Internal environment is the immediate surrounding of the body cells while external environment is the immediate surrounding of the organism

37.

- Radiation;
- Conduction;
- Convection;
- Evaporation ;

38. a) A – capsular space/ Bowmans capsule;

B – Descending wing of loop of Henle;

D – Glomerula

b) Urea;

39. Ovary; accept ovules

Anthers;

40. a) Detoxification;

b) Liver;

c) Prevents ammonia from accumulating to toxic levels; which would affect body functions;

d) Urea;

e) Excess amino acids are broken down to form amino group; which is combined with hydrogen atom to form ammonia;

f) It is transported to the kidney; through the renal artery where it is excreted

41. a) platelets exposed to air rupture on damage tissues to release thromboplastin/(enzyme) /thrombokinase; Thromboplastin neutralizes heparin; and activates prothrombin to thrombin;

throbin activates the conversion of fibrinogen to fibrin; which forms meshwork of fibres on the bruised surface;

b) blood clotting is the conversion of soluble blood protein into a mass of tangled threads of insoluble protein; while haemogglutination is the clumping together of red blood cells; c)haemophilia;

42. a)i)glucose is completely reabsorbed at proximal convoluted tubule back to blood stream;

ii) Protein has molecules hence not ultrafiltered (from glomerulus) to proximal convoluted tubule);

b) Create a steep diffusion gradient; hence higher rate of reabsorption of useful Substances-glucose/amino acids/sodium and chloride ions from the nephron tubules back to the blood stream;

c) -antidiuretic hormone;

-Aldosterone;

d) nephritis; kidney stones

43. (a) Nephron;

(b) (i) D = Afferent arteriole;

M = Efferent vessel;

(ii) Q = Aldosterone ; G - ADH/ vasopressin.

(c) Red blood cells/white blood cells/ plasma proteins;

(d) This shows that reducing sugar (glucose) was present in urine; the person is likely to be suffering from Diabetes mellitus;

44. (b) (i) Blood sugar level increased as a result of the glucose being absorbed in the ileum; by
diffusion / or active transport;

(ii) – The blood sugar level dropped as a result of the conversion of glucose to glycogen;
(and fats) by influence of insulin;.
- There was also an increased rate of respiration reducing the blood sugar level;

(c) 90 mg/100ml of blood;

(d) Person B has a defect in the pancreas; He did not produce enough insulin to control the
blood sugar level;

(e) By administration of insulin;

(f) - A constant level of blood sugar ensures optimum levels of metabolism;
- High level will increase the osmotic pressure and that affect metabolism;
- Low levels reduce energy supply in the body tissues and affect metabolism;

(g) - Glucose is used for respiration;
- Glucose was lost in urine;

45. a) Axis

Scale

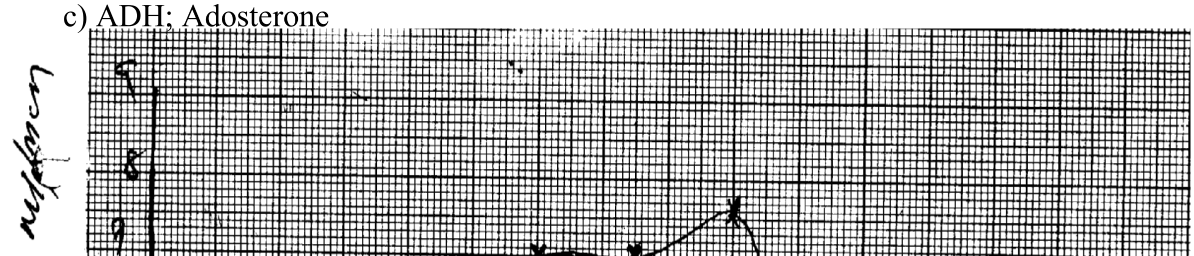
Plotting

b) i) The rate increases with time;
Because a lot of acid been drunk;
Very little ADH or No ADH produced yet;
No reabsorption taking place;

ii) The rate remain constant
Pituitary not stimulated to produce ADH
Nephrone, less permeable
No water being reabsorbed back to blood;

iii) The rate reduces with time;
Little water remaining in blood; due to a lot of water lost through urine;
No water being taken

c) ADH; Adosterone



Urine output ml/mcm

Time in minutes

46. (a) Explain how urea is formed in the human body:

- Excess amino acids are deaminated and the converted into urea in the liver

(b) Describe the path taken by urea from the organ where it is formed until it leaves the human body

- Urea from the liver is carried through hepatic vein into post/in prior vena cava; right auricle, right ventricle; pulmonary artery into lungs; Pulmonary vein , left auricle; left ventricle; aorta renal artery; glomerulus's; into Bowman's capsule; kidney tubules ascending and descending); collecting tubule ureter; into urinary bladder , urethra and out of the body in the form of urine.

47. a) B- Bowman's capsule
C- Loop of Henle
D- Distal convoluted tubule

b)

- have numerous/ many microvilli; to increase surface area for reabsorption
- coded to slow down filtrate for reabsorption
- have many/ numerous mitochondria to provide energy for reabsorption

c) - Active transport

d) Afferent arteriole is wider than the efferent arteriole

(d) Storage of vitamins (e.g. vitamin A, B2 & D)
- Storage of mineral salts (e.g. Potassium and Iron)
- Storage of Blood
- Manufacture of R.B.C
- Manufacture of plasma proteins (Albumen, fibrinogen & Globulin)
- Regulation of amino acids (deamination)
- Regulation of lipids
- Regulation of body temperature (thermoregulation)
- Destruction of worn out R.B.C
- Elimination of sex cells

48. a) X- Thromboplastin

Y- Fibrin
Z-- Thrombin

b) Promotes wound healing; stops further loss of blood/ bleeding; prevents entry of pathogens/ injection

c) Blood contain leparin/ anti clotting factor eight; that inhibits blood coagulation

49. a) Response of an endotherm to heat grain

i) Subcutaneous of fat little/ localized: to encourage heat loss/ not to impede heat loss

ii) Hair is lowered/ lies flat; by relaxation of erector pilli muscles; insulator reduced/ little air trapped; heat readily lost (by radiation and convector)

iii) Sweating/ panting occur
Evaporation of water absorbs latent heat of vaporization; leaving a cooling effect

iv) Cutaneous/ superficial blood vessels dilate;
Blood flows near skin surface facilitating heat loss

v) Metabolic rate falls/ BMR falls
Less heat generated to avoid overheating

b) Response of Endotherm to heat loss

i) Subcutaneous/ Adipose fat insulates; facilitating heat conservation

ii) Hair raised/ erects; by contraction of erector pili muscle; Insulator increased/ traps air; facilitating heat conservation

iii) Cutaneous/ superficial blood vessel vasoconstrict blood flows deep in the dermis; conserving heat

iv) Sweating/ panting stops; little heat is conserved

v) Extra heat is produced; by increase in metabolic rate of liver/ muscles/ shivering/ goose pimples/ animals become more active

50. (a) (i) Efférent arteriole/vessel; (I mark)
(ii) Loop of Henle: (Rj. Wrong spelling) (I mark)
(b) (i) Small sized; Few; (2mrks)
(ii) Large sized: Many: (2marks)
(c) (i) Glucose:
(ii) Diabetes mellitus: (Rej; wrong spelling)

51. High body temperature above normal: sweat glands: produce sweat: water in the sweat evaporates/ sweat evaporates: absorbing latent heat of vaporization produces a cooling effect.

Hairs lie flat; due to relaxation of erector pilli muscles: no/little air is trapped: [fins increased heat loss from the body; Blood arterioles/vessels; vasodilate/dilates: more blood floss to the skin hence more heat is dispersed by radiation and convection: when the body temperature is low below normal; sweat glands produce less/no sweat: no latent heat is absorbed/more heat is retained in the body; The hairs stand upright/erect: to trap air between them: that insulates the body against at loss; more heat is retained in the body; Blood vessels/arterioles constrict/vasoconstrict: less blood flows to the skin: reduces heat loss/ more heat is retained in the body;

Subcutaneous fat/ adipose [issue; beneath the skin insulates the body against heat loss: more heat is retained in the body: 22 marks

9. Ecology

1. a) Capture –recapture method;

b) Calculate the population of grasshoppers using the above data

$$\frac{FM \times SC}{MR} = \frac{36 \times 45;}{4} = 405;$$

2. a) Help to breakdown dead organic matter hence reducing bulk; in the recycling of Nutrients;

b) Regulate the predator – prey population;

3. a) Grass____________ grasshoppers _____________ birds;

b) Not all the energy is transferred from one trophic level to another; some is lost as heat, some is used up during metabolism and some is lost when organisms die and decay;

4. Autecology is the study of population / study of members of a species;

Biomass is the quantity of matter of a given type of organisms at a given trophic level;
Or the dry weight of an organism;

5. – Availability / adequate food supply ;

- Absence of predations ;
- Absence of disease; (mark the first two pts

6. (a) Habitat – physical location with asset of condition where an organism lives; while niche is the exact place where an organism occupy and its role in the habitat;
(b) Producers have a greater biomass than primary consumers since they start the food chain.
Inter-trophic energy losses occur in form of heat;
(c) It is non-toxic; It's organism specific;

7. Reduce oxygen supply and hence suffocation and death of plants and animals, clog respiratory surfaces (gills and stomata) leading to death;

8. (a) Food web;
(b) Three;
(c) Sun

9. a) Microscopic plants- mosquito larvae- small fish- large fish- crocodiles

b) Large fish;

10. a) Owl is nocturnal , white mice are easily seen and predated on, black mice camouflaged/ not easily predated on;
b) (Theory of) Natural selection;

11. a) Capture recapture method

b) i) $P = \frac{FM \times SC}{MR}$

$= \frac{725 + 974;}{139}$

= 5080;

Where FM – First marked

SC – Second recapture

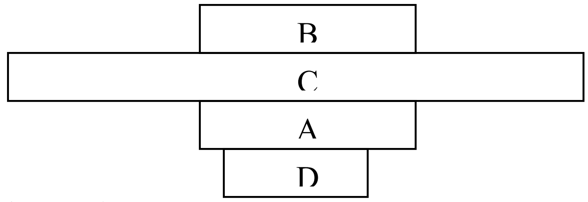
MR – Marked recapture

P - Population

ii) – No fish moves in or out of the area between counts ;
- The marked fish mix freely with other fish populations;
- Marking does not expose the fish to predation ;
- No variation in population size ;

12. D $\longrightarrow$ A $\longrightarrow$ C $\longrightarrow$ B

(b) - Correct label;
- A,B same size;
- C-largest;
- D- smallest;



13. - Protects delicate internal parts from mage;
- prevents excess loss of water (desiccating);
- provides surfaces for attachment of body muscles / organs;

14. a) Grass $\longrightarrow$ Grasshopper $\longrightarrow$ Guinea Fowl;

Grass $\longrightarrow$ Termites $\longrightarrow$ Guinea Fowl;

b) - Leopards will decrease;
- Gazelles will also decrease;

c) Grass;

15. Population — all members of one species occupying a particular habitat at a given time;

Community — all organisms belonging to different species that interact in the same habitat;

16. - lay down two ropes parallel to each other a meter apart; count the number of shrubs between the two ropes at marked points; and record the number; repeat the process several times;

Obtain average number; calculate area of the belt transect.

17. a) Population = $\dfrac{FM \times SC}{Mr}$

$$P = \frac{10 \times 50}{4} = \frac{500}{4}$$
$$= 125;$$

b) No entry or exit of fish;

Tags did not influence the general behavior of fish

18. - they decompose organisms; aid in nutrient circulation

19. i) Accumulation of CO_2 in the atom

ii) Increase in environmental temperature
- Eratic weather changes

20. - Enzymes amylase digests starch to maltose

- Mucus lubricates food

21. Due to (stiff) competition for available resources which leads to elimination/exclusion;

22. a)feeding level;
b)quaternary consumer;
c)sun/source of energy;

23. Adaptive radiation/divergent evolution;

24. i)crab pop= $\frac{\text{number marked in } 1^{st} \text{ catch x total no. in } 2^{nd} \text{ catch}}{\text{Number marked(recaptured)in second catch.}}$

$= \frac{400 \times 360}{90}$

$=1600$;

ii) Capture mark release recapture/
Capture-recapture /capture release /recapture;

25. (a) Suck small crawling insects (from tree trunks):

(b) Catching (flying) insects in grass:

26. (a) Used for the collection of flying specimens such as butterflies;
(b) Used for sucking small insects from barks of trees and under stones;
✓
(c) Used for trapping crawling insects such as termites;
✓

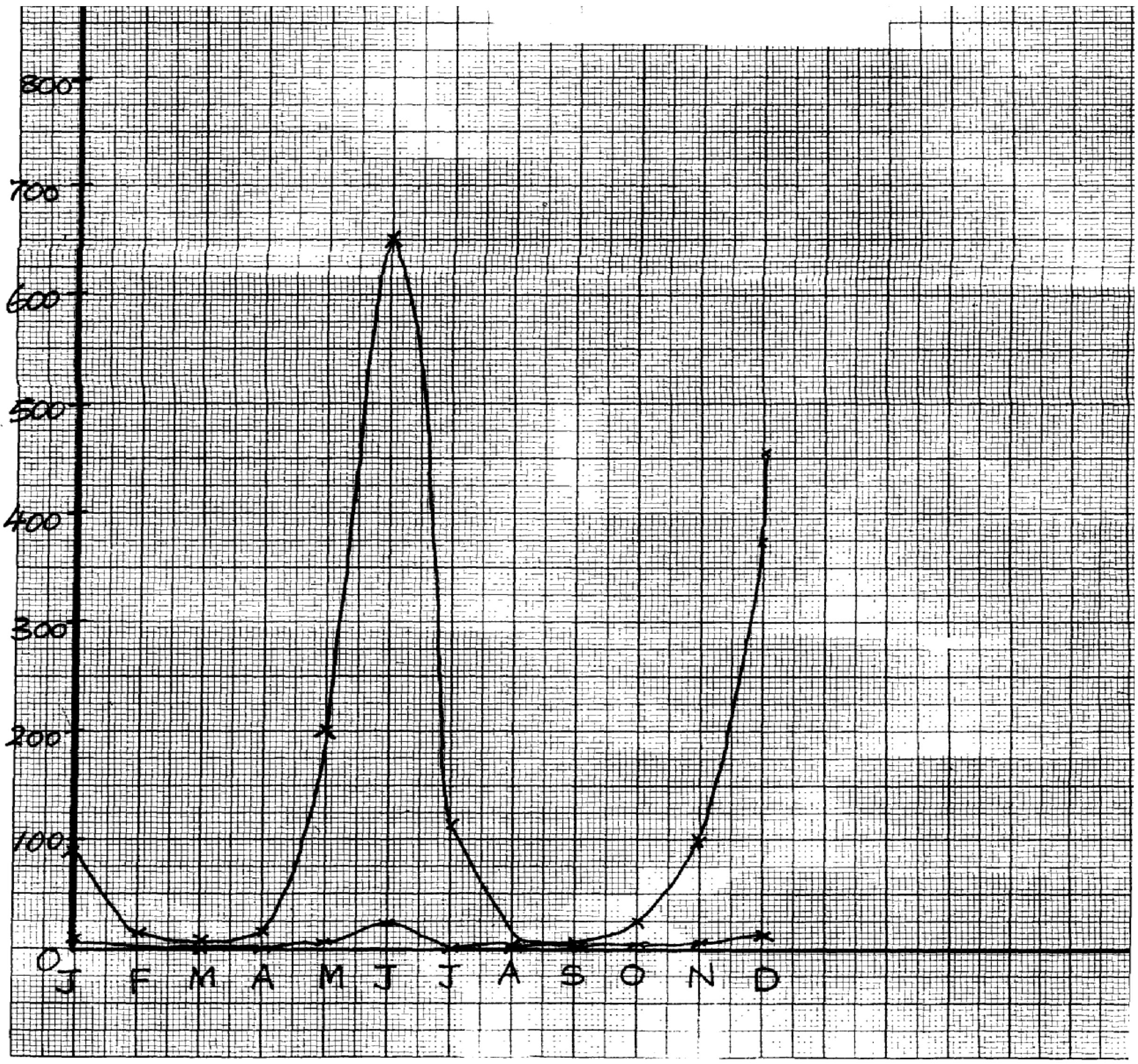
27. 1. Competition;
✓
2. Emigration;
✓
3. Predation;
✓ 4. Parasitism;

28. (a) Biotic and abiotic factors (2x ½ =1mk)
(b) - Feacal analysis
- Type of dentition type of beak (2 x1=2mks)

29. X – denitrifying bacteria/
Y – Animals/ herbivores; accept primary consumers
Z – Nitrogen fixing bacteria (in soil) accept Azotobacter

30. a) Check graph

- Labelling axes;;
- Scale
- Plotting;
- Joining (smooth contineas);
- Identifying the graph;



b) i) The population of locusts increase with increase in that the amount of rainfall;

ii) – Increased amount of food;

- Improve breeding conditions;

c) - The population of both decreases

- Less food availability for locusts and hence crows;

d) i) Quadrat method;

ii) total counts

e) i) locusts ____ primary consumers;
Crows ______ secondary consumers;
ii) Grass ____ Locusts ________ crows;

f) - Grass would increase;
- Crows would reduce;

g) Wild animals are browzers hence obtain food while cows are grazers hence lack grass

h) Biomass is the total dry weight of organisms at a particular trophic level;

31. (a) (i) Antelope A;
(ii) Reason- Rate of multiplication /reproduction is higher in species A than B;
(b) (i) Sigmoid curve /ogive/s-shaped curve;
Accept any one correct
(ii) PQ- Lag phase /slow growth phase; QR- Exponential/log / rapid growth phase;
RS – Deceleration phase ST- Stationary/constant growth phase;
(c) (i) Q and R
Marked with rapid population growth rate; many mature reproducing organisms/individuals/antelopes;
Absence of environmental resistance;
(ii) S and T - Growth rate stagnant/birth rate equals to death rate; the ecosystem has attained its carrying capacity/environmental resistance (density dependent) have set-in;
(d) (i) Interspecific;
(ii) Thin and tall; yellow/pale green; low yield
(e) By occupying different (ecological) niches;
(f) Move swiftly to escape predators; camouflage to avoid noticed by predators; Eyes on the side of the head to give them a wide field of view enabling them to keep track of their enemies;
(g) Capture –recapture method,; direct count,
Aerial photography

32. Water- The availability of adequate amounts of water lead to plant growth which provides food for animals. In aquatic environment, wateris a medium in which gametes are released thus lead to continuity in procreation.
Temperature- Influences the rate of enzyme catalyzed reactions. Therefore, it exerts an influence on almost all activities of plants and animals such as respiration, photosynthesis, growth, transport e.t.c.
Light-Is necessary in plants for photosynthesis as it influences flowering of a wide variety

of plants, affecting opening and closing of stomata, affect the rate of transpiration.

Salinity- Is the salt content of eater. It varies in aquatic habitat. Fresh water organisms suffer the risk of loosing water.

Humidity – Determines the amount of water loss from a bodies animals and organs of plants; high humidity means less evaporation; and low humidity means high rate of evaporation and transpirations;

pH – It determines if water habitat is acidic or alkaline; PH has a great influence on physiological function of organisms affects enzyme concern reactions since enzymes operate within a narrow pH ranges

Wind- Wind came physical damage to plants; increase rate of transpiration as air blows away; causes migration of insects; wind having gases may acid rain in a region; wind is an agent of pollination and dispersal;

33. (a)

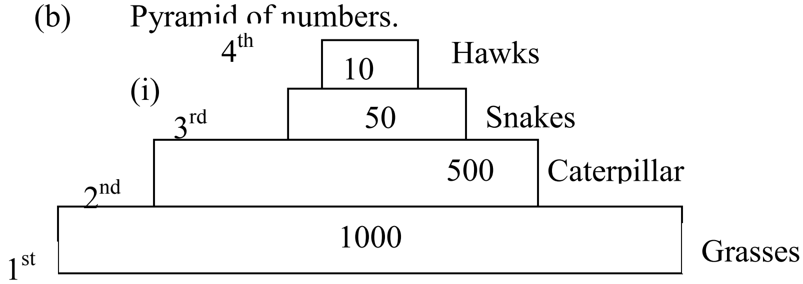
Grasses → Caterpillar → Frogs → Snakes → Hawk

Grasses → Squirrel → Hunting dogs → Hawk → Vulture

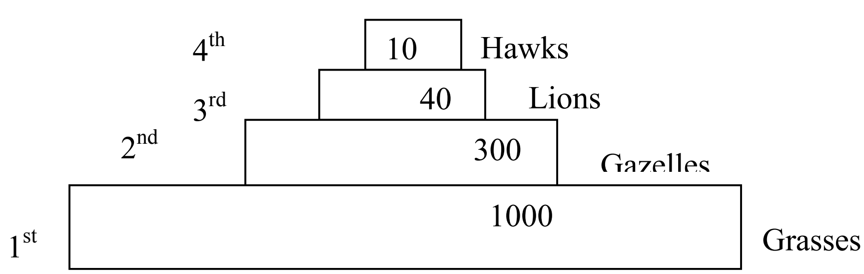
Grasses → Elephant → Vultures

Grasses → Caterpillar → Snake → Hawk

(b) Pyramid of numbers.



Or;



(ii)

(c) Effects of removing the hunting dogs.
- Increase in number of gazelles and squirrels due to reduced predation leading to increased pressure upon the grass;

(d) During transfer of energy at each feeding level, some amount of energy in form of heat is lost only about 10% would be transferred from the grains to steers and out of the 10 % about 1 kg would be transferred to man. The rest would be lost as heat or ingestible material.

34. a) i) Slugs; mice;/ Amphids/ caterpillar

ii) Primary consumers;

b) i) plants _________ mice _______ snakes __________ Hawks;

Plants ________ Caterpillar ________ insectivorous birds _______ hawks

c) Plants ;_________ directly obtain energy from the sun

Hawks – Loss of energy in form of heat; through process of respiratal/ defaecation/ excretion

35. a) A lot of food causes population increase due to high rate of reproduction and immigration resulting in completion for food/ death/ emigration; reducing population; little food leads to competition; leading to emigration/ death; reducing population
b) Energy from the sun is trapped by green plants; during photosynthesis; producing chemical energy/ carbohydrates/ food
Green plants are producers/ 1st trophic level; Green plants are eaten by herbivores which are primary consumers/ occupy the second trophic level, when plants dies and animals die organisms die; saprophytic fungi/ bacteria/ micro organisms feed on them; thus decomposing them into smaller/ simpler substances/ they are decomposers/ detritivores; At all levels some energy is lost; through respiration

36. a) A- Ovary

B- Oviduct/ fallopian tube
C- Uterus/ uterine wall
D- Cervix

b) Produce ova

Produce femme hormones/ Estrogen and progesterone

c) - Highly vascularized to supply nutrients to foetus/ drain away excretory wastes

- Inner wall lined with Endometrium for implantation of fertilized egg/ zygote

- Muscular for peristalisis to expel menses during menstruation/ parturition

- Great capacity to expand during gestation to accommodate developing foetus

d) -copulation/ Achieve orgasm in Human male followed by ejaculation

- birth canal

37. a) use the capture -recapture method; capture the grass hoppers; count; and mark using
permanent ink; record; releases; and allow time(1-24hrs);recapture and count the marked and unmarked;
Total population is equal to the number marked and unmarked grasshoppers in the second sample X number of marked grasshoppers in the first sample ; divided by number of grasshoppers marked in the second sample that were recaptured;

Acc $P = \frac{FMxSC}{MR}$

where FM-1st captured
SC-2nd capture(marked and unmarked
MR-marked recaptured
(rej. ½ mark i.e. 10/2=5) acc specified distance apart e.g. 3m apart

b. run two ropes parallel to each other a meter apart; counts of shrub are made between the two ropes at marked points/whole belt (and recorded);report the process severally(at least 3 times);calculate shrub area of the belt transect; calculate shrub population for whole area;
Rej all shrubs counted

NB shrub pop=$\frac{\text{average shrubs per transect x total area of grassland}}{\text{Average area of belt transect}}$ (max 3)

38. (a) (i) Phytoplanktons:

(ii) Hawk; and water snake:

(b) - Decrease in phytoplanktons:
- Increase in population of small fish:

(c) Hawk;- Top predator amount of energy decreases in successive trophic level/energy is lost through respiration; undigested/unconverted food:

(d) Residue is poisonous to man;

-Kill non- targeted organism / Beneficial organisms:
-Remains for along time in the ecosystem / pollutes environment:

(e) (i) Causes decomposition/Recycling of nutrients:

(ii) Root nodules: have bacterial / Rhizobium sp: to convert free nitrogen: into nitrates in the soil;

(f) Capture - recapture: capture release recaptures:
(g) Manufacture food: (OWTTE) to be used by themselves: and all other organisms in the ecosystem (awls)

39. Broad/ wide lamina: to Provide a large surface area to trap maximum sunlight or photosynthesis;
- Thin lamina; to reduce the distance covered b\ light and carbon (iv) oxide: to reach the photosynthetic cells/ palisade cells;
- Cuticle; is transparent to allow light reach photosynthetic cells:
- Waterproof climatic cuticle: to reduce water loss/Transpiration:
- Numerous stomata: efficient gaseous exchange: palisade (mesophyll) cells: have numerous chloroplasts: for maximum photosynthesis: spongy mesophyll cells: are irregular in shape creating large air spaces between: for efficient /free circulating air; Lear veins; have x 1cm 1r transport of water and mineral salts: and phloem for transport of manufactured food;
- Leaf mosaic: to maximum trapping of sunlight for photosynthesis:
- Guard cells: to control opening and closing stomata: Guard cells have chloroplasts for photosynthesis:

10. Reproduction in (a) plants (b) animals

1. a) i) integuments ;

ii) Primary endosperm nucleus;

b) This is fruit development without fertilization;

2. - Secretion of progesterone and oestrogen;

- Controls exchange of material between maternal and foetal blood;
- Prevents entry of pathogens from the maternal to the foetal circulatory system;

3. – Sexual intercourse with infected persons;
- Transfusion with infected blood;
- sharing contaminated needles;
- Infected mother to child through breastfeeding;
- Contact with infected blood/body fluids through cuts or wounds; (mark the first 3 points)

4. (a) Par thenocapy;
(b) Ethylene;
(c) Promoted differentiation of adult features;

5. – Site for fertilization;
- Conducts on a from ovary to the uterus;

6. (a) X – Polar nuclei; Y – Egg cell;
(b) – Results to variation; that makes the plant to be adapted for survival;

7. (a) Chiasma; reject – chiasmata

(b) (i) Provide a chance for the exchange of genes (along the portion of chromosome);
(ii) Meiosis;

8. (a) When they can freely interbreed to produce fertile/viable offspring;

(b) Is the occurrence of two distinct reproductive forms in the life cycle of an organism; the diploid sporophyte phase and the haploid gametophyte phase;

9. (a) Acquired characteristics are not inherited/inherited characteristics are found in reproductive cells only;
(b) Mutations bring about variation which when advantageous can be passed on from one generation to the next; and this can lead to emergence of new species;

10. (a) Gaseous exchange; means through which foetus get nutrients from the mother; offers a means for elimination of wastes by the foetus; supplies antibodies to the embryo from the mother; secrets progesterone hormone that maintains pregnancy;
(b) because testerrone is transported through the blood

11. – Protandry

- Protogyny;
- Self sterility/incompatibility

12. – Ability to pollinate;
- Ability to photosynthesis;
- Ability to disperse seeds/fruits;
- Ability to absorb water and mineral salts from the soil;

13. (a) Fusion of one male nucleus with an egg cell to form a diploid zygote; and fusion
of the other male nucleus with two polar nuclei to form tripoid endosperm;
(b) – Are brightly coloured to attract insects
- Have seed coat that is resistant to digestive enzymes
- Have hooks for attachment to passing animals
- Are freshly/succulent to attract insects

14. a) Oxytocin;

b) Progesterones;
on different individual plants;
-some plants are self –sterile in their pollen grains transferred to stigmas in the same plant fail to germinate;
-in some plants stamens and carpel on the same plant mature at different times;
-in many plants the stigmas are located higher than the anthers;

15. -some plants are dioecious which means that staminate and distillate flowers are borne

16. a) A – Has umbilical vein and artery to supply foetus with nutrients and removal of waste
products; √

B – Protects embryo from shock/regulate temp. of developing embryo/ suspends and supports
embryo;
b) Foetus head is turned towards the cervix; √
c) To supplement iron synthesized by the mother since it (iron) is needed for haemoglobin
formation in the foetus; √

17. i) Marginal; √
ii) Free central; √

18. a) Cypsela b) Animal

19. i) Production of the hormones progesterone and oesterogene continues;

ii) These hormones inhibit the production of follicles;
Stimulating hormone (FSH) and lutenising hormone (LH);

iii) This inhibits the maturation of more follicles;

20. a) It brings about useful variations which make the off springs better adapted for survival b) i) 33;
ii) 11;

21. a) A – Antipodal cells; B - Embryo sac; D- Synergid ;
b) Double fertilization

22. They cannot freely interbreed to produce a viable /fertile offspring OR- do not have hereditary
distinction to interbreed to produce a fertile viable offspring;

23. Adverse temperature , wind/air current, pH, light noise ;

24. (a) (i) Epigynous –a condition where other floral parts arise/positioned above the ovary
/inferior ovary

(ii) Staminate flower – Male flower (accept – has stamen only / male parts only);
(b) Meninges;

25. (a) Yeast ; (b) Budding;

26. – Through breast feeding if mother's nipple and baby's mouth have rushes/wounds
- During delivery;
- During pregnancy;

27. a) Production of spindle fibres
b) i) Absorbs light energy; which is used to break down water molecules into O_2 gas and
H^+/ atoms ;
ii) Glucose;

28. a) Prophase I ; Reject prophase alone
b) i) There is crossing over of genes that leads to variations;
ii) Leads to formation of gametes;
Brings about genetic variation;
It helps retain a constant diploid chromosomal constitution in a species at fertilization;

29. a) Stamens hanging outside the flower; large anthers loosely attached to flexible filaments;

Large amounts of small; light and powdery pollen grains to be easily blown by the coid;

b) Monoecious plants have both male and female flowers bone on separate plants;

30. a) Inability of seeds to germinate despite all the conditions necessary for germination are provided;

b) Scarification;

Increase the concentration of hormones which stimulate germination/ increase auxin conc;

Allow the embryo to mature before planting seeds;

Remove germination inhibitors;

31. a) Allows the adult to reproduce;

Allows the species to disperse in order to colonize new habitats;

b) Leads to the formation of the larval cuticle;

32. - Hot water kills organisms in the water;
- Reduces oxygen content in the water leading to suffocation;
- Chemicals in the element may lead to entrophication;

33. - Chances of fusion of gametes are low
- Large amounts of gametes are produced leading to wastage
- Changes of survival of the young ones are low since there is lack of parental care

34. - Allow nutrients to pass from mother to Foetus

- Allow diffusion of excretory products from Foetus to mother's blood for excretion
- Produce hormones Oestrogen & Progesterone / that retains pregnancy.
- Prevents passage of foreign particles e.g. pathogens.

35 . a) i) prophase I

ii) Chiasmata Formation / cross over

b) _Ovary
- Anthers

36. - Ensures no competition for dispersal;

- Survival of pupa stage;

37.

Mitosis	Meiosis

- Two diploid daughter cells are formed - Four haploid daughter cells are formed;
- No crossing over - There is crossing over because of chiasmata; formation;
- Takes place in one cell division - takes place in two cell division;
- Leads to growth - leads to gamete formation
- Takes place in somatic cells - takes place in reproductive cells;

38. Due to (stiff)competition for available resources which leads to elimination/exclusion;

39. i) healing and repair of the uterine wall following menstruation stimulates the pituitary
Gland to secrete L.H; (award any one)
ii)-cause ovulation
-changes the remnants of graafian follicle to corpus uteum;
-initiates secretion of progesterone; (award any one.
iii)-accelerates growth and maturity of graafian follicle;
-stimulate the graafian follicle to secrete oestrogen; (award any one.)

40. (a) A - Syncarpous: B- Apocarpous; Rj: Wrong spellings
(b) A fused ovaries B — separate ovaries:
(c) Hinder self pollination? fertilization:

41.

Sperm	Ovum
- Spear shaped. - Posses a tail. - Has acrosome . No vitelline membrane.	- Spherical shaped - No tail - No acrosome - Has vitelline membrane.

42. (a) anthers; (b) – tube nucleus;
- Generative nucleus;

43. (a) – Metaphase 1; rej. Metaphase.
(b) - Homologous chromosome arranged on the equator;.
✓
- Spindle fibres formed and attached at the centromere of the chromosome;

44. Progesterone;

45. - Seed dormancy allows the plant to escape harsh conditions of the environment
- It also allows time for the seed to disperse;
- Seed dormancy allow time for the seed to fully mature (after ripening period);

46. (i) - A fruit has two scars while a seed has a single scar
- Fruits are covered by epicarp while seeds have seed coats/testa
(ii) Biological control helps to prevent pollution f the environment

47. (a) Site for sperm formation (b) For nourishment of sperm cells /support

48. (a) Ovary; anther (b) Small/light/smooth

49. - Self sterility;
- Dioecious plants;
- Protandry and protogyny;

50. In birds the embryo develops externally. It is totally dependent on food stored in the egg for its nourishment; In mammals the embryo receives nourishment from the mother through the placenta

51. Pollination is the transfer of pollen grains from an anther to a stigma;
Fertilization is the fusion of the nucleus of a male gamete with the nucleus of a female gamete to form a zygote;

52. a) Water dispersal

- Such seeds and fruits enclose air in them to lower their density for buoyancy;
- They are fibrous/ spongy to lower the density for buoyancy;
- Have impermeable seed coat or epicarp to prevent water from entering during flotation so as to avoid rotting;
- The seeds can remain viable while in water and only germinate while on a suitable medium;

Wind dispersal - They are light; and small; to be easily carried by wind currents due to lower density;
- Have developed extension which create a larger surface area; so as to be kept afloat in wind currents e.g. * Parachute like structures;
* Wing like structures;

Animal dispersal - Brightly colored to attract animals
-Fleshy to attract animals;
- Some have hook like structures to attach on animals fur

Self dispersal - They have weak lines on the fruit wall along which they burst open to release seeds, which get scattered. This occurs when temperature changes suddenly

b)
- The zygote formed when egg nucleus fuses with one male nucleus develops into the embryo of a seed
- The triploid nucli develops into the primary endosperm of the seed
- The inner and outer interguments develops into the seed testa
- The ovary wall differentiates into epicarp, mesocarp and endocarp forming a fruit
- The ovule then develops into a seed

- The corolla dries up and withers away
- The calyx may persists shortly as it photosynthesis but afterwards , shrivels, dries and withers away
- The Androecin shrivels,dries and withers away
- The stigma together with the style shrivels, dries and withers away

53. Wind dispersal.

- Parachute of hair, increase surface area to be carried by wind /float
- Wing like structures, increase surface area to be carried by wind /floats.
- Small/light, seed/fruits to be carried by wind have censor mechanism/split open particularly and shaken by wind to throughout the seeds.

Animal dispersal
- Juicy/succulent/fleshy, to attract animals; hooked; to stick on animals bodies and be carried away.
- Hard seed coat; to resists digestive enzymes. Hence come out a long with feaces/dropping of animals.
- Brightly coloured; to attract animals that carry them away.
- Scented; to attract animals that eat and scatter their seeds.

Water dispersal;
- Fibrous fruit wall/mesocarp with air spaces to store air hence make them buoyant/float in water;
- Air floats make them buoyant/float on water.
- Self dispense mechanism
- Fruits dry and crack/open violently along the lines of weakness throwing away the seeds.

54. (a) Pituitary gland

(b) (ii) Testerone
(iii) Follicle stimulating hormone
(v) Leutinising hormone
(c) Sterility/lack of spermatogenesis. Failure of secondary sexual characteristics.
(d) Inhibit production of F.S.H
Inhibit production of L.H

55. (a) I – F.S.H (Follicle stimulating Hormone);

II- Lutenizing Hormone (LH);
III. – Androgen/Testerone/male Hormone
(b) Progesterone;- brings about protogenetion/development/thickening of uterine wall;
(c) A – Inhibition of L.H
B – Stimulation of L.H
(d) – Growth of hair on the armpit and pubic region; - Development of pimples on the face;

56. (a) Role of spleen in human defense mechanism:-

- Form lymphocytes which ingest pathogens present in the blood;

- Produce antibodies; which neutralizes poisons produced by the pathogens

(b) Ways of controlling HIV spread:

- Testing and transfusing blood free form the HIV
- Avoid sharing of cutting instruments (OWTTE) any two

(c) Meaning of the word Acquired Immuno Deficiency Syndrome:

- Development of lack of immunity system resulting to various chains of infections

(d) Reason for encouraging vaccination prevent/control infection which is better/cheaper than treatment

(e) Is acquired when an individual is infected and naturally produces immunity and recover from the infection

57. Seeds and fruits are adopted to the various methods of dispersal:-

Water dispersed fruits and seeds;

- Mesocarp fruits has air spaces thus light/buoyant to float; therefore carried away by water; seeds are protected from soaking by water proof pericurp / testa;

Animal dispersed fruits/seeds;

Presence of hooks for attachment to animals thus carried away to other places; fruits are also brightly coloured;

-Succulent; aromatic /scented to attract animals; the seed coats are hard and resistant to digestive enzymes; the seeds are therefore dropped away in feaces/droppings'

Self dispersed seeds/fruits/explosive mechanism;

- The dry pods/fruits splits along line of weaknesses/sutures; scattering seeds away form parent plant;

Wind dispersed fruits/seeds;

- censer mechanism; open/split; to disperse the seeds. Perforated capsule is usually loosely attached to stalk / long stalk is swayed away by wind scattering seeds;
- Presence of hairs /wing-like structures/floss/extension which increase surface area for buoyancy; making it easy for fruits/ seeds to be blown away;
- Fruits /seeds are light due to small size; therefore, easily carried away by wind;

58. a) A- Ovary

B- Oviduct/ fallopian tube
C- Uterus/ uterine wall
D- Cervix

b) Produce ova
Produce femme hormones/ Estrogen and progesterone

c) - Highly vascularized to supply nutrients to foetus/ drain away excretory wastes

- Inner wall lined with Endometrium for implantation of fertilized egg/ zygote
- Muscular for peristalisis to expel menses during menstruation/ parturition
- Great capacity to expand during gestation to accommodate developing foetus

d) -copulation/ Achieve orgasm in Human male followed by ejaculation
- birth canal

59 . a) chorion; Rej Amnion/Amniotic membrane.

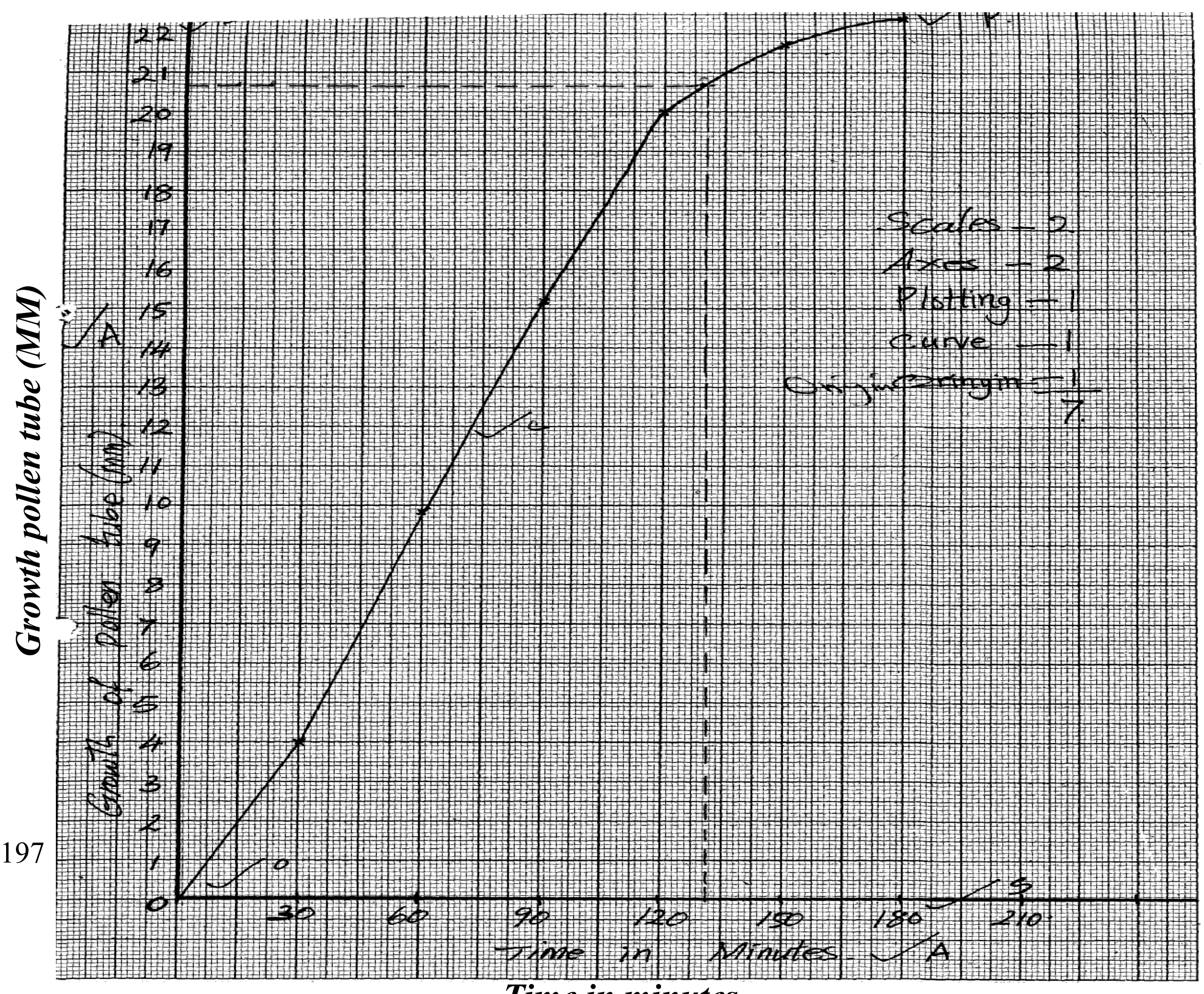
b) i) A: (umbilical Artery; Rej Arteriole
B: (umbilical vein; Rej venule

ii) More food nutrients; more oxygen in umbilical vein/less food nutrients; more excretory products in umbilical Artery;
Rej.(ii)if (i) is wrong
Rej oxygenated/deoxygenated

c) highly vescularized;
-large surface area; acc. Numerous villi for large surface area
-presence of secretory cells/are glandular; any 2 Rej. Source of hormones.

d) cushion /absorbs shock/buoyancy;

60. (a)



b) i) 30min/aftter every 30min; Rej if no units
ii) 20.4 -20.8mm;
iii)105min-106min; Rej after 105/106 min.
iv) 0 + 120minutes
growth fast/growth rapid /rate of growth rapid/growth rate pattern rapid;
Rej. Exponential growth
reason: pollen tube young/has enough nutrients in culture;
to 180 minutes- grows slowly /rate of growth decline /decrease/growth rate pattern
decrease;
reason: pollen tube mature/old/has exhausted nutrients;
v) directs role gametes/nuclei/nucleus to ovules; Rej. Ovary/pollen grains for male gametes.

c) integument develop/changes to-seed coat/testa;
zygote-embryo;
triploid nucleus-endosperm;
ovary wall- pericarp;
ovary- fruits;
ovules-seeds;
corolla/petals/style/stamens/filament-dry out /fall off /wither(losing a scar);
calyx may persist(dry up &fall off) Rej. die/disappear.

11. Growth and development in (a) plants (b) animals

1. a) Moulting hormone/ ecdysone

   b) It allows growth to take place; since growth can not take place in the presences of the

2. a) Long sightedness/ hypermetropia ;

   b) Convex/ converging lenses;

3. (a) – Excretion;

- Osmo-regulation;

(b) – Glucose
- Amino acids;

(c) – Nephritis;
- kidney stones /Gall stones;
- Hepatitis A and B; (mark first 2 pts (2mks)

4. (a) Intermittent growth curve;

(b) Moulting;

(c) Ecdysone;

5. Natural immunity is inherited /transmitted from parent to offspring; Acquired immunity is developed after suffering from a disease or through vaccination;

6. A – Cell elongation/expansion ;

B – Cell division/multiplication ;

C – Cell différentiation/maturation ;

7. Continuous variation shows gradation in characteristic with intermediate; discontinuous shows distinct characteristics between organisms with no intermediate groupings;

8. a) to investigate the effect of the force of gravity on the growth of a seedling (shoot and root);

b) Force of gravity cause accumulation of auxins on the lower side of the seedling
- Higher concentration of auxin will promote growth in the shoot but inhibit growth in the Roots;
- There will be more/ faster growth on the upper side of the root than on the lower side hence the downward bending;
-There will be more/ faster growth on the lower side of the shoot than on the upper part hence the upward curvature;

9. i) Between xylem and phloem;

10. Growth – Increase in size of an organism or its parts due to synthesis of protoplasm

Development – Differentiation and formation of various tissues to perform specialized functions;

11. - Reduce competition between the young ones (larvae) ;

- Avoid predation of the young ones as they are different ;
- The pupa stage can withstand harsh environment by being inactive;

12. Disadvantages of exoskeleton;

– Limits growth
- heavy to the insect;

13. Primary growth results form the activity of primary/embryonic tissues/apical meristems and lead to increase in height, while secondary growth result from activities of secondary meristems; /cambium and leads to increase of girth/diameter /circumference;

14. i) — Oxidizes food to release energy needed for germination;

ii) — Stores food for the seed;
- Stores enzymes;

15. .- Selective weeding

- Ripening of fruits
- Parthenocorpy
Reject Prunning of coffee and tea

16. - -Contraction of muscles

- Formation of bones

17. allow growth to the place;
(ii) Grain/cotyledon remains underground below the soil level: (I mark)

18. (a)(i) Hypogeal;
(b) Photosynthesis; OWTTE
-Gaseous exchange; accept. Transpiration.

19. (a) Effect of unilateral/unidirectional light on shoots:
(b) Seedling/shoots growth towards light' growth curvature towards light;

20. (a) Intermittent growth;
(b) Moulting /ecdysis;
(c) Ecdysone rej. Moulting hormones;

21. (a) Divide giving rise to more vascular tissues – phloem and xylem; hence leading to secondary growth/thickening of the stem;

(b) They lack vascular cambium;

22. - It has chondrionic villis to increase surface area for excahgne of materials
 - Has thin epithelium for rapid exchange of exchanged substances
 - Has counter current flow of foetal and maternal blood to enhance speed diffusion gradient.
 - Highly vascularised (dense network of capillaries) for faster transport of exchanged material

23. (a) For oxidation of stored food;
(b) Breakdown and oxidation of food

24. (a) (i) osmotaxi/cremotaxis

(ii) phototaxis
(i)Sensory neuron
(ii) Direction of nerve impulse
(iii) Schwarn cell
(iv) insulate the axon/Speed up transmission of impulses

(c)

25.

Reflex action	Conditioned reflex action
Single stimulus to bring about response. Simplest form of behaviouir and is independent of experience Sensory and mother component are the same at all times	Repeatal stimulus to bring about response Involves modifications of behaviour and dependent experience . Primary sensory component is repeat by a sensory component bat the motor. Component remains unchanged.

(a) To absorb carbon (IV) oxide;

(b) to provide moisture to germinating seeds;
(c) (i) (Left – right direction ⟶);
(ii) Oxygen in the tube is taken up by the seeds for germination; the Carbon (IV Oxide
Produced during respiration and the one in the tube reacts with potassium hydroxide
pellets; lowering the pressure inside the set-up; the higher pressure from outside the tube
forces the dye in the direction shown;

26. (i) Lag phases; Dry mass increases slowly; became in plant has not developed

leaves; for photosynthesis hence is depending on stored food;

(ii) Expontial phase; Rapid growth /increase in dry mass, leaves developed;

photosynthesis taking place leading to accumulation of food and rapid cell

division / plant adapted to the environment

(iii) Death phase/ senescence; Negative growth/decrease in dry mass as some tissues die after reaching maximum maturity; Fall in photosynthesis activity; toxic wastes poison tissues;

(c) (i) When dry mass was first recorded/at germination

(ii) Dry mass would decrease first because food is oxidized to produces energy; water and carbon dioxide/utilized in respiration;

(d) (i) Harvest every week about five seedlings; dry in oven to a constant dry mass; Calculate the average mass for one seedling and record the results.

(ii) Advantage; Dry mass is not affected by environmental conditions while fresh weight is dependent on the amount of water in the plant which fluctuate with environmental factors affecting transpiration rate.

27. a)klinostat/clinostat;

b)i)the radicle remains /grow horizontally;

ii)rotation of klinostal causes uniform distribution of auxins/ indoleacetic acid; hence uniform growth/clongations (no curvature formed);

c)the experiment repeated but with stationary klinostat;

d)-(tropism)enable plants to get water-hydrotropism;

-chemotropism aids plants in fertilization and nutrients absorption;

-thigmotropism enable weak plants to obtain support

-phototropism enable plants to obtain light for photosynthesis;

-geotropism enables the roots grow down the soil towards the centre of the earth thus providing support to the plant

28. a)graph

-axes have to be labelled- ½ mk@-1mk No axes marking stops there.

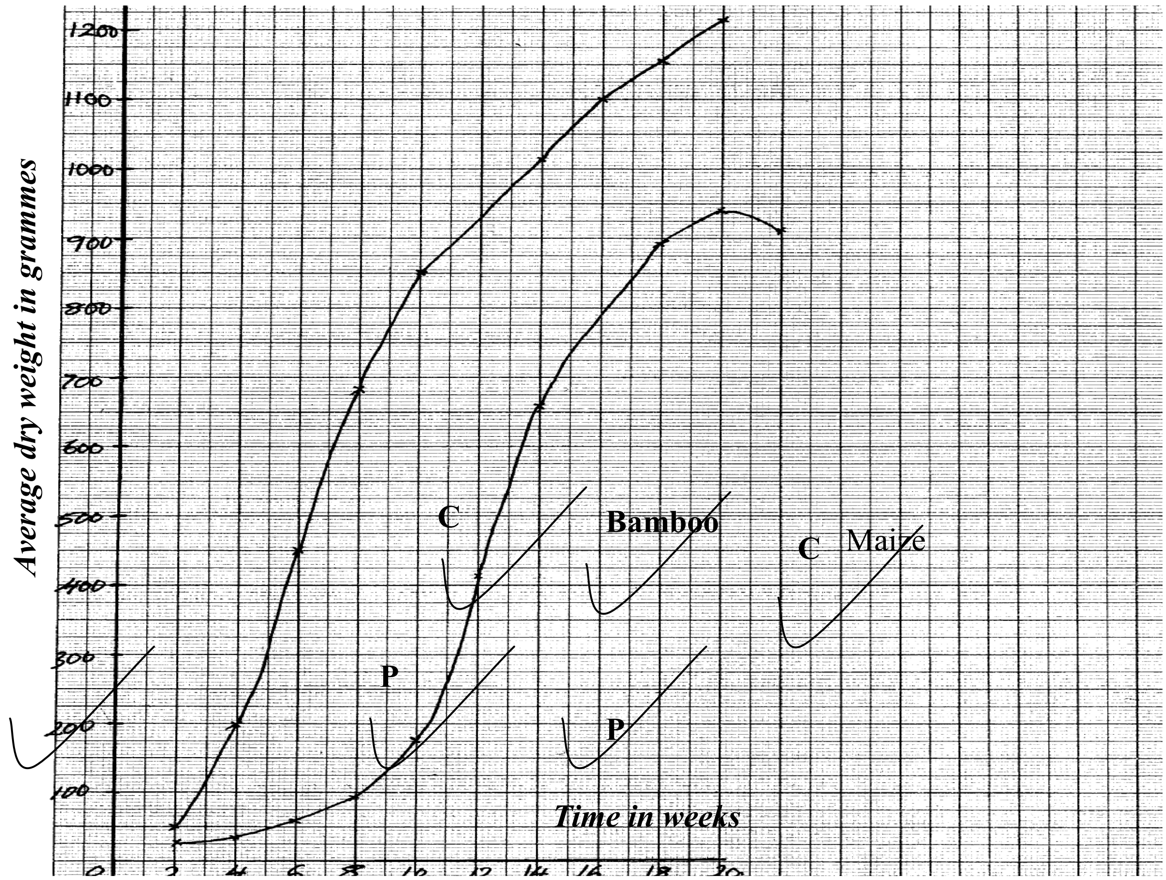
-scale-should be appropriate and workable- ½ mk @-1mk

-plotting correctly-1mk@*RCH*∗wrong scale stop marking.

-curve-smooth and not extrapolated beyond 3 small squares- ½ mk@-1mk

-identity- ½ mk each-*RCH*

Note/ -axes reversed-award only for identity
-no origin-award only for one scale/vertical one.



b) i)bamboo;
ii)(bamboo)have higher average weigh
c) i)average height of maize plant between weeks 14 and 18 constant (at 2.1m); maximum
height attained; average weight increased; because there was slight increased in the girth;
ii) dry weight represents the actual dry matter/fresh weight includes weight of water;
iii)-average height was determined by measuring the length; of the plants at various intervals;
-average dry weight was determined by heating the plants to exclude all the water; and then
taking their dry weights;
d) both height and weight are used to show rates of growth;
e) lacks cambium(tissue)hence no secondary thickening;

29. a) Fusion of an egg cell nucleus with sperm cell nucleus; to form a zygote
b) i) Meiosis
ii) In the testis/ testes/ ovary/ ovary
c) i) There is increased blood supply causing thickening of the uterine walls;

ii) Capillaries break up/ endometrium is lost with some blood/ menstruation occurs

d)

- Large number/ numerous blood vessels to increase surface area for exchange of materials
- Thin membrane for faster diffusion across it
- Has villi to increase surface area for diffusion
- Special cells to produce hormones
- Membrane selectively allows materials across it

30. a) $\frac{\text{Number of seeds that germinate}}{\text{Number of seeds planted}} \times 100$ = % seed germination

b) Seeds dry mass would have resulted in death of embryo thus no germination

c)i) Mean seedling fresh mass include the mass of water that has not resulted from growth

ii) At regular intervals of time; uproot seedlings (say five each) dry to constant weight, and record

d) Directly proportional / Increase in seed mass results in increase in % seed germination, survival and seedling fresh mass

e) Embryo well developed/ Embryo very mature;

- Large food reserves for growth and development

30. a) Directly proportional/ increase in enzyme concentration results in increase in reaction rate

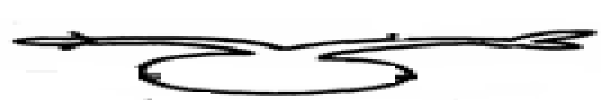
b) i) Increase in substrate concentration results in increase in reaction rate
Increase in concentration results in more active sites occupied by substrate molecules, resulting in higher turn over

ii)A rate of reaction constant/ does not change
Active sites fully occupied

iii) Sharp decrease in reaction rate
Enzymes denatured

c) PH/ Enzyme inhibitors/ Enzyme co- factors

31. a)



b) the rotation of the machines/ klinostat ensured equal distribution of Auxins in the seedling (upper & inner) side;

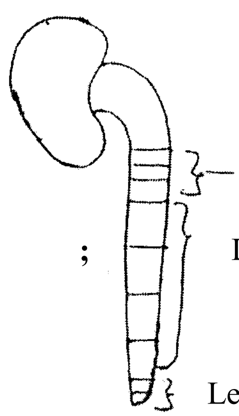
c) Klinostat;
d) radicle grow dominants; plumule grow upwards;
e)

STIMULI	RESPONSE (name)
light	Phototropism;
gravity	Geotropism;

32. (a) (i) Carbam inohaemoglohin:
(weak) Carbonic acid: (2marks)
(ii) Oxyhaemoglobin; (I mark)
(b) Secretes pleural fluid:
- Makes lungs air tight:(OWTTE) (2marks)
(c) Carboxyhaemoglobin doesn't dissociate readily (OWTTE):
Hence its formation reduces the capacity of haemoglobin to carry oxygen to time lungs
hence resulting in death: (2marks)
(d) Cuticle: lenticels: (Both to be correct to score I mark) (I mark)

33. (a) Region of rapid growth / cell elongation in a radicle: root

(b)



;

Region with least increase in marks

Least increase in marks

Least increase in ma mrks

(c) Dense cy1oplasms
Lack cell vacuoles
Thin cell walls
(d) -Presence of germination inhibitors / abscicic acid:
- Low concentration of hormones / Enzymes/ gibberclic acid:
- Impermeable seed coats to water and oxygen:
- Embryo not fully developed:

12. Genetics

1. a) BB;
b) AA;

2. a) Black mice are better adapted camouflage with the environment hence less are eaten by the

owls compared to the white mice which are easily seen;

b) Theory of natural selection;

3. - Heterostly – stigma located above anthers;
- Self sterility or incompatibility – pollen grain from the same plant do not germinate
- Protandry – Male parts mature before female parts;
- Protogyny – Female parts mature before male parts;

4. (a) Complete dominance is when an alliele completely surprises another intermediate fruits;
Incomplete dominance is when heterozygous organisms show an intermediate trait;
(b) Genetic recombination's of alleles reading to variations; Independent assortment of chromosomes;
Random fusion of gametes; mutations;
Environment (may either enhance or suppress expression of a gene);

5. (a) C-A – G – U – C _ A ;
(b) – Stones genetic information (in a coded form);
- enables transfer of genetic information unchanged to daughter cells through replication);
- Translates genetic information into characteristic of an organism 9thorugh protein synthesis);

6. Ability to pollinate; response to stimuli (tactic) nastic or tropics); Ability to exploit localized nutrients an ability to photosynthesize; Ability to disperse seeds/fruits, propagation;

7. (a) Glucose;

(b) The person was a sufferer of diabetes mellitus;
(c) Pancrease;

8. Continuous variation shows gradation in characteristic with intermediate; discontinuous shows distinct characteristics between organisms with no intermediate groupings;

9. -mutation;

-intermixing of genes already in the population through sexual reproduction recombination;
-crossing over during prophase of meosis I
-interdependent assortment of chromosomes, during metaphase of meosis I

10. i) Substitution;

ii) Deletion;
iii) Inversion;

11. i) C G G A T C T A G T G;

ii) C G G A U C U A G U G;

12. a) Continuous ;

b) Nutrition/ environment; genes;

13. a) Father X^HY ;

Mother X^HX^h ;

b) i) Genes found in the same chromosome and usually transmitted together;
ii) Across to determine an unknown genotype involving use of a recessive parent;

14. a) Colour blindness; haemophilia;

Sickle cell anaemia;
b) Part of X chromosome has homologous portion on the Y chromosome therefore if the X has
the recessive trait, it will show on the male phenotype ;

c) The son inherits the X chromosome from the mother while the daughter inherits the
X chromosome from the father;

15. (a) Inversion ;
- ustard gas;
- ionizing radiation;
- gamma rays;
- X- rays ;

16. (a) Ribonuclei acid /RNA

- Because it has uracil / presence of uracil;

G C A G

- S – P- S- P – S – P - S – P / G C A G

17. (a) Due to co-dominance /partial dominance/incomplete dominance/(Acc. equal dominance)
(b) Red: 2Pink : white – 1: 2:1 (Acc. 1RR: 2RW: 1WW) mark as a whole;
(c) Why women should drink extra milk;
(i) Bore formation for infants ;
(ii) pressure on bladder by the enlarging uterus;

18. a) Genes which are located on the sex- chromosomes and therefore are transmitted along with them
Example Haemophilia; colour blindness;
b) Where more than two genes control a particular characteristic/ trait;
Example ABO blood group system;

19. a) Parental Genotype Rr, Rr ;

b) Red: white;
119/41; 41/41;
2.90: 1
3: 1;

20. (i) Y – Chromosome-hairy pinna, pre-mature boldness; ; (any one)
(ii) X – Chromosome- haemophilia (bleeders disease); colour blindness; (any one)

21. The Gene that determine the growth of long hair on pinna is sex linked and an Y-chromosomes; V hence can only be inherited by males as a single gene and it expresses itself out phenotypically

22. Due to crossing over: that results in exchange of genetic materials between homologous chromosomes;

23. (a) Co dominance/ incomplete dominance:
(b) 1 Red flowered; 2 pink flowered; I white flowered: for ratio for phenotype)

24. (a) Albinism;
(b) Makes skin supple;
- Kills bacteria/ a mild antiseptic;

25. - Change in base sequence of the DNA;
✓ 1.

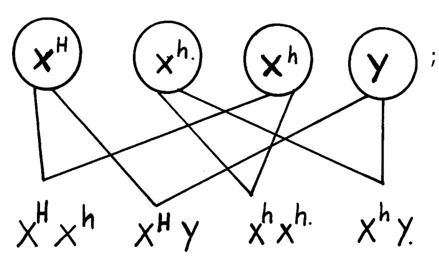
26. (i) Sudden and spontaneous change in structure of chromosome and DNA which is inherited
(ii) Chemical ionizing radiations, Uv light, extreme temperature or some virus

27. (a) GCCTATG – DNA
GCCUAUG- MRNA
(b) Ribosome;

28. a) Parental phenotype Pink flight feathers X White flight feathers
Parental genotype X^RX^r X X^rY
Parental gametes R X r X r X r
Fusion Rr R r r r
F1 genotypes XX XY XX XY;
b) incomplete;
c) i) Ribonucleic acid;
ii) has uracil base;
ii) – 3;
- There are three codons;

29. A – X^hY;
B – X^HY;
F – X^HX^h;

X^H X^h; X X^h;Y;

(b)


(c) Albinism; sickle cell anaemia; colour blindness; chondrodystrophic dwarfism;

30. (i) Father Mother
X^HY X^HY^H
Since father cannot have the recessive gene ad fail to be affected. The mother must be a carrier
on her second X chromosome for a male son to be haemophiliac.

(ii) Parental phenotypes mother carrier, father normal

Parental genotypes X^HX^h X^HY

X^H X^h X^H Y

X^HX^h X^HX^H X^HY H^hY

Carrier female Normal female Normal male Haemophiliac male

31. a) the two genes that control flower colour ,that is the gene for red flowers and the

one for white are codominate;

b) F1 phenotype pink flowers pink flowers

F1 genotype RW X RW ;

Gameter R W R W;

Fussion

F2 genotypes RR $\underbrace{\text{RW RW}}$ WW;

F2 phenotypes red

Flowered pink white flowered ;

Flowered

c)genotypic ratio= 1RR:2RW:1WW/RR:RW:WW=1:2:1 ;
Phenotypic ratio=1 red flowered:2 pink flowered:1white flowered ;
Notes: i) there must be cross on genotype
ii) gameter should be circled
d) recessive gene expressed it self only underlined homozygous condition while dominant gene expresses it self in both homozygous and heterozygous conditions;

32. (a) (i) Male and female flowers are separate hence cross pollination is made possible. ♂ ♀

T T t t

(ii) 1 Yellow : 3 Purple

Rej.: 15 yellow : 45 Purple

F1 genotype

(b) Let letter T represents purple maize grain

Let letter t represent yellow maize grain

♂ ♀

Parental phenotype: Purple Yellow

Parental phenotype: Purple Yellow

Genotype: TT x tt

Genotype: TT x Tt

F2 genotype

Gametes: T T t t

Gametes:

Tt Tt Tt ✓1

(c) Gene for purple grain;

(d) (i) Finger prints are used to identify criminals;

(ii) Blood groups are used to settle parental disputes;

33. a) Aa; Aa; because of one child - 4

b) AA; Aa; because of cross between parent 1 and 2
c) Lethal genes easily inherited;
d) Sickle celled anaemia; colour blindness; haemophilia

34. (a) Y – Blood group A^{+};

Z – Blood group B^{-}

(b)

Parental phenotypes	♂ Rhesus (+ve)		Rhesus (-ve)
Parental phenotypes	Rh Rh	X	Rhd Rhd;

Parental gametes: Rh^D Rh^D Rh^d Rh^d

Fusion;

F1 offspring; Rh^DRh^d Rh^DRh^d Rh^DRh^d Rh^DRh^d;

c) All Rhesus positive/ all Rh^DRh^d;

(d) None

35. – By keeping their mouth open/panting; to lose heat over surface area of the tongue by evaporation;

-Basking; to gain heat by conduction;
- Shivering; to generate heat through increased metabolism;
- Physical activity (e.g. running); to generate heat through metabolism;
- Hibernation; to increase metabolism;
- Putting on warm clothes when it is cold; to retain the heat energy;
- Reduction of physical activity; to reduce the metabolic rate;
- Migratory behaviour to cooler environment; to reduce the body temperature;
- Moving into water when it is hot; to cool the body;
- Staying around fire place; to gain heat by convection;
- Taking hot drinks; to warm the body;

36. a) Parental genotypes

i) Woman/ O^X - AO
ii) Man/ O - BO

b

	A	O
B	AB	BO
O	AO	OO

c) Cases of disputed paternity settlement

- Determining compatible blood groups in blood transfusion

d) i) Corresponding complementary DNA strand GAA;
ii) Corresponding RNA CUU

iii) Nitrates/ sulphites/ hydroquinone/ gamma/ beta/ alpha/ x-rays/ UV light/ hydrogen peroxide

37. let R rep. gene for Red flowers

W.rep gene for white flowers

a) parental phenotype Pink flowered Pink flowerd

genotype RW X RW ;

gametes R W R W ;

RR RW WR WW;

red pink white

b) phenotypic ratio 1Red:2Pink:1White;

genotypic ratio: 1RR:2RW:1WW;

c) $\frac{2}{4}$ x 480;=240

38.

Parental genotype; ***B B*** ***B b***

Gametes ; ***B*** ***B*** ***b*** ***b***

♀ \ ♂	***B***	***Bb***
b	***Bb***	***Bb***
b	***Bb***	***Bb***

Fl generation Award for punnet Square and genotypes

(b) (i) IBB : 2Bb: lbb

(1 mark for ratio, 1 mark Par genotype)

(ii) 3 B lack: I brown

(iii) 24;

39. (a) Homologous structures:

Structures of common embryonic origin modified to perform different functions;

Example: Eye structure in man and octopus/ wings in birds and insects (I mark)

Analogous structures Example

(b) They undergo mutations: resulting in new forms that rcsis selection resistant to drugs;

(c) (i) Failure of chromosomes to separate during anaphase I resulting in gametes with an extra chromosome and others with less chromosomes: (I mark)

(ii) Downs syndrome / Klinefelters syndrome/ Turners syndrome: any I (1mk)

40. a) Homologous structures have a common embryonic origin but are modified to Perform different functions; while analogous structures have different embryonic origin but are modified to perform similar functions;

b) Nictitating membrane; post anal tail; body hair;

41. a) Pentadactyl limb structure of mammals; beaks of birds; feet of birds;

b) - Missing links between fossils because some parts or whole organisms were not fossilized

- Some parts were distorted during fossilization hence may give wrong impression of structures;

- Some structures have been destructed by geological activities;

42. Camouflage is the conceal/ element of identity of an organism by resembling the color of the environment while mimicry is the imitation of non- living organisms to conceal identity

43. Light energy splits water molecules; into hydrogen ions and oxygen atoms;

44. (a) Caecum/ Rumen/ pauch;

(b) Closes to prevent food from moving up the oesophagus;

45. (a) – the soft bodied organisms fail to fossilize;

- Human activities interfere with fossilization;

Earth movements e.g. volcanic eruptions interfere with fossilization; (mark any first2 pts

(b) – They resembled from neck downwards;

-They walked upright;

- The shape of the skull suggested they were able to speak;

46. a i)vestigial structures are those structures that have ceased to be functional over along period of time and hence reduced in sizes

ii)-appendix;

-caecum

-coccyx or tail/tail bone;

- Nictitating membrane/semi - lunar fold at the corner of the eye;

-ear muscles

- Body hair;

b) Disease causing organism mutates; and became resistant;

47. Struggle for existence –environmental pressure on the population in order to survive;

Survival for the fittest-advantageous variations an individual possesses to make it survive;

48. Secretion of antidiuretic hormone; rearbsorption of salts at the loop of Henle;

49. -Divergent evolution refers to a situation where by organisms that are believed to have had a common ancestral origin have homologous structures which have been modified to suit different environments;

50. a) Allows survival of organisms with better qualities / traits / characteristics; eliminates organisms with unfavorable characteristics/ traits;

b) Divergent;

51. Evidence does not support Larmarks theory

Acquired characteristics are not inherited/;

Inherited characteristics are found in reproductive cells ;

52. (a) Vestigial structures

(i) Are those structures that have ceased to be functional over a long period of time hence reduced in size;

(ii) Appendix/coccyx/tail/ nictitating membrane semilun fold at the corner of the eye/caecum/ear muscles, body hairs;

(b) Disease causing micro-organisms mutate and become resistant;

53. a) The gradual emergence of complex life forms from pre-existing simple forms over along period of time ;

b) Nature selects those organisms with structures that are well adapted to survival in the environment. These structures are passed to their offspring; organisms with structures that are poorly adapted perish ;

54. The insecticide kills most of the insects when introduced; those that survive; give rise to a new generation of flies that are resistant to insecticide.

55. - Most organisms especially soft-bodied ones do not form fossils;

- Most fossils have not yet been discovered;
- Exposed fossils are usually destroyed by physical and chemical weathering;
- Earth movements e.g. volcanicity, earthquakes, tsunami do destroy fossils;
- Most animals are prayed upon;

56. -Fossil records/paleontology ;

-Comparative anatomy/taxonomy;
-Comparative embryology;
-Geographical distribution;
-Cell biology;

-Comparative cellulogy/immunology; (award 1st three 3mks)

13. Evolution

1. a) Homologous structures have a common embryonic origin but are modified to Perform different functions; while analogous structures have different embryonic origin but are modified to perform similar functions;
   b) Nictitating membrane; post anal tail; body hair;

2. a) Pentadactyl limb structure of mammals; beaks of birds; feet of birds;

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4. Light energy splits water molecules; into hydrogen ions and oxygen atoms;

5. (a) Caecum/ Rumen/ pauch;
   (b) Closes to prevent food from moving up the oesophagus;
6. (a) – the soft bodied organisms fail to fossilize;
   - Human activities interfere with fossilization;
   Earth movements e.g. volcanic eruptions interfere with fossilization; (mark any first2 pts
   (b) – They resembled from neck downwards;
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   - Body hair;
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17. -Fossil records/paleontology ;

-Comparative anatomy/taxonomy;
-Comparative embryology;
-Geographical distribution;

-Cell biology;
-Comparative cellulogy/immunology; (award 1st three 3mks)

18. Nature selects organisms that are well adapted and allows them to survive: but rejects those that are poorly adapted they perish/die/become eliminated;

19. (a) The genotype of an organism is not changed by characteristics acquired during the life/ phenotypically acquired characteristics do not affect the genotype of an individual

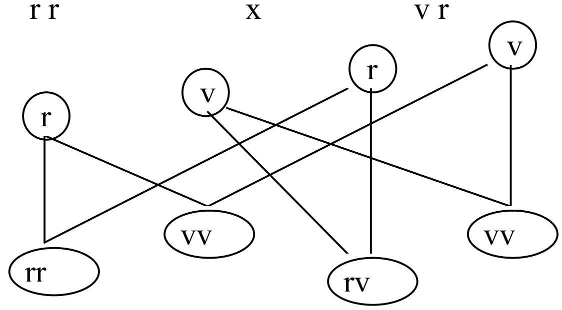
(b) - Missing links (due to decomp0osing of savaged form)
- Distortion of parts (some parts were flattened);
- Geographical activities (e.g. earthquake, faulting, erosion) (any 2)

20. (a) White flowers.

(b) The white flowers were fewer that is the ratio of ¼ of the total flowers.

Parental phenotype	white flowers		white flowers
Parental genotypes	r r	x	v r

Gametes



(c) A cross between unknown genotype with a homozygous recessive/double recessive genotype

(d) - Low mental capability
- Short/stubby fingers
- Slit eyes

21. (a) Emergence of new life forms//species//organisms; from pre-existing forms gradually over a long period of time;

(b) Fossil records//Palaeontology;
These are remains of organisms preserved in some naturally occurring materials e.g. sedimentary rocks for many years; They give direct evidence of the type of organisms that existed at a certain geological time//show a gradual increase in complexity/morphological changes of organisms over a long period of time e.g. skull of man

Geographical distribution;
present continents are thought to have been a large land mass joined together; continental drift led to isolation that lead to different patterns of evolution; e.g. camels of Africa resemble the Ilamas of S. America// tiger of Asia resemble jaguars of S. America // unique Marsupials of Australia;
(accept any valid example)

Comparative Embryology;

Vertebrate embryos show morphological similarities in their early development; suggesting these organisms have a common origin; Accept – embryos of mammals /reptiles/ amphibians compared to show the similarities;
<u>Cell Biology// Cytology;</u>
Occurrence of cell organelles e.g. Mitochondria
Cytoplasm nucleus// Accept any correct organelle; point towards a common ancestor;
<u>Comparative serology;</u>
Analysis of blood proteins and antigens / Rh factor/ blood group /haemoglobin reveal phylogenetic blood group/haemoglobin reveal phylogenetic relationships; Those species that are more close phylogenetically related contain more similar blood protein;// Antigen-antibody reactions/serological tests/experiments with serum reveal some phylogenetic relationship depending on the level of precipitation.
<u>Comparative anatomy/taxonomy;</u>
- Members of a phylum show similarities indicating common ancestry; These organisms have similar functions e.g. presence of digestive, urinary, nervous systems e.t.c;
- Homologous structures like pentadactyl limbs in different animals like monkey and rats have similar borne arrangement hence same origin but modified to perform different functions// adaptive radiation//divergent evolution; vestigial organs//coccyx Appendix;
- Analogous structures like wings of birds and wings of insects with different embryonic origin but perform same function//convergent evolution; (maximum 18mks)

N/B- Mention of each evidence 1mk each
- It is muscular//Has cardiac muscles which are myogenic;//capapble of contracting and relaxing without nervous stimulation to ensure the heart beat without stopping;
- Supplied by vagus and sympathetic nerves; which control the rate of heart beat depending on body's physiological requirement;
- Has tricuspid and bicuspid valves//arteria ventricular valves; to prevent back flow of blood into wrong directions;
- Has semi lunar valves at the base of pulmonary artery and aorta; to prevent back flow of blood into right and left ventricles respectively;
- Presence of valve tendons attached to the walls //arteria ventricular walls; prevent arteria ventricular valves // tricuspid and bicuspid valves from turning inside out;
- Supplied by coronary artery; to supply food and oxygen t the cardiac muscles for their pumping action;
- Coronary vein; draws away metabolic wastes;
- Heart is enclosed by pericardial membrane; which secrete fluids which lubricates//reduces friction on the walls as it pumps;
- Pericardial membrane is lined with a layer of fat to act as shock absorber; hold the heart in position; checks over dilation of the heart;

- The heart is divided into two by (artria ventricular) septum; which prevents mixing of oxygenated and deoxygenated blood;
- The sino-artria node// pace maker; initiates a wave of excitation leading to contraction and relaxation of cardiac muscles;
- The artria –ventricular node; in the heart spread out waves of excitation through out the heart

The structure tied to function wrong function cancel the mark of the structure. Correct structure minus function do not qualify for a mark

22. (a) Nature or the environment selects those individuals that are sufficiently adapted; and rejects those that are not adapted;

(b) Adaptation by natural selection.

- Individuals of the same species show variations.
- The variations are caused by genes that can be passed on from parents to the off springs (inherited);
- Some of these variation become more suitable or favorable or advantageous in the prevailing environmental conditions;
- Because organisms usually produce more off springs than the environment can support; competition for resources sets in;
- This leads to struggle for existence;
- Individuals with more favorable characteristics/ adaptations/ gene mutation have better chance of survival in the struggle;
- Hence they reach reproductive age, reproduce and pass on favorable characteristics to the off spring;
- Those with less favorable characteristics or adaptations fail to reach sexual maturity; they die young;
- Examples of natural selection include- malarial parasite/plasmodium which has developed strains that are resistant to anti-malarial drugs;
- Sickle cell trait; the homozygous die young and the heterozygous are resistant to malaria.

(c) – Convergent evolution.

- This is a phenomenon where structures from different embryonic origins are modified to perform the same function. E.g. wings of birds and those of insects, eyes of human beings and those of octopuses;

- Divergent evolution.

- This is a phenomenon where one basic structural form is modified to give rise to various different forms which perform different functions. E.g. pentadactyl limbs of vertebrates, shapes of beaks in birds;

(d) Evidences to show that evolution has taken place. (Any 4)

i) Fossil records.√

ii) Comparative anatomy. .√

iii) Comparative embryology. .√

iv) Geographical distribution (continental drift).√

v) Vestigial organs.√

vi) Cell biology. √

(i) Fossils records;

Fossils are remains of dead organisms preserved naturally. They indicate that organisms have evolved from simple life forms to most complex forms. Fossils of human beings indicate that the modern human being has a highly developed brain and uses speech for communication unlike the early human being. Of horses show that the modern horse is 1.5 m high, lives in dry grassland, teeth are adapted for chewing and it stands on one digit whose distal end is converted into hoof.

(ii) Comparative Anatomy;

This involves comparing the form and structure of different organisms.Some groups organisms show basic structural similarities suggesting common or related ancestry showing divergent evolution.

Other groups of organisms show morphological similarities but are found to have different ancestry showing convergent evolution;

(iii) Vestigial Organs;

Some structures have ceased to be functional and have reduced in size; such structures are called vestigial structures. Examples include the appendix and the tail in human beings; reduced wings in flightless birds, nictitating membranes in mammalian eyes and lack of visible limbs in pythons.

(iv) Geographical distribution;

- Its believed that long ago the land was one mass which later drifted apart to form the current continents. This is called the continental drift.

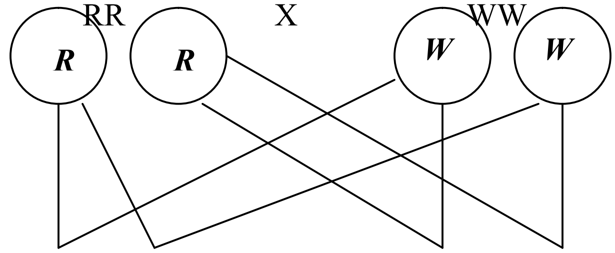
- Regions with similar climatic conditions and lie in the same latitude have flora and fauna that are not identical. This indicates that they have evolved differently; e.g. Amazon forest of South America has long tailed monkeys, panthers and jaguars while similar African forests have short tailed monkeys, leopards and cheetahs.

(v) Comparative embryology;
Studies show that embryos of fish, birds, amphibians, reptiles and mammals are morphologically similar during the early stages of development but with time they develop and change to look like their parents;

(vi) Cell biology;
- Cells of higher organisms show basic similarities in their structure and function; e.g. the presence of cell membranes and organelles such as mitochondria, ribosomses and golgi bodies.
- Higher plant cells have cellulose cell walls, chloroplasts and starch showing evolution from a common ancestry.
- The blood pigment, haemoglobin is common in vertebrates and invertebrates.

23. a) organic evolution is the process by which changes in the genetic composition occur in response to environmental changes
RCH
b) within the population some individual posses the gene for resistance to the antibiotic or it develops the genes by mutation ; such genes lead to production of enzyme which neutralize the antibiotic; the resistance forms survive the antibiotic hence transmit their advantages genes to their offspring; thus a new population of resistance strains is established(e mergence of new species(speciation)
c) fossil records;
-remains of ancestral forms that were accidentally preserved in some naturally occurring materials
-they give direct evidence of the type of animals and plants that existed at a certain geological agp
-the fossils records also show gradual increases in complexity of organism over time
e.g. evolution of man
-by comparing fossils of different organism its possible to tell the phylogenic relationship between the organism

24. RR X WW

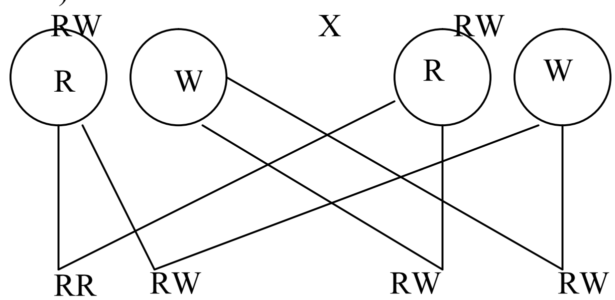


;

;

RW *RW* *RW* **RW**

b)



Red Pink Pink White ;

1 : 2 :

c) Gene for red colour and white colour in flowers are co dominant/ equal dominance/ none is dominant/ recessive

14. Irritability and sensitivity in (a) plants (b) animals

1. - To prevent excessive water loss/desiccation;
   - Provide surface area for muscle attachment;
   - Support and protect inner delicate tissues;
2. a) Phototropism;

   b) Auxins migrate from the side of the shoot that is exposed to light towards the darker side;
   The higher concentration of auxins on the darker side stimulates rapid growth hence the shoot bends towards the light source;
3. a) Iris;

   b) Circular muscles relax; radial muscles contract widening the pupil;
   c) Adjustment of the eye structure to bring an image from a near or far object into sharp FREE on the retina;
4. a)

Taxes	Tropisms
- Locomotory responses - Fast response - No hormones involved	- Growth responses - Slow response - Influenced by hormones

   (b) – Escape from harmful conditions/stimuli;
   - Move in search of food/nutrients;
5. (a) (i) Motor neurone;

   (ii) Cell body located in the central nervous system;
   (b) Arrow head towards terminal dendrites

(c) C- Protection/insulate axon;
D- Speeds up transmission of an impulse

6. (a) Due to the difference in atmospheric pressure and the pressure inside the ear;
(b) Swallowing; yawning;

7. (a) Photosynthesis;

(b) Night on the list side makes the auxin to move/migrate/Diffuse to the dancer side;
there auxin cases faster elongation/growth I the dark side; Hence curvature/bending;

8. – Colour blindness;
- Haemophilia;
- Sickle cell anemia;
- Albinism

9. a) short-sightedness/myopia;
b) This defect can be corrected by wearing glasses with concave (diverging) lenses; these bend light rays outwards before they reach the eyes enabling them to be FREEed ion the retina;

10. a)

Arteries	Veins
- Thick muscular - No valves (except pulmonary artery and aorta at the base - Narrow (small) lumen	- Thin muscular walls - valves present; - Wide lumen (large) lumen;

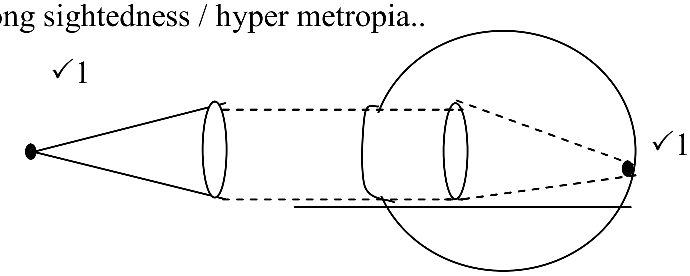
b) Arteriosclerosis; reject Artheroma

11. a) Retina;

b) Cones and rods;

12. a) Long sightedness / hyper metropia..

b)



c) Stereoscopic vision/ binocular vision/ improved visual acuity; gives a wider angle of vision; if
one is damaged man is not blinded;

13. Water proof – Prevent water from reaching the inner cells

Has Karatin – For protection against mechanical injury

14. i) Equalizes the pressure between the outer ear and the middle ear;
ii) Transmits and amplifies vibrations from the ear drum to the oval window;

15. a) – Conditioned reflex requires repeated stimulus to bring about response while simple reflex requires single stimulus to bring about response;
- Conditioned reflex requires behaviour modification hence experience while simple reflex involves direct action and is independent of experience;
b) It has a long axon to transmit nerve impulse myelin sheath and rod of ranvier for faster impulse transmission;

16. a) i) Iris;
ii) Optic name;
b) Circular muscles of the iris;
Radial muscles contract;
The size of the pupil enlarge to allow more light to enter;

17. Chemotropism; Reject chemotrophism
Negative photo taxis; Reject photo taxis alone

18. i) Thigmotropism / 1-laptotropism

ii) High concentration of auxin on side away from contact surface; promotes faster growth of this side; causing tendril to curl round the object.

19. Thigmotropism / haptotropism;

20. a)Hormone/growth substance /IAA; which inhibits the development/growth of lateral shoots/buds/causes apical dominance; /removal of the terminal buds cause the growth/development and sprouting of lateral buds ; 2mks
b)The pruning of coffee /tea/ledge; etc Rej. Pruning alone/trimming ;

21. cerebrum/cerebral hemisphere/cerebral cortex;

22. a)long sightedness/hyper netropia;
b)the eye ball too short/eye lenses are unable to FREE because they are flat//thin/weak; hence unable to FREE the image on the retina OR the eye are unable to commodate/change their focal length; 2
c)by wearing convex/biconvex lens/converging lenses; 1mk

23. a)A-relay/intermediate /associates;
B-motor neurone/efferent neurone;
b) Impulse initiates release of acetyl choline /transmitter substance (at the end of the sensory neurone);acetyl choline which diffuses across the gap; generate an impulse

in the next neurone; (Rj. Message for impulse)

24. (a) -Hearing;
-Body balance (and posture);
(b) Coiled to accommodate many sensory cells:
- Filled with endolymph to transmit (sound) vibrations.
- Has sensory hairs/cells to generate nerve impulses when stimulated:

25. Used in the transmission of nerve impulse.
✓
- For respiration;
✓

26. - Proper functioning of the nervous system and alimentary canal;
✓

27. (a) Enables the organism to escape from injurious stimuli/seek favourable habitats;
(b) Cerebrum

28. The conified layer of the epidermis consist of dead cells which form a tough outer coat;
that protects the skin against mechanical damage/bacterial infection/ water loss;
Sebaceous glands produce an oily secretion sebum which give hair its water repelling property; that keeps the epidermis sapple and prevents it from dyring/sebum too prevents bacterial attack due to its antiseptic property;
Has blood vessels; that dilate and contract;
In hot conditions, they dilate; increasing blood flow near the skin surface enhancing blood flow near the skin surface; minimizing heat loss;
Has hairs; stand during cold weather thus trapping a layer of air which prevents heat loss; In hot weather they i.e close to the skin surface; to enhance heat loss to the atmosphere.
Hair follicle; has many sensory neurons which respond to movements of the hair; increasing sensitivity of the skins. Has subcutenous layer; contains fat whihch acts as a heat-insulating layer and a fuel storage;
Has malpighian layer; consists of actively dividing cells tht contain fine granules of melanin; that prevents the skin against ultraviolet light rays from the sun;

29. a) i) Myopia/ short sightedness
ii) Long eyeball/ too long eye ball
b) Use of concave/ diverging lens; to diverge the rays from image to FREE onto retina
c) For colour reception/ vision
For vision in bright light/ day
d) Retina has many rods; to perceive / enable organism see in dim light/ darkness

30. Water proof – Prevent water from reaching the inner cells

Has Karatin – For protection against mechanical injury

31. i) Equalizes the pressure between the outer ear and the middle ear;
ii) Transmits and amplifies vibrations from the ear drum to the oval window;

32. a) – Conditioned reflex requires repeated stimulus to bring about response while simple
reflex requires single stimulus to bring about response;

- Conditioned reflex requires behaviour modification hence experience while simple reflex involves direct action and is independent of experience;

b) It has a long axon to transmit nerve impulse myelin sheath and rod of ranvier for faster impulse transmission;

33. a) i) Iris; ii) Optic name;

b) Circular muscles of the iris; Radial muscles contract;
The size of the pupil enlarge to allow more light to enter;

34. Chemotropism; Reject chemotrophism
Negative photo taxis; Reject photo taxis alone

35. i) Thigmotropism / 1-laptotropism

ii) High concentration of auxin on side away from contact surface; promotes faster growth of this side; causing tendril to curl round the object.

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b)The pruning of coffee /tea/ledge; etc Rej. Pruning alone/trimming ;

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b)the eye ball too short/eye lenses are unable to FREE because they are flat//thin/weak; hence unable to FREE the image on the retina OR the eye are unable to commodate/change their focal length; 2

c)by wearing convex/biconvex lens/converging lenses; 1mk

40. a)A-relay/intermediate /associates;
B-motor neurone/efferent neurone;

b) Impulse initiates release of acetyl choline /transmitter substance (at the end of the sensory neurone);acetyl choline which diffuses across the gap; generate an impulse in the next neurone; (Rj. Message for impulse)

41. (a) -Hearing;
-Body balance (and posture);

(b) Coiled to accommodate many sensory cells:
- Filled with endolymph to transmit (sound) vibrations.
- Has sensory hairs/cells to generate nerve impulses when stimulated:

42. Used in the transmission of nerve impulse.
✓

43. - Proper functioning of the nervous system and alimentary canal;
✓

- For respiration;
✓

44. (a) Enables the organism to escape from injurious stimuli/seek favourable habitats;
(b) Cerebrum

45. (a) Positive phototropism; reject phototropism only
(b) Positive geotropism; reject geotropism only
(c) Thigmotropism

46. - comified layer made of dead cells to protect from mechanical damage, also protect *KKE*
from desiccation/dehydration.
- Granular with living cells to replace the worn out layer.
- Malphigian layer – cells divide to form new epidermis.
- Malphigian cells with melanin pigment which protect from c ultra violet rays from the sun.
- Blood vessels to supply oxygen and nutrients. Remove CO_2 and nitrogenous wastes.
- Superficial blood vessels/arterioles dilate. When it is hotg. So that more blood flows near the skin surface for more heat loss.
- Superficial blood vessels consmet/vasoconstriction. When it si cold. So that less blood flows near the skin surface to reduce heat loss.
- Erector pili muscle contract when it is cold to raise hair/hair stands upright. To trap more a ir to reduce heat loss/insulate.
- Erctor pili muscles relax when it is hot to make hair lie flat. On the skin to trap less air hence reduce insulation/increase heat loss.
- Sweat glands excrete excess water, mineral salts traces of live and lactin acid.
- The water in sweat evaporates carrying away latent heat of vaporization to lower the body temperature.
- Subcutaneous layer/dispose tissue insulation the bod y/ reduce heat loss.
- Sensory nerve ending which are sensitive to touch/pain/heat cold.

47. Conjuctiva – protects eyeball from mechanical injury

Cornea – Allows light to pass through
Iris – Controls amount of light passing through
Retina – Where image is formed
Forea – Where image is formed
Sctera – Protect the eye ball; give it shape
Choroid – Absorbs stray lights
Provide nourishment to the eye
Aqueous/ vitreous humour – refract light into the eye towards retina maintain shape of eye ball
Ciliary body – Controls curvature of the lens
Rods – Perceive light of low intensity
Cones – Perceive light of high intensity

15. Support and movement in (a) Plants (b) animals

1. a) have closely packed cells which when turgid provide mechanical support;
   b) Their walls are thickened with cellulose which offer mechanical support;
   c) Consists of dead cells thickened with lignin;
2. a) Lumbar vertebra;

   b) - Broad neural spine;
      - Large and broad centrum;
      - Broad and long transverse processes;

   c) Passage of spinal cord;
3. (a) Cervical vertebra;
   (b) R – (Facel) for articulation with the next verterbra;
   S- (Transverse process) for attachment of muscles;
   (c) Treck region/ cervical region;
4. (i) Humerous; Scapula;

   (ii) Synovial fluid; Lubrication of bones/prevent friction;
5. – Endoskeleton;

   - Hydrostatic skeleton;
   - Exoskeleton;

6.

Muscle cell	Palisade cell
- Lacks chloroplast - lacks cell wall - small in size - presence of centrioles - tiny and numerous	- Has chloroplast; - has cell wall; - large in size ; - lack of centrioles; - large central cell vacuole ;

7. a) Ulna;

b) i) Humerus;
ii) Hinge

8. a) Exoskeleton;

b) Supports body tissue and organs, protects inner parts, reduces water loss/ evaporation, helps
in movement/ attachment of muscles;

9. a) Provide mechanical strength / support/ it is a strengthening tissue;

b) Xylem vessels and tracheids have lignified walls; to provide support;
one is damaged man is not blinded;

10. a) Tendons are structures which attach skeletal muscles to bone while ligaments are structures
that hold two bones together;
b) Use of turgor pressure / turgidity; use of tendrils and climbing stems; tissue distribution in
stems (parenchyma) sclerenchyma / collenchyma); use
of xylem (thickened tracheids & vessels) ; use of spines and thorns e.g roses.

11. a) i) Lignin;

ii) Phloem;
b) Growing areas of root, stem/ shoot, meristems ;
Storage organs – Fruits, seeds, stems, roots, leaves;

12. - Maintain shape of the body ;

- Protect delicate organs of the body e.g. heart, brain;
- Place/ area of attachment for other organs of the body;

13. Capitulum

14. (a) Scapula;

(b) (i) Humerous;
(ii) Ball and socket joint ;
For muscle attachment;

15. a) Femur; ;
b) Reloic girdle/ pubis of peloic girdle;
c) Hinge joint;

Rj. thoracic alone or vertebra alone
b)X-neutral spine;
W-centrum;

16. (a) Axis;
(b) Fits in the neural canal of atlas to permit for turning of the head:

17. a) - Sclerenchyma;
- Xylem;
- Collenchyma;
Accept Parenchyma
b) i) X – Biceps;
Y- Triceps;
Reject Flexor and Extensor
ii) X (Biceps) relaxes; as Y (Triceps) contracts;
c) Hinge joint;

18. a)locomotion enables animal to move from one place to another in search of food; mates;
to escape from predators; to disperse/avoid unfavourable environments;
b)-have streamline body which reduces fiction; the scales overlap backwards and he lies flat close to the body, thus enhancing he streamline shape;
-they have air-filled swim bladder tat helps them to maintain a density that is equal to that of the surrounding water; helping the fish to float; (making forward movement easy)
-tail fin long/has large surface area to increase the amount of water displaced resulting in an increase in forward thrust;
-they have strong tail muscles which enable the tail to move from side to side against water;(pushing the fish to move forward)
-the have paired pectral and pelvic fins; which are used for steering; for bringing about downward or upward movement; as breaks//for braking; and for preventing pitching;
-they have unpaired fins, dorsal and anal fins; which increases the vertical surface area preventing fish from rolling or yawing;
-the fish has inflexible head which help, fish to maintain forward thrust;
-have fleshable backbone onto which myotomes are attached; the muscles contract and relax to bring about undulation movement;
-fish also secretes mucus which covers body and reduces friction during movement;

16. Human health

1. a) Vibrio cholerae;

   b) i) Pig;
      ii) Fresh water snail;
   c) Injected by a female anopheles mosquito;
2. (a) Fever;

   (b) Plasmodium;
   (c)- Uses of vaccines;
   - Sleeping under treated mosquito nets
   - Getting rid of stagnant water and bushes around residential areas;
   - use of ant malarial drugs; (any 1st correct)
3. <u>Salmonella</u> <u>typhii</u>

4. a)plasmodium parasite;

   b)-drainage of poles that act as breeding grounds for mosquitoes;
   - pools that cannot be drained should be sprayed with oil or insecticides to destroy mosquito
     larvae;
   - fish that feed on mosquito larvae may be introduced into such pools;
   - tall grass and bushes which be cleared near human dwelling;
   - visitors to areas where malaria is prevalent should take anti-malaria drugs;
5. (a) (i) <u>Salmonella</u> <u>typhii;</u>

   (ii) <u>Entamoeba</u> <u>histolytica;</u>
   (b) Malaria ;
6. a) Protoctista;
   b) Cholera;
      Syphilis;
   c) Use of condoms;
      Abstinence;
      Faithfulness to one partner;
7. To know HIV status; so as to take appropriate measures. If positive start medication/negative
   avoid infection;
8. a) malaria;
   b) Amoebic dysentery/ Amoebiasis;

SECTION III- QUESTIONS
CONFIDENTIAL INFORMATION TO SCHOOLS AND PRACTICALS

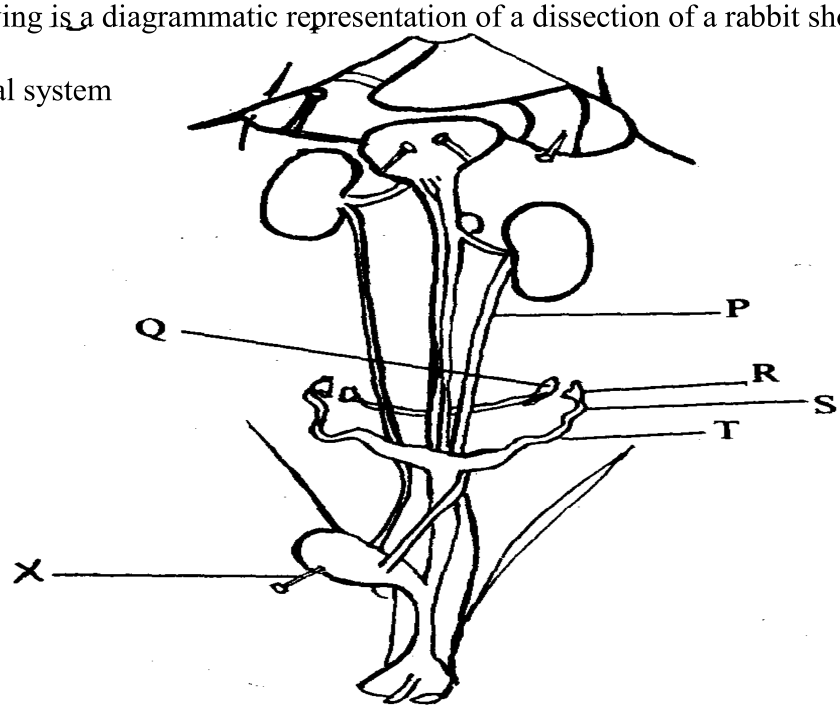
KAKAMEGA CENTRAL DISTRICT

Each candidate will require the following:-

- *Solution V- sucrose laboratory chemicals*
- *Solution W – Glucose*
- *Means of heating*
- *2 test tubes*
- *Benedicts solution (10mls)*
- *Dilute HCl (5mls)*
- *Sodium hydrogen carbonate solution*
- *Access to water*
- *A dropper*
- *Measuring cylinder (to measure 10mls)*

1. a) Identify solutions **V** and **W** by carrying out the food tests as indicated in the table below
b) Which of the two solutions **V** and **W** would you recommend for a person who needs an
immediate source of energy? Give a reason for your answer.

2. The following is a diagrammatic representation of a dissection of a rabbit showing the
urinogenital system



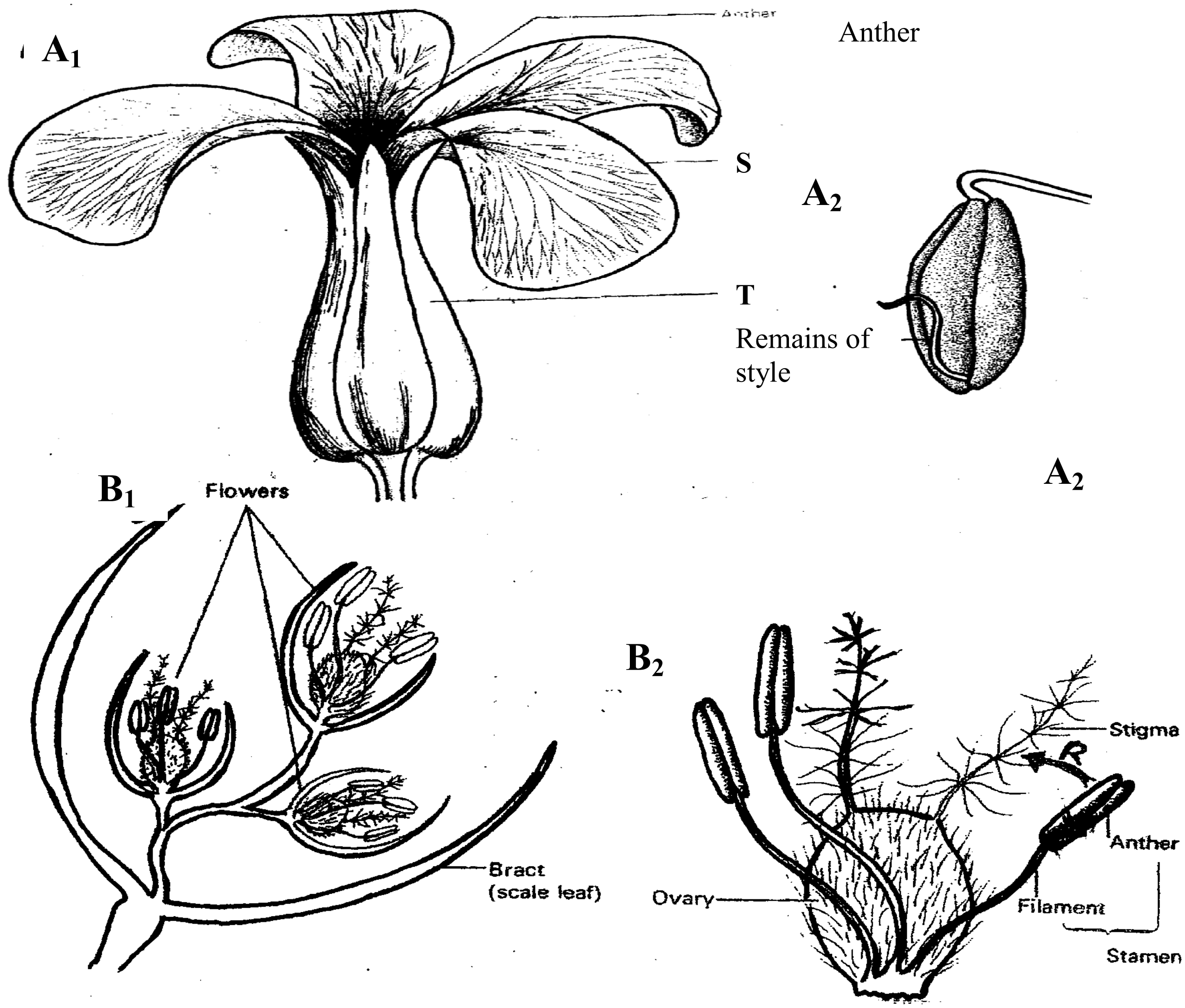
a) In the table below, name the structures labeled **P, Q, S** and **T**. For each of the structures,
state **one** function

b) i) Identify the sex of the rabbit that was dissected

ii) Give **two** reasons for your answer in **b(i)** above

c) Name the instrument labeled **x** in the diagram above

3. Study the photographs below and use them to answer the questions that follow;



a) Using the number of flowers arising from the shoot of each plant, state the flowers

labelled **A_1** and **B_1**

b) Name the class of the plant from which each of the flowers was obtained. Give a reason

for your answer in each case :

c) Name the parts labelled **S** and **T**

d) What type of ovary is shown in flower **B_1**? Give a reason for your answer.

e) i) Name the agent responsible for the process represented by the arrow labelled

R in **B_2**

ii) Give a reason for your answer in **e (i)** above

iii) List **two** other features (not shown in the photograph) expected of such flowers as **B_1**

f) i) Name an agent that brings about a similar process as the one shown by the

arrow in **B_2** for **A_1**

ii) Give a reason for your answer in **f(i)** above.

g) What is the likely agent of dispersal of the specimen labelled **B_2**?

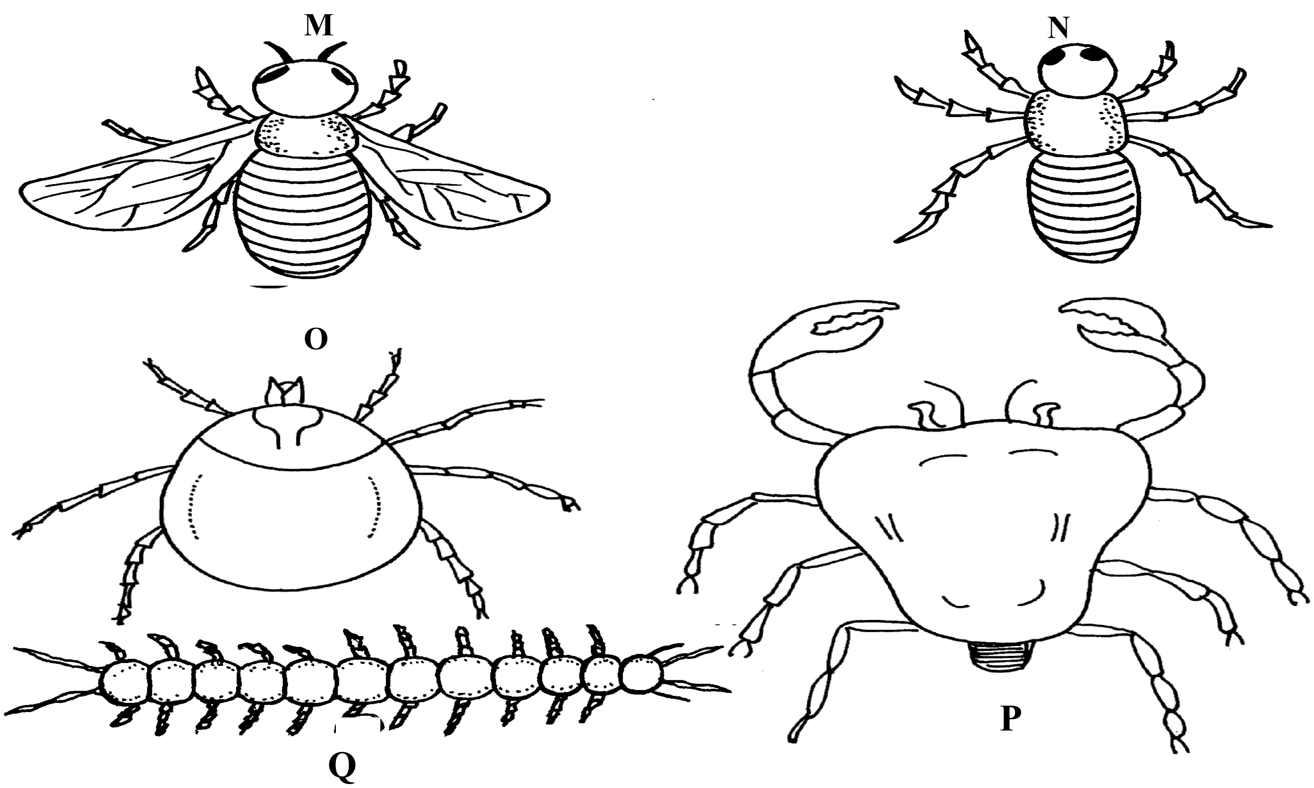
CONFIDENTIAL INFORMATION TO SCHOOLS AND PRACTICALS

KAKAMEGA EAST DISTRICT

Each candidate requires the following:-

- *One large ripe tomato labelled D_1*
- *One ripe orange/lemon fruit labelled D_2*
- *5ml DCPIP (1g of DCPIP dissolved in $1000cm^3$ of distilled water)*
- *Four clean test tubes*
- *Three droppers*
- *Scalpel blade*

1. Study the diagrams **M, N, O, P** and **Q** below representing organisms in the environment and
use them to answer the following questions:-



(a) With the reasons, identify the phylum to which they belong:-
(b) Identify the classes of the following organisms:-

M .... **N** ....
O .... **P** ....
Q ....

(c) Give **two** reasons for identifying the classes of organisms **M** and **P**

(d) What is the economic importance of the class to which **M** belongs?

2. You are provided with specimen **D_1**. Make a vertical (*longitudinal section through it to obtain two equal halves)*

(a) (i) Draw and label one half of **D_1**

(ii) Calculate the magnification of your drawing (show your working)

(iii) With reasons, identify the type of fruit **D_1**

(b) Squeeze juice from **D_1**, into a beaker. Label two test tubes **A** and **B.** In each test tube put $1cm^3$ of DCPIP

(i) To test tube **A** add the juice drop by drop shaking well after each drop. Record the number of drops required to decolourize DCPIP in the table below.

TEST TUBE	**No. of drops required to decolourize DCPIP**
A	
B	

(ii) Identify the food substance being tested

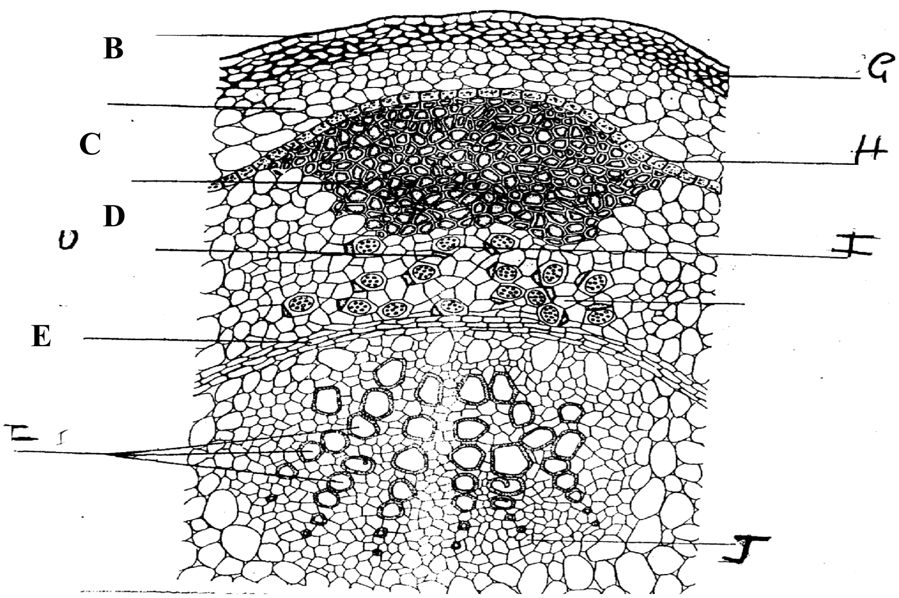
(iii) Which of the specimens **P_1** and **P_2** has high of the food substance being tested above?

(iv) What is the value of the food substance above to a growing baby?

(c) Boil the remaining juice extracted from **D_1** in the boiling tube for one minute and cool it.

Using a dropper, add the boiled juice into another test tube labelled **B**. Containing $1cm^3$ of DCPIP. Record the number of drops required to decolourize the DCPIP in the table above. What is the effect of boiling the juice?

3. The diagram below represents a cross-section of a plant stem. Study it carefully and answer the que at follow:-



(a) Identify letters that represent tissues responsible for support and name the tissues

(b) State **two** ways in which the tissues named in **(a)** above offer support

(c) (i) Identify the part labelled **H**

....

(ii) What is the role of this part?

(d) (i) If the plant from which the section had been obtained was placed in water containing

eosin dye, which part would you expect to be stained with the dye?

(ii) Name **three** forces which help water containing this dye (eosin) to pass through the dyed tissue

(e) (i) Name the tissues labelled **I**

....

(ii) What is the name of the cell **C** seen adjacent to tissue **I**?

(iii) State the function of this cell **C**

CONFIDENTIAL INFORMATION TO SCHOOLS AND PRACTICALS

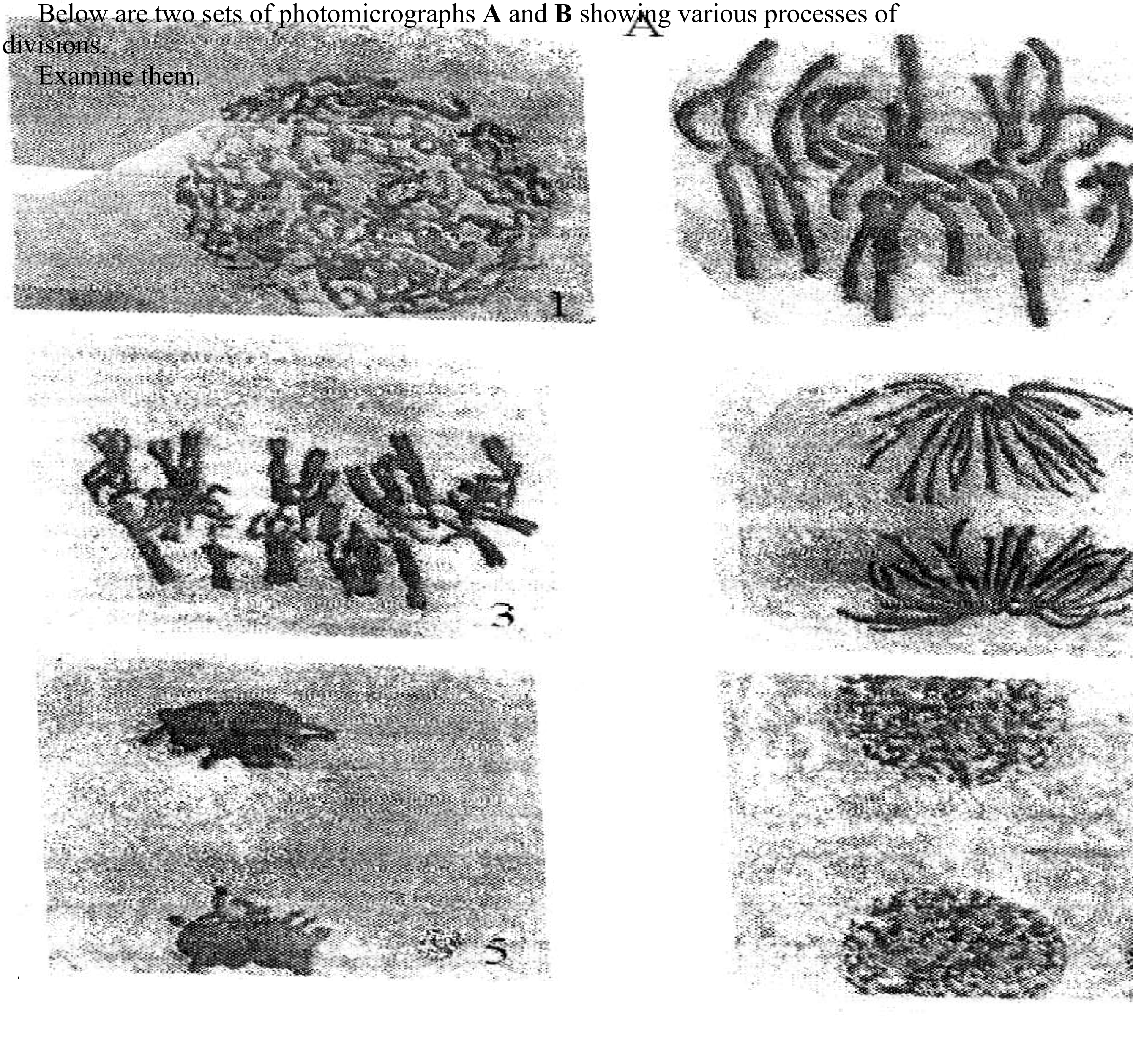
MIGORI DISTRICT

Each candidate should be provided with the following:

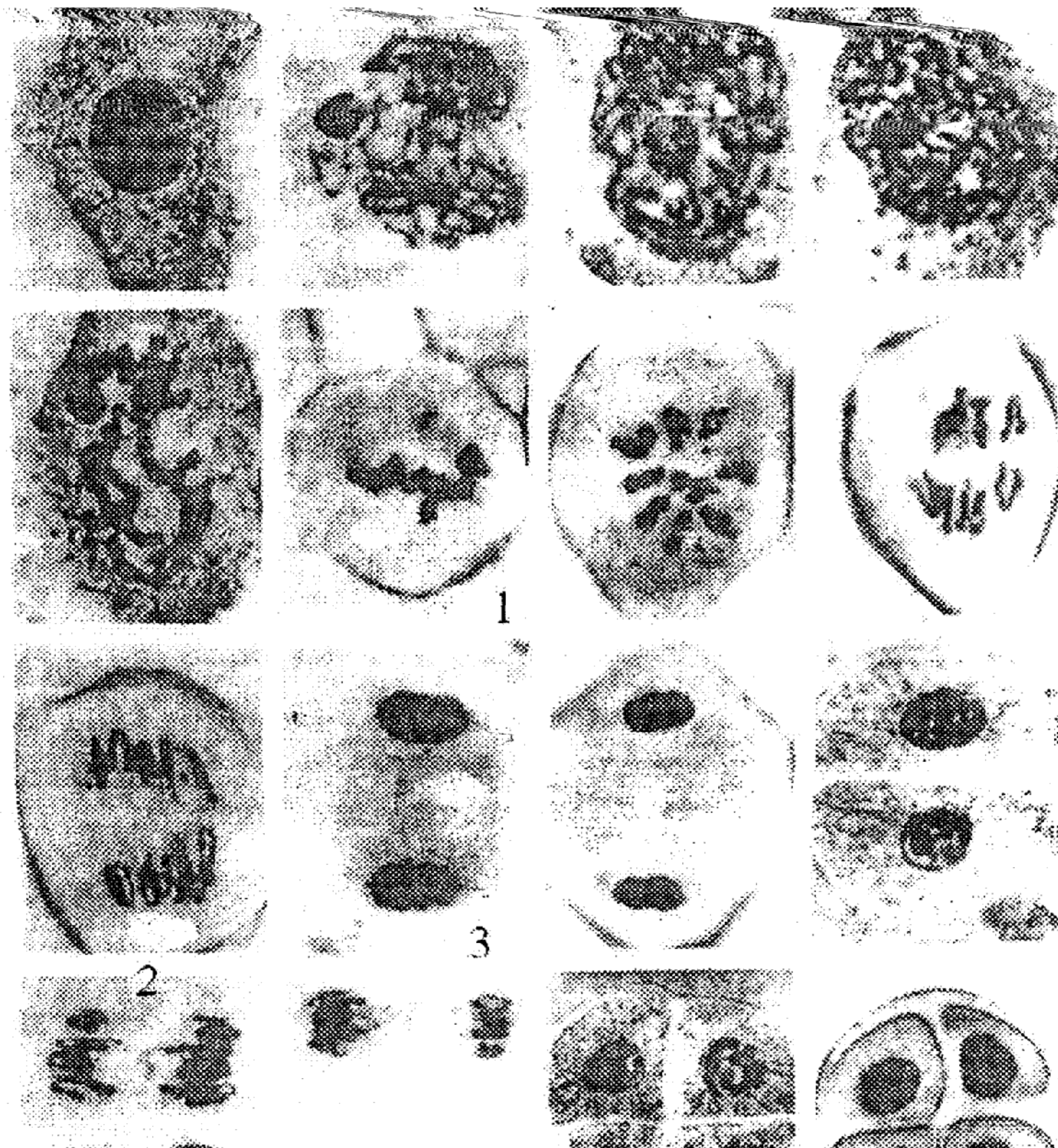
- *Irish potato (one large one each – N)*
- *75 ml of conc. brine labelled L_1.*
- *75 ml of distilled water labelled L_2.*
- *Potato borers.*
- *Six test tubes.*
- *Iodine.*
- *Benedict's solution.*
- *Sodium hydroxide.*
- *10% copper II sulphate.*
- *Means of heating.*
- *Means of timing.*
- *A ruler.*

Q.1 Below are two sets of photomicrographs **A** and **B** showing various processes of cell divisions.

Examine them.



B



Q. (a) Using observable features only, identify the type of cell division represented by the
photomicrographs in set **A** and set **B**. Give a reason in each case.

Cell division in set **A**
Reason:
Cell division in set **B**.
Reason:

(b) Name the division process represented by number 3 and 4 in photomicrographs of set A and
number 1 and 3 in photomicrographs in set B. **Complete** the table below.

(c) Name **one** region in higher pants where the cell division represented by photomicrographs
set **A** and **B** occurs.

(d) Describe the process that is taking place at photomicrographs set **A** number **3** and
photomicrograph set **B** number **2**.

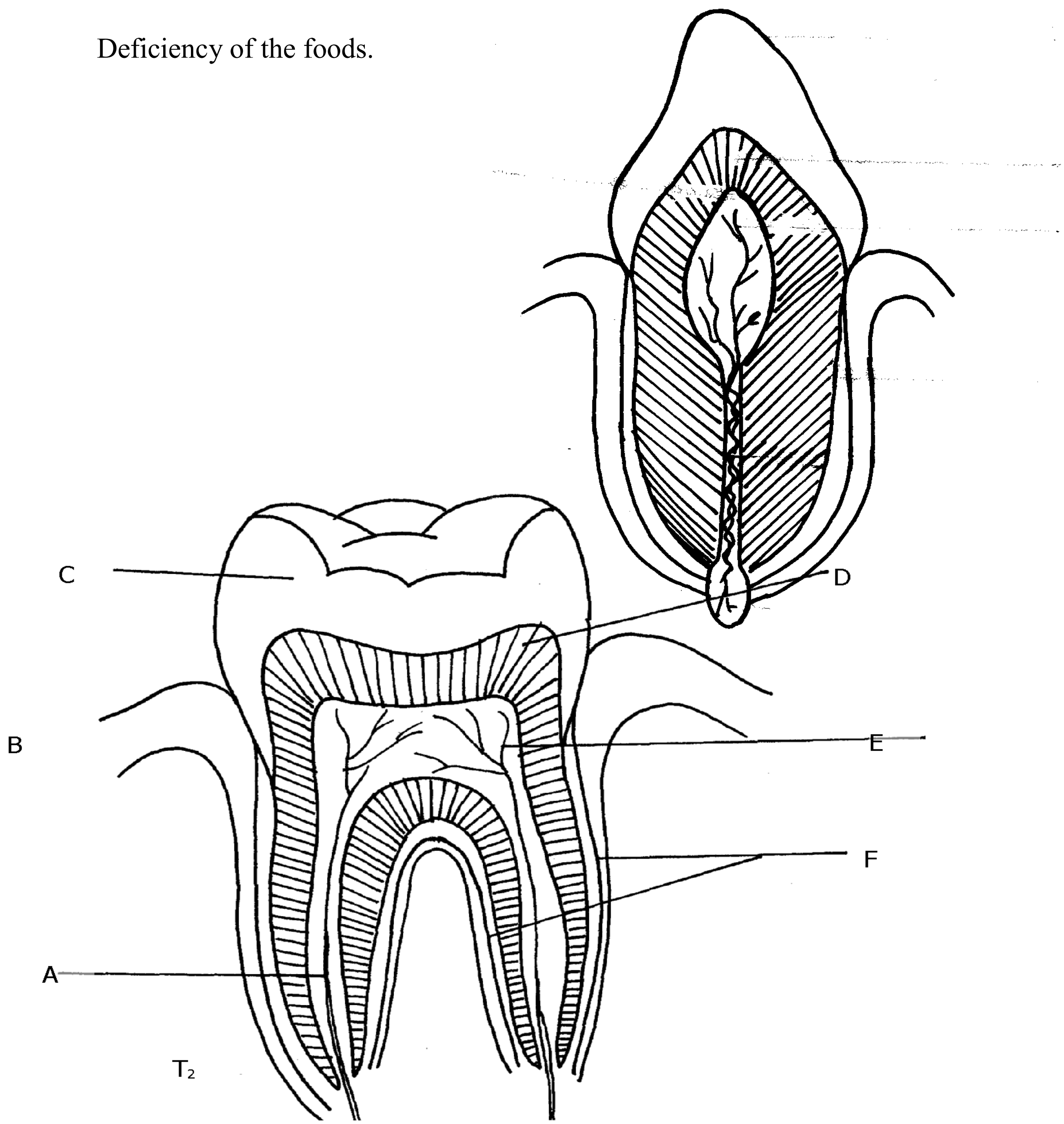
(e) State the **importance** of each of the cell division in **A** and **B** in the bodies of living
organisms.

2. You are provided with specimen **N.** You have also been provided with a cork borer bore out
three (3) pieces each measuring 5 cm. Take each piece and place into the test tubes labelled
A, B and C separately.
Fill test tube A with solution labelled L_1.
Fill test tube **B** with solution labelled $\mathbf{L_2}$.
Leave test tube labelled **C** empty (**Do not pour anything into it.)**
(a) (i) Remove the pieces and dry each using blotting paper and measure its

length. Record in the
table below.
(ii) Account for the observation made in the measurements of each piece after 30 minutes
above.
(b) (i) Crush the remaining tissue into a paste and carry out food tests on it using the reagents
provided.:

(ii) What would imbalances of such food substances cause in the diet?
Excess of the foods.

Deficiency of the foods.



3. Study the photographs provided above and answer the questions below.
(a) Give the identity of **T_1** and **T_2.**

(b) How is each specimen adapted to its function?

(c) Label the parts of **T_2** marked **A** to **F**

(d) State the effect of too much sugar in the diet on specimen **T_1** and **T_2** in humans.
(e) (i) What is the name of the gap found between **T_1** and **T_2** in herbivores.

(ii) State the function of the gap named in **e(i)** above.

CONFIDENTIAL INFORMATION TO SCHOOLS AND PRACTICALS

NYAMIRA DISTRICT

Each Candidate should be provided with the following:-

REQUIREMENT

- *2 Boiling tubes*
- *2 test tubes*
- *Test tube rack*
- *Means of heating*
- *1% copper sulphate solution*
- *2M sodium hydroxide solution*
- *Iodine solution*
- *Mortar and pestle*
- *Scalpel*
- *20% Hydrogen peroxide solution*
- *Fresh potato*
- *Droppers*
- *100ml beaker*

Schools should also have ordinary laboratory apparatus in addition to those listed above

1. You have been provided with specimen **Q** which is a fresh potato, liquid R (Hydrogen peroxide

and reagents 1% copper sulphate, 2M sodium hydroxide and iodine solution. Use them to carry out the tests below:

(a) Using a scalpel, cut two small cubes measuring 1cm x1cm from the fresh potato. Place one of the cubes in boiling water for 10minutes,then remove the cube and let it cool. Place it in a boiling tube and label it **A**.

Place the fresh piece of potato cube in another boiling tube labelled **B** and then add equal amounts of hydrogen peroxide to each test tube at the same time. Write your observations.

Observations:

(a) (i) Boiling tube **A**

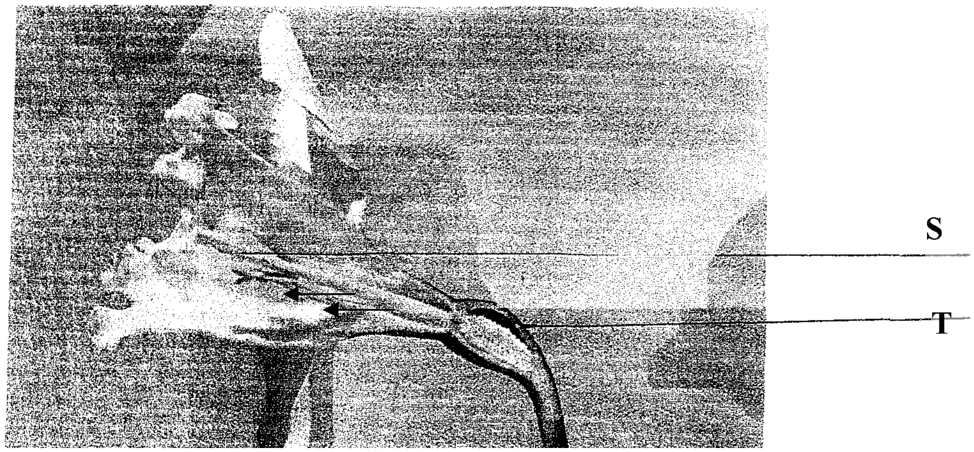
(ii) Boiling tube **B**

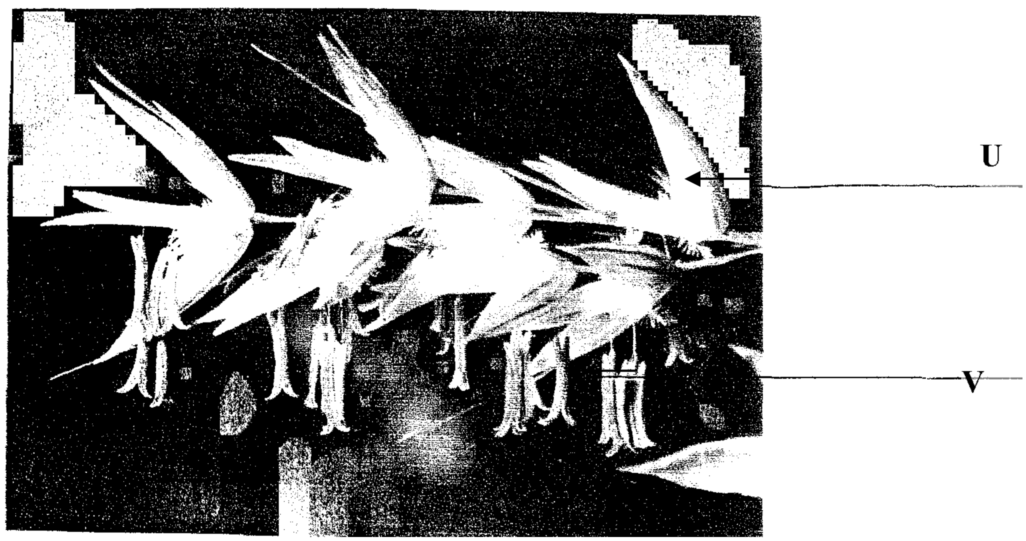
(b) Explain your observations **in (i)** and **(ii)** above

(c) Crush a small piece of the remaining potato in a motar. Add a little amount of distilled water to make a mixture. Use it to carry out food tests below:

2. **X** and **Y** are specimens obtained from plants. Study them carefully and then use them to answer

questions that follow:-





(a) Label the parts:-

S...

T..

U..

V...

(b) State with reasons the mode of pollination for specimen

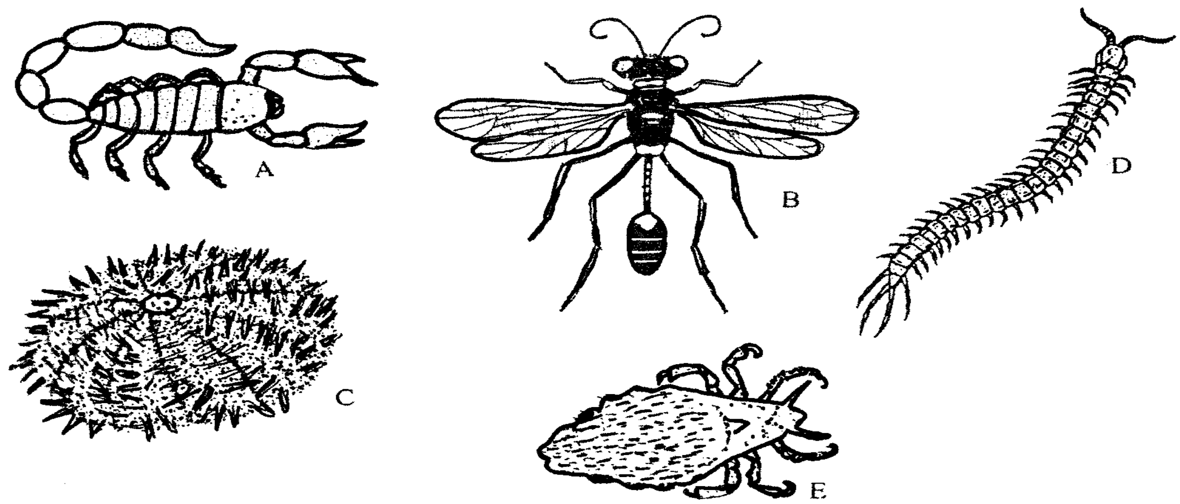
X

Y:

(c) Name the part of specimen **X** that develops into a fruit

3. You are provided with photographs of animals which belong to the same phylum. Study the

photographs and the dichotomous key below to enable you identify the taxonomic group to
which each animal belongs:-



1. (a) Jointed legs present ....Go to 2
   (b) Jointed legs absent ....Go to 7
2. (a) Three pairs of legs ....Go to 3
   (b) More than three pairs of legs ....go to 5
3. (a) Wings present ....Go to 4
   (b) Wings absent................................***Anoplura***
4. (a) One pair of wings ....***Diptera***
   (b) Two pairs of wings ....***Hymenoptera***
5. (a) Four pairs of legs ....***Arachnida***
   (b) More than ten pairs of legs................Go to 6
6. (a) One pair of legs in each body segment.........***Chilopoda***
   (b) Two pairs of legs in each body segment..........***Diplopoda***
7. (a) Body partially closed in a shell ....***Mollusca***
   (b) Body surface has spiny projections ....***Echinodermata***

(a) Using the key, identify organisms **A** to **E** giving the sequence of steps followed to arrive at the identity of each organism

(b) (i) Using observable features only, state the phylum to which the organisms on the photograph belong:

(ii) State **one** observable feature that enables you to arrive at the answer in **(b) (i)** above

CONFIDENTIAL INFORMATION TO SCHOOLS AND PRACTICALS

SOTIK DISTRICT – 1ST EXAM

1. egg albumen
Pineapple juice mixture (10ml) labelled Z
Iodine solution
Ethanol
Distilled water
DCPIP
Benedict's solution
Source of heat (hot water bath)
Four (4) test-tubes
2. Bougainvillea leaf –P
Kikuyu grass leaf -Q
Hand lens

3. Hand lens
Freshly killed housefly
Safety pin/pair of forceps

1. (a) You are provided with the solution labelled **Z**. Using the apparatus and the reagents provided, carry out the tests for the various food substances

Food	**Procedure**	**Observation**	**Conclusion**

(b) State the organ(s) which produce(s) enzyme(s) which are required to digest the contents of solution **Z** completely

(c) Name the end products of digestion of solution **Z**

(d) Give **two** functions of the products named in **(c)** above in the human body

2. You are provided with the specimens **P** and **Q**:

(a) (i) What is the mode of nutrition for the organisms represented by the above specimens?

(ii) Give a reason for your answer in **(a) (i)** above

(iii) Write an equation for the physiological process involved in the mode of feeding in **(a)(i)** above

(b) Draw and label specimen **P**

(c) State **three** observable differences between specimens **P** and **Q**

(d) Name the trophic level of the organisms from which the specimens were obtained in the ecosystem

(e) Explain the role played by the organisms in the ecosystem

(f) Which features adapt specimen **Q** to enabling the organism from which it was detached to live in its habitat?

3. Using a hand lens, study the specimen provided and answer the questions that follow:

(a) Give the phylum and the class to which the specimen belongs

(b) State **two** characteristics which are unique to members of the class suggested in **(a)** above

(c) Using the observable features only, explain how the specimen is adapted to living in its habitat.

CONFIDENTIAL INFORMATION TO SCHOOLS AND PRACTICALS

UGENYA - UGUNJA DISTRICT

Each student should be provided with the following:-

- *Onion bulb*
- *Iodine solution (5ml)*
- *Cover slip (1pc)*
- *Microscope slide (1pc)*
- *Means of labeling*
- *Hydrogen peroxide – 5ml per student*
- *Test tube (4)*
- *Distilled water*
- *Saturated sodium Chloride solution – Liquid H – 5ml per student*
- *Blotting paper (1pc)*

- *Means of timing*
- *Pestle and Mortar*
- *Piece of liver*
- *Wooden splint*
- *Benedicts solution – 5ml*
- *Scalpel blade*
- *Means of heating*
- *Boiling tube (1)*
- *Glass rod*
- *A pair of forceps*
- *Microscope (one for a group of five)*

N/B – Provide a medium power objective lens of x10 and eye piece lens of x10 or x15.

1. You are provided with a portion of an onion bulb. Remove one fleshy leaf from the onion bulb, peel the epidermis from the inner surface of the leaf and place it on a drop of iodine solution on a glass slide. Place a cover slip on the epidermis. Drain the excess iodine solution by use of a piece of blotting paper from the edge of the cover slip then leave the set up for one minute.
Place a drop of **liquid H** at the edge of the cover slip. Leave the set up for **5 minutes** then drain excess liquid from the opposite of the slip using a blotting paper. Observe under medium power of the light microscope provided.
(a) **Draw** and **label** two neghbouring cells

(b) Account for the results in **(a)** above

(c) Using a pestle and mortar, crush two fleshy leaves of the onion bulb, add 4mls of distilled water and stir. Decant into a test tube and label the resultant filtrate as **solution J_1** and retain the residue.

Using the reagents provided, carry out food tests on **solution J_1** and fill the table below:

FOOD SUBSTANCE	PROCEDURE	OBSERVATION	CONCLUSION

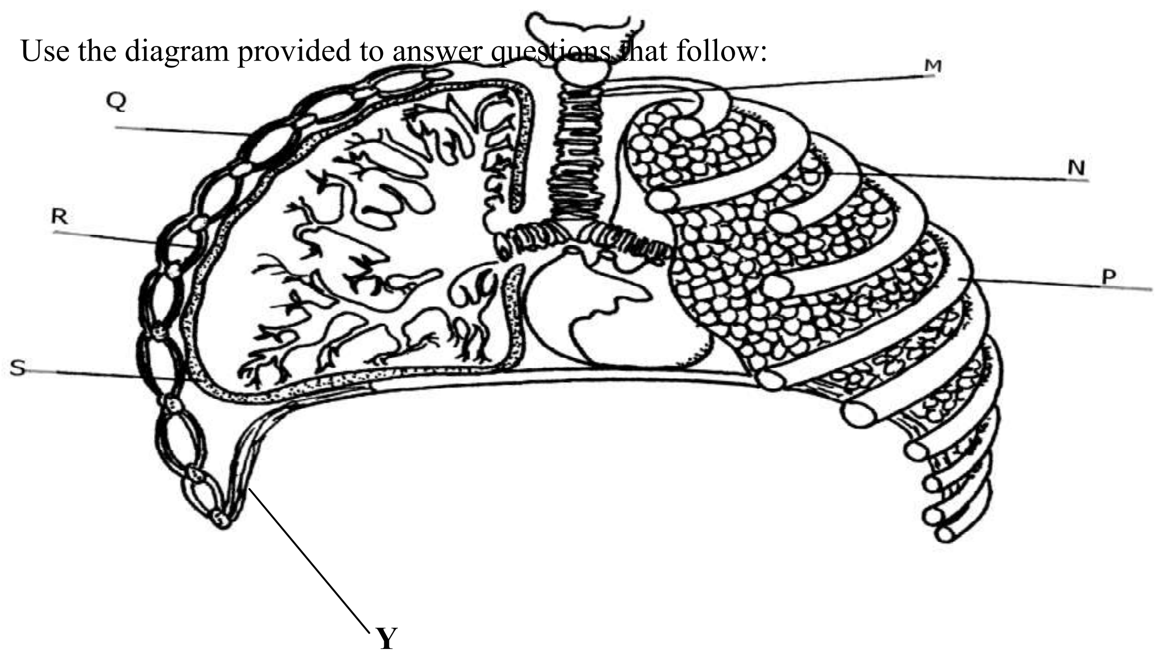
(d) Label one test tube as **J_2** and another as **K** . Add 2mls of Hydrogen peroxide to each of the test tubes.

(i) Into the test tube labelled **J_2**, place the entire residue obtained in (c) above and immediately introduce a glowing splint. Record your observations in the table below. Into the test tube labelled **K**, place the piece of liver provided then immediately introduce a glowing splint into the mouth of test tube and record your observations in the table below.

(ii) Name the enzyme responsible for the reactions in the test tubes above

(iii) Explain the significance of the difference in the observations in part (i) above

2. Use the diagram provided to answer questions that follow:



(a) Name the bones that articulate with the structure labelled **P**

(i) Dorsally ....

(ii) Ventrally...

(b) Give **three** adaptations of structure **M** to its functions

(c) (i) Name the fluid found within the part labelled **S**

(ii) State the function of the fluid named in (c) (i) above

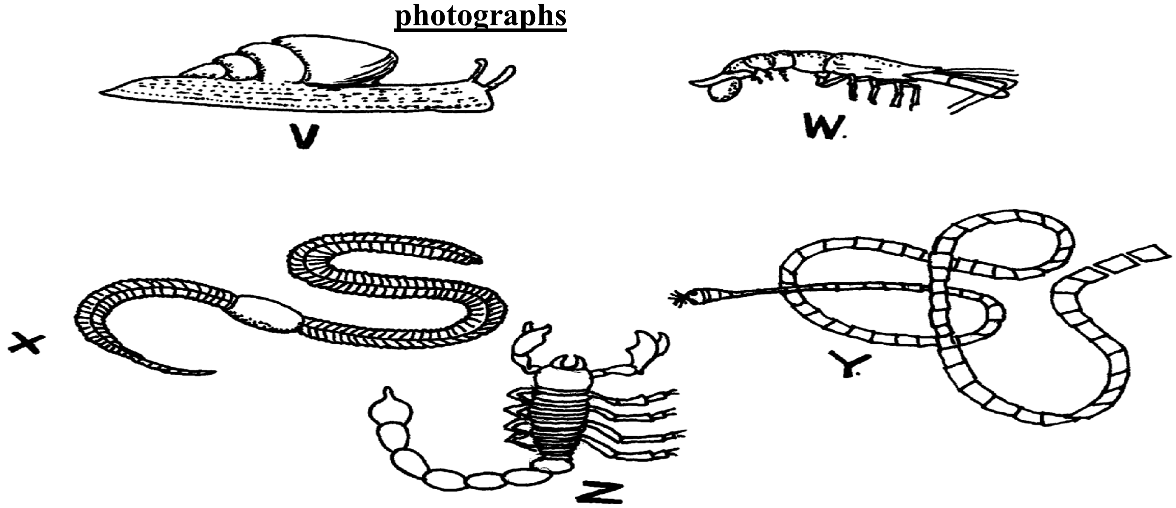
(d) Identify the parts labelled: **Q** & **R**

(e) State **two** changes that take place in the organ labelled N when the structure **Y** contracts

(f) How is large surface area achieved for efficient functioning of the organ labelled **N**?

3. A student collected a number of invertebrates whose photographs appear below. He constructed a Dichotomous key as shown below to enable him place each specimen into its taxonomic group

photographs



DICHOTOMUS KEY

1. (a) Organisms with a flat body	Go to 9
(b) Organisms without a flat body	Go to 2
2. (a) Organisms having a body in a shell	***Mollusca***
(b) Organisms without a shell	Go to 3
3. (a) Organisms having a segmented body	Go to 4

(b) Organisms with a body not segmented	***Nematoda***
4. (a) Organisms having jointed appendages	Go to 6
(b) Organisms without jointed appendages	Go to 5
5.(a) Organisms with a long cylindrical body	***Annelida***
(b) Organisms having a short stout body	***Trematoda***
6. (a) Organisms with antennae	Go to 7
(b) Organisms lacking antennae	Go to 8
7. (a) Organisms with a pair of antennae	***Insecta***
(b) Organisms with more than one pair of antennae	***crustacea***
8. (a) Organisms with pincer-like mouth parts	***Arachnida***
(b) Organisms with sucking mouth parts	***Acarina***
9. (a) Organisms having a ribbon like body	***Cestoda***
(b) Organisms with circular body	***Crinoidea***

(a) Using the dichotomous key, identify the taxonomic group of each of the five specimens shown in the photographs. In each case show in sequence, the steps in the key that you have followed to arrive at the identity of each specimen.

(b) Name a pathogen that attacks human beings and is associated with the organism labelled **V**

CONFIDENTIAL INFORMATION TO SCHOOLS AND PRACTICALS

NDHIWA DISTRICT

Each candidate will require the following:-

- *Winged cockroach lablled R*
- *Tick labelled N*
- *Soldier termite labelled P*
- *Adult housefly labelled Q*
- *10ml 20 volume hydrogen peroxide*
- *Pestle and mortar*
- *Spatula preferably a wooden one*
- *Scalpel*
- *Ruler caliberated in centimeters*
- *A medium size irish potato tubers labelled L*
- *Measuring cylinder (to measure up to 4mls)*
- *2 boiling tubes*
- *3 test-tubes*
- *Means of labeling (two each)*
- *A maize seedling with opened coleoptile*
- *Green leaves and should also have remains of grain labelled K seedlings grown in the sand on trays in plastic containers give good specimens*
- *Distilled water*
- *Iodine solution*
- *Benedicts solution*
- *Means of heating*

1. You are provided with specimens labelled **N, P, Q** and **R**. Using the following characteristics and

in the order given only

- Number of legs
- Presence of wings
- Number of wings

a) Construct a three –step dichotomous key. Use the given letters for identification

(Specific names not required)

b) i) Using observable features only, state the phylum to which specimen **R** belongs

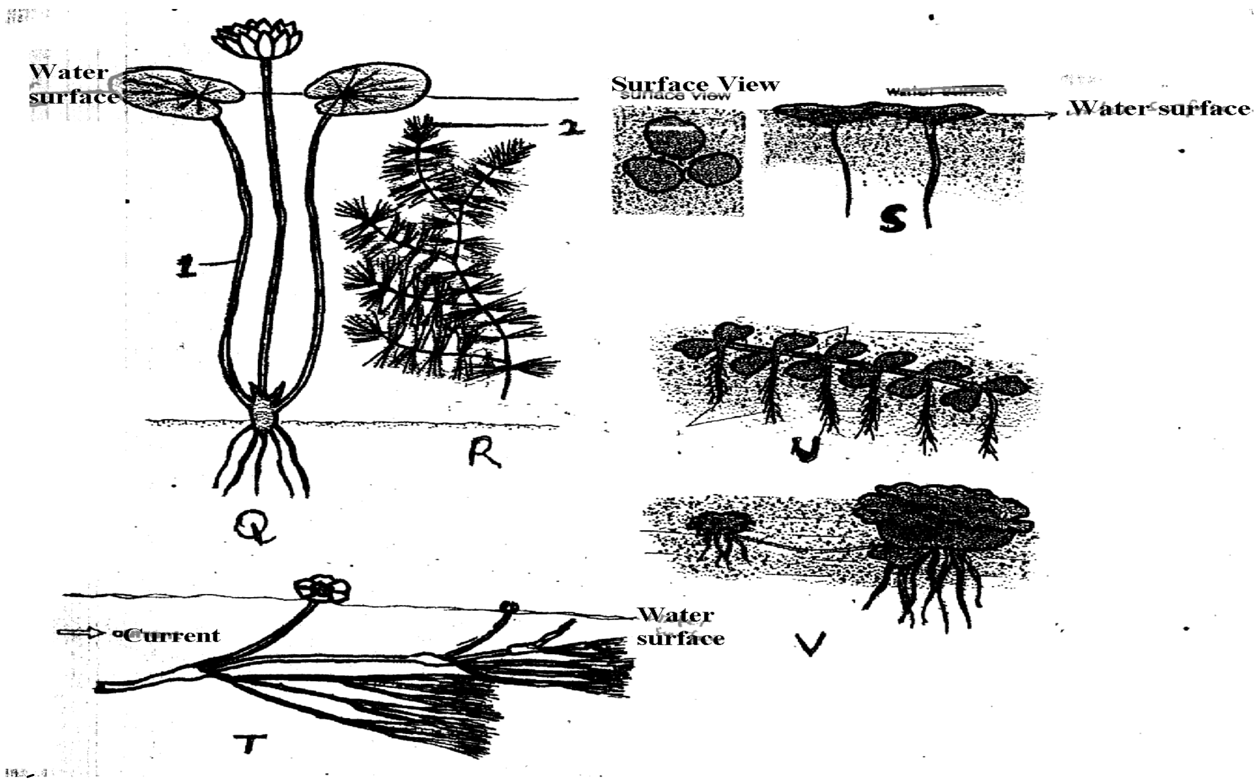
ii) Give **three** reasons for your answer in **(b) (i)** above

c) i) Using observable features only, classify specimen **N** in its class

ii) Give **four** reasons for your answer in (**c) (i)**

2. Study the photographs of some hydrophytes shown below. They show various adaptations

they have to overcome problems they are exposed to due to the nature of their habitats



i) What are hydrophytes?

ii) Name the structures of plants labelled **1** and **2**

iii) State **two** problems which hydrophytes are faced with in their habitat

iv) With reference to the photographs, how are the hydrophytes adapted to solve each of the problems you have stated in **part 2 (iii)** above?

v) State **two** internal adaptive features of the plants not shown in the photographs above that enables them to live in their habitat

vi) What type of hydrophytes do the following plants represent?

R .... **S**......................................

3. You are provided with a specimen labelled **L** and hydrogen peroxide

a) Cut two equal cubes whose sides are about 1cm from specimen **L**. Place one of the cubes into a boiling tube labelled **A**. Crush the other cube using pestle and mortar. Place the crushed material in another boiling tube labelled **B**.

To each boiling tube add 4ml of hydrogen peroxide

i) Record your observation

ii) Account for the result in **(a) (i)** above

iii) Write an equation for the break down of hydrogen peroxide

b) You are provided with a specimen labelled **K**. Separate the roots and leaves from the remains of the grain. Crush the roots, leaves and the remains of the grain separately. To each crushed materials add 1ml of water. Put the extract from the materials into separate test tubes and label them using the reagents provided. Test for the food substances in each of the extracts. Record the procedure, observation and conclusions in the table below:-

c) Account for the results obtained in **(b)** above

i) Roots

ii) Remains of grains

iii) Leaves

CONFIDENTIAL INFORMATION TO SCHOOLS AND PRACTICALS

MUMIAS DISTRICT

<u>Requirements :-</u>

- *Unripe pawpaw fruits (one pawpaw- ten students)*
- *Beaker (4)*
- *Razor /scalpel*
- *Ruler*
- *Solution G – distilled water*

- *Solution H – salt solution of different concentration namely 10%, 20%, 60%*
- *Labels*

1. You are provided with specimen **D** and two solution **G** and **H.**

Cut five longitudinal strips of the specimen **D** peelings of approximately 0.5cm width,
0.5cm breadth and 5cm length.

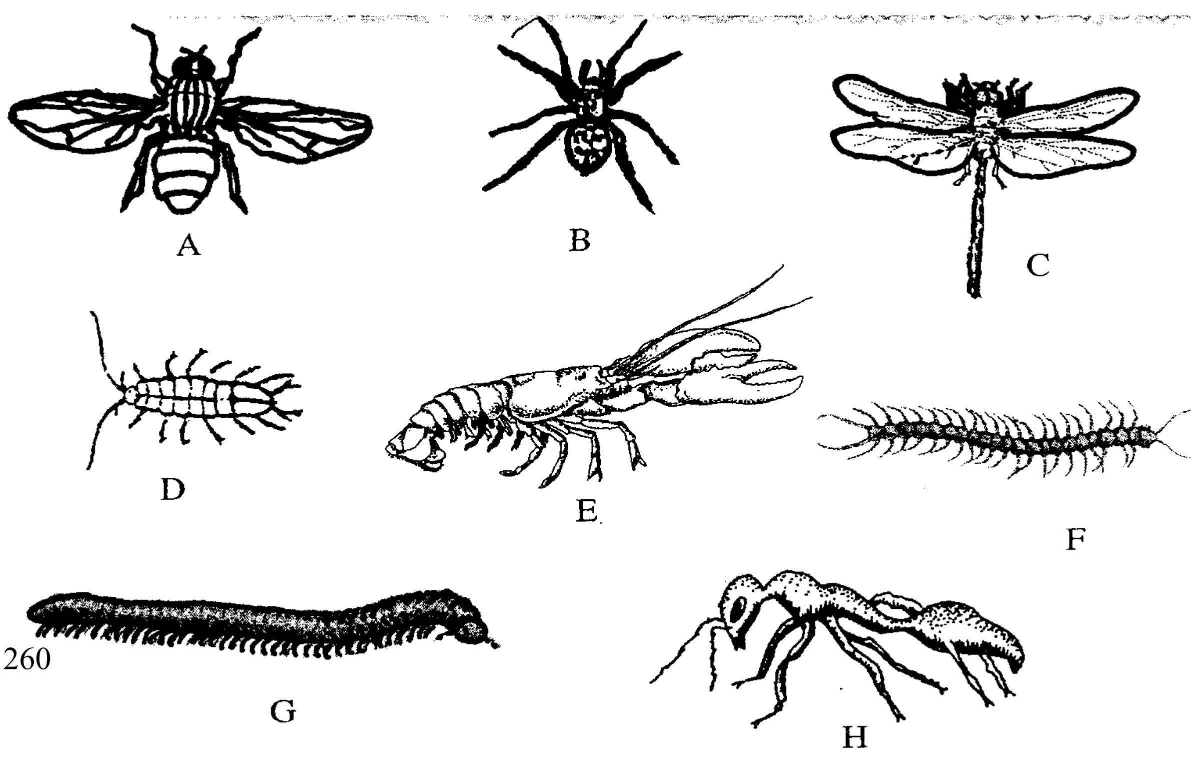
Place one strip in a beaker having solution **G**.
Place other strips in separate beakers containing different concentration of solution H as indicated
in the table below:

Beaker	Solution
1	**Solution G**
2	10% solution **H**
3	20% solution **H**
4	60% solution **H**

Leave the set-up for 30minutes
(i) Record your observations in the table below:
(b) Account for the observation in trips **1, 2** and **4**

(c) Suggest the identity of solution **G** and **H**

2. During a biology lesson, students made drawings of invertebrates shown below. Use the dichotomous key provided below to identify the organisms;



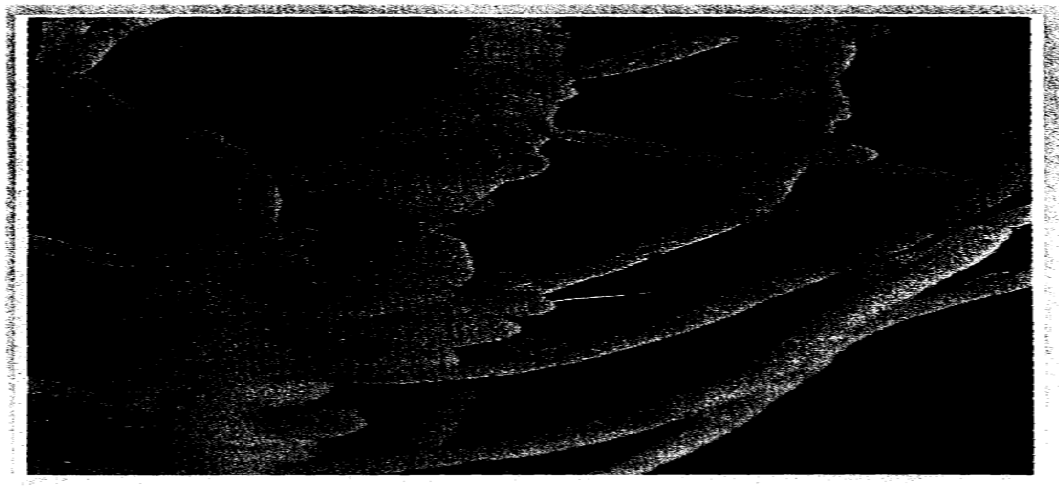
1. a. Animal with wings ……………………………….go to 2
   b. Animal without wings ……………………………go to 3
2. a. With one pair of wings………................................Housefly
   b. With pairs of wings ………………………………..Dragonfly
3. a. With three pairs of legs …………………………...Ant
   b. With more than three pairs of legs……………..…go to 4
4. a. With four pairs of legs …………………………...Spider
   b. With more than four pairs of legs ………………..go to 5
5. a. With two pairs of antennae …………………..…...go to 6
   b. With one pair of antennae………………………...go to 7
6. a. With six pairs of legs……………………………..Water slater
   b. With ten pairs of legs …………………………......Fresh water shrimp
7. a. With a body ……………………………..….….millipede
   b. With a dorso-ventrally flattened body…………….Centipede

(a) Complete the steps **2(b)** and **7(b)** by filling in the key above

(b) Complete the table to identify the organisms:

(c) State the classes of specimens **B, C, E** and **G**

3. The photograph **Z** below is apart of a plant. Examine it



(a) Label any **three** parts of the plant part in photograph **Z**

(b) Name the type of organisms that is associated with this part of the plant

(c) Photograph **Z** was taken from a special type of plant. What is the name of this group

of plants?

(d) Photograph **Z** exhibits a certain phenomenon;

(i) Name the phenomenon

(ii) State the significance of this phenomenon named in **d(i)** above

(iii) What is the product of this phenomenon?

(iv) Name t**wo** organisms that covert the product of the phenomenon in **d(i)** above into the raw material

CONFIDENTIAL INFORMATION TO SCHOOLS AND PRACTICALS

KISUMU WEST DISTRICT

Each candidate should have:-

- *Starch suspension labelled Liquid X*
- *Iodine solution*
- *Benedict's solution*
- *2M hydrochloric acid (1ml)*
- *2 Droppers*
- *Measuring cylinder (10ml size)*
- *Means of heating/Bunsen burner*
- *5 test-tubes*
- *Water in a small beaker*
- *Thermometer*
- *Test-tube holder*
- *3 boiling tubes*
- *Tripod stand and gauze*
- *3 labels*
- *White tile*
- *Water bath*
- *Diastase/ amylase* **enzyme (0.5g per student)**

N/B: -Liquid X is prepared by dissolving 5g of soluble starch in 50ml of distilled water.

Thorough stirring is required whenever it is being used.

1. You are provided with liquid **X** and substance **Q**
(a) Place three drops of liquid **X** onto a white tile. Add four drops of iodine solution and record

your observation.

(b) Pour 2ml of liquid **X** into a test-tube. Add equal amounts of Benedict's solution and boil the mixture. Record your observation

(c) Label three boiling tubes as set-ups **A**, **B**, and **C**. Place 3ml of liquid **X** into each of the set-ups.

Divide substance **Q** into three equal portions.

To set-up **A**, add one portion of substance **Q** and shake.

- Place the second portion of substance **Q** into a test tube. Add 1ml of water to it and boil for four minutes. Add it to set-up **B** and shake.
- To set –up **C**, add the third portion of substance **Q**. Add 8 drops of 2M hydrochloric acid and shake.

Place the three set-ups in a warm water bath maintained at 37°C for 40minutes.

Cool the set-ups by dipping the boiling tubes in cold water

Place 2ml of the contents of each set-up into three separate test tubes. Add equal amount of Benedict's solution to each of the three test-tubes and boil.

Record your observations :-

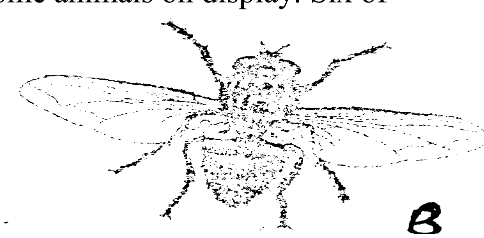
(d) Account for your observations in the set-ups:-

(e) Give the most likely identity of substance **Q**

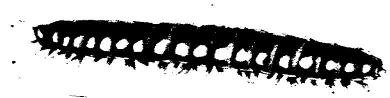
(f) Why was the water bath maintained at 37°C

2. During a visit to a museum, students were shown some animals on display. Six of the animals are shown in the photographs below:


A


B


C


D

(a) Using observable features only, classify the animals, **A, B** and **E** into their respective classes.

Give a reason for your answer in each case

(b) State **one** morphological difference between **C** and **E**

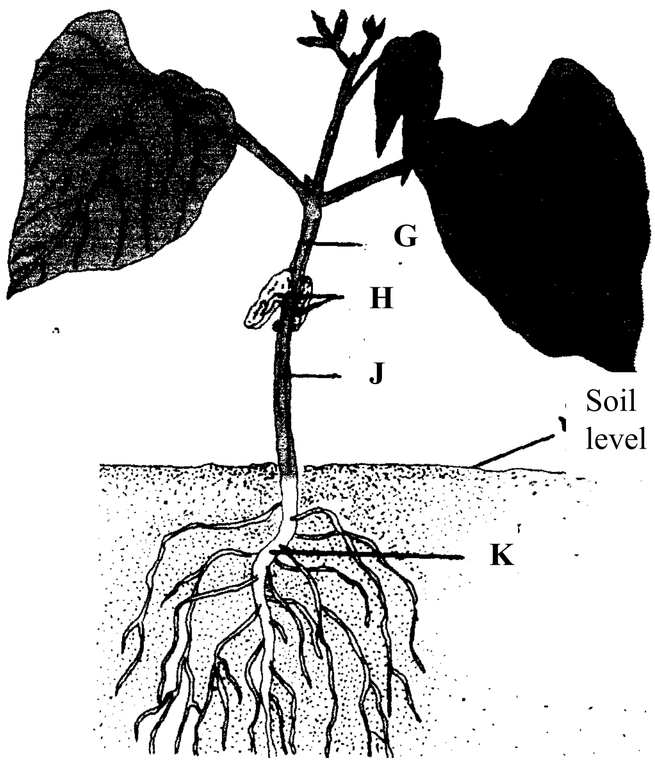
(c) The dichotomous key constructed below can be used to identify some of the animals viewed in the museum:-

1. (a) Has jointed legs ....go to 2
   (b) Lacks jointed legs ....go to 3
2. (a) Has five or less pairs of legs .... go to 4
   (b) Has more than five pairs of legs .... go t o 5
3. (a) Has bilateral symmetry..EUNICE
   (b) Has radial symmetry..LUDIA
4. (a) Has five pairs of legs ....CANCER
   (b) Has four pairs of legs ....LACTRODECTUS
5. (a) Has I pair of legs per body segment ....SCOLOPENDRA
   (b) Has 2 pairs of legs per body segment ....SIGMORIA

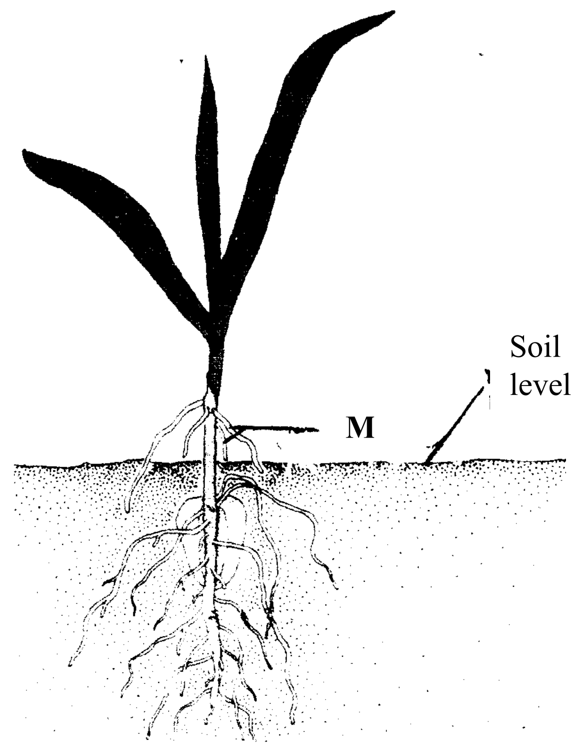
Use the dichotomous key above to identify animals labelled **C, D** and **E**. In each case show in

sequence the steps followed (e.g. 1b, 2b, 3a e.t.c.) to arrive at the identity of each animal

3. Below are photographs **I** and **II** of young plants.



PHOTOGRAPH I



PHOTOGRAPH II

(a) With a reason in each case, name the class to which the plants belong:

(i) Plant in photograph **I**

....

Reason...

(ii) Plant in photograph **II**

....

Reason...

(b) Identify the parts labelled **G, J** and **M**

(c) State **two** functions of the part labelled **H**

(d) (i) Name the swellings that would be developed in the roots of the plant in **photograph I** later in its life

(ii) Which organism would be found in the swellings in **(d)(i)** above?

(e) (i) State the type of germination exhibited by the plant in **photograph II**

(ii) Give a reason for your answer in **(e) (i)** above

CONFIDENTIAL INFORMATION TO SCHOOLS AND PRACTICALS

TRANS NZOIA WEST DISTRICT

Requirements for each candidate:

- *L_1- solution of egg albumen – 20ml*
- *L_2 – solution of starch and glucose – 20ml*
- *Visking tubing (10cm long)*
- *Thread*
- *250ml beaker*
- *Stirring rod*
- *Iodine solution*
- *Benedicts solution*
- *Source of heat*
- *4-test tubes*
- *1-boiling tube*
- *Test-tube holder*
- *Test-tube rack*

1. You are provided with liquids labeled L_1, and L_2 and a piece of visking tubing. Spare about

10ml of each of the liquids for part (a) of this question

Using a piece of thread, tightly tie one end using the visking tubing

Open the other end of the visking tubing and half fill it with liquid L_1. Tightly tie this end.

Ensure there is no leakage at both ends. Immerse the tubing in a beaker containing liquid L_2

a) Using the Iodine and the benedict's solutions provided, test for the food substances in

liquid L_1 and L_2. Record your observations in the table below :-

After at least 30 minutes, remove the visking tubing from the beaker and wash the outside of

the tubing thoroughly to remove traces of liquid L_2

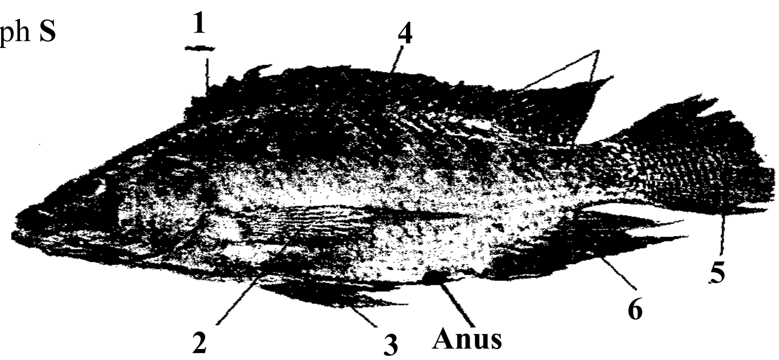
b) Using the same reagents, test for food substances in liquid L_1 in the visking tubing.

Record your observations in the table below:

c) Account for the results obtained after carrying out tests on liquid L_1 before and after

immersion into liquid L_2

2. Below is photograph **S**



a) i) Name the class to which the organism in the photograph belongs:

ii) Give **three** observable reasons for your answer in **(a) (i)** above

b) State **two** functions of the part labeled **1**

c) Name the fins on the specimen that:

i)Enable the specimen to balance, brake and change direction

ii) Prevent the fish from rolling and yawing

d) Measure in millimeters the length of the:

i) Photograph **S** from the tip of the mouth to the tip of the tail

Length…………………… mm

ii) Photograph **S** from anus to the tip of the tail.

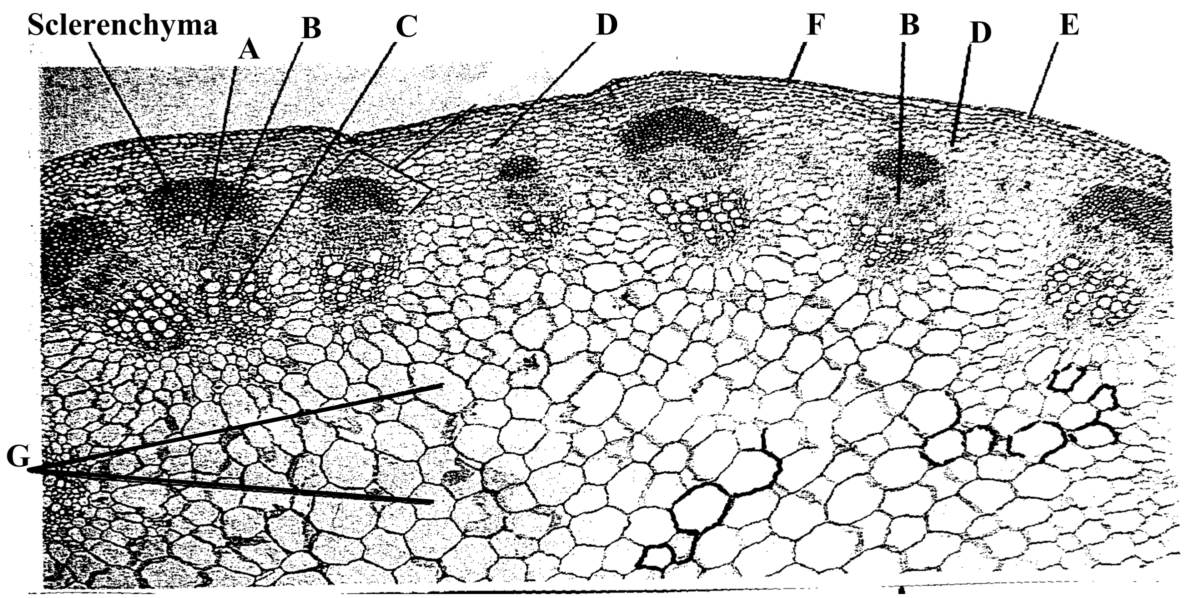
Length……………………….… mm

iii) Using the measurements in **d (i)** and **d (ii)** above, calculate the tails power

iv) State the significance of tail power in specimen **S**

e) Other than structures in **(c)** above, state **two** observations of the animals in photograph **S**
to locomotion in water

3. Study the photomicrograph **M** which shows plants tissues



a) Name the parts labeled **A -G**

b) State the function(s) of tissues labeled **A, B, C**

c) Name the cell types found in parts labeled **D** and **G**

d) How are sderenchyma cells adapted to their function?

e) Distinguish between the section above and the one from the root of the same plant

CONFIDENTIAL INFORMATION TO SCHOOLS AND PRACTICALS

RACHUONYO DISTRICT

Each school will need 20g of yeast powder for every to candidates picked and sealed in polythene papers and labeled substance K.

To schools - Substance K is to reach school on the morning of the examination packed substance 'K' delivered to schools on the day of Biology examination

Each candidate will require the following:-

- *4 test-tubes with tightly fitting corks*
- *6ml of 10% glucose in a test tube*
- *3 labels per candidate*
- *6ml of distilled water in a test tube*
- *Source of heat*
- *2cm by 2cm liver piece*
- *Dilute hydrogen peroxide (20 volumes)*
- *About 2g of substance K*
- *Substance K to be provided by RAEC on the morning of examination day)*

1. During a visit to a museum, students were shown ten specimens of invertebrates on display.

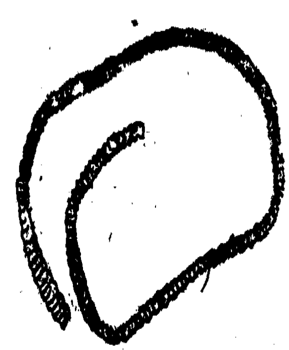
The teacher provided a dichotomous key to enable them classify each specimen on display.

Five of the specimens are shown in the photographs below:

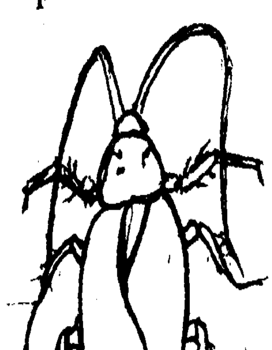
E

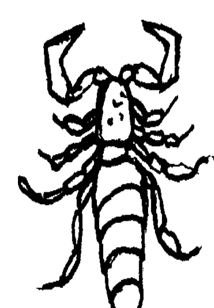
F

G



H





DICHOTOMOUS KEY

1.	a. Animal with flattened body	Go to 9
	b. Animal without flattened body	Go to 2
2.	a. Animal with body in shell	Mollusca
	b. Animal without shell	Go to 3
3.	a. Animal with segmented body	Go to 4
	b. Animal with body not segmented	Nematoda
4.	a. Animal with jointed appendages	Go to 6
	b. Animal without jointed appendages	Go to 5
5.	a. Animal with long and cylindrical body	Annelida
	b. Animal with short stout body	Trematoda
6.	a. Animal with antennae	Go to 7
	b. Animal without antennae	Go to 8
7.	a. Animal with one pair of antennae	Insecta
	b. Animal with more than one pair of antennae	Crustacea
8.	a. Animal with pincer like mouth parts	Arachnida
	b. Animal with sucking mouth parts	Acarina
9.	a. animal with long ribbon like body	Cestoda
	b. Animal with circular body	Crinoidea

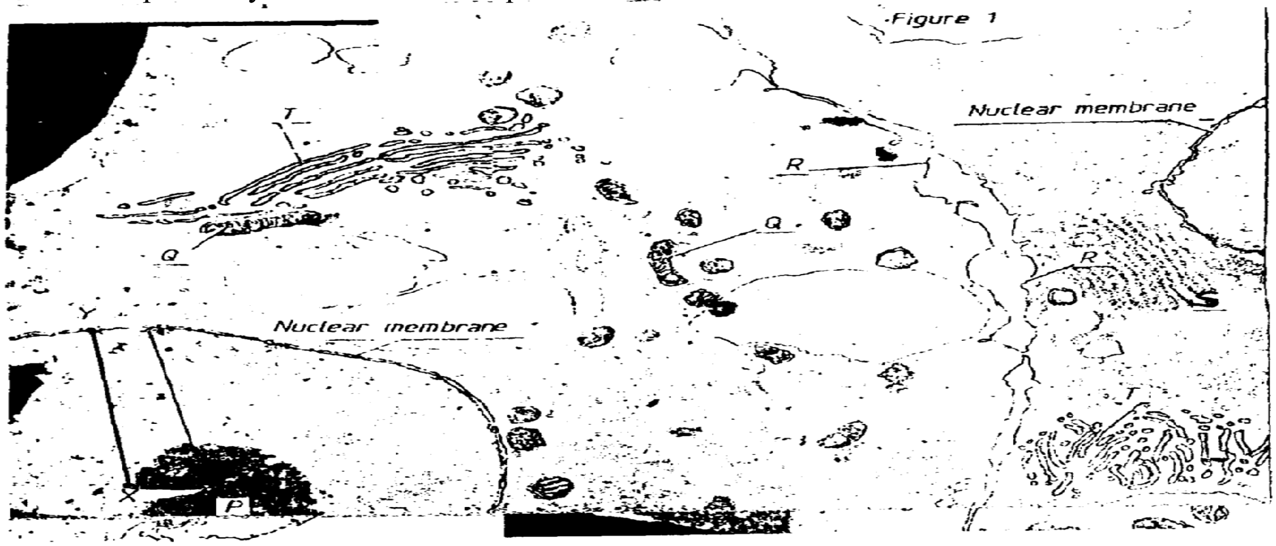
(a) Use the dichotomous key to identify the taxonomic group of each of the five specimens in the photographs. In each case, show the sequence of steps e.g. 1a, 2b, 7b e.t.c. in the key that you followed to arrive at identity of each specimen

(b) State the phylum to which animal **G** belong

(c) Apart from jointed appendages, state 2 other distinguishing characteristics of the phylum named in **B** above

2. The photomicrograph below represents parts of 2 adjacent cells as seen under an electron

microscope. Study it and answer the questions that follow:-



(a) Use the table below to name **P, Q, R, S** and **T.** For each organelle, state one function

(b) The magnification of the cells in this micrograph is x 10,000. Use a ruler to measure the

radius of the nucleus between **X** and **Y** in millimeters. Calculate the actual radius of the

nucleus before magnification in mm

Length……………………………………mm

Actual radius of nucleus

3. You are provided with 10% glucose solution and substance **K.** place equal amounts of the

glucose solution in test tubes labeled 1, 2 and 3. Divide the substance **K** into 3 equal portions.

To one portion, add 2ml of water and boil, cool it down. Pour this mixture into test tube 1

Add another portion of substance **K** to test-tube 2 and shake.

Put 2ml of distilled water in test tube 3. Close the 3 test-tubes tightly using well fitting corks, and allow the set-ups to stand for at least 20 minutes

(a) Record your observation: (i) Test tube 1

(ii) Test tube 2

(iii) Rest tube 3

(b) (i) Name the process being investigated in the experiment

(ii) Write down an equation for this process

(iii) In which organelle does the process take place

(c) (i) Suggest the identity of substance **K**

(ii) Account for the results in test-tube 1

(d) Cut a small piece of liver 2cm by 2cm. drop it into the test tube containing dilute hydrogen peroxide. Leave for 2 minutes

(i) State your observation

(ii) Account for your observations

CONFIDENTIAL INFORMATION TO SCHOOLS AND PRACTICALS

KAKAMEGA NORTH DISTRICT

Each candidate to be provided with:

- *Scalpel blade*
- *A ripe tomato labelled specimen Q.*
- *A mortar and a pestle*
- *Filter paper,*
- *Means of heating*
- *DCPIP*
- *3 droppers*
- *50cm^3 beaker*
- *4 A freshly killed soldier termite labelled specimen R*

- *4 A freshly killed maize weevil labelled specimen. S*
- *4 A hand lens.*

1. You are provided with specimen labelled **Q**. Examine it and;

(a) (i) Name the part of a plant specimen **Q** is

(ii) Give a reason for your answer in **(a)(i)** above

(b) (i) Name the likely mode of dispersal of specimen **Q**

(ii) Give **two** reasons for your answer in **(b)(i)** above

(c) Make a transverse section through specimen **Q** to obtain two equal halves

(i) Draw and label one of the halves of specimen **Q**.

(ii) Crush one f the halves of specimen **Q** in a mortar using a pestle to obtain a paste. Gently decant the juice into a boiling tube.

d) Using the reagents provided test for the food substances present in the juice extracted.

Record your procedures, observations and conclusions in the table below.

Food substance	**Procedure**	**Observations**	**Conclusion**

2. You are provided with specimens labelled **R** and **S.** Examine then;

(a) (i) Name the phylum to which the specimen **R** and **S** belong

(ii) Give **three** reasons for your answer in **(a)(i)** above.

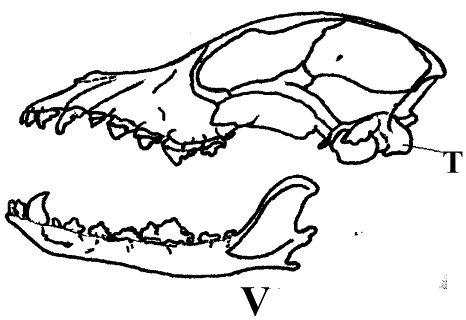
(b) (i) Name the class to which specimens **R** and **S** belong

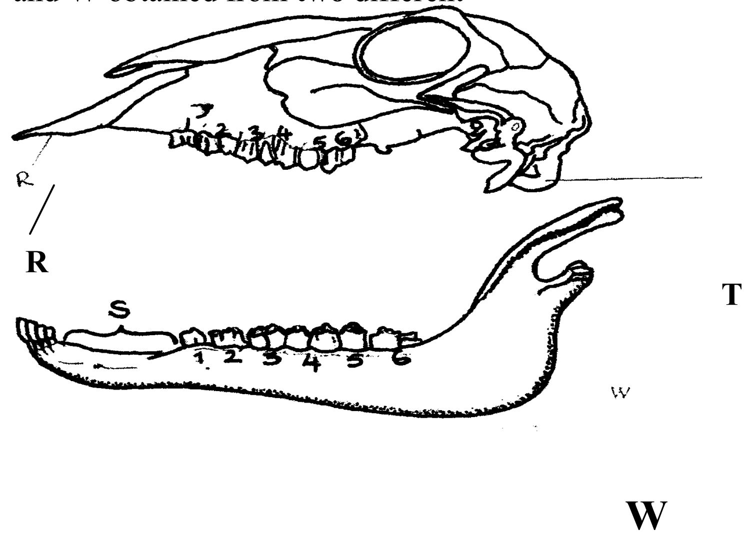
(ii) Give **three** reasons for your answer in **(b)(i)** above.

(c) State **two** observable differences between specimens **R** and **S**.

3. The drawings below illustrate two skulls **V** and **W** obtained from **two** different mammals.

Examine them.





(a) State the mode of feeding of the organism from which each of the skulls was obtained.

Give **two** reasons in each case.

(b) Label canine on drawing **W** and carnassial teeth on drawing **V**

(c) State the function of each of the following labelled parts on the drawing **R** & **S**

(d) Write down the dental formula of the organism from which skull **W** was obtained

(e) State **four** observable differences between the skulls **V** and **W**.

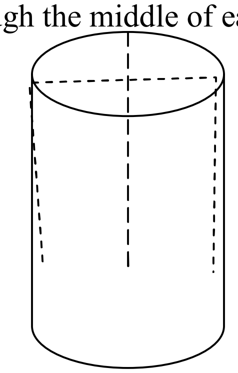
(f) (i) Name the part labelled **T**

(ii) Name the vertebra that articulates with the part labelled **T**.

CONFIDENTIAL INFORMATION TO SCHOOLS AND PRACTICALS

SOTIK DISTRICT- 1ST EXAM

1. You are provided with two pieces of plant material labelled specimen Q. Using a scalpel cut two slits half way to obtain four flaps through the middle of each piece as shown in the diagram below:-



Place one piece in the solution labelled **M_1** and the other in the solution **M_2** . Allow the set up to stand for 30minutes

(a) After 30minutes remove the pieces and press them gently between the fingers

(i) Record your observations M_1 M_2

(ii) Account for the observations in A above

(b) Examine the pieces

(i) Record other observations besides those made in (a) (i) above

(ii) Account for the observations in (b)(i) above

2. You are provided with specimen labelled **K**

(a) (i) Name the class to which the specimen belongs

(ii) Give **three** reasons for your answer in a (i) above

(b) What term is used to describe the shape of the specimen?

(c) Name and draw the fins on the specimen that;

(i) Enable the specimen to balance, brake and change direction

(ii) Prevent the fish from rolling and yawing

3. You are provided with a specimen labelled

(a) (i) What part of a plant is specimen **T?**

(ii) Give a reason for your answer in a(i) above

(b) (i) Cut a transverse section through specimen **T** (i) Draw and label one of the cut surfaces (ii) State the magnification of your drawing

(iii) State the type of placentation of specimen **T**

(c) Name the agent of dispersal of specimen **T**

(d) State how specimen **T** is adapted to its mode of dispersal

CONFIDENTIAL INFORMATION TO SCHOOLS AND PRACTICALS

BUTERE EAST DISTRICT

Each candidate should be provided with the following material/apparatus for the practical:-

- *Medium sized Irish potato (1 piece each) labeled Q.*
- *Mortar and pestle.*
- *Scalpel*
- *Distilled water*
- *Cotton thread (20 cm long).*
- *Visking tube 15 cm long.*
- *100 ml beaker.*
- *Stirring rod.*
- *Iodine solution.*
- *Means of timing.*
- *Photomicrographs labeled M and N.*
- *Transparent ruler graduated in mm*
- *Specimen K – Medium sized orange (should be moderately ripe and juicy)*
- *Test tubes (3 per candidate) in a test tube rack.*
- *DCPIP solution.*
- *Iodine solution.*
- *Benedicts solution.*
- *Means of heating water bath.*
- *Test tube holder.*
- *3 droppers.*
- *10 ml measuring cylinder.*

1. You are provided with a specimen labeled Q. Slice off about 2 cm thick disc from the

specimen. Peel it. Place the piece into a beaker and mash it into a paste using pestle and mortar.

Add 20 ml of distilled water and stir. Tie one end of the transparent visking tubing provided.

Decant the extract into the tubing and tie the other end tightly.

Ensure there is no leakage at both ends of tubing. Rinse the outside of tubing with water.

Immerse the tubing with its contents in a 100 ml beaker containing iodine solution. Allow to stand for 20 minutes..

a) Record your observations in the table below.

(b) Account for the results obtained from (a) above.

(c) What is the significance of the process being investigated to plants?

2. Study the micrographs M and N show forms of a sexual reproduction in a certain group of

organisms. Study them and answer questions that follow.

(a) (i) State the kingdom to which the two specimens belong.

(ii) Give reason for your answer.

(b) (i) What types of asexual reproduction are represented by the two specimens?

(ii) State an example in each case of an organism that uses the type of reproduction named

above.

(c) Consider the point marked X and Y. Measure the distances between the two in cm.

(i) Distance ____________________________________ cm

(ii) If the magnification of N is x60 of the actual specimen. What is the size of the actual

specimen in micrometers? (Show your working)

What is the economic importance of the activities of **M** and **N**?

3. You are provided with the specimen labeled **K**. Make a transverse section of the specimen

(a) Draw and label the transverse section of the specimen.

(b) Which type of fruit is specimen **K**?

(ii) How is the specimen adapted for dispersal by the agent named in c (i) above?

(d) Squeeze the juice from the specimen **K** into a small beaker. Using the reagent provided to

test for the food substances in the juice. Record the substances, procedures, observations and

conclusions in the table below.

CONFIDENTIAL INFORMATION TO SCHOOLS AND PRACTICALS

TRANS MARA DISTRICT

Each student should be provided with:-

1. *Specimen P – fresh piece of mammalian lungs*
   Specimen Q – fresh piece of mammalian trachea
   - *Petri dishes –*
   - *A white tile*
   - *A scalpel blade*
   - *A hand lens*
   - *Ruler*

2. *Specimen R – fresh peas in a pod (Legume)*
   Specimen S – An orange/lemon (ripe)

1. (a) You are provided with specimen **P** and **Q** which were obtained from the same animal.

Examine them carefully and answer the questions that follow:

(b) Which organ system were the specimen **P** and **Q** obtained from

(c) State the functions of **P** and **Q** in the organ system named in (b) above

(d) State **four** adaptations in each one of specimen **P** and **Q** to their functions

(e) Using a scalpel cut and draw a well labelled transverse section of specimen **P**

2. You are provided with specimen labeled **R** and **S**. Use them to answer the questions that follow:

(a) State the type of fruit labelled **R** and **S**

(b) (i) Draw a plan diagram of the longitudinally cut surface of specimen **R**

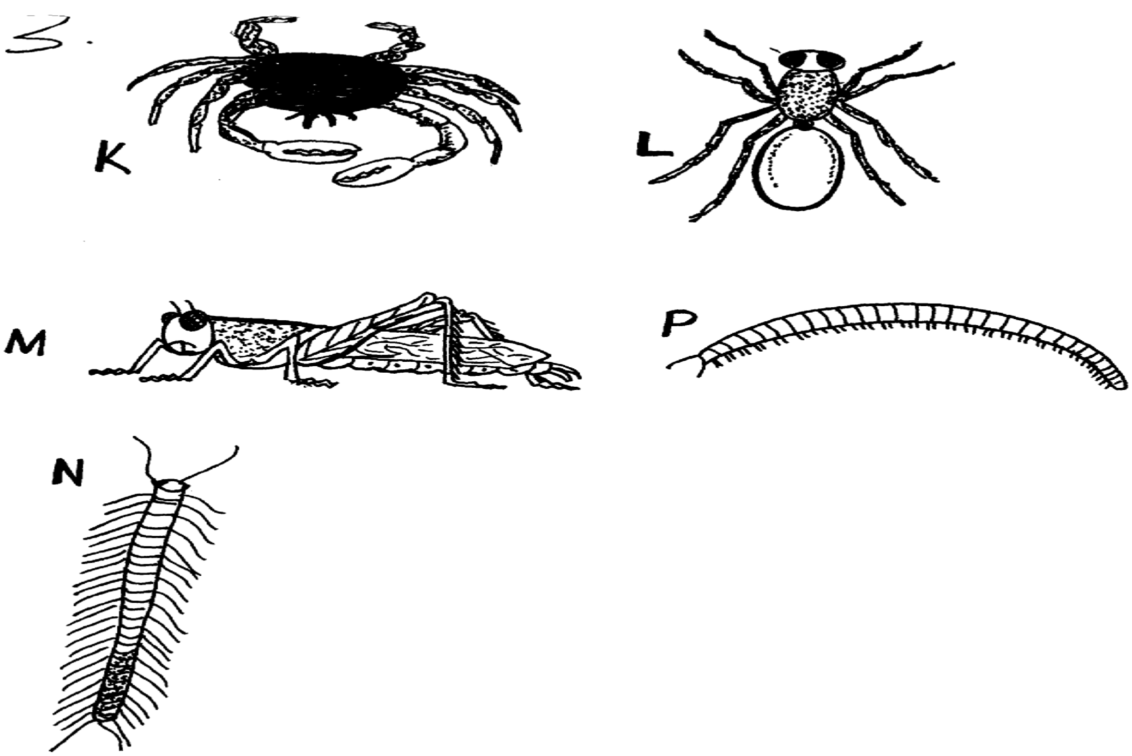
(ii) Work out your magnification

(iii) State the placentation of specimen **R**

(c) (i) State the method of dispersal of specimen **R** and **S** giving reasons for each case. Fill your answers in the table below

(ii) Give **one** advantage of the method of dispersal of specimen **S** and one disadvantage of dispersal of specimen **R.**

3. You are provided with photographs of specimen **K, L, M, N** and **P.** using observable features only, answer the questions that follow:



(a) (i) State the phylum of the organisms

(ii) Give **two** reasons for your answer in (a) (i) above

(b) With reasons give the class of :

(i) Specimen **K**

Reason

(ii) Specimen **N** .

Reason

(c) (i) State **two** ways by which specimen **M** is adapted to locomotion

(ii) Identify the type of growth that occurs in members of specimen **M**

(iii) Name the hormone responsible for metamorphosis in specimen **M**

(d) State **two** economic importance of specimen **P**

CONFIDENTIAL INFORMATION TO SCHOOLS AND PRACTICALS

SOTIK DISTRICT

1. *Each candidate should have one fruit of;*
   - *Black jack labeled P*
   - *Tomato (ripe one – money maker variety) labeled Q*
     - *Bean/pea (mature one) labelled R*
   - *Sonchus /fleabane/dandelion labeled S*

2. *Each candidate should have access to :-*
   - *DCPIP solution*
   - *Ethanol*
   - *Benedict's solution*
   - *Iodine solution*
   - *Hot water bath*
   - *Clean water*

   N/B Use clean droppers where applicable to minimize contamination of reagents.

3. *Each candidate should be provided with;*

   (i) 4 test-tubes in a test tube rack.

   (ii) Razor blade (or they can be asked to bring theirs).

1. You are provided with specimens labeled **P, Q, R** and **S**

   (a) State the type of fruits represented by each of the specimens

   (b) Explain how specimens **R** and **S** are adapted to their agents of dispersal

   (c) Cut the specimen **Q** transversely in the middle. Draw a well labeled diagram of the face of the cut surface of one of the halves

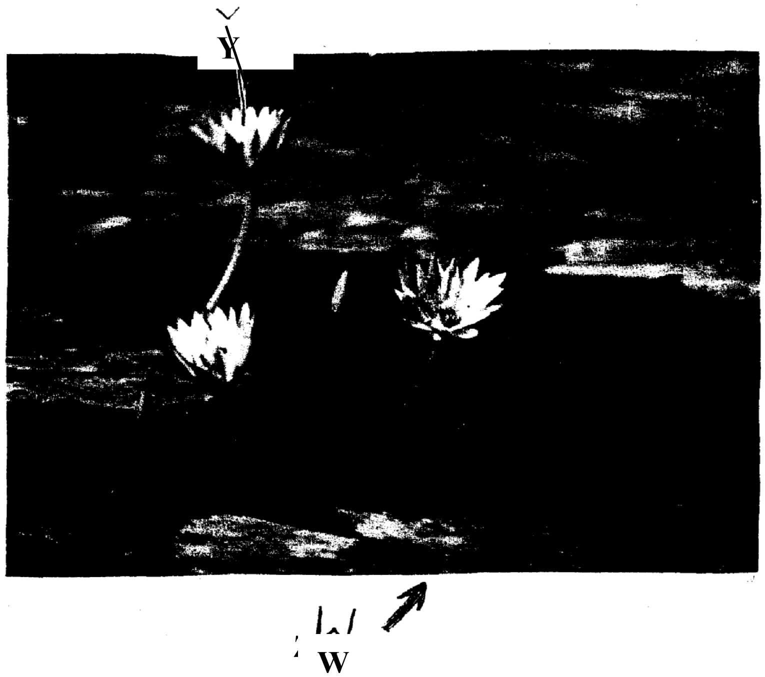
(d) State the placentation of specimen **Q**

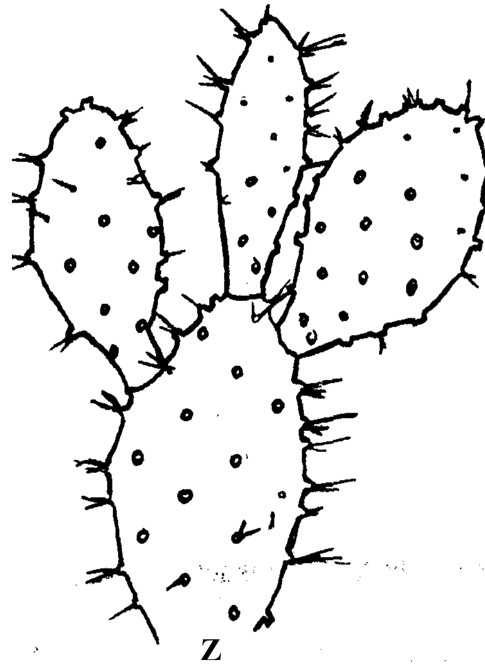
2. (a) Wash of the halves of specimen **Q** and place it in a mortar and grind it using the pestle to obtain its juice. Then add clean water enough to fill a test tube and shake. Then decant the juice into a clean test tube. Using the apparatus and the chemicals provided subject

(b) Explain how digestions of the components of the food sample are digested in the ileum of a mammal

(c) What is the importance of specimen **Q** in the human diet?

3. The photographs **W** and **Z** below are of plants obtained from different habitats





(a) Suggest the possible habitat of specimen **W**

(b) (i) Name the structure labeled **Y** in specimen **W**

(ii) State the function of the structure named in **(b) (i)** above

SECTION III- MARKING SCHEME

KAKAMEGA CENTRAL DISTRICT

1. (a)

Food substance	Solution	Procedure	Observation	Conclusion
Reducing sugar	V	To 1 ml of food substance add 1 ml of benedicts solution. Place in a warm/ hot water bath/ heat boil;	Blue colour retained;	Reducing sugar absent;
	W		Blue- green – yellow- orange; Acc – brown - final colour	Reducing sugar present
Non- reducing	V	To 1 ml of food substance	Blue- green yellow-	Non reducing

sugar		add 3 drops of dilute hydrochloric acid. Boil cool add sodium hydrogen carbonate till fizzing stops. Add 1 ml of Benedicts solution and place in a warm/ hot water both/ heat/ boil	orange Acc- final colour only - brown - Reject brick red	sugar present;
	W		Blue colour retained;	Non-reducing sugar absent

b) Solution W;
Reason – it is a reducing sugar that is absorbed directly and used in respiration to produce energy

2. a)

Structure	Name	Function
P	Ureter	Transports urine from kidney to urinary bladder
Q	Ovary	Oogenesis/ formation and release of ova i.e. ovulation
S	Fallopian tube/ oviduct	Passage of ova
T	Uterus	Pregnancy ACC implantation/ development of the embryo

b) i) Female;
ii) – Has ovaries/ oviducts
- Has uterus
- has a vestibule
- has a clitoris

c) Dissecting pin;

3. a) A1 – solitary
B1 – Inflorescence

b)

	Class	Reason
A1	Dicotyledonae	Has four petals
B1	Monocotyledonal	Has three stamens

c) S – Petal
T- Sepal, Accept Calyx

d) Type – superior;
Reason – above receptacle;

e) i) Wind;
ii) Feathery stigma; (to increase surface area for trapping pollen grains from the air)
iii) Small, light pollen grains

- Flower not scented;
- Inconspicuous flower; Mark first 2

f) i) Insects
ii) Tubular/ bell shaped corolla
Accept large petals

g) Animal;

KAKAMEGA EAST DISTRICT

1. (a) (i) Arthropoda; (Rej. Antrhopoda/Arthropods)
(ii) Have segmented bodies;
- Have appendages;
- Have exoskeleton;

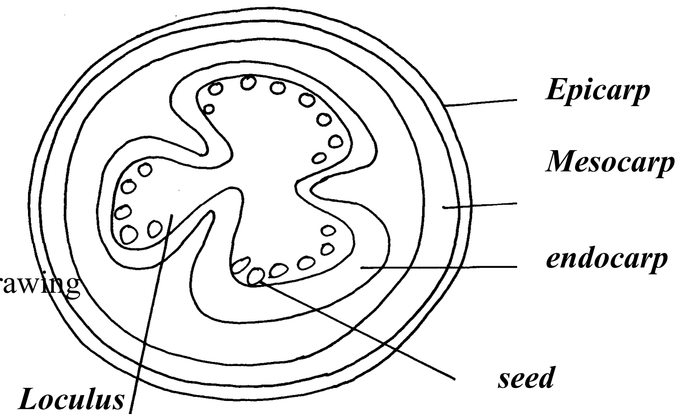
(b) M – Insecta; rej. Insects
N – insecta;
O – Arachnida; rej arachnids
P – Crustacea; rej. Crustacean
Q – Chilopoda;

(c) M – 3 pairs of limbs;
- body divided into three parts/head thorax and abdomen);
- Has a pair of wings;

P – Body divided into two parts/celaphalothorax, abdomen;
- Have carapace/hard outer shell);
- Have two pairs of antennae;
- Have a specialized limb for feeding/defence/chelicera; (any two marks)

(d) – Vectors of disease causing micro-organisms /pathogens;
- bee make honey;
- are pollinating agents;
- destroy timber e.g. termites
- Crop pests e.g. weevils

2. a (i)



(ii) Magnification = length of drawing
Length of image

x ½ upto x 1½ ;
(iii) Berry – many seeds
- Endocarp fleshy/juicy; (any one)

(b) (i)

A	40;
B	Numerous Not decolourite DCPIP;

(ii) Ascorbic acid/vitamin C;

(iii) Prevents scurvy/bleeding of gums hence weak teething

(c) Boiling vapourises the vitamins/vitamin C;

3. (a) U – scelerenchyma
F- xylem tissues;

D – phloem tissues;

AT- parenchyma tissue J

(b) D – carry substances hence remain turgid; to offer supper
F- are strengthened with lignin;

T- remain turgid to outer supper (any two)_
U – provide rigidity of the (2mks) stem;

(c) (i) H – Endodermis

(ii) Demarcates cortex from central cylinder;
(d) (i) F;
(ii) Root pressure;
Transpiration pull;

Capillarity/cohesion and adhesion; rej adhesive and cohesive

(e) (i) Phloem (tissue);

(ii) Companion cell;

(iii) has mitochondria which provide energy for translocation n the phloem;

MIGORI / NYATIKE DISTRICT

1. (a) Set A Mitosis;

Reason:- Two (daughter) cells formed;

Set B Meiosis
Reason-Four (daughter) cells formed;

(b) A 3 Metaphase;

4 Anaphase;

B 1 Metaphase 1;

3 Telophase1;

(c) Set A – shoot tip/root tip/cambium /flower buds/apical meristems/cambium meristems;
Set B – Anther /ovary;

(d) Set A; number 4- Chromosome align at the equator

Set B: - number 2: Homologous chromosomes separate and move (migrate towards the opposite poles;

(e) Set A- Results in growth;
Set B- Gamete formation/gamete variation

2. (i) A = 3mm;

B = 7mm
C = 5mm;

(ii) A- The solution (L_1) is hypertonic to the cell sap of the potato tissue; water is drawn out of them by osmosis; the cells become plasmolysed and flaccid and they shrink /decreases in length;

B- The solution L_2 is hypotonic to the cell sap of the potato tissue; they gain water by osmosis; and become turgid. They cause the tissue to increase in length;

C - No change; control experiment

(b)

FOOD SUBSTANCE	PROCEDURE	OBSERVATION	CONCLUSION
STARCH	Add 3 drops iodine. Solution to food substance to be tested	Colour changes to blue-black (black)	Starch present
REDUCING SUGAR	Add on equal amount of Benedict's solution to the food substance and heat to boil	The colour changes from blue to green to yellow to orange and brown precipitate formed	Reducing sugar present
PROTEIN	To about $2cm^3$ of food substance add $1cm^3$ of NaOH solution . Add 1-2 drops of copper sulphate	Purple or violet colour formed	Proteins present

(b) Obesity;

Marasmus in children and muscle wasting in adults

3. (a) T_1 – incisor; (tooth)

T_2 – Molar; (tooth)

(b) Incisor (T_1) – sharp /wedge-shaped; for cutting;
Molar (T_2) – broad surface with cusps; for grinding

(c) A – Nerve

B- Pulp cavity

C – Enamel
D – Dentine
E – Blood vessel
F – Periodontal membranes

(d) Cause bacteria to grow and produce acids which cause tooth decay;

(e) (i) Diastema;

(ii) Allows movement of tongue when cutting grass and turning food in the mouth

NYAMIRA DISTRICT

1. (a) (i) No gas produced

(ii) Gas produced

(b) (i) In boiled potato cube, enzyme catalase is denatured hence no reaction when

(ii) Fresh potato cube had an enzyme catalase which broke/decomposed hydrogen peroxide to water and oxygen, hence production of gas.

(c)

Food substance	Procedure	Observation	Conclusion
Starch	To fold substance add iodine solution	Blue colour formed	Starch present
Proteins	To food substance as sodium hydroxide and copper II sulphate solution	Light green mixture	Proteins absent

2. (a) S – Style

T – Ovary

U – Anthens

V – Petal

(b) X – Mode- Insect

Reasons – Brightly coloured to attract insects.

Anthers inside the flower to be reached by pollinating agent.

Stigma is above the anthers to pick pollen from the incoming pollinating agent

Y- Mode –wind

Reasons:- Long anthers exposes outside the flower to be easily reached by the flower for pollen grain to be easily blown by wind.

3. (a)

Organism	Steps followed	Identify
A	1a, 2b, 5a	Arachnida
B	1a, 2a, 3a, 4b	Hymenoptera
C	1a, 7b	Echinodermata
D	1a, 2b, 5b, 6a	Chilopoda
E	1a, 2a, 3b	Anaplura

(b) (i) Arthropoda

(ii) Segmented body

SOTIK DISTRICT 2ND EXAM

1(a)

FOOD	PROCEDURE	OBSERVATION	CONCLUSION
STARCH	To 2mls of Z is added 2 drops of iodine solution	The colour of the mixture remained brown	The solution does not contain starch
REDUCING SUGAR	2ml of solution Z is mixed with 2ml of benedict's solution and heated	The colour of the mixture turned blue	Absence of reducing sugar
VITAMIN C	2ml of DCPIP is placed in a test tube and solution Z is added drop wisely	The DCPIP is decolorized loses its colour	Vitamin C is present
LIPID	2ml of solution Z is put in test tube and ethanol added until it clears then distilled water is added	White colour develops	Lipids present

NB/

-exact quantities must be mentioned

-correct order of chemicals

-correct colours

b) Pancreas reject pancrease

c) Fatty acids and glycerols (Reject if one is missing

d)-can be broken down; to liberate energy; for cellular functions

-can be used to synthesize; structural components of the cell; OWTTE

2. a) i) autotrophic (nutrition)- (1mk) Reject. Antotropism

ii) They are green hence has chlorophyll for photosynthesis;

iii) Carbon (iv) oxide+ water sunlight $\xrightarrow[\text{Chlorophyll}]{}$ glucose +oxygen

$$6\ CO_2+6\ H_2O \longrightarrow C_6H_{12}O_6+6\ O_2$$

NB.-chemical symbols must be correct

-the equation must be balanced

b) NB.-all rules for drawing apply

-1 mark for accuracy (1x1=1mk)

-max of 3 correctly labelled parts (3x1=3mks)

P	Q

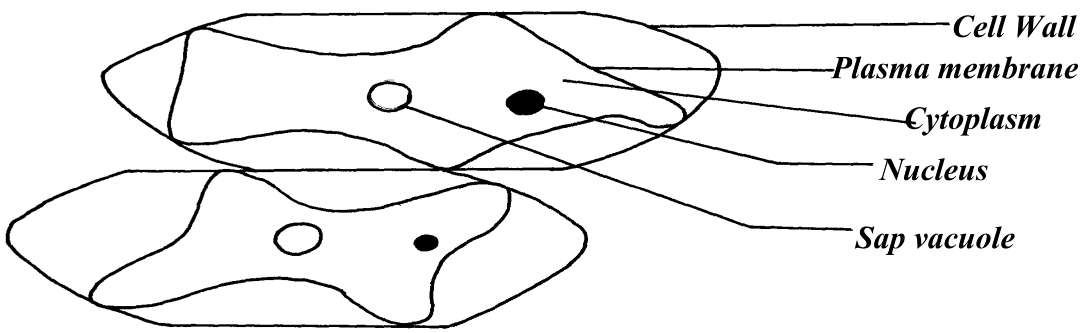
i)has broad lamina ii)has hairless lamina iii)lamina has network of veins	-has narrow lamina -has hairy lamina -lamina has parallel veins OWTTE

3. a)Phylum :Arthropoda. Reject. Anthropoda/Arthropod
   Class: insecta. Reject insect
   b)-the body has 3 pairs of leg (jointed legs/6(jointed legs)
   -the body is divided into 3 clear body parts (i.e. head, thorax and abdomen)

   Reject if no mention of distinguished parts
   ACC. Any other correct feature max
   c)-has very large compound eyes; for spotting; food, enemies, mates etc
   -has (6) muscular legs; for efficient locomotion; in search of food, mates etc
   -proboscis; for efficient sucking of liquid food; OWTTE STK

UGENYA- UGUNJA DISTRICT

1. a)



D – 1 ½ P - 1
Mag: x100 or x150 L – $^{4}/_{2}$

Cl – ½

Mag – ½

b) Liquid H is hypertonic/ highly concentrated; to cause a high osmotic pressure; water molecules are drawn from the onion epidermal cells by osmosis; excess water loss results in the plasma membrane detaching from the cell wall/ hence the cell is plasmolysed; (OWTTE)

c)

FOOD SUBSTANCE	PROCEDURE	OBSERVATION	CONCLUSION

Starch ✓	- Place 2 ml of J_1 in a test-tube. - Add 2 – 3 drops of ✓ iodine solution. - Shake and observe.	• Solution changes from Brown to Blue- ✓ black /Black	• Starch present. ✓
Reducing Sugar ✓	- Place 2 ml of J_1 in a test-tube. - Add equal amount ✓ of Benedict solution. - Heat to boil.	• Solution changes from blue, green, yellow ✓ then brown.	• Reducing sugar present. ✓

d) i)

TEST TUBE	OBSERVATIONS
J_2	- Glowing splint does not rekindle/ relight; or glowing splint relights /rekindles slowly; √1
K	- Glowing splint relights/rekindles faster; √1

ii) Enzyme catalase;

iii) The liver has more catalase enzyme; since it undertakes the role of detoxification in mammals;

3. i)

ORGANISM	STEPS FOLOWED	IDENTITY
V	1b, 2a;	Mollusca;
W	1b, 2b, 3a, 4a, 6a, 7b;	Crustacea;
X	1b, 2b, 3a, 4b, 5a;	Annelida;
Y	1a, 9a;	Cestoda
Z	1b, 2b, 3a, 4a, 6b, 8a;	Arachinida;

ii) Schistosoma sp / S. haematobium / S. japonicum;

rej. – when not underlined separately.

- Wrong spelling.

- Lower case "S" for 1st letter.

NDHIWA DISTRICT

1. (a) With six legs ....go to 2
   (b) With eight legs ....N

. (a) With wings ....go to 3
  (b) without wings ....P

(a) With one pair of wings ....R
(b) With two pairs of wings ....Q

(b) (i)Arthropoda
(ii) – Presence of exoskeleton
- Segmented body
- Jointed appendages/legs, limbs)

(c) Arachnida
- 8 legs /4 pairs of legs
- Two body parts
- Lack of wings
-No antenna

2. (i) Plants which normally grow n fresh water/plants which normally grow in very we places
(ii) Part 1 – leaf stalk

part 2 – leaves

(iii) - Low O_2 concentration
- Low light intensity
- Low mineral salt concentration content

- A lot of water
- Waves and currents

(iv) – Some (emergent) have broad leaves with numerous stomata on the upper surface to increase transpiration
- Highly dissected leaves to increase surface area for absorption of maximum light CO_2 for photosynthesis and gaseous exchange
- - Flowers are raised above the water to allow pollination
- - Some (floating) have log fibrous roots to absorb mineral salts
- - Long leaf stalk to expose the eaves above water for photosynthesis

(v) - Many stomata on the upper surface to increase transpiration

- Numerous and sensitive chloroplasts that photosynthesize at low light intensity
- Large air filled issues /aerenchyma for buoyancy and gaseous exchange (store O_2 for respiration)
- Poorly developed vascular bundles to discourage water absorption

(vi) R -Submergent
S- Floaters

3. (a) (i) A - less/few bubbles/slow effervescence/fizzing/froth/foam
B- Rapid fizzing / bubbles

- (award 1mk for bubbling /effervescence in both A and B- Reject if bubbling only appears in either A or B)

✓ ✓
(ii) Large surface area in B than in A for enzymatic activity in

- Part (iii) tied to (i)

- Bubbles due to enzymatic activity (award only 1mk)
Hydrogen Peroxide water + Oxygen
- Chemical symbols alone or words alone

- Wrong enzyme, means wrong commitment

Accept:-H_2O_2 + catalase $\longrightarrow$ H_2O O_2 + Catalase

(b)

	Procedure	Observation	Conclusion
Roots	Add one 2/3 drops of iodine (soln.) Accept add iodine or any other measurements	No color change/colour of iodine / brown /yellow colour	Starch absent
	To 1m of extract , add 1/2ml /equal amounts of	Blue to green	Traces of reducing sugars
	Benedict's solution heat to boil	Yellow – orange/brown	Reducing sugar present ICCP simple sugar
Remains of grains	Add drops of iodine		Starch present
	Add drops of Benedict's solution	Green to yellow to orange/brown	Reducing sugar present
Leaves	Add iodine	No change	Starch absent

	Add Benedict's	- Green to yellow to orange to brown	Reducing sugars present

rej. – if for starch is written under procedure

– brick red

(c) Roots- Presence of reducing sugars translated from the remains ✓ of grain/as leaves; for provision of energy /respiration/growth & ✓ development /metabolic activities e.g. active transport;

- Absence of starch because roots are not storage organs.

Remains of grains ✓

- Presence of reducing sugars translocated from the leaves/ hydrolyzed starch; ✓
- Presence of starch because grain is a storage organ/some starch had not been hydrolyzed for germination /growth;

Leaves - Presence of reducing sugars due to photosynthesis;

-Absence of starch because reducing ✓ sugars has not been converted ✓ to starch;

MUMIAS DISTRICT

1. a) Observations

Strip in beaker	Observation
1	Inside of the peeling curves outwards
2	Remained straight
3	Inside of the peeling inwards
4	Inside of the peeling curves inwards, more than in 3

b) Accounting for 1, 2 and 4

1: The cells of the inside of the peelings have cell sap which is hypertonic to solution S; hence draws in water by osmosis; and (swells up to) become turgid; leading to more increase in length of that side and curvatime on peeling sides

2: The cells of the inside of the peeling have cell sap which is atomic solution H; hence no net osmosis

3: The cells of the inside of the peeling have cell sap which is hypotonic to solution H, and lose water by osmosis to become flaccid; this side shrinks hence curvature inwards

c) Solution G – Distilled water

Solution H – Concentrated solution

2. a) steps

2b – two

7b- cyndrical

b)

specimen	steps	Identify
A	1a, 2a	Housefly
B	1b, 3b, 4a	Spider
C	1a,2b	Dragon fly
D	1b, 3b, 4b, 5a, 6a	Waters/ ater
E	1b,3b, 4b, 5a, 6b	Fresh water shrimp
F	1b, 3b, 4b, 5b, 7b	Centipede
G	1b, 3b, 4b, 5b, 7a	Millipede
H	1b, 3a	Ant

c)

	Class
B	Arachnida
C	Insecta
E	Crustacea
G	Diploda

3. a) Legume stem;
Roots
Nodule

b) Rhizobium bacteria bacteria/ nitrogen fixing bacteria

c) Leguminous plants

d) i) symbiosis

ii) Rhizobium bacteria which lives in the root nodules of leguminous plants fix free nitrogen in the soil into nitrates ; which are absorbed by plant proteins; bacteria benefit from shelter and carbohydrates provided by the plants;
this relationship enables plants to thrive on nitrogen deficient soils

iii) Nitrate

iv) Pseudommonas denitrificans
Thiobacillus denitrificans

KISUMU WEST DISTRICT

1. (a) Blue black/black dark blue colour is formed
   (b) No colour change/colour of Benedict's solution remains;
   Rej: No change /no reaction/ no observation /nothing happens
   (c) Set-up A- colour changes from blue to green to yellow to orange/brown;
   Set-up B: No colour change/ colour of Benedicts' solution remains;
   Rj- No change/no reaction/no observation/nothing happens
   Set-up C- No colour change/colour of Bendict's solution remains;
   Rj- No change /no reaction/ no observation/ nothing happens
   (d) Set-up A – Enzyme amylase/diastase/invertase (in Q); digests /hydrolisis/breaks down/ converts starch (in liquid X); to reducing sugar/maltose;
   Set-up B: boiling denatures/destroys enzymes amylase/diastase/invertase; henc starch is not converted to reducing sugar/maltose;
   Set up C:- Hydrochloric acid provides unfavourable PH for enzyme amulase diastase/invertase; hence starch is not converted to reducing sugar/maltose;
   (e) Enzyme amylase/diastase/invertase;
   (f) To provide optimum temperature for reaction of enzyme amylase/diastase;

2. (a) (i) Chilopoda; Rj wrong spellings of classes but award marks for reasons
   Reason – One pair of legs per body segment;
   -Dorsoventrally flattened body; (consider first one only)
   (ii) Insecta;
   Reason- Body is divided into three parts/regions;
   - Three pairs of legs;
   - presence of wings;
   (iii) Arachnida;
   Reason:- Four pairs of limbs/legs;

	C	E
-	5pairs of legs	4Pairs of legs
-	Has antennae	Lacks antennae
-	Lacks chelicerae /pedipalps	Has chelicerae/pedipalps
-	Has carapace	Lacks carapace

(c)

ANIMAL	STEPS FOLLOWED	IDENTITY

C	1a, 2a, 4a;	CANCER;
D	1, 2b, 5b;	SIGMORIA;
E	1a, 2a, 4b;	LACTRODECTUS;

3. (a) (i)Dicotyledonae; Rej; wrong spellings of classes

Reason: - Net-veined leaves

- Tap root system; Rj: tap roots (Mark first one

(ii) Monocotyledonae;

Reason :- Parallel –veined leaves;

-Fibrous root system; Rj – Fibrous roots (mark first one)

(b) G – Epicotyl; Rj wrong spellings

J- Hypocotyl;
M – Prop roots;

(c) Stores food during germination

- turns green and carries out photosynthesis;

(d) (i) root nodules;

(ii) Rhizobium bacteria

(e) (i) Hypogeal germination;
(ii) Cotyledon remains in the soil;

TRANS NZOIA WEST DISTRICT

1. a)

Liquid	Food substance	Procedure	Observation	Conclusion
L1	Starch	To 2ml of L, in a test tube add afew drops of iodine solution; Reject if heating is done	No observable color change	Starch absent
	Reducing sugar	To 2 ml of L, in a test tube add an equal amount of Benedicts solution and heat/ immerse in a warm water bath	No observable color change	Reducing sugar absent
L2	Starch	To 2 ml of L in a test tube add a few drops of iodine solution	Color changes to blue- black	Starch present
	Reducing sugar	To 2 ml of L add an equal amount of Benedicts solution and heat/ immerse in a water bath	Color turns from blue – green – yellow – orange/ red	Reducing sugar present

b)

Food substance	Procedure	Observation	Conclusion
Starch	To 2 ml of L, in a test tube add a few drops of iodine solution	No observable color change	Starch absent
Reducing sugars/ simple sugar	To 2 ml of L in a test tube add an equal amount of Benedicts solution and heat/ immerse in a warm water bath	Color turns from blue – green – yellow to orange/ red	Reducing sugars present

For procedure and food substance mark once

c) - On immersing the visking tubing containing L1 into solution L2, a concentration gradient was created

- The reducing sugars/ simple sugars in L2 moved by diffusion; through the visking tubing into L1, due to their small size; hence their presence in L1, at the end

Starch absent in L1, because the molecules are too large to pass through the tiny pores of the visking tubing

2. a) i) Pisces; Reject Pieces/ fish

ii)
- Presence of scales
- Presence of fins
- Presence of operculum

b)
- Protection of gills
- Gaseous exchange

c) i) 2 / pectoral fins

3 / pelvic fins

ii) 4 / dorsal fin

5 / candal fin

6 / anal/ ventral fin

d) i) 64 mm

ii) 29-30 mm

iii) $\frac{\text{Length from anus to tip of tail}}{\text{Length from tip of mouth to tip of tail}} = \frac{30/29}{64} = 46.9\% / 45.3\%$

64% must be there to score

iv) The high tail power creates enough force to enable the fish to push forward

e) - Streamlined body for easy movement/ reduce water resistance

- Scales on the body overlap facing the posterior end for easy movement/ to reduce resistance from the water
- Massive head prevents the fish from being deflected from its oath when swimming

3. a) A– Phloem
B. – Vascular cambium
C. – Xylem
D. – Cortex
F. – Collenchyma
G. – Epidermis
H. – Pith

b) A – Trans location
B – Divides to give rise to new tissues (for secondary growth)
C – Transport of water; and mineral salts

c) Parenchyma

d) - Cell wall thickened with lignin; for strengthening/ mechanical support

e)

Root	Stem
Star- shaped Xylem with phloem in the arms	Vascular bundles arranged in a ring
No pith	Pith present
Presence of root hairs	Absence of root hairs

RACHUONYO DISTRICT

1. a)

ANIMAL	IDENTIFY	STEPS
E	Mollusce	1b, 2a
H	Annelida	1b,2b,3a,4b,5a
J	Cestoda	1a, 9a
M	Insecta	1b,2b,3a,4a,6a,7a
N	Arachnida	1b,2b,3a,4a,6b,8a

Each correct identify 1mk

Each correct step 1mk

Reject wrong order of steps

Reject wrong spelling for identify

Identify tied to steps (if steps is wrong then reject identity)

b) Arthropoda√ 1mk Rj wrong spellings

c) Segmented body√/ Exoskeleton (made of chitia)√

2 a)

ORGANELLE	NAME	FUNCTION
P	nucleolus√	Manufacture ribosomes
Q	Mitochondrion√ Acc.mitochondria	Site for respiration √
R	Cell membrane/plasmolemma/ √plasma membrane	Control entry and exit of √ material into cell
S	Rough endoplasmic reticulum√	Transport proteins√
T	Golgi body/apparatus√	Packaging and transport of glycol Proteins√

Rj. Wrong spelling for name but mark function of right

b) Length – 31mm

c) actual radius = $\frac{\text{photomicrograph length}}{\text{Magnification}}$

$= \frac{31}{10,000}$ √

= 0.0031mm√

3. a) i) No bubbles/no effervescence/no observable change

ii) Bubble/effervescence/foam/increase in warmth

iii) No bubbles/no effervescence/no observable change

b)i) fermentation (anaerobic) respiration

ii)$C_6H1_2O_6 \longrightarrow 2C_2H_5OH + CO_2$ +Energy

Glucose $\longrightarrow$ ethanol +carbon (iv) oxide +energy

iii) mitochondrion/mitochondria

c) i) yeast/enzyme/catalyst (

ii) Boiling denatures yeast/enzyme; hence glucose not broken down/respired; to

produce Carbon (IV) oxide gas (which forms bubbles)

d) i) bubbles/effervescence/foam

ii) Catalyst (in liver); breaks down hydrogen peroxide; to form water and oxygen

(which forms bubbles);

1. (a) (i) M_1 – stem firm/hard/stiff/rigid/tough

M_2- stem soft/flexible/flabby Reject flaccid/weak

(ii) Solution M1 is hypotonic (to cell sap) /cell sap is hypertonic (to L_1)

Solution M_1 is less concentrated/more dilute than (cell sap; hence water moved into the

(stem) cells/osmosis occurs; cells become turgid

M_2- Solution M_2 is hypertonic/more concentrated than cell sap more dilute/cell sap

more dilute/ cell sap less concentrated (than L_2); water moves out of the

cell/osmosis occurs; makes the cell flaccid

(b) (i) M_1 (slit opens wider/widens/strip separates; and the bends/outwards or backwards or curved; M_2- strips) remains closed together/slits remains closed/strips shrinks or shrunk;

(ii) In M_1 cells in inner surface/cut surface /cutical cells enlarge more/; because they took in more water (by osmosis) than outer cells /outer surface/epidermal surface (which have

cuticle) OR

M_2 Cells of inner surface/cut surface/cutical cells shrunk; because they lost more water (by osmosis) than outer cells/epidermal cells which have cuticle

N/B mark only once i.e. for M_1 or M_2

2. (a) Pisces reject- spelling mistakes

– Pisces/fish or pisces(fish)

(ii) Presence of scales

- presence of fins
- - presence of lateral line
    - -presence of gills
    - Presence of operculum

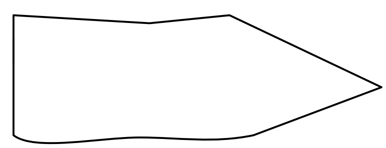
(i) is tied to (ii), so if (i) is wrong reject reasons even if correct

(b) Streamlined;

(c) (i) pectoral fin

right identity-

Right drawing-



pelvic fin

N/B - Reject wrong drawing if identity is wrong

- Reject wrong fins among the right ones

- The shape should be continuous

- Accept spines are single lines e.g.

(ii) Dorsal fin

N/B- Spines must be present to award a mark

Anal fin(Ventral fin)- Right identity-

-drawing

Caudial (Tail fin) N/B- The identity must be correct to award drawing mark

3. (a) (i) Fruit;

(ii) Two scars/point of attachment at receptacle and to the remains of style;

(b) (i) Drawing -3maks Mark clockwise

Label $^5/_2$ = max 2

<u>drawing mark 1</u>

Given when there is continuous double outline of epicarp

<u>drawing mark 2</u>

Given when endocarp with seeds is present

loculi with juice sac is present

<u>drawing mark 3</u>

When placenta is centrally located and not shaded

(b) (ii) x ½ - x 3; N/B Reject X , x signs

(iii) Axial/Axile/Central; Reject mistakes- free central

(c)Animal, accept- man, human being

Reject- human alone, animal dispersal

(d) - Seeds are hard/slimy/slippery (coat) to prevent digestion;

- It is scented /sweet smell to attract the agent;

- It is brightly coloured to attract the agent;

- It is succulent / juicy to attract the agent; Mark any three correct

BUTERE EAST DISTRICT

1. a)

	Extract Inside tubing	Iodine solution outside the tubing
Before experiment	White suspension	Brown /Yellow.
After experiment	- Blue/ Black /Blue – black colour observed - The level increased/size of Viking tubing increased.	- Brown colour of iodine retained. - Level of iodine reduced.

b) – The extract inside the tubing contains starch, A blue – black was observed due to diffusion of iodine from the beaker across the Viking tubing membrane; since iodine has a low molecular size;

- Iodine solution retained the brown colour because starch molecules in the extract are large; in size and could not pass through the pores of the viking tubing membrane; into the beaker.

c) i) Gaseous exchange;

ii) Transpiration;

iii) Translocation of sugars;

2. a) i. Fungi

ii. Reproduce asexually by budding or sporulation.

b) i) M - Budding
N – Sporulation
ii) M – Yeast

N – Rhizopus / Bread mould/mucor.,

c) i) 8.7 cm
ii) Linear Magnification = $\frac{\text{Linear dimension of the Image}}{\text{Linear dimension of actual object}}$
Linear dimension of Image = 8.7 cm x 10000 µm = 8700 µm

$$X60 = \frac{8700}{x}\ \mu m$$

$$x = \frac{8700}{60}\ \mu m$$

$$X = 1450\ \mu m$$

d) M – Used in baking Industry and brewing Industry.

N – Causes decay of dead organisms releasing nutrients.

- Causes food decay.

3. a)

b) A berry/ hesperidium.

c) i) Animal dispersal.

ii) – Succulent endocarp/Juicy endocarp.

- Scented

- Bright colour exocarp.
- Seeds resistant to digestion.

d)

Food Substance	Procedure	Observation	Conclusion
Reducing sugar	- Put test substance in the t. tube - Add benedict soln. - Boil	- Colour changes to yellow/orange/red	- Reducing sugar present
Vit. C Ascorbic acid	- Put 2cm^3 of given vol. of DCPIP in t. tube. - Add juice/test substance.	- DCPIP decolourised	- Vit. C present.
Protein	- Put cm^3 of juice in a t. tube. - Add 2 cm^3 of	- No colour change	- Protein absent.

	NaOH - Add $2cm^3$ of $CuSO_4$		

TRANS MARA DISTRICT

1. (a) P – Part of mammalian lung

Q – Part of mammalian trachea

(b) Respiration/breathing system

(c) P- This is where diffusion/gaseous exchange occurs

Q – Allows passage of air into the lungs

(d) - It is elastic to allow stretching or expansion

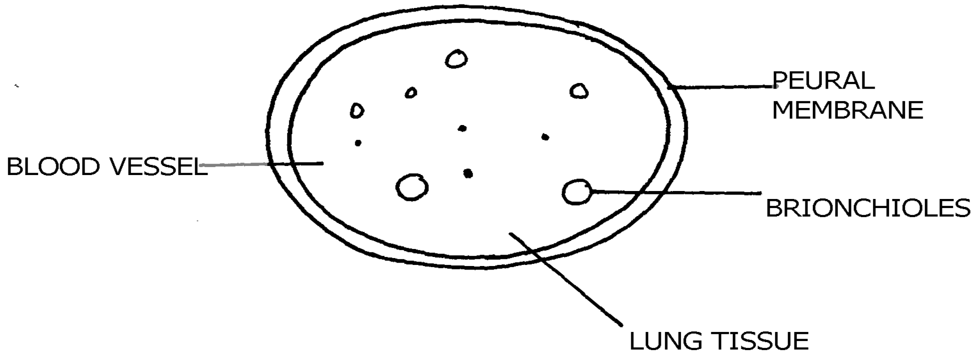
- Has numerous blood vessels to facilitate efficient transportation of gases
- Presence of bronchiole for passes of air in and out
- Presence of pleural membrane that produces pleural fluid thus reducing friction
- Presence of spongy air spaces /alveoli to increase the surface area for gaseous exchange

Q – Rigid, firm/hard rings of cartridge to prevent collapsing/keeps it open to allow passage of air.

- Presence of muscles between the rings/cartilage to allow for movement

- Mucus living to trap foreign particles/filter air
- Cartilage rings are C-shaped to allow room for expansion

- To score a mark; the feature is tied to a function

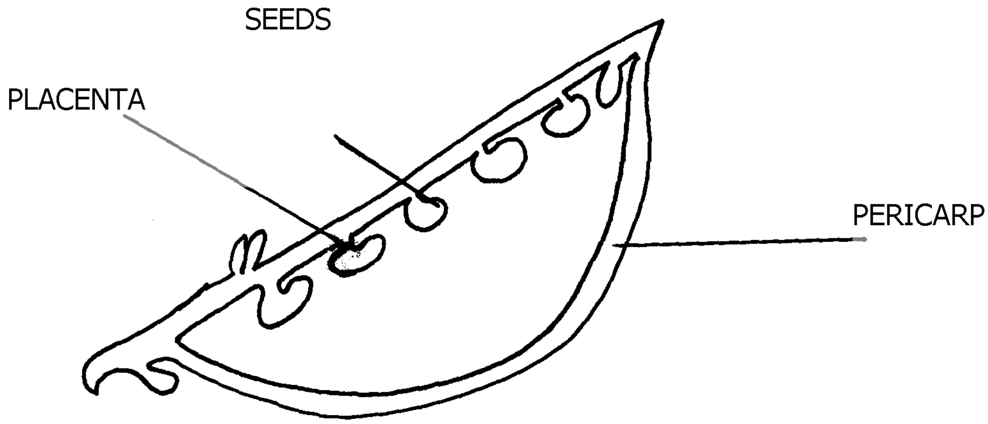


N/B- The score for the drawing – the drawing should have continuous outline (double) no shading and proportional in pencil.

- To score the label mark, the label line should not cross, no arrows

2. (a) specimen – R- Legume, S-Berm

(b) (i)



(ii) Magnification = $\frac{\text{length of drawing}}{\text{Length of specimen}}$

$= \frac{X}{Y}$

(iii) Marginal placentation

(c) (i)

Specimen	Method of dispersal	Reasons
R	Self explosion	- has line of weakness - splitting line
S	Animal/man Reject: bird	- has brightly coloured skin to attract animal
		- succulent - has sweet smell scent

(ii) S – Can be dispersed over a long distance hence low chances of overcrowding

R- Dispersed over a short distance hence high changes of overcrowding

3. (a) (i) Phylum arthropoda

(ii) – Have segmented bodies

- Posses jointed limbs and appendages

(b) (i) K – cross – crustaceae

Reasons – has two pairs of antennae

- has forked appendage

(ii) N – class –chilopoda

Reason – All many segments with one pair of legs per segment

(c) (i) – Has two pairs of using for flying

- Has powerful (muscular) hind limb for hopping/jumping

- Intermittent growth

(ii) – moulting/ecdysome hormone

(d)- Enhance nutrient cycling/humus

- Aeration of soil

SOTIK DISTRICT

1. a)

SPECIMEN	TYPE OF FRUIT
P	Cypsella
Q	Berry
R	Legume
S	Cypsella

b) **R** has two lines of weaknesses (sutures) along which it splits; to release seeds by explosive mechanism;

Accept Self dopersal for explosive mechanism

Reject self explosion or self explosive mechanism

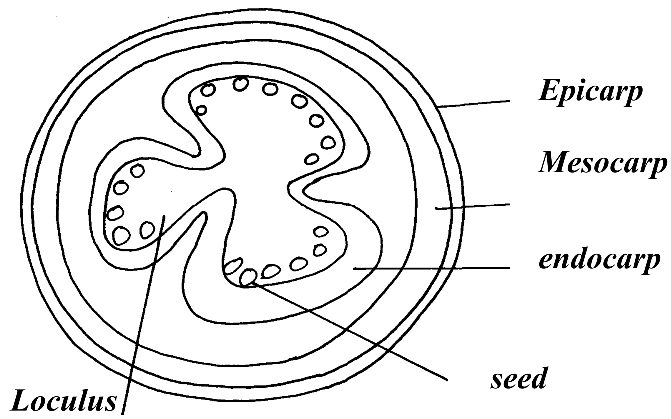
Agent = 1 mk, adaptation = 1 mk; reason = 1 mk

S is very small; with pappus; making it light; to float easily to be dispersed by wind;

Agent = wind 1 mk Any one adaptive feature & explanation

Adaptation = 1 mk Reason = 1 mk

c)



d) Axile placentation

2. a)

FOOD	PROCEDURE	OBSERVATION	CONCLUSION
Lipids	$4cm^3$ of food sample mixed with $4cm^3$ of ethanol then add clean water(1 mk)	No change in colour	Absence of lipids
Reducing sugar	$2cm^3$ of food sample in mixed with $2cm^3$ of Benedicts solution and heated in the hot water bath(1mk)	The colour changes from blue to brown 1mk	Presence of reducing sugar 1 mk
Ascorbic acid (Vitamin C)	$2cm^3$ of DCIP is put in a test tube. Add food sample droplisively (1mk)	The DCIP is decolorized (1mk)	Ascorbic acid (Vitamin C) present(1 mk)
Starch	$2cm^3$ of food sample placed in a test tube and four drops of iodine solution added (1 mk)	The colour of the solution turned brown 1 mk	Starch is absent 1 mk

b) Since a carbohydrate is present, maltase; acts on maltose; producing glucose; or lactase;
acts on lactose; into galactose and glucose; i.e. enzymes in ileum; product;

c) Provides simple carbohydrates e.g. glucose which can be broken down in body cells to liberate energy; or it is a source of vitamin C which is necessary for proper development of epithelial tissues controlling scurvy;OWITTE Any two fully explained answers

3. a) i) W – Aquatic (water)

ii) Z – Desert or semi- desert or dry land

b) i) Y- Flower

ii) Sexual reproduction Reject reproductive alone

c) Observable features apply i.e. has thick succulent stem; for storage of water; and Respiratory; its leaves are reduced into spines; to lower the SA for transpiration; or for Protection against herbivores

FORM 1 WORK

CHAPTER ONE

INTRODUCTION TO BIOLOGY

1. Write three major differences between plants and animals.

2. List the use of the energy obtained from the process of respiration.

3. State three characteristic similar in plants and animals.

(Section A)

4. Motor vehicles move, use energy and produce carbon dioxide and water. Similar characteristics occur in living organisms yet motor vehicles are not classified as living. List other characteristics of living things that do NOT occur in motor vehicles.

CHAPTER TWO

CLASSIFICATION I

INTRODUCTION

1.

a) What is meant by the term binomial nomenclature? (1mk)

b) Give two reasons why classification is important (2mks)

2. Explain the following terms; (3mks)

a) Classification

b) Taxonomy

c) Binomial nomenclature

3.

a) State three characteristics of Monera that are not found in other kingdoms (3mks)

b) Name the class to which a termite belongs (1mk)

4. Ascaris lumbricoides is an example of an endoparasite. The name Ascaris refer to

5. Blackjack (Bidens pilosa) belongs to the family compositae. What does pilosa stand for? (1mk)

6. Define the term species. (1mk)

7. Distinguish between Taxonomy and taxon. (1mk)

CHAPTER THREE

THE CELL

1. Which organelle would be abundant in?

   Skeletal muscle cell ___________________

   Palisade cell ___________________

2. State the functions of the following organelles.

   Lysosomes ___________________

   Golgi apparatus ___________________

3. State the functions of the following organelles;

   Goigi apparatus ________________

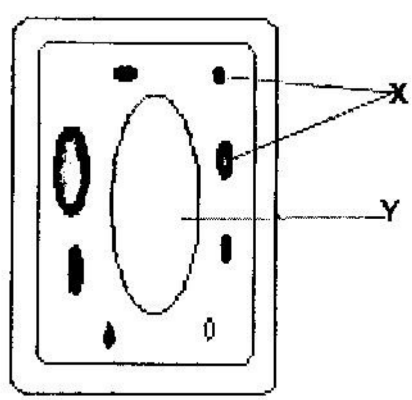
   Ribosomes cell ________________

4. Name the organelles that perform each of the following functions in a cell.

   Protein synthesis ___________________

   Transport cell secretions ___________________

5. The diagram below represents a cell.



a) Name the parts labeled x and y

X _______________

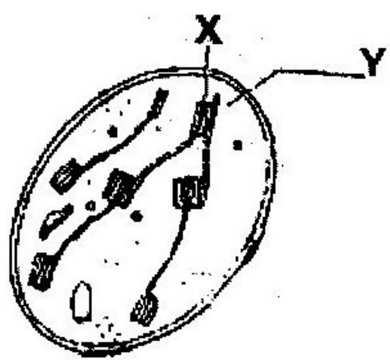
Y _______________

b) Suggest why the structures labeled x would be more on one side than the other side.

6.

a) State the function of cristae in mitochondria (1mk)

b) The diagram below represents a cell organelle



(i) Name the part labeled Y (1mk)

(ii) State the function of the part labeled X (2 mks)

7.

a) What is the formula for calculating linear magnification of a specimen when using a hand lens? (1mk)

b) Give a reason why staining is necessary when preparing specimens for observation under the microscope. (1mk)

8.

State three functions of Golgi apparatus. (3mks)

9. Name two structures found in plant cell but are absent in animals cell.

10. Write the role of the following parts of a microscope

i) Nerve cell

ii) Palisade cell

iii) Root hair cell

iv) Red blood cell

11. The diameter field of view of a light microscopic is 3.5mm. Plant cells lying of the diameter are 10. Determine the size of one cell microns (1mm = 1000µm)

12. Define the following

i) Tissue

ii) Organ

iii) Organ system

CHAPTER FOUR

CELL PHYSIOLOGY

1. The table below shows the concentration of some ions in pond water and in the cells sap of an aquatic plant growing in the pond.

Ions	Concentration in pond water (parts per million)	Concentration in cell sap (parts per million)
Sodium	50	30
Potassium	2	150
Calcium	1.5	1
Chloride	180	200

a) Name the processes by which the following ions could have been taken up by this plant. (2mks)

i) Sodium ions

ii) Potassium ions

b) For each processes named in (a) (i) and (ii) above, state one condition necessary for the process to take place. (2mks)

2. Explain how water in the soil enters the root hairs of a plant. (4mks)

3. Explain how drooping of leaves on a hot sunny day is advantageous to a plant. (2mks)

4. a) What is diffusion? (2mks)

b) How do the following factors affect the rate of diffusion?

i) Diffusion gradient (1mk)

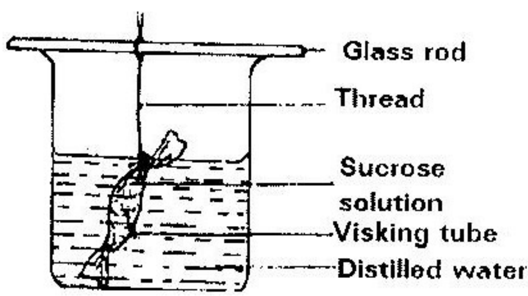
ii) Surface area to volume ratio (1mk)

iii) Temperature (1mk)

c) Outline 3 roles of active transport in the human body (2mks)

5. State the importance of osmosis in plants (3mks)

6. An experiment was set up as shown in the diagram below.



The set up was left for 30 minutes.

a) State the expected results. (1mk)

b) Explain your answer in (a) above. (3mks)

7. Explain why plant cells do not burst when immersed in distilled water. (2mks)

8. Distinguish between diffusion and osmosis. (2mks)

9. Define the following terms in relation to a cell

a) Isotonic solution

b) Hypotonic solution

c) Hypertonic solution (3mks)

10. Addition of large amounts of salt to soil in which plants are growing kills the plants. Explain (6mks)

11. Explain why

a) Red blood cells burst when placed in distilled water while plant cells remain intact.

b) Fresh water protozoa like amoeba do not burst when placed in distilled water. (2mks)

CHAPTER 5

NUTRITION IN PLANTS

1. An experiment was carried out to investigate the rate of reaction shown below.

   Sucrose → Fructose + Glucose

   For the products; fructose and glucose to be formed, it was found that substance K was to be added and the temperature maintained at 37^0C. When another substance L was added, the reaction slowed down and eventually stopped.

   a) Suggest the identity of substances K and L. (2mks)

   K ___________________________

   L ___________________________

   b) Other than temperature state three ways by which the rate of reaction could be increased. (3mks)

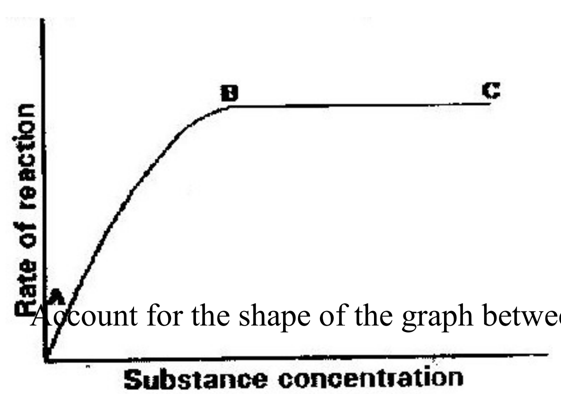
   c) Explain how substance L slowed down the reaction. (1mk)

2. State the role of light in the process of photosynthesis. (2mks)

   Name one product of dark reaction in Photosynthesis (1mk)

3. State one effect of magnesium deficiency in green plants.

4. The graph below shows the effect of substrate concentration on the rate of enzyme reaction.



   a) Account for the shape of the graph between

i) A and B (3mks)

ii) B and C (2mks)

b) How can the rate of reaction be increased after point B? (1mk)

c) State two factors that affect the rate of enzyme reaction. (2mks)

5. a) State the function of co-factors in cell metabolism. (1 mk)

b) Give one example of a metallic co-factor. (1 mk)

6. Name two mineral elements that are necessary in the synthesis of chlorophyll.

(2mks)

7. What is the role of the vascular bundles in plants nutrition? (3mks)

8. Describe what happens during the light stage of photosynthesis. (3mks)

9. Photosynthesis takes place in two stages. Name the part of the chloroplast where

i) Light stage occurs

ii) Dark stage occurs (2mks)

b) How is dark stage dependant on the light stage of photosynthesis? (2mks)

10. A solution of sugarcane was boiled with hydrochloric acid; sodium carbonate was heated with Benedict's solution. An orange precipitate was formed.

a) Why was the solution boiled with hydrochloric acid? (1mk)

b) To which class of carbohydrates does sugarcane belong?

c) Name the type of reaction that takes place when:

i) Simple sugars combine to form complex sugar. (1mk)

ii) A complex sugar is broken into simple sugar. (1mk)

d) State the form in which carbohydrates are stored in:

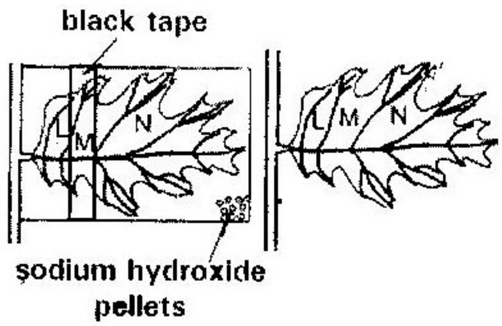
i) Plants

ii) Animals (2mks)

11. i) Name structural units of lipids (1mk)

ii) State three important functions of lipids in living organisms. (3mks)

12. The diagram below shows an experiment carried out to investigate photosynthesis in a potted plant which has been kept in the dark for 48 hours.



The setup was left in the sunshine for 6 hours. The leaf was tested for starch using iodine solution at the end of the experiment.

a) What would be the colours of the regions of the leaf marked L. M and N? (3mks)

b) What is the function of the sodium hydroxide pellets? (1mk)

CHAPTER SIX

NUTRITION IN ANIMALS

1. a) Name the bacteria found in the root nodules of leguminous plant. (1mk)

   b) State the association of the bacteria named in a) above with the leguminous plants. (1mk)

2. a) State the function of co-factors in cell metabolism.

   b) Give one example of metallic co-factor.

3. Name the disease in humans that is cause by lack of vitamin C. (1mk)

4. Name a disease caused by lack of each of the following in human diet;

   Vitamin D (1mk)

   Iodine (1mk)

5. Explain how birds of prey are adapted to obtaining their food. (2mks)

6. Explain biological principles behind the preservation of meat by;

   i) Salting

   ii) Refrigeration

   iii) Canning (3mks)

7. State one similarity and one difference between parasitic and predatory modes of feeding (3mks)

8. In an investigation, the pancreatic duct of a mammal was blocked. It was found that the blood sugar regulation remained normal while food digestion was impaired. Explain these observations. (3mks)

9. Give a reason why lack of roughage in diet often leads to constipation.

10. a) What does the term digestion mean? (2mks)

b) Describe how the mammalian small intestine is adapted to its function. (18mks)

11. State the role of vitamin C in humans. (2mks)

12. a) Distinguish between the terms homodont and heterodont. (1mk)

b) What is the function of carnasial teeth? (1mk)

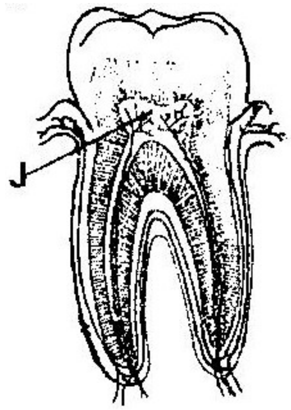
c) A certain animal has no incisors, no canines, 6 premolars and 6 molars in its upper jaw, in the lower jaw there are 6 incisors, 2 canines, 6 premolars and 6 molars. Write its dental formula.

13. a) State two functions of bile juice in the digestion of food. (2mks)

b) How does substrate concentration affects the rate of enzyme action? (1mk)

14. Name the end-products of the light stage in photosynthesis. (2mks)

15. The diagram below represents a section through a human tooth.



a) i) Name the type of tooth shown.

ii) Give a reason for your answer in (a) (i) above. (1mk)

b) State a factor that denatures enzymes. (1mk)

16. a) Name a fat soluble vitamin manufactured by the human body. (1mk)

b) State two functions of potassium ions in the human body. (2mks)

17. a) The action of ptyalin stops at the stomach. Explain. (1mk)

b) State a factor that denatures enzymes. (1mk)

c) Name the features that increase the surface area of small intestines. (2mks)

18 Define the following terms (5mks)

a) Ingestion

b) Digestion

c) Absorption

d) Assimilation

e) Egestion

19 Explain the role of the following organs in the digestion of food in a mammal.

a) Salivary glands

b) Pancrease

c) Liver (3mks)

20 State any three functions of the mucus, which is secreted along the wall of the alimentary canal. (3mks)

21. Explain why the digestion of starch stops after food enters the stomach. (3mks)

22. Give an account of the adaptation of a named herbivore to its mode of feeding. (3mks)

23. What are the contents of gastric juice and what is their role in digestion. (6mks)

24. Liver damage leads to impaired digestion of fats . Explain the statement. (3mks)

25. For each of the following nutrients give one example of a good source and one example of its role in the body.

Nutrient	Food source	Role in the body
Vitamin A		
Iron		
Iodine		
Vitamin D		
Protein		

(10mks)

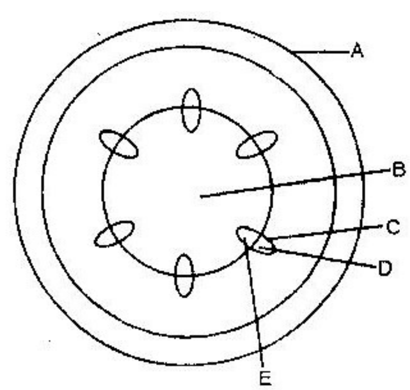
FORM TWO WORK

CHAPTER 1

TRANSPORT IN PLANTS

1. In an experiment, a leafy shoot was set up in a photometer and kept in a dark room for 2 hours. The set up was then transferred to a well-lit room for 2 hours.
   a) What was the aim of this experiment? (1mk)
   b) Explain the results which would be expected in each of the two experiments conditions. (3mks)
2. Explain how drooping of leaves on a hot sunny day is advantageous to plant. (2mks)
3. Explain how environmental factors affect the rate of transpiration in flowering plants. (20mks)

4. The diagram below represents a transverse section of a young stem.



a) Name the parts labeled A and B (2mks)

A___________________

B____________________

b) State the functions of the parts labeled C, D and E

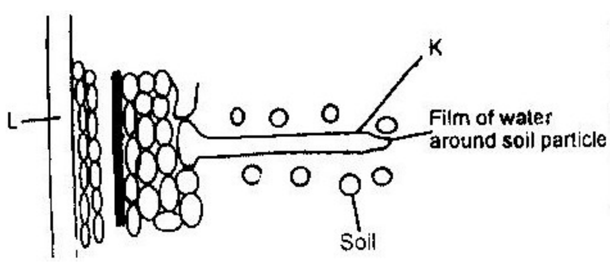
C________________

D_______________

E_______________

c) List three differences between the section shown above and one that would be obtained from the root of the same plant (3mks)

5. The diagram below represents the pathway of water from soil into the plant.



a) Name the structures labeled K and L

K_______________

L_______________ (2mks)

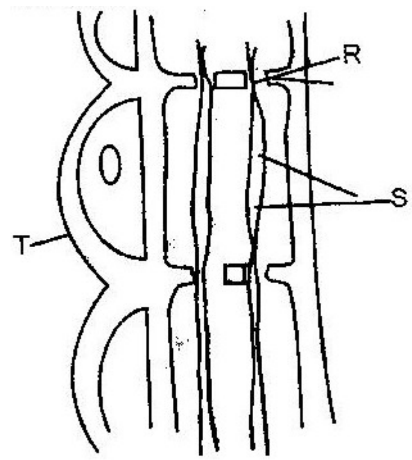
b) Explain how water from the soil reaches the structure labeled L. (5mks)

c) Name the process by which mineral salts enter into the plant. (1mk)

6. State two ways in which xylem are adapted to their function. (2mks)

7. What makes young herbaceous plant remain upright? (2mks)

8. The diagram below represents part of phloem tissue



a) Name the structures labeled R and S and a cell labeled T.

R ____________________

S ____________________

Cell T ____________________ (3mks)

b) State the function of the structure labeled S. (1mk)

c) Explain why xylem is a mechanical tissue (2mks)

9. Name the

a) Material that strengthens xylem tissue. (1mk)

b) Tissue that is removed when the bark of a dicotyledonous plant is ringed. (1mk)

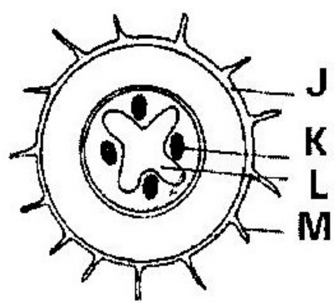
10. How are xylem vessels adapted for support? (1mk)

11. What is the role of vascular bundles in plant nutrition? (3mks)

12. a) Name two tissues which are thickened with lignin. (2mks)

b) How is support attained in herbaceous plant? (1mk)

13. The diagram below represents a transverse section through a plant organ.



a) From which plant organ was the section obtained? (1mk)

b) Give two reasons for your answer in (a) above. (2mks)

c) Name the parts labeled J, K and L (3mks)

J___________________________________

K___________________________________

L___________________________________

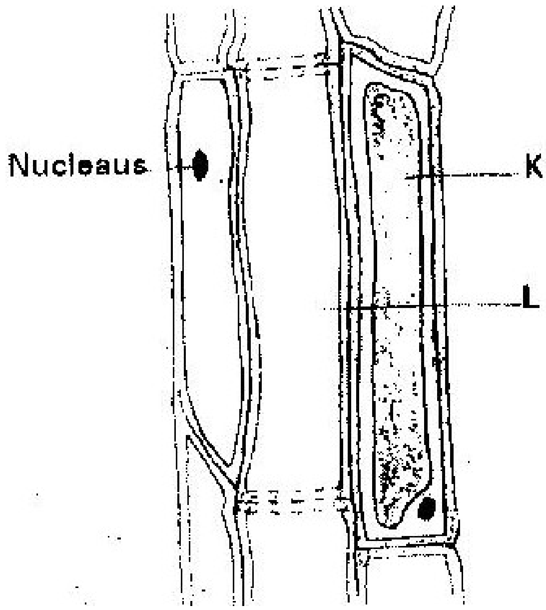
d) State two functions of the part labeled M. (2mks)

14. Describe how water moves from the soil to the leaves in a tree. (20mks)

15. State two ways in which the root hairs are adapted to their function.

(2mks)

16. The diagram below represents a plant tissue.



17. In an experiment to determine the effect of ringing on the concentration of sugar in phloem, a ring of bark from the stem of a tree was cut and removed. The amount of sugar in grammes per $16cm^3$ piece of bark above the ring was measured over a 24 hour period. Sugar was also measure in the bark of a similar stem of a tree which was not ringed. The results are shown in the table below

Time of the day	Among of sugar in grammes per 16 cm^3 piece of bark	
	Normal stem	Ringed stem
06 45	0.78	0.78
09 45	0.80	0.91
12 45	0.81	1.01
15 45	0.80	1.04
18 45	0.77	1.00
21 45	0.73	0.95
00 45	0.65	0.88

.

a) Using the same axes, plot a graph of the amount of sugar against time (6mks)

b) At what time was the amount of sugar highest in the;

i) Ringed stem (1mk)

ii) Normal stem (1mk)

c) How much sugar would be in the rigged stem if it was measured at 03 45 hours. (2mks)

d) Give reasons why there was sugar in the stems of both trees at 06 45 hours. (2mks)

e) Account for the shape of the graph for the tree with ringed stem between:

i) 06 45 hours and 15 45 hours (3mks)

ii) 15 45 hours and 00 45 hours (2mks)

f) Other than sugars name two compounds that are translocated in phloem. (2mks)

18. Explain why plants shed off their leaves. (2mks)

19. a) What is the importance of transpiration to plants?

b) Give adaptive features which enable a plant to reduce the loss of water.

CHAPTER 2

TRANSPORT IN ANIMALS

1. People can die when they inhale gases from burning charcoal in poorly ventilated rooms. What compound is formed in the human body that leads to such deaths? (1mk)

2. Explain why blood from a donor whose blood group is A cannot be transfused into a recipient whose blood group is B. (2mks)

3. State one difference between closed and open circulatory systems. (1mk)

4. a) Give an example of a phylum where all members have
   i) Open circulatory system
   ii) Closed circulatory system (2mks)

   b) What are the advantages of the closed circulatory system over the open circulatory system? (5mks)

5. Explain two ways in which mammalian erythrocytes (red blood cells) are adapted to their function (2mks)

6. a) i) Name the blood vessels that link arterioles with venules. (1mk)

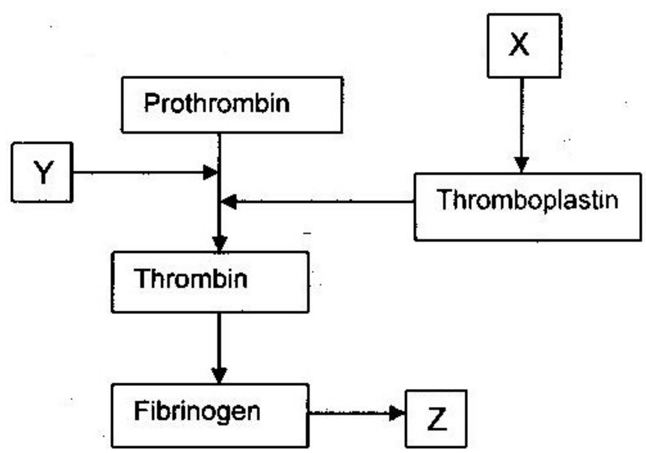
   ii) Explain four ways in which the vessels you named in (a) above are suited to carrying out their functions. (4mks)

   b) State two ways in which the composition of blood in the pulmonary arterioles differ from that in the pulmonary venules. (2mks)

7. Why would carboxyhaemoglobin lead to death? (2mks)

8. Explain how the red blood cells of mammals are adapted for efficient transport of oxygen. (2mks)

9. The chart below is a summary of the blood clotting mechanism in man.



Name

i) The blood cells represented by X

ii) Metal ion represented by Y

iii) The end product of the mechanism represented Z

10. a) How can excess bleeding result in death? (2mks)

b) Name the process by which the human body naturally stops bleeding. (1mk)

c) How can low blood volume be brought back to normal? (2mks)

11. a) Name one defect of the circulatory system in humans. (1mk)

b) State three functions of blood other than transport. (3mks)

12. a) What prevents blood in veins from flowing backwards? (1mk)

b) State two ways in which the red blood cells are adapted to their function. (2mks)

13. State one way by which HIV/AIDS is transmitted from mother to child. (1mk)

14. Explain how the various components of blood are adapted for their function. (20mks)

15. Distinguish between blood, plasma, serum, tissue fluid and lymph. (10mks)

16. a) A patient whose blood group is A died shortly after receiving blood from a person of blood group B. Explain the possible cause of death of the patient. (2mks)

    b) A person of blood group AB requires a transfusion.

    i) Name the blood groups of the possible donors (2mks)

    ii) Give reasons for your answer in (i) above. (2mks)

17. Differentiate between active immunity and passive immunity. (2mks)

18. Explain why a person can catch a cold several times in a year but only catches measles once in his or her lifetime. (2mks)

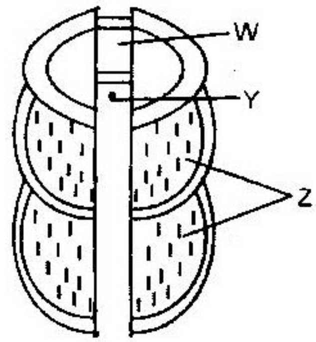
19. Most carbon dioxide is transported from tissues to the lungs within the red blood cells and not in the blood plasma. Give two advantages of this mode of transport. (2mks)

20. What is the importance of tissue fluid? (2mks)

CHAPTER 3

GASEOUS EXCHANGE

1. Discuss how gaseous exchange occurs in
   a) Terrestrial Insects (9mks)
   b) Bony fish (11mks)
2. a) Explain how mammalian lungs are adapted for gaseous exchange. (8mks)
   b) Describe how carbon dioxide is produced by
      i) Respiring muscle cells reaches the alveolar cavities in mammalian lungs.
      ii) Respiring mesophyll cells of flowering plants reaches the atmosphere. (12 mks)
3. a) Describe the path taken by carbon dioxide released from the tissues of an insect to the atmosphere.
   b) Name two structures used for gaseous exchange in plants. (2mks)
4. Why are gills in fish highly vascularized? (1mk)
5. Describe the
   a) Process of inhalation in mammals. (10 mks)
   b) Mechanism of opening and closing of stomata (10 mks)
6. Name three sites where gaseous exchange takes place in terrestrial plants. (3mks)
7. How is aerenchyma tissue adapted to its function? (2mks)
8. The diagram below represents a part of the rib cage.



a) Name parts labeled W, Y and Z.

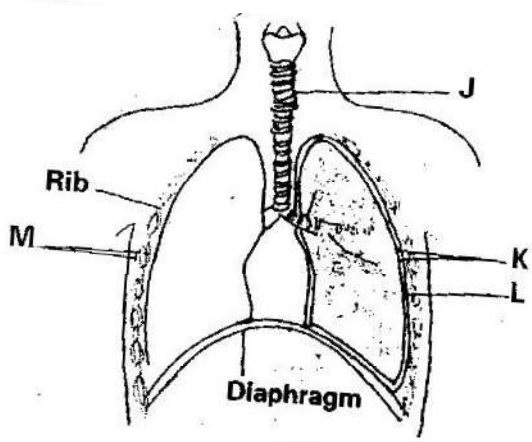
b) How does the part labeled Z facilitates breathing in? (1mk)

9. State two ways in which floating leaves of aquatic plants are adapted to gaseous exchange. (2mks)

10. a) Name two structures for gaseous exchange in aquatic plants. (2mks)

b) What is the effect of contraction of the diaphragm muscles during breathing in mammals? (3mks)

11. The diagram below represents some gaseous exchange structures in humans.



a) Name the structure labeled K, L and M (3mks)

b) How is the structure labeled J suited to its functions? (3mks)

c) Name the process by which inhaled air moves from the structure labeled L into blood capillaries. (1mk)

d) Give the scientific name of the organism that causes tuberculosis in humans. (1mk)

12 State three factors that make alveolus adapted to its function. (3mks)

13. Explain how the alveoli are ventilated.

14. Explain why water logging of the soil may lead to death in plants. (2mks)

15. Write three advantages of breathing through nose than through mouth. (3mks)

16. State and explain ways the leaves are adapted for gaseous exchange (4mks)

17. Name three gaseous constituents involved in gaseous exchange in plants. (3mks)

18. Name three sites of gaseous exchange in frogs. (3mks)

19. Name the main site of gaseous exchange in

a) Mammals

b) Fish

c) Leaves

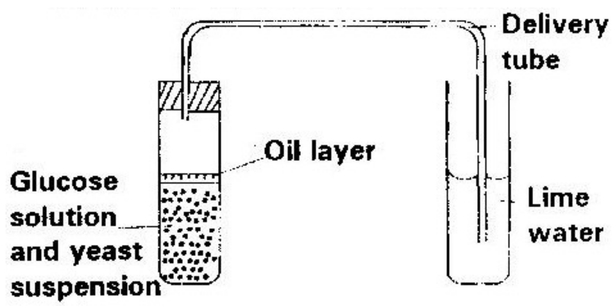
d) Amoeba (4mks)

20. Name the physiological process by which gas exchange takes place at the respiratory surface in animals and plants (1mk)

CHAPTER FOUR

RESPIRATION

1. The diagram below shows a set up that was used to demonstrate fermentation.



Glucose solution was boiled and oil added on top of it. The glucose solution was then allowed to cool before adding the yeast suspension.

a) Why was the glucose solution boiled before adding the yeast suspension? (1mk)

b) What was the importance of cooling the glucose solution before adding the yeast suspension? (1mk)

c) What was the use of the oil in the experiment? (1mk)

d) What observation would be made in test tube B at the end of the experiment (1mk)

e) Suggest a control for this experiment (1mk)

2. Give two reasons why accumulation of lactic acid during vigorous exercise lead to an increase in heart beat. (2mks)

3. A process that occurs in plants is represented by the equation below.

$C_6H_{12}O_6 \longrightarrow 2C_2H_5OH) + (2CO_2) + \text{Energy}$

Glucose Ethanol Carbon Dioxide

a) Name the process (1mk)

b) State the economic importance of process name in (a) above. (1mk)

4. Other than carbon dioxide, name the other products of anaerobic respiration in plants. (2mks)

5. Name the substance which accumulates in muscles when respiration occurs with insufficient oxygen. (1mk)

6. a) In what form is energy stored in muscles? (1mk)

b) State the economic importance of anaerobic respiration in plants. (2mks)

7. State four ways in which respiratory surfaces are suited to their function. (4mks)

8. a) A dog weighing 15.2kg requires 216kJ while a mouse weighing 50g requires 2736KJ per day. Explain. (2mks)

b) What is the end product of respiration in animals when there is insufficient oxygen supply? (1mk)

9 a) Name the products of anaerobic respiration in:

i) Plants (1mk)

ii) Animals (1mk)

b) What is oxygen debt? (1mk)

10. $5C_{51}H_{98}O_6 + 145O_2 \longrightarrow 1\ O_2CO_2 \quad + 98\ H_2O$ + energy

The above equation shows an oxidation reaction of food substances.

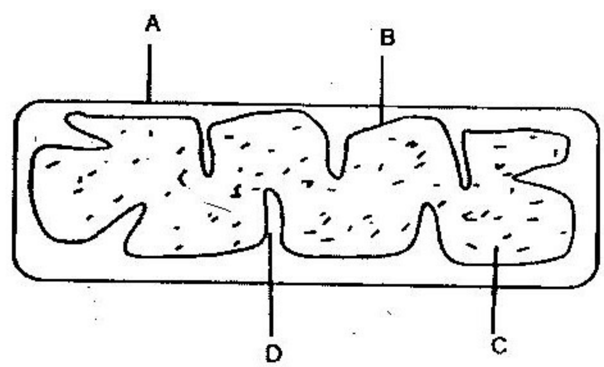
a) What do you understand by the term respiratory quotient? (1mk)

b) Determine respiratory quotient of the oxidation of food substance. (2mks)

c) Identify the food substances. (1mk)

11 Write differences between aerobic respiration and photosynthesis. (4mks)

12. Below is a diagram of an organelle that is involved in aerobic respiration.



a) Name the organelle (1mk)

b) Name the parts labeled A, B, and C. (3mks)

c) What is the purpose of the folding labeled D? (1mk)

d) Give the chemical compound which is formed in the organelle and forms the immediate source of energy.

CHAPTER 5

EXCRETION AND HOMEOSTASIS

1. In an investigation the pancreatic duct of a mammal was blocked. It was found that the blood sugar regulation remained normal while food digestion was impaired. Explain these observations. (2 marks)
2. (a) Explain why the body temperature of a healthy human being must rise up to 39^0c on a humid day. (2 marks)

   (b) In an experiment a piece of brain was removed from rat. It was found that the rat had large fluctuations of body temperatures suggest the part of the brain that had been removed. (1 mark)
3. (a) Explain why sweat accumulates on a person's skin in a hot humid Environment. (2 marks)

   (b) Name the specific part of the brain that triggers sweating. (1 marks)
4. Explain why some desert animals excrete uric acid rather than ammonia. (2 marks)
5. State the role of the following hormones in the body

   (a) Insulin (3 marks)

   (b) Antidiuretic Hormone (3 marks)
6. What osmoregulatory changes would take place in a marine amoeba if it was transferred to a fresh water environment?
7. Name two components of blood that are not present in glomerular filtrate. (2 marks)

8. How would one find out from a sample of urine whether a person is suffering from diabetes mellitus? (2 marks)
9. When is glycogen, which is stored in the liver, converted into glucose and released into the blood? (2 marks)
10. A person was found to pass out large volumes of dilute urine frequently. Name the

(a) Diseases the person was suffering from (1 marks)

(b) Hormone that was deficient (1 mark)

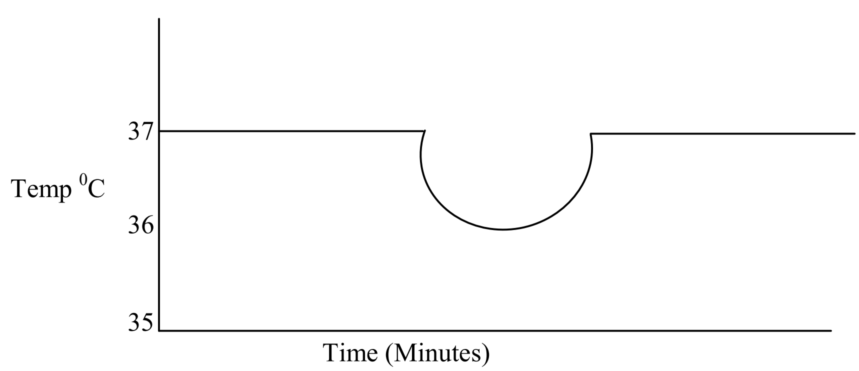
11. State the importance of osmoregulation in organisms (2 marks)

12. What happens to excess fatty acids and glycerol in the body? (2 marks)
13. Give reasons for each of the following

(a) Constant body temperature is maintained in mammals (1 mark)

(b) Low blood sugar level is harmful to the body (2 marks)

14. The temperature of a person taken before during and after taking a cold bath. The results are shown in the graph



(a) Explain why the temperature fell during the bath (2 marks)

(b) What changes appeared in the skin that enabled the body temperature to return to normal. (2 marks)

15. (a) Name the fluid that is produced by sebaceous glands (1 mark)

(b) What is the role of sweat on the human skin? (2 marks)

16. State the role of insulin in the human body? (1 mark)

17. Describe how the human kidney functions. (20 marks)

18. (a) What is the meaning of the following terms:

(i) Homeostasis (1 mark)

(ii) Osmoregulation (1 mark)

19. (a) Explain what happens to excess amino acids in the liver of humans. (3 marks)

(b) Which portions of the human nephrons are only found in the cortex? (3 marks)

(c) (i) What would happen if a person produced less antidiuretic hormone? (1 mark)

(ii) What term is given to the condition described in (c) (i) above? (1 mark)

20. Define the following terms

(a) Excretion

(b) Secretion

(c) Egestion (3 marks)

21. Name the components of blood that do not enter the renal tubule in mammals

(2 marks)

22. The table below shows the approximate percent concentration of various components in blood plasma entering the kidney glomerular filtrate and urine of a healthy human being.

Component	Plasma	Glomerular	Urine Filtrate
Water	90	90	94
Glucose	0.1	0.1	0
Amino Acids	0.05	0.05	0
Plasma proteins	8.0	0	0
Urea	0.03	0.03	2.0
In organic ions	0.72	0.72	1.5

(b) Name the process responsible for the formation of glomerular filtrate.

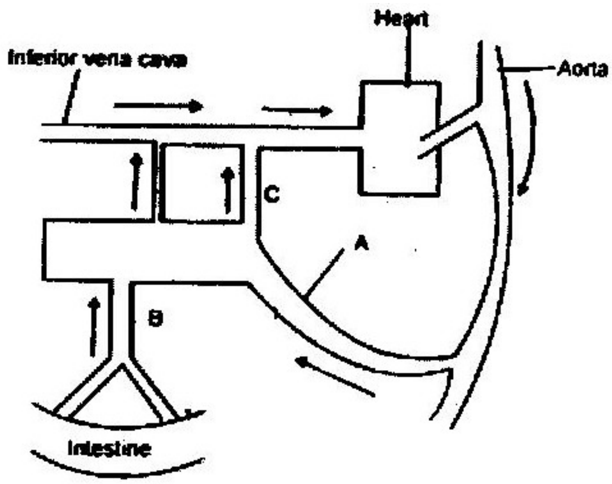
(c) What process is responsible for the absence of glucose and amino acids in urine?

(d) Explain why there are no plasma proteins in the glomerular filtrate

(e) Besides plasma proteins what other major component of blood is absent in the glomerular filtrate.

(f) Why is the concentration of urea in urine much higher than its concentration in the glomerular filtrate?

23. When the environmental temperature is very high, some animals urinate on their legs or lick the sides of their body. How does this help in temperature regulation?
24. Fish are able to use more of their food intake for growth than mammals. Suggest an explanation for this.
25. Explain the term negative feedback
26. Study the diagram below and answer the questions that follow.



(a) Name the blood vessels labeled A, B and C.

(b) If the animals has recently fed on a diet which is rich in proteins and carbohydrates in which of the vessels labeled A, B, and C would you expect to find the highest concentration of:

(i) Glucose

(ii) Amino acids

(iii) Carbon (IV) oxide

(iv) Oxygen

(v) Urea

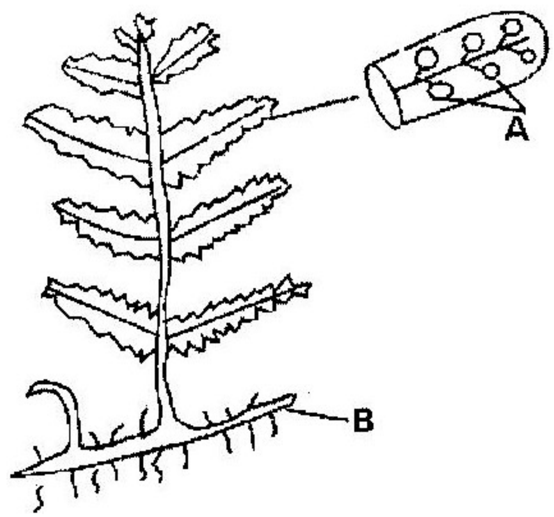
(c) During fasting, the level of blood glucose in vessels C may be higher than the level in vessel B explain

FORM 3 WORK

CHAPTER 1

CLASSIFICATION II

1. State two ways in which some fungi are harmful to man (2 marks)
2. The diagram below represents a fern



Name

(a) Parts labeled A and B (2 marks)

(b) The division which the plant belongs (1 mark)

3. An organism with an exoskeleton, segmented body, two pairs of legs per segment, a pair of eyes and a pair of eyes and a pair of short antennae belongs to the phylum (1 mark)
4. When are two organisms considered to belong to the same species? (2 marks)
5. A student caught an animal which had the following characteristics;
   - Body divided into two parts
   - Simple eyes
   - Eight legs

The animal belongs to the class (1 mark)

6. Below is a list of organisms, which belong to classes insecta, myriapoda and arachnida. Tick, centipede, praying mantis, tsetse fly. Millipede and spider. Place the organisms in their respective classes in the table below. Give reasons in each case.

Class	Organisms	Reasons
Insecta		
Myriapoda		
Arachnida		

7. State two characteristics features of members of division bryophyte (2 marks)
8. State two ways in which some fungi are beneficial to humans (2 marks)
9. Other than having many features in common state the other characteristics of species (1 mark)
10. Beside the abdomen, name the other body part of members of arachnida (1 mark)
11. Name the phylum whose members possess notochord. (1 mark)
12. Name the class in the phylum arthropoda which has the largest number of individuals (1 mark)
13. To which class does an animal with two body parts and four pairs of legs belong? (1 mark)
14. (a) Name two organisms that cause food spoilage (2 marks)

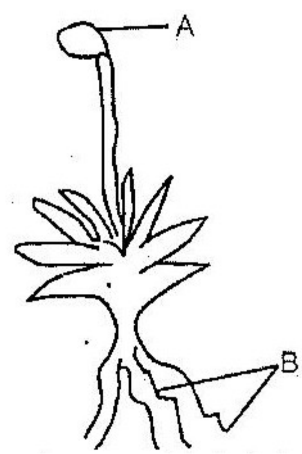
(b) Name two methods of food preservation and for each state the biological principal behind it. (2 marks)

15. (a) List two characteristics that mammals share with birds (2 marks)

(b) State two major characteristics that are unique to mammals (2 marks)

16. What two characteristics distinguish animals in phylum chordata? (2 marks)

17. The diagram below shows a plant



(a) Name the parts labeled A and B (2 marks)

(b) Name the division to which the plant belongs (1 mark)

(c) Which is the dominant generation of the plant in the diagram? (1 mark)

(d) State three characteristics of the organisms in the division named (b) above?

18. What three characteristics are used to divide the arthropods into classes? (3 marks)

19. (a) Write two differences between algae and fungi (2 marks)

(b) Give the economic importance of algae (1 mark)

20. (a) What is alteration of generations? (3 marks)

(b) Name two divisions in plant kingdom that shows alternation of generation

(2 marks)

21. (a) A millipede, grasshopper and crayfish all belong to phylum arthropoda. Mention three major characteristics that they have in common. (3 marks)

(b) The specific name of Irish potato is solanum Tuberrasum

(i) Identify two errors that have been made when writing the name

(ii) What is the species name of Irish potato?

(c) An ecologist came across a plant with the following characteristics, green in colour, non- flowering, compound leaves and sori on the underside of the leaflets. State the probable division of the plant. (1 mark)

22. An organism with an exoskeleton, segmented body, two pairs of legs per segment, a pair of eyes and a pair of short antennae belongs to the class (1 mark)

23. List the main characteristics that are used to sub- divide arthropods into classes

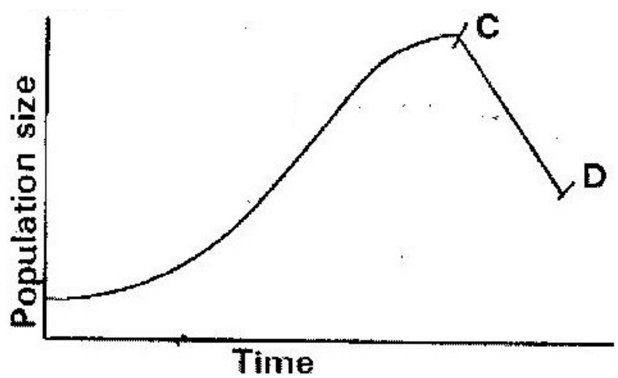
(2 marks

24. Name the main method of reproduction among bacteria. (1 mark)

CHAPTER 2

ECOLOGY

1. State how excessive use of pesticides may affect soil fertility
2. The graph below represents a population growth of a certain herbivore in a grassland ecosystem over a period of time.



Suggest three factors that could have caused the population change between C and D

(3 marks)

3. A biologist carried out a study to investigate the growth of a certain species of herbivorous bony fish and the factors influencing plant and animal life in four lakes A, B, C and D. The lakes were located in the same geographical area.
   Two of the lakes A and B were found to contain hard water due to presence of high content of calcium slats. The mean body length of 2 year old fish, amount of plant life and invertebrates biomass in each lake were determines. The data was as shown in

Lakes	Mean body length (cm)	Type of water	Amount of plant life	Invertebrate biomass (g/cm^3)			
				Insects	Snails	Crabs	Worms
A	31.2	Hard	1050	11	300	10	180
B	28.6	Hard	950	72	100	9	90
C	18.4	Soft	1.2	97	0	2	20
D	16.3	Soft	0.5	99	0	1	10

(a) Describe the procedure that may have been used to determine the mean body length of the fish (6 marks)

(b) What are the likely reasons for the difference in the mean body length of the fish living in lakes A and D? (4 marks)

(c) Suggest one reason for the absence of snails in lakes C and D? (1 mark)

(d) (i) Name any six abiotic (physical) factors that are likely to influence the plant and animal life in lake A. (3 marks)

(ii) Explain how each of the factors named in (i) may influence the plant and animals life in Lake A. (6 marks)

4. During an ecological study of a lake a group of students recorded the following observations.

(i) Planktonic crustaceans feed on planktonic algae

(ii) Small fish feed on planktonic crustaceans worms and insect larvae

(iii) Worms feed on insect larvae

(iv) Bird species feeds on small fish planktonic crustaceans and worms

(v) Insect larvae feed on small fish

(a) From this record of observations construct a feed web (5 marks)

(b) From the food web you have constructed in (a) above isolate and write down a food chain that ends with

(i) Bird species as a secondary consumer (1 mark)

(ii) Large fish as tertiary consumer (1 mark)

(c) The biomass of the producers in the lake was found to be greater than that of primary consumers. Give an explanation for this observation? (1 mark)

(d) Using either the observations recorded by the students or the food web you have constructed name (1 mark)

(i) Two organisms that compete for food in the lake. (2 marks)

(ii) The source of food the organisms in d (i) above compete for (1 mark)

(e) (i) State three ways by which many may interfere with this lake ecosystem (3 marks)

(ii) Explain how each of the ways you have states may affect life in the lake? (6 marks)

5. In an investigation, a student collected two plants A and B. Plant A had hairy leaves and epidermis. Leaves of plant B

   (i) Plant A (1 mark)

   (ii) Plant B (1 mark)

6. An investigation was carried out between 1964 and 1973 to study the changes of fish population in a certain small lake. Four species of fish A, B, C and D were found to live in this lake. In 1965 a factory was built near the lake and was found to discharge hot water into the lake raising the average temperature from 25^0C to 30^0C. In 1967 sewage and industrial waste from a nearby town was diverted into the lake was stopped. The fish population during the period of investigation is shown in the table below.

Fish species	Fish populations during the period						
	1964	1966	1969	1970	1971	1972	1973
A	6102	223	26	106	660	4071	7512
B	208	30	11	22	63	311	405
C	36	100	0	0	0	0	0
D	4521	272	23	27	79	400	617

(a) (i) In which year were the fish populations lowest?

(ii) State the factors that might have caused the lowest fish populations during the year you have stated in (a) (i) above (3 marks)

(iii) Explain how each factor you have stated in (a) (ii) above could have brought about the changes in fish populations (11 marks)

(b) (i) What is the difference in the rate of population recovery of species A and

D? (1 mark)

(ii) Suggest two biological factors that could have led to this difference

(2 marks)

(c) (i) State a method that might have been used to estimate the fish population in the lake (1 mark)

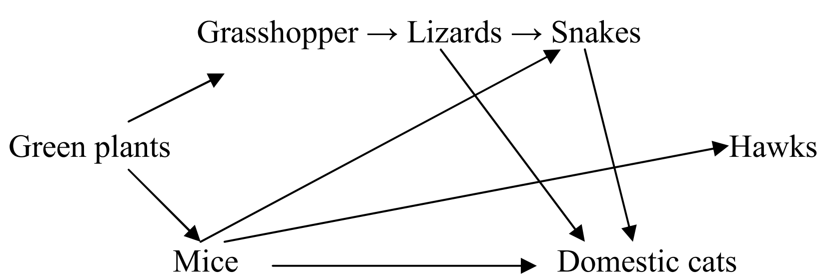
(ii) State one disadvantage of the method you have stated in (c) (i) above

(1 mark)

7. Industrial wastes may contain metallic pollutants. State how such pollutants may indirectly reach and accumulate in the human body if the wastes were dumped into rivers.

8. State three measures that can betaken to control infection of man by protozoan parasites (3 marks)

9. The chart below shows a feeding relationship in a certain ecosystem



(a) Construct two food chains ending with a tertiary consumer in each case (2 marks)

(b) Which organisms has the largest variety of predators in the food web? (1 mark)

(c) Name secondary consumers in the food web (2 marks)

(d) Suggest three ways in which the ecosystem would be affected if there was prolonged drought (3 marks)

10. To estimate the population size of crabs in a certain lagoon, traps were laid at random. 400 crabs were caught, marked and released back into the lagoon. Four days later, traps were laid again and 374 crabs were caught. Out of the 374 crabs, 80 were found to have been marked.

(a) Calculate the population size of the crabs in the lagoon using the formula below

$$N = \frac{n \times M}{M}$$

Where

N= Total population of crabs in the lagoon

n= Total number of crabs in the second catch

M= Number marked crabs during the first catch

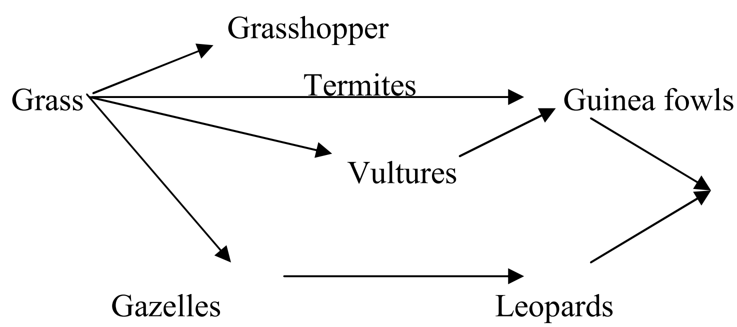
M= Number of marked crabs in the second catch. (2 marks)

(b) State two assumptions that were made during the investigation (2 marks)

(b) What is the name given to this method of estimating the population size?

(1 mark)

11. The figure below represents a feeding relationship in an ecosystem



(a) Write down the food chains in which the guinea fowls are secondary consumers

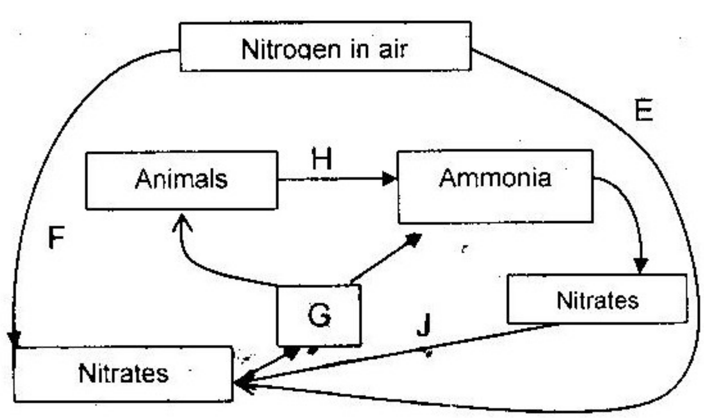
(1 mark)

(b) What would be the short term effects on the eco- system of lions invaded the area? (3 marks)

(c) Name the organisms through which energy from the sun enters the food web.

(1 mark)

12. The diagram below represents a simplified nitrogen cycle



(a) Name the organisms that causes processes E and J (2 marks)

(b) Name the processes represented by F and H (2 marks)

(c) Name the group of organisms represented by c (i)

13. (a) Distinguish between a community and a population (2 marks)

(b) Describe how a population of grasshopper in a given area can be estimated (5 marks)

14. Explain how the various activities of man have caused pollution of air (20 marks)

15. Explain how birds of prey are adapted to obtaining their food (2 marks)

16. (a) Name the crop infested by phytophthora infestants and the disease it causes

Crop -

Disease -

(b) State four control measures against the diseases (4 marks)

17. Explain why the carrying capacity for wild animals is higher than for cattle in a given piece of land (2 marks)

18. (a) What is meant by

(i) Autecology (1 mark)

(ii) Synecology (1 mark)

(b) The number and distribution of stomata on three different leaves are shown in the table below

Leaf	Number of stomata	
	Upper epidermis	Lower Epidermis
A	300	0
B	150	200
C	02	13

Suggest the possible habitat of the plants from which the leaves were obtained.

(3 marks)

Leaf Habitat

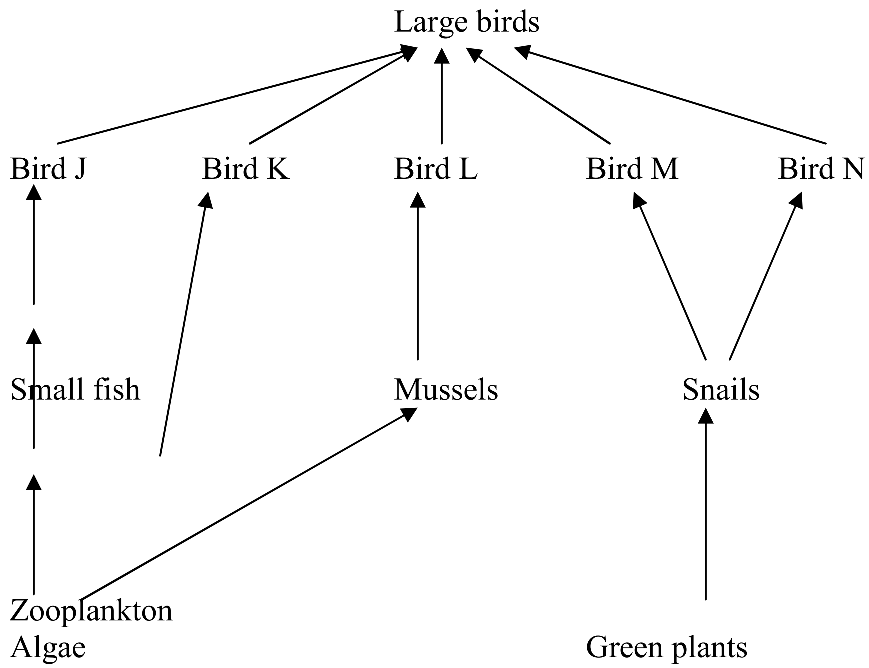
A ______________________

B ______________________

C ______________________

(c) State the modification found in the stomata of leaf C

19. After an ecological study of feeding relationships students constructed the food web below



(a) Name the process through which energy from the sun is incorporated into the food web (1 mark)

(b) State the mode of feeding of the birds in the food web (1 mark)

(c) Name two ecosystem in which the organisms in the food web live(2 marks)

(d) From the information in the food web construct a food chain with the large bird as a quarter – nary consumer (1 mark)

(e) What would happen to the organisms in the food web if bird N migrated?

(f) Not all energy from one trophic level is available to the next level. Explain (3 marks)

(g) (i) Two organisms, which display a role in the ecosystems, are not included in the food web. Name them. (1 mark)

(ii) State the role played by the organisms named g (i) above. (1 mark)

(h) (i) State three human activities that would affect the ecosystems (3 marks)

(ii) How would the activities stated in h (i) above affect the ecosystems?

(3 marks)

20. How is aerenchyma tissue adapted to its functions? (2 marks)

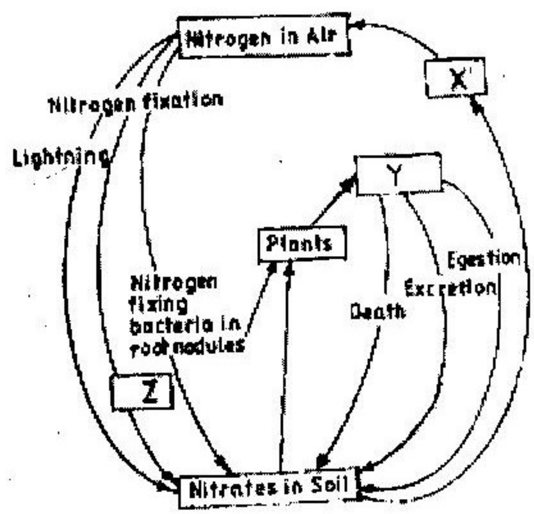
21. Explain how abiotic factors affect plants (20 marks)

22. What is the importance of the following in an ecosystem? (3 marks)

(a) Decomposers

(b) Predation

23. Chart below represents a simplified nitrogen cycle



What is represented by X, Y and Z? (3 marks)

24.

(a) Distinguish between pyramid of numbers and pyramid of Biomass

(2 marks)

(b) Give three reasons for loss of energy from one trophic level to another in a food chain. (3 marks)

(c) Describe how the belt transect can be used in estimating the population of a shrub in a grassland (2 marks)

25.

(a) Distinguish between population and community (2 marks)

(b) Name a method that could be used to estimate the population size of the following organisms

(i) Fish in a pond (1 mark)

(ii) Black jack in a garden (1 mark)

26.

State two ways in which schistosoma species is adapted to parasitic mode of life

27.

Describe causes and methods of controlling water pollution (20 marks)

28. (a) What is biological control of population growth? (2 marks)

(b) Describe one example where biological control has been used successfully (2 marks)

(c) Explain why the number of predators in nay ecosystem is less than the number of their prey (1 mark)

29. Suggest reasons to account for the following observations.

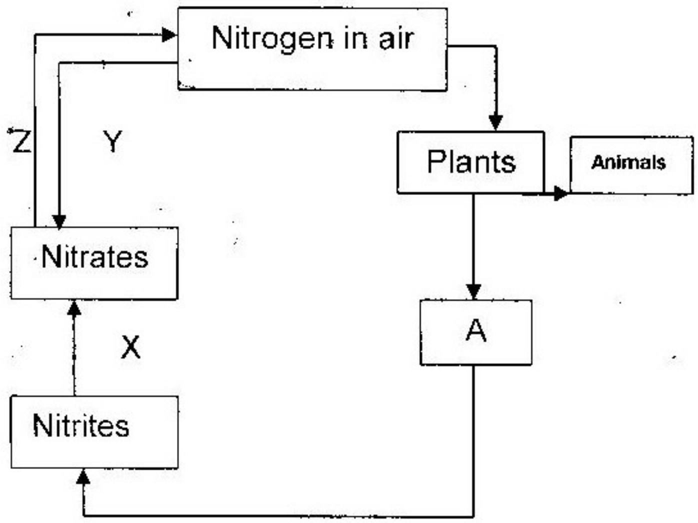
(b) Antelopes are more commonly found in open grassland while giraffe while giraffes are commonly found in wooded areas. (2 marks)

(b) In the savannah there is a wider variety of herbivores in wooded areas than in open grassland (1 mark)

(c) Removal of predators for an herbivore may in the long run lead to a decrease in its population

30. Explain why primary productivity decreases with depth in aquatic environments. (2 marks)

31. The following is a simplified drawing of nitrogen cycle.



(a) Identify the compound named A (1 mark)

(b) Name the processes

X ________________

Y ________________

Z ________________

(c) In what form is nitrogen found in plants and animals?

32. An investigation was carried out to study the type of food eaten by birds found in forest and savannah in a certain area. The table below compares the feeding habitats of the birds found in a closed forest area and an open dry savannah of the area.

Diet	Percentage of birds	
	Forest	Savannah
Insects only	60	50
Vertebrates	10	10
Seeds	5	20
Fruits	25	10
Other plant materials	5	5
Number	120	60

(a) Work out the difference in the number of bird species the feed on:

(i) Fruits found in forest and savannah (2 marks)

(ii) Seed found in forest and savannah (2 marks)

(b) State two factors that may cause this difference in (a) above (2 marks)

(c) In another investigation two vertebrate species from the savannah were counted and recorded on monthly basis as shown below.

Year	Month	Species A	Species B
1998	July	96	240
1998	August	79	590
1998	September	75	900
1998	October	87	750
1998	November	-	230
1998	December	99	80
1998	January	129	200
1998	February	96	330
1998	March	99	300
1998	April	79	320
1998	May	135	90
1998	June	104	450

(i) Which species show more fluctuation in numbers? (1 mark)

(ii) Suggest an explanation of this (3 marks)

(d) Suggest two ways by which the savannah environment can be destroyed and how it can be conserved (4 marks)

CHAPTER 3

REPRODUCTION IN PLANTS AND ANIMALS

1. At what stage of mitosis do chromosomes replicate to form daughter chromatid? (1 mark)
2. Fill in the blank spaces in the statement below

   After fertilization of an ovule _________ develops into a testa and _________ develops into a testa and _____________ develops into endosperm. (2 marks)
3. State the difference between the composition of maternal blood entering the placenta and material blood leaving the placenta (3 marks)
4. After four months of pregnancy the ovaries of a woman can be removed without terminating pregnancy. However during the first four months of pregnancy the ovaries must remain intact if pregnancy is to be maintained. Explain these observations (3 marks)
5. Name two mechanisms that prevent self pollination in flowers that have both male and female parts (2 marks)
6. State three characteristics that ensure cross pollination takes place in flowering plants (3 marks)
7. Give a reason why it is necessary for frogs to lay many eggs (1 mark)
8. A flower was found to have the following characteristics
   - Inconspicuous petals
   - Long feathery stigma
   - Small light pollen grains

   (a) What is the likely agent of pollination of the flower? (1 mark)

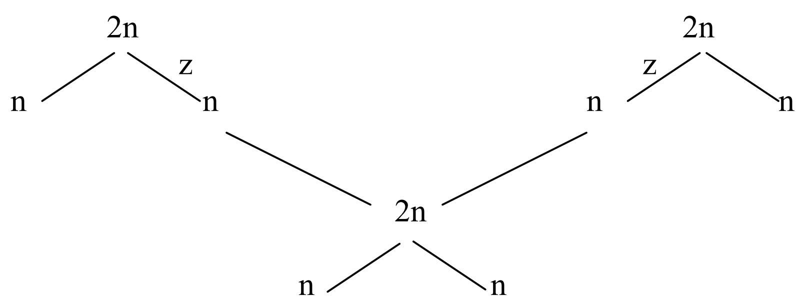
(b) What is the significance of the long feathery stigma in the flower (1 mark)

9. State two ways by which the human Immuno Deficiency virus (HIV) is transmitted other than sexual intercourse? (2 marks)

10. Explain why sexual reproduction is important in organisms (3 marks)

11. State two disadvantages of self- pollination (2 marks)

12. The chart below shows the number of chromosomes before and after cell division and fertilization in a mammal.



(a) What type of cell division takes place at Z? (1 mark)

(b) Where in the body of a female does process Z occur? (1 mark)

(c) On the chart indicate the position of parent and gametes (2 marks)

13. (a) What is meant by the terms

(i) Epigynous flower (1 mark)

(ii) Staminate flower (1 mark)

(b) How are the male parts of wind- pollinated flowers adapted to their function? (4 marks)

14. Name the part of a flower that developed into:

(a) Seed (1 mark)

(b) Fruit (1 mark)

15. (a) State two processes which occur during anaphase of mitosis (2 marks)

(b) What is the significance of meiosis? (2 marks)

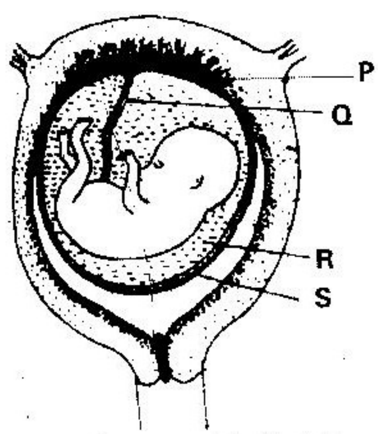
16. (a) Explain how the following prevents self- pollination:

(i) Protoandry (1 mark)

(ii) Self- sterility (1 mark)

(b) Give three advantages of cross- pollination (3 marks)

17. The diagram below represents a human foetus in a uterus



(a) Name the part labeled S (1 mark)

(b) (i) Name the types of blood vessels found in the structure labeled Q

(2 marks)

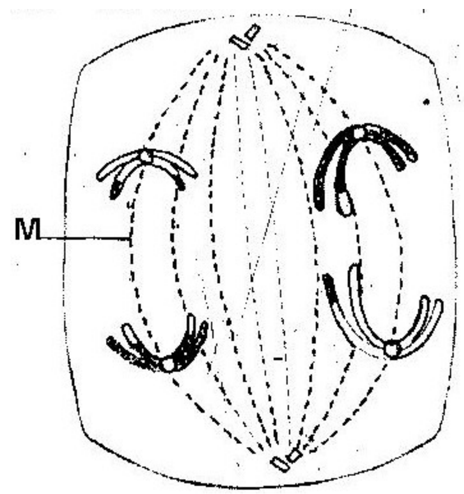
(ii) State the difference in composition of blood in the vessels named (b) (i) above (2 marks)

(c) Name two features that enable the structure labeled P carry out its function

(2 mark)

(d) State the role of the part labeled R (1 mark)

18. The diagram below represents a stage during cell division



(a) (i) Identify the stage of cell division (1 mark)

(ii) Give three reasons for your answer (a) (i) above (2 marks)

(b) Name the structure labeled M (1 mark)

19. State two disadvantages of sexual reproduction in animals (2 marks)

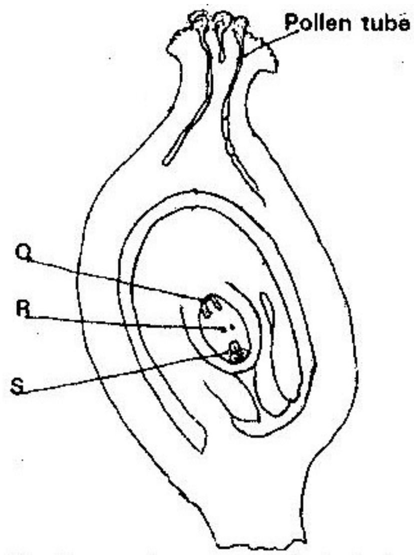
20.

(a) What is meant by the following terms?

(i) Protandry (1 mark)

(ii) Self- sterility (1 mark)

(b) The diagram below shows a stage during fertilization in plant



(i) Name the parts labeled Q, R, and S (3 marks)

(ii) State two functions of the pollen tube (2 marks)

(c) On the diagram, label the micropyle (1 mark)

21. (a) Describe how insect pollinated flowers are adopted to pollination

(6 marks)

(b) Describe the role of each of the following hormones in the human menstrual cycle.

(i) Oestrogen

(ii) Progesterone

(iii) Luteinizing hormone (3 marks)

22. Describe the role of hormones in the human menstrual cycle (20 marks)

23. What part does the placenta play in the

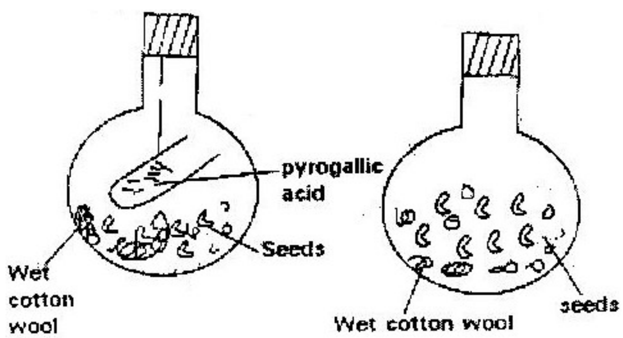
(i) Nutrition of the embryo

(ii) Protection of the embryo (4 marks)

CHAPTER FOUR

GROWTH AND DEVELOPMENT

1. Explain why several auxiliary buds sprout when a terminal bud in a young tree is removed.
2. Account for loss in dry weight of cotyledons in a germinating bean seed.
3. What is the effect of gibberellins on shoots of plants?
4. A student set up an experiment as shown in the diagram below

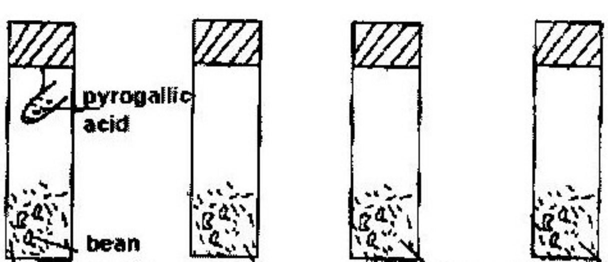


The set up was left at room temperature for a week

(a) What was the aim of the experiment?

(b) What would be the expected results at the end of the experiment?

5. State two advantages of metamorphosis to the life of insects
6. During germination and early growth, the dry weight of endosperm decreases while that of the embryo increases. Explain
7. In an experiment, a group of student set up four glass jars as shown in the diagram below jar A, B and C were maintained at 25^0C for 7 days. While Jar D was maintained at 0^0 c for the same period of time.



(a) What was this set up supposed to investigate?

(b) Why was pyrogallic acid included in glass jar A?

(c) Explain why glass jar C and D were included in the experiment

(d) What result would you expect in glass jar A and B at the end of the experiment?

(e) State two artificial ways of breaking seed dormancy

8. Removal of the apical bud from the shrub is a practice that results in the development of the lateral buds which later from the branches.

(a) Give reasons for the development of the lateral braches after the removal of the apical bud

(b) Suggest one application of this practice

(c) What is the importance of this practice?

9. In an experiment some germination seeds were placed in large airtight flask and left for four days

(a) Suggest the expected changes in the composition of gases in the flask on the fifth day

(b) Give four reasons for your answer in (a) above

(c) Name two factors that cause dormancy in seeds

10. (a) Distinguish between epigeal and hypogeal germination (1 mark)

(b) Why is oxygen necessary in the germination of seeds? (2 marks)

11. An experiment was carried out to investigate the effect of hormones on growth of lateral buds of three pea plants

The shoots were treated as follows:

Shoot A- Apical bud was removed

Shoot B – Apical bud was removed and gibberellic acid placed on the cut shoot

Shoot C- Apical bud was left intact.

The length of the branches developing from the lateral buds were determines at regular intervals

The results obtained are as shown in the table below

Time (days)	Length of branches in mm		
	Shoot A	Shoot B	Shoot C
0	3	3	3
2	10	12	3
4	28	48	8
6	50	90	14
8	80	120	20
10	118	152	26

(a) Using the same axes, draw graphs to show the length of branches against time (8 marks)

(b) (i) What was the length of the branch in shoot B on the 7th day? (1 mark)

(ii) What would be the expected length of the branch developing from shoot A on the 11th day? (1 mark)

(c) Account for the results obtained in the experiment (6 marks)

(d) Why was shoot C included in the experiment? (1 mark)

(e) What is the importance of gibberallic acid in agriculture? (1 mark)

(f) State two physiological processes that are brought about by the application of gibberellic acid on plants. (2 marks)

12. (a) State tow environmental conditions that can cause seed dormancy

    (b) Name the part of a bean seed that elongates to bring about epigeal germination

(1 mark)

13. (a) "True growth is not simply an increase in size" State four different ways in which true may be defined.

    (b) State two external factors, which influence growth in plants and describe one effect of each.

    (c) Fill in the spaces in the following table, which refers to hormones involved in growth processes.

Name hormone	Site of hormone production	Effect
	Thyroid gland	
		Maturation of Graafin follicles
Auxins		
Gibberellins		

14. Seedling from 100g of maize seed was grown in the dark for 10 days. The seedlings were then analyzed and compared with 100g of ingeminated maize. The following results were obtained.

	Dry mass of ingeminated seeds	Dry mass of seedling after 10 days
Cellulose Starch Other organic Material Ash	2g 63g 13g 2g	5g 9g 27g 4g
Total dry mass	80g	45g

(a) Why is dry mass used for comparison?

(b) How would one ensure that the drying process had been completed

(c) Account for the decrease in the total dry mass of the seedlings

(d) Why did the seedling contain more cellulose than the underminated seeds?

(e) What is the most likely source of the carbon used to form this new cellulose?

FORM 4 WORK

CHAPTER 1

GENETICS

1. The figure below is a structural diagram of a portion from a nucleic acid strand.

--- S --- P --- S --- P --- S --- P --- S

C G U C

(a) Giving a reason, name the nucleic acid to which the portion belongs.

(2 marks)

Name ___________________________

Reason ___________________________

(b) Write down the sequence of bases of a complimentary strand to that shown above (1 mark)

2. State two structural differences between ribonucleic acid (RNA) and deoxyribonucleic acid (DNA) (2 marks)

3. Name a disorder of human blood that is caused by mutation (1 mark)

4. State the function of deoxyribonucleic acid (DNA) molecule (1 mark)

5. Give a reason why it is only mutation in genes of gametes that influence evolution (2 marks)

6. In an experiment, red flower were crossed with plants with white flower. All the plants in the F1 generation had pink flowers.

   (a) Give a reason for the appearance of pink flower in the F1 generation (1 mark)

(b) If the plants from F1 generation were selfed, state the phenotype ratio of the F2 generation (2 marks)

7. State two characteristics that researchers select in breading programmes. (2 marks)

8. Give an example of sex- linked trait in humans on; (2 marks)

Y chromosome __________________

X chromosome __________________

9. In an experiment, a variety of garden peas having a smooth seed oat was crossed with a variety with a wrinkled seed coat. All the seeds obtained in the F1 had a smooth seed coat. The F1 generation was selfed. The total number of F2 generation was 7324.

(a) Using appropriate letter symbols, work out the genotype of the F1 generation. (4 marks)

(b) From the information above, work out the following for the F2 generation

(i) Genotype ratio (2 marks)

(ii) Phenotype ratio (1 mark)

(iii) Wrinkled number (1 mark)

10. In a certain plant species, some individual plant may have white, red or pink flower. In an experiment a plant with white parent plant were pure lines. All the plants from F1 generation were pink. Using letter R to represent the gene for red colour and letter W for white colour;

(a) Work out the genotype of F1 generation (3 marks)

(b) If the plants from F1 generation were selfed, what would be the phenotypic ratio of the F2 generation? (3 marks)

(c) What is the genetic explanation for the absence of plants with red and white in the flower F1 generation? (2 marks)

11. In a breeding experiment, plants with red flower were crossed. They produced 123 plants with red flowers and 41 with white flowers.

(a) Identify the recessive character. Give a reason

(b) What were the genotypes of the parent plants that give rise to the plants with red and white flowers?

(c) If the white flowers were selfed, what would be the genotypes of their offspring?

12. (a) Name two disorders in humans caused by gene mutation (2 marks)

(b) Describe the following chromosomal mutations

(i) Inversion (2 marks)

(ii) Translocation

(c) In mice the allele for black fur is dominant to the allele for brown fur. What percentage offspring would have brown fur from a cross between heterozygous black mice and brown mice? Show your working. Use letter B to represent the allele for black colour. (4 marks)

13. (a) What is meant by the term allele? (1 mark)

(b) Explain how the following occur during gene mutation

(i) Deletion (1 mark)

(ii) Inversion (1 mark)

(c) What is a test- cross? (1 mark)

14. In maize the gene for purple colour is dominant to the gene for white colour. A pure breeding maize plant with purple grains was crossed with a heterozygous plant.

(a) (i) Using letter G to represent the gene for purple colour, work out the genotypic ratio of the offspring (5 marks)

(ii) State the phenotype of the offspring (1 mark)

(b) What is genetic engineering? (1 mark)

15. Define the following terms as used in genetics.

(i) Alleles

(ii) Genotype

(iii) Phenotype

16. A farmer mated his dark red cow with a white bull. The cow gave birth to a light red calf

(a) State why the calf is light red and not dark red or white

(b) If a light red bull is mated with a dark red cow, work out using appropriate letter symbols the probability of getting a light offspring

17. (a) What is meant by linked genes?

(b) (i) In fruit flies (Drosophila) the gene for red eyes ® is dominant over the one for white – eye (r). If a true breeding white – eyed male, all the offspring will be red eyed. However, if a true – breeding white-eyed female is mated with a true- breeding red- eyed male, all the female offspring will be red – eyed. Explain this apparent contradiction.

(ii) Work out the ratio of the expected phenotypes if a red- eyed female offspring from the cross- described in (i) above is mated with red- eyed males.

18. (a) Explain the term variation with reference to the study of genetics.

(b) Using relevant examples distinguish between discontinuous variation and continuous variation

(c) What is the importance of genetic variation?

(d) Describe one example where genetic variations has helped a species to survive

19. The diagram below shows the base sequence of part of a nucleic acid strand. Observe it and answer the questions that follow

G T T A G C T G A

(a) What do the letters G, T , C and A represent?

(b) Giving your reasons state whether it is part of DNA or an RNA strand.

(c) Show the complementary DNA strand

(d) Show the complimentary RNA strand

20. In human couples the sex of a baby is determined by the man. Explain this statement.

CHAPTER 2

EVOLUTION

1. State the difference between Lamarckian and Darwinian theories of evolution
2. Two populations of the same species of birds were separated over a long period of time by an ocean. Both populations initially fed on insects only. Later it was observed that one population fed entirely on fruits and seeds. Although insect were available. Name this type of evolutionary change.
3. Explain why Lamarck's theory of evolution is not accepted by biologists today
4. State three pieces of evidence that support the theory of evolution. (3 marks)
5. state two advantages of natural selection to organisms (2 marks)
6. Give a reason why each of the following is important in the study of evolution

   (i) Fossils records

   (ii) Comparative anatomy
7. Describe how natural selection brings about adaptation of a species to its environment (6 marks)
8. Explain how the process of evolution may result to the formation of new species
9. What is meant by
   (a) organic evolution (1 mark)
   (b) continental drift (1 mark)
10. Explain continental drift as an evidence of evolution (3 marks)
11. (a) What is a test- cross? (2 marks)

    (b) Give a reason why organisms become resistant to drugs (1 mark)

12. Distinguish between the following terms

    (a) Homologous structures

    (b) Analogous structures (4 marks)

13. (a) What is meant by natural selection?

    (b) Explain the role played by mutation in evolution (5 marks)

14. Define the following terms

    (a) Hybrid

    (b) Hybrid vigour

15. The peppered moth exists in two varieties, which are genetically controlled. The dark variety is found predominantly in industrial cities and the white variety is found predominantly in rural areas. Explain how this pattern of distribution supports the theory of evolution by natural selection (6 marks)

16. Explain what is meant by the following concepts

    (a) Special creation (2 marks)

    (b) Organic evolution (2 marks)

CHAPTER 3

RECEPTION, RESPONSE AND CO-ORDINATION

1. State one structural and one functional differences between motor and sensory neurons

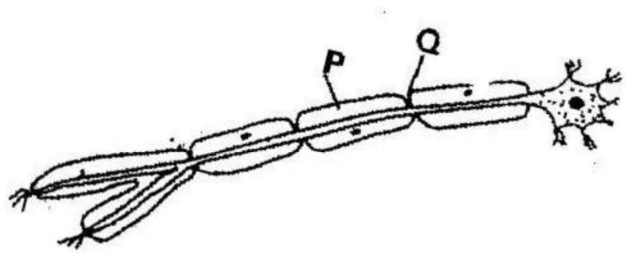
   Structural differences

   Functional differences (2 marks)

2. The table below shows two mammalian hormones. For each hormone, state the site of production and its function in the body.

Hormone	Site of production	Function
Oestrogen		
Aldosterone		

3.



   (i) With an arrow, indicate on the diagram the direction of the impulse through the neurone (1 mark)

   (ii) State the functions of parts labeled P and Q (2 marks)

4. (a) How are structures of the human eye adapted to their functions (14 marks)

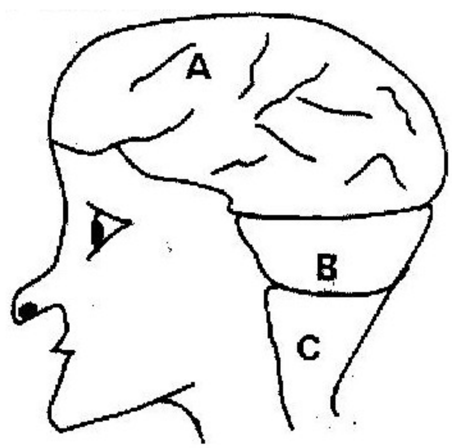
(b) State three defects of the eye and how each can be corrected (6 marks)

5. State the changes that occur in a nerve axon to produce an action potential

(3 marks)

6. In an accident a victim suffered brain injury. Consequently he had loss of memory. Which part of the brain was damaged? (1 mark)

7. The diagram below shows surface view of a human brain



(a) Name the parts labeled B and C (2 marks)

(b) State three functions of the part labeled A (3 marks)

(c) State what would happen if the part labeled B was damaged. (1 mark)

8. What is the function of the following cells in the retina of the human eye?

(2 marks)

(a) Cones

(b) Rods

9. (a) State the functions of the following parts of the mammalian ear

(i) Tympanic membrane (3 marks)

(ii) Eustachian tube (1 mark)

(iii) Ear ossicles (2 marks)

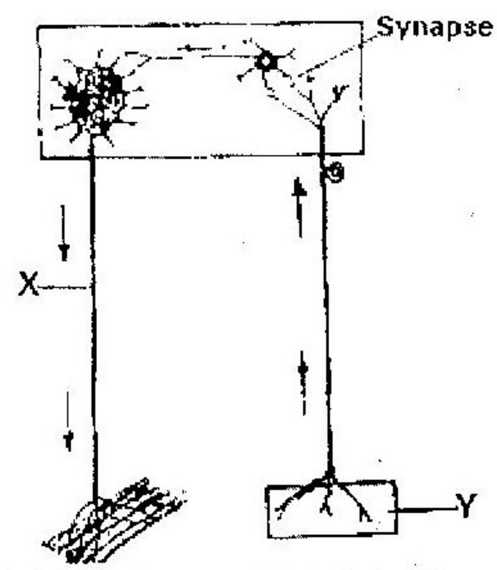
(b) Describe how semi- circular canals perform their functions (2 marks)

10. State the importance of tactic response among some members of Kingdom Protista? (1 mark)

(a) What name is given to response to contact with surface exhibited by tendrils and climbing stems in plants? (1 mark)

(b) State three biological importances of tropisms to plants (3 marks)

11. The diagram below represents a reflex arc in human



(a) Name the parts labeled X and Y (2 marks)

X ___________________________

Y ___________________________

(b) Name the substance that is responsible for the transmission of an impulse across the synapse (1 mark)

12. (a) State the function of the ciliary muscles in the human eye. (1 mark)

(b) State two functional differences between the rods and cones in the human eye

(2 marks)

13. State the function of each of the following parts of human ear (4 marks)

(a) Ear ossicles

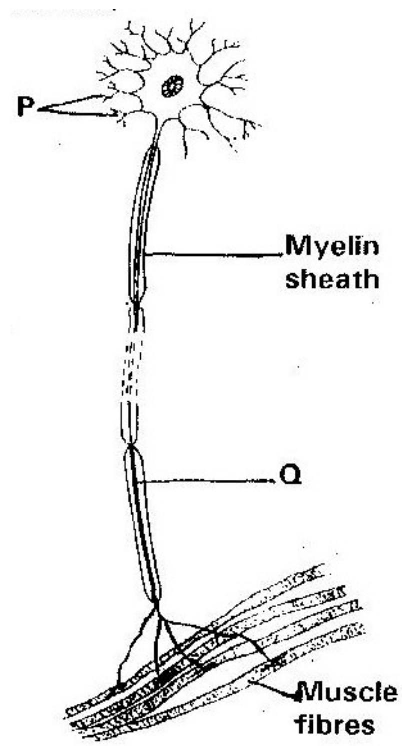
(b) Cochlea

(c) Semi- circular canals

(d) Eustachian tube

14. (a) Where in the human body are relay neurons found? (1 mark)

(b) The diagram below represents a neurone



(i) Name the neurone (1 mark)

(ii) Name the parts labeled P and Q (2 marks)

15. (a) Name the hormone that is responsible for apical dominance (1 mark)

(b) What is thigmotropism? (1 mark)

16. Describe the structure and functions of the various parts of the human ear

(20 marks)

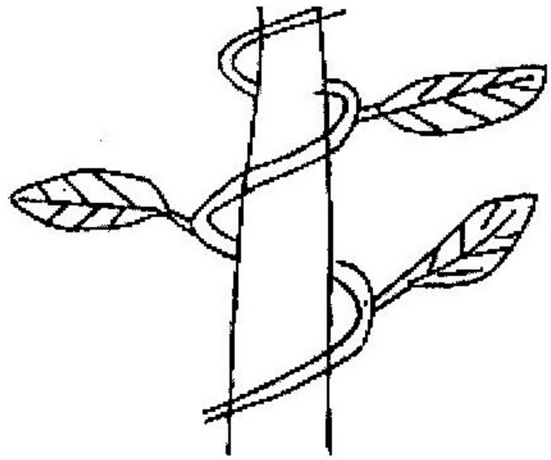
17. Nocturnal animals such as the owl are capable of seeing fairly well at night

What two retinal adaptations have made this possible? (2 marks)

18. State two functions of the human ear? (2 marks)

19. State four differences between co- ordination of the human eye's internal response to light and that of tropic movement of the flowering plant in response to light. (4 marks)

20. The figure below shows a stem of a plant growing round a tree trunk



(i) What is the name of the response, which causes the twisted growth?

(1 mark)

(ii) Explain how the twisting process is accomplished (2 marks)

(iii) Identify the state of leaves if the plant is autotrophic (2 marks)

21. Euglena is positively phototactic. Of what biological significance is this characteristics? (1 mark)

22. State the function of acetylcholine (2 marks)

23. Where in the human body is the relay neurone located? (1 mark)

24. State three effects of nicotine to human health (3 marks)

25. state the part of the eye involved in

(i) Colour vision

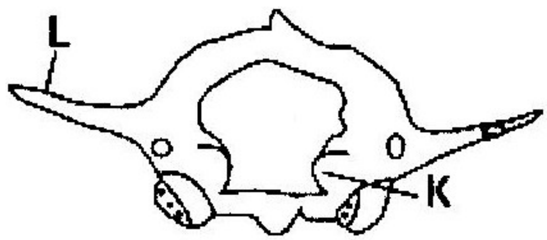
(ii) Maintaining shape of the eyeball

(iii) Change in diameter of the lens

CHAPTER 4

SUPPORT AND MOVEMENT IN PLANTS AND ANIMALS

1. The diagram below represents in a mammalian bone



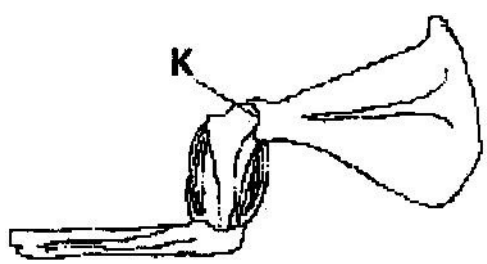
(a) State the function of the part labeled K and L (2 marks)

(b) State the region of the body in which the bone is found (1 mark)

2. State two ways in which skeletal muscle fibres are adapted to the function

(2 marks)

3. The diagram below shows the arrangement of bones and muscles in a human arm.



(i) Name the parts of the bone labeled K (1 mark)

(ii) How do the muscles work to extend the arm? (3 marks)

4. State three structural differences between biceps muscles and muscles of the gut

	Biceps	Gut muscles
(i) (ii) (iii) (iv)		

5.

(a) Name the bone (1 mark)

(b) Name the type of joint formed by the bone at its anterior end with the adjacent bone (1 mark)

6. Give a reason why the lumbar vertebrae have long and abroad transverse processes (2 marks)

7. (a) Name the three types of skeletons found in multicellular animals (3 marks)

   (b) Describe how the cervical, lumbar and sacral vertebrae are suited to their functions (17 marks)

8. A bone obtained from a mammal is represented by the diagram below

(a) Name the bone (1 mark)

(b) Which bones articulate with the bone shown in the diagram at the notch?

(2 marks)

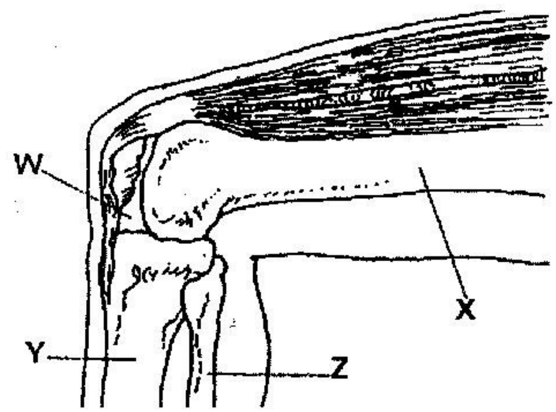
9. (a) Name the cartilage between the bones of the vertebral column

(1 mark)]

(b) State the function of the cartilage in (a) above (1 mark)

10. How are xylem vessels adapted for support? (1 mark)

11. The diagram below represents bones at a joint found in the hind limb of a mammal



(a) Name the bones labeled X, Y, and Z (3 marks)

X _______________________

Y _______________________

Z _______________________

(b) (i) Name the substance found in the place labeled W (1 mark)

(ii) State the function of the substance named in (b) (i) above

(c) Name the structure that joins the bones together at the joint (1 mark)

(d) State the differences between ball and socket joint and the one illustrated in the diagram above (1 mark)

(e) Name the structure at the elbow that performs the same functions as the patella (1 mark)

12. (a) State a characteristic that is common to all cervical vertebrae

(b) Name two tissues in plants that provide mechanical support (2 marks)

13. (a) Name the three types of muscles found in mammals and give an example of where each on of them is found

(b) State the difference between ball and socket and hinge joint (1 mark)

14. State three functions of an insects exoskeleton (3 marks)

15. State the function of the following fins of a fish

(a) Dorsal fin (1 mark)

(b) Pectoral and pelvic fins (1 mark)

(c) Caudal fin (1 mark)

16. State the diagnostic features of the cardiac muscles (3 marks)

The following figure is a part of a pelvic girdle known as the innominate bone

(a) Make a complete drawing of the girdle (1 mark)

(b) Name the bones that articulate with the pelvic girdle. In each case name the part that articulates with (2 marks)

17. Distinguish between tendons and ligaments (2 marks)

18. Explain what antagonistic muscles are and give an example (4 marks)

19. (a) Name three types of strengthening tissues found in plants (3 marks)

(b) Explain how the tissue in (a) above are adapted to their functions (3 marks)

20. (a) Name the three main types of joint (3 marks)

(b) Give an example of where each type of joint name in (a) above is found in the human body (3 marks)

21. What makes young herbaceous plant remain upright? (2 marks)]

22. Name three types of muscles found in the human body, state where each type is located and how each is adapted to its functions. (12 marks)

KCSE SAMPLE PAPERS

KCSE SAMPLE PAPER 1

TOTAL MARKS: 80

<u>Answer all questions</u>

1. Name the structures used for locomotion in each of the following organism

   (a) Euglena (1 mark)

   (b) Paramecium (1 mark)

2. (a) What is sex - linkage? (2 marks)

   (b) Name one sex- linked trait in human beings (1 mark)

3. Blackjack (bidens pilosa) belongs to the family compositae. What is the plants

   (a) Genus (1 mark)

   (b) Species (1 mark)

4. Name two metabolic waste products in

   (a) Birds (2 mark)

   (b) Plants (2 marks)

5. State the adaptations of seed to dispersal by wind (3 marks)

6. State the importance of the growth of pollen tubes in flowering plants

   (1 mark)

7. State three structural differences between DNA and RNA in living cells

   (3 marks)

8. (a) State two differences between meiosis and mitosis (2 marks)

   (b) State two processes that takes place during interphase (2 marks)

9. Name two parts in the human body with cilia (2 marks)

10. The diagram below represents a closed stoma

(a) Identify the cells labeled A and B (2 marks)

(b) Name the excretory product in plants which is excreted through the stomata

(1 mark)

(c) State one adaptation of the guard cell to its function (1 mark)

11. Name two organisms that form the biological environment of a malaria parasite

(2 marks)

12. Name the organs of the mammalian body that are responsible for production of gametes (2 marks)

13. List three adaptations of fruits that are dispersed by animals (3 marks)

14. The equation below show what happens in cellular respiration

$C_{18}H_{38} + 26O_2 \rightarrow 18CO_2 + 18H_2O$ + Energy

(a) Name the type of respiration shown and where it occurs in a cell

(2 marks)

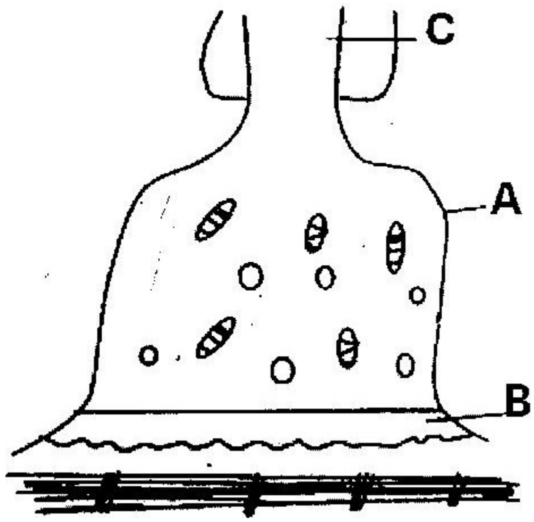
(b) Determine the respiratory quotient of the process (2 marks)

(c) What food substrate is broken in the respiration? (1 mark)

15. List two features of the small intestine that increase its surface area

(2 marks)

16.



The diagram above shows synapse at a neuromuscular junction

(a) Name the parts labeled A and B

(b) State the function of the part labeled C

17. Explain why food is stored in an insoluble form in the cells of living things

(2 marks)

18. State the observation made in germinating seeds when the

(a) Hypocotyle elongates

(b) Epicotyle elongates

19. State the differences between assimilation and absorption of food nutrients

(2 marks)

20. State three homeostatic function of the liver

21. Differentiate between lactic fermentation and alcoholic fermentation

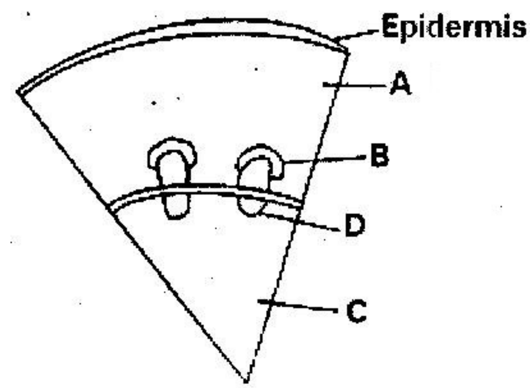
(4 marks)

22. Water logging can cause the death of some plants. Explain (2 marks)

23. Distinguish between plasmolysis and haemolysis (2 marks)

24. John and grace who are siblings are both normal and so are their parents but they have a haemophilic brother. Give the genotypes of their parents. (2 marks)

25. The diagram below shows a section of a dicotyledonous stem.



Name the tissues labeled A and D and state the function of each.

26. Name the organism that causes each of the following diseases

(a) AIDS

(b) Bilhazia

(c) Cholera (3 marks)

27. List three examples of gaseous exchanges surfaces in animals (3 marks)

28. state the significance of photosynthesis (3 marks)

29. Explain the meaning of each of the following

(a) Continental drift

(b) Fossils (2 marks)

30. Green plants grow towards a source of light

Name this type of response (1 mark)

KCSE SAMPLE PAPER 2

TIME 1 ¾ HOURS

This paper consists A and B. Answer ALL the question in section A in the spaces provided. In Section B answer question 6 (compulsory) and either question 7 or 8 in the spaces provided.

SECTION A (40 MARKS)

Answer all the questions in this section

1. (a) What is gene linkage (1 mark)

(b) Haemophilia is sex linked trait

(i) If a normal woman but a carrier for haemophilia marries a normal man work out the phenotype of the offspring using a genetic cross. (5 marks)

(ii) Why is haemophilia, more common defect in males than in females? (1 mark)

(iii) Other than haemophilia state any other sex linked defect in man (1 mark)

2. The table below shows the percentage composition by volume of inhaled and exhaled air

Gas	Inhales air %	Exhales air %
Oxygen	21	16
Carbon (IV) Oxide	0.04	4.0
Nitrogen	79	79

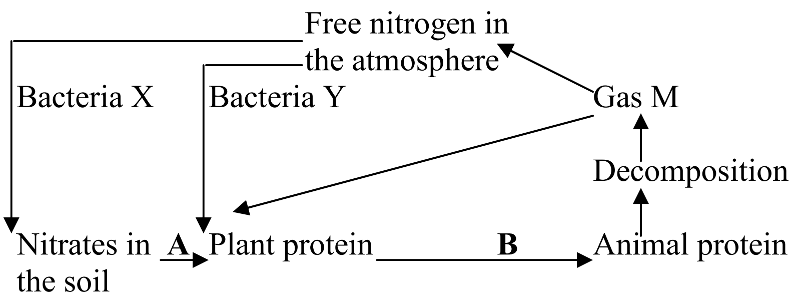
(a) (i) By what percentage is the carbon (IV) Oxide concentration in exhaled air higher than inhaled air? (3 marks)

(ii) Explain the differences in the composition of the gases between inhaled and exhaled air. (3 marks)

(b) State two ways in which leaves of plant are adapted for gaseous exchange?

(2 marks)

3. The chart below represents the flow of nitrogen in the ecosystem



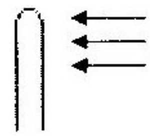
(a) Name the bacteria labeled X and Y (2 marks)

(b) Name the gas M

(c) Name processes A, B and C

(d) State the bacteria involved in the process named C

4. An experiment was set up to investigate the effect of unilateral light in growth of maize seedlings. The diagram in the table below represents experimental set up at the beginning.

Experimental set up	Beginning of experiment	Expected results
C		
D		
B		

(a) Using diagrams complete the table to show the expected results in experimental set up. (3 marks)

(b) Account for your results in experimental set up (3 marks)

(c) Explain the purpose of experimental set up B and C (2 marks)

5. (a) What is internal fertilization? (1 mark)

(b) Suggest two disadvantages of internal fertilization in most mammals (2 marks)

(c) State two roles of placenta in mammals (2 marks)

(d) Mention one role played by each of the following hormones in human menstrual cycle

(i) Follicle stimulating hormones (FSH)

(ii) Oestrogen

(iii) Luteinizing Hormone (LH)

SECTION B (40 MARKS)

Answer question (compulsory) in the spaces provided either question 7 & 8 in the spaces provided after question 8.

6. The following data are results of making daily growth measurement ion an organism over a period of 24 days during its development.

Day	Width of head (mm)	Length of hind femur (mm)
1.	3.0	7.0
2.	3.5	7.5
3.	4.0	8.0
4.	4.0	8.0
5.	4.0	8.0
6.	4.0	9.2
7.	4.0	10.5
8.	4.4	12.0
9.	4.7	12.0
10.	5.0	12.0
11.	5.0	12.0
12.	5.0	12.0
13.	5.0	12.0
14.	5.0	12.0
15.	5.0	13.3
16.	5.0	14.8
17.	5.7	16.4
18.	6.4	18.0
19.	7.0	18.0
20.	7.6	18.0
21.	7.6	18.0
22.	7.6	18.0
23.	7.6	18.0
24.	7.6	18.0

(a) Using a suitable scale draw graphs of width of head and length of femur against time.

Draw the graphs on the same axis. (8 marks)

(b) (i) Name the growth pattern represented by the graph (1 mark)

(ii) With reference to your graph identify the phylum to which the organisms belong. Give a reason for your answer (2 marks)

(c) Account for the length of hind femur between

(i) Day 3 and day 7 (3 marks)

(ii) Day 7 and day 10 (2 marks)

(d) State two hormones involved in the growth pattern represented by the graphs (2 marks)

(e) State two advantages of metamorphosis in organisms (2 marks)

7. Describe how water and mineral salts move from soil until they reach the leaves in a tall plant. (20 marks)

8. (a) Describe the following terms:

(i) Secretion

(ii) Excretion

(iii) Egestion (3 marks)

(c) Explain how the mammalian kidney is adapted to its functions. (17 marks)

KCSE SAMPLE PAPER 3

(PRACTICAL)

(TOTAL MARKS 40)

You are provided with specimen

A. Onion bulb

B. Cockroach

- Iodine solution
- Benedict's solution
- Means of heating
- Hind legs

1. Make a longitudinal section through specimen A

(a) Make a labeled diagram of the specimen (3 marks)

(b) (i) Using a pestle and a mortar, crush some leaves of one half of the specimen. Using the reagents provided tests for food substances present in extract. Record your finding in the following table.

Food	Procedure	Results	Conclusion

(ii) Account for the result in (b) (i)

(c) (i) Cut six leafstalks into 4 cm long pieces make a slit 2 cm long at one end of each of the stalks. Immerse the stalks into tube- tubes with the following liquids.

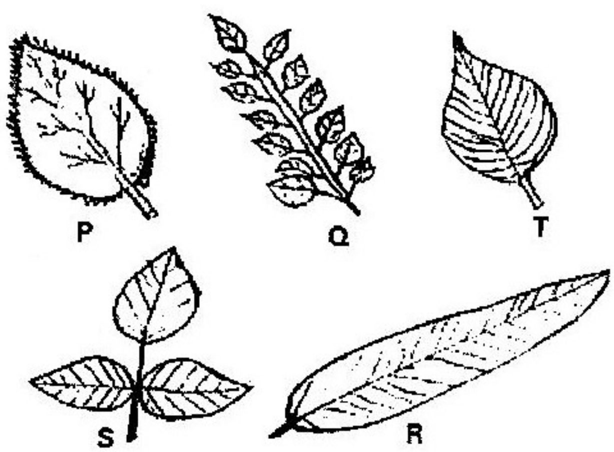
- Distilled water
- 20% salt solution
- Extract obtained in (b) above

Leave the set – up for 30 minutes. Account for the result.

(ii) Of what significance are the results obtained in distilled water?

(d) How do the part of the plant adapt to its way of life? (2 marks)

2. Study specimens below and answer the questions that follow



A dichotomous key was constructed identify the plant leaves as follows

1. (a) Simple leaf...Go to 2

   (b) Compound leaf...Go to 4

2. (a) Leaf has serrated margin ....Hibiscus

   (b) Leaf has smooth margin ....Go to 3

3. (a) Leaf has smooth margin ....Mango

   (b) Leaf is ovate ....Morning glory

4. (a) Leaf has many leaflets ....Nandi Flame

   (b) Leaf has 3 leaflets..Bean

Identify steps followed for each of the leaves and state its identity

Leaf	Steps followed	Identity
P Q R S T		

3. Using the hand lens to study specimen B.

   (a) Identify the Phylum and class of which the specimen belongs. Give two reasons for each.

   Phylum ___________________________

   Reasons

   (i)

   (ii)

Class ___________________________

(i)

(ii)

(b) Draw a well labeled diagram of specimen B (3 marks)

(c) State three difference between specimen B and man (2 marks)

(d) How is specimen B adapted to living between cracks (2 marks)

KCSE SAMPLE PAPER 231/3

(PRACTICAL)

1. You are provided with specimens labeled J1, J2, K1 and K2. Examine them.

J1 – Bean seedling with a curved end

J2 – Bean seedling with straightened end

K1 – maize seedling with coleoptile intact

K2 - Maize seedling with first foliage leaves

(a) With a reason name the order to which specimens J1 and J2 and K1 and K2 belong (4 marks)

J1 and J2 ___________________________

Reasons ____________________________

K1 and K2 __________________________

Reasons ____________________________

(b) (i) Name the curved part of specimens J1 (1 mark)

(ii) What is the importance of curvature? (1 mark)

(c) Explain how the curved part in J1 will straighten so that the stem will look like that of J2 (4 marks)

(d) Name the part that protects the plumule in specimen K1 and K2(1 mark)

(e) (i) Which of the two types of seedlings may form swelling on the roots later in its life?

(ii) What is the name of the swellings? (1 mark)

(iii) Name the organisms that would be found in the swellings (1 mark)

(iv) Explain the relationship that exists between the named organisms and the plant. (3 marks)

(f) (i) Name the structures found on the stem just below the leaves of specimen J2. (1 mark)

(ii) State two functions of the structure named in (f) (i) above (2 marks)

(g) (i) State the type of germination exhibited by specimen K1 and K2 (1 mark)

(ii) Give a reason for your answer in (g) (i) above (1 mark)

(h) Name the root system found in specimens

J1 and J2 (1 mark)

K1 and K2 (1 mark)

2. You are provided with specimens labeled M and N which were obtained from an animal. Examine them.

M - Trachea (part of trachea)

N - Part of lung

(a) Identify the specimen of (2 marks)

Identity of M___________________

Identity of N___________________

(b) Name the part of the body from where each of the specimens was obtained.

Specimen M ___________________

Specimen N ___________________

(c) What is the relationship between specimen M and N terms of their functions. (2 marks)

(d) For each specimen name observable feature and state how each feature adapts the specimen to its function? (10 marks)

Specimen	Feature	How feature adapts to its function
M N		

3. You are provided with specimen labeled Q and R. Examine them

   Q - Lumbar vertebrae

   R - Cervical vertebrae

   Both obtained from the same mammal

(a) Identify the specimens and each case give two reasons for your answer. (6 marks)

(i) Specimen Q ___________________

Reason 1 ___________________

2 ___________________

(ii) Specimen R ___________________

Reason 1 ___________________

2 ___________________

(b) State four ways in which specimen R is adapted to its functions

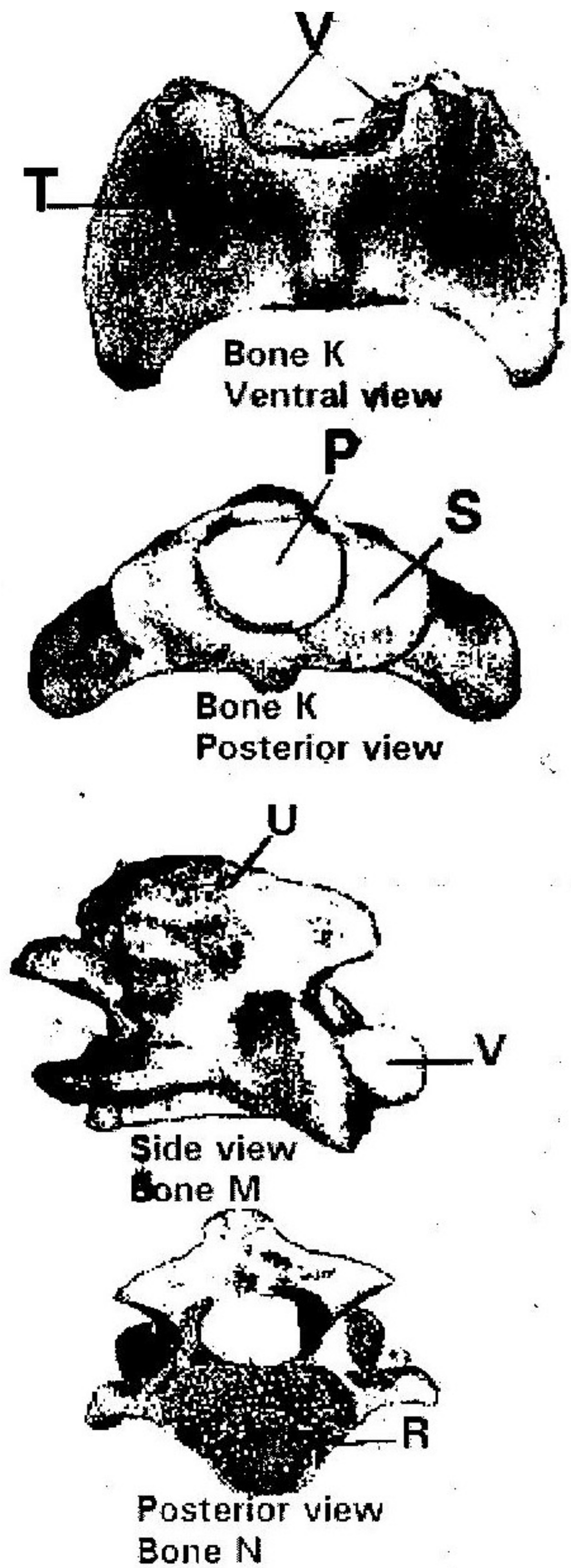
(4 marks)

(c) State four differences between specimens Q and R (4 marks)

(d) Draw and label the anterior view of specimen R

BIOLOGY PAPER 3

1. The photographs below are of bones obtained form the same region of a mammalian body. Photographs labeled K are different views of the same bone while M and N are views of different bones.



(a) Name the region from which the bones were obtained (1 mark)

(b) Identify the bones (3 marks)

K ____________________________

M ____________________________

N ____________________________

(c) State three characteristics features of the bone on photographs labeled K.

(3 marks)

(d) Name the structures that fit in the opening labeled P in the photographs of bone K.

(2 marks)

(e) State the functions of the parts labeled S and T in photographs of bone K

(2 marks)

S ____________________________

T ____________________________

(f) Name the structures that articulate with the parts labeled V in the photographs of bone K. (1 mark)

(g) Name the parts labeled U and V in the photograph of bone M and R in the photograph of bone N (3 marks)

U ____________________________

Y ____________________________

R ____________________________

2. You are provided with two pieces of plant material labeled specimen D. Using a scalpel cut a slit halfway through the middle of each piece as shown in the diagram below.

   Place one piece in the solution labeled L_1 and the other in the solution labeled. Allow the set up to stand for 30 minutes.

   (a) After 30 minutes remove the pieces and press each gently between the fingers.

   (i) Record your observations

   L_1 ______________________________ (1 mark)

   L_2 ______________________________ (1 mark)

   (ii) Account for the observation in (a) (i) above (2 marks)

3. You are provided with three sets of seedlings labeled A, B and C. Examine them.

   (a) State the conditions under which each set was grown (3 marks)

   Set A ______________________________

   Set B ______________________________

   Set C ______________________________

   (b) State four differences between the seedlings in set A and B (4 marks)

   (c) (i) Give a reason why plants exhibit the phenomena named in (c) (i) above (1 mark)

   (d) (Name the response exhibited by the seedling in set C. (1 mark)

   (e) Explain how the response named in (d) above occurred (3 marks)

ANSWERS TO TOPICAL QUESTIONS

FORM 1 WORK

CHAPTER 1 – INTRODUCTION TO BIOLOGY

1.

Plants	Animals
- Manufacture their own food - Grow continuously throughout their life - Slow in responding to stimuli	- Cannot manufacture their own food - Stop growing when they reach maturity - Fast in responding to stimuli

2. - Growth of the organism

   - Movement of the organism

   - Maintenance of metabolic activities

3. They grow
   - They respire
   - They have gaseous exchange
   - They reproduce
   - They excrete
   - They respond to stimuli
   - They have nutrition

4. They reproduce

   They grow

   They respond to stimuli

CHAPTER 2- CLASSIFICATION 1

1.

(a) Binomial nomenclature is a system of naming organisms by giving them two scientific names, the generic and the specific names.

(b)
- It makes it easier to identify an organism.
- It is easier to describe an organism as it is based on characteristics of the organism
- Large number of organisms are divided into smaller groups depending on characteristics
- The whole world uses the same groupings, so everyone understands each other.

2. (a) Classification – placing of animals and plants into group according to their similarities in structure, physiological processes and ancestry.

(b) Taxonomy - scientific study of classification.

(c) Binomial nomenclature- system of naming using two names. The first part of the name represents the genus (generic name) while the second part refers to species or is the specific name.

3.

(a) - Nucleus not organized
- Organelles not bound by membrane
- Absence of mitochondria

(b) Class insecta

4. Genus

5. Specific name
6. Species – A species is the smallest unit of individual organisms which has hereditary distinction from that of any other group and whose members naturally interbreed to produce fertile offspring
7. Taxonomy- Scientific study of classification
8. Taxon- Each group of classification

CHAPTER 3 – THE CELL

1. (a) - Secretion of useful substances
   - Formation of secretory vesicles
2. (a) - Destroying old and worn out organelles
   (b) - Secretion reticulum (rough)
   - Formation of secretory vesicles
3. (a) - Mitochondrion
   (b) - Chloroplast
4. - Ribosomes
   - Endoplasmic reticulum (rough)
5. (a) - X- chloroplasts
   - Y - Vacuole
   (b) In dim light. They move to the upper part of the cell in order to receive enough sunlight for photosynthesis
6. (a) - Increase surface area for attachment of respiratory enzymes hence increasing rate of respiration.

(b) (i) Stroma

(ii) Absorb sunlight used for light stage of photosynthesis

7. (a) $\text{Drawing Magnification} = \dfrac{\text{Length of the drawing}}{\text{Length of the object}}$

(b) It is adding a dye to the specimen to make the features clearer and distinguishable

.

8. - Form vesicles that ransport materials to other parts of the cell e.g. proteins
-Transport secretions to the cell surface for secretion e.g. enzymes and mucus.
- They form lysosomes

9. - Cell wall
- Large vacuole
- Chloroplast
- Starch granules

10. (i) Reflect light from the source to the microscope/specimen
(ii) Regulate amount of light entering the microscope/reaching specimen.
(iii) Move body tube up and down in order to obtain a rough FREE of the image of specimen.

11. It is the ability to differentiate two structures or organelles lying close

12. (a) A cell is structurally and physiologically modified in order to perform a

particular function.

(b) (i) Presence of dendrites to receive impulses

(ii) Presence of chloroplasts to trap sunlight

(iii) Elongated and no cuticles in order to absorb water

(iv) Biconcave shape to increase surface area for diffusion of oxygen/haemoglobin.

13. 1 mm = 1000μm

3.5 mm = 3500 μm

10 cells = 3500 μm

1 cell = $\frac{3500}{10}$ μm

1 cell = 350 μm

14. (i) Made of several specialized cells grouped together and perform particular function.

(ii) Made of a group of specialized tissues grouped together performing a particular function

(iii) It is made of several organs that perform a particular function.

CHAPTER 4

CELL PHYSIOLOGY

1. a) i) Diffusion

   ii) Active transport

   b) Diffusion-A concentration gradient between sodium ions in sap and those in the pond.

   Active transport-energy in form of ATP must be available/Oxygen and food in the living tissue for respiration provide energy.

2. A film of water surrounds the soil particle. Root hairs of the plants penetrate between the soils particles/are close to the soil particles; cell sap of the root hair cells is more concentrated in solutes/has less water than the soil solution. Thus water moves into root hair cell by osmosis i.e across the cell a wall and the semi permeable membrane.

3. The leaves expose a smaller surface area to the sun. Thus reducing transpiration/excessive water loss.

4 a) Diffusion is defined as the net movement of a substance from a region where its concentration is high to a region where its concentration is low.

   b) i) Diffusion gradient-the greater the diffusion gradient, the greater the rate of diffusion

   ii) Surface area to volume ratio-the greater the S.A.V.R the higher the temperature the greater the rate of diffusion.

   iii) Temperature –The higher the temperature the greater the rate of diffusion

c) i) Absorption of mineral salts from the soil by root hairs

ii) Re-absorption of glucose molecules in the kidney tubule.

iii) Absorption of digested food in the ileum e.g glucose, amino acids.

5. i) Uptake of water from the soil into root hairs of plant roots

ii) Movement of water from the veins of leaves through the leaf cells to the atmosphere during transpiration.

6. a) The visking tubing was fully filled with solution. Level of water in beaker decreased .

b) Sucrose solution in visking tubing created high concentration gradient.

-Water molecules moved from distilled water to the visking tubing by osmosis.

7. -Plant cells have cells membrane and cell wall. When the cell is placed or immersed in distilled water, the water is absorbed by osmosis. As cell becomes turgid, the cell created an inward force, wall pressure that prevents the cell from bursting.

8.

Diffusion	Osmosis
• Involves movement of particles of molecules of liquid or gas. • It may be through a membrane or in air. • Not affected by PH changes.	• Involves movement of solvent • It takes place through a semi-permeable. • Rate affected by pH changes.

9. a) Isotonic solution- a solution which has the same concentration as the cell sap.

b) Hypotonic solution- a solution which is less concentrated than the cell sap.

c) Hypertonic solution- A solution which is more concentrated than the cell sap.

10. Plants normally grow in soils whose solute concentration is lower than that of the cell sap. This enables the plants to take up water by osmosis. Addition of large amounts of salt to the soil increases the solute concentration of soil water beyond that of the cell sap. The result is that the plants lose water to the soil by osmosis. Since water is very important for maintaining the structural and metabolic activities of plants, its deficiency leads to death of the plants.

11. a) The red blood cells take in water by osmosis. They swell and

exert pressure on the fragile plasma membrane which then breaks. Plant cells take in water and swell but do not burst. This is because their tough cell wall can only stretch to a limited extent. Once fully stretched, the cell wall resists further expansion of the cell and no more water is taken up.

b) Fresh water protozoa take in water by osmosis. The excess water is then actively pumped into the contractile vacuole which discharges the water to the outside.

CHAPTER -5

NUTRITION IN PLANTS

1. a) K- Enzyme , sucrose, invertase

   L- Inhibitor

   b) Additional of sucrose/substance, Addition of enzyme, Optimum PH, Removal of products.

   c) Complete with substrate for active site of the enzyme.

2. a) Split water molecules/photolysis

   b) Glucose

3. Yellowing of leaves/stunted growth/chlorosis/lack of chlorophyll.

4. a) i) A and B -more active sites of enzymes available for a large number of molecules of substrate. There is increase in rates of reaction

   ii) B-C

   - Enzyme/substance are in equilibrium. All active sites are occupied hence rate of reaction is constant.

   b) Raising concentration of enzymes

   c) PH, temperature, inhibitors/cofactors.

5. a) Substances that activate enzymes

   b) Iron/Magnesium/Zinc/Copper.

6. - Magnesium,
   - Nitrogen
   - Iron

7. Xylem

- Transport water to photosynthesizing cells from stem
- Offer support to the lamina for maximum exposure to sun-light.

Phloem

- Transport manufactured food away from the leaf to create high concentration gradient.

8. Takes place in the grana of the chloroplast. Light is absorbed and used to split water molecules into hydrogen ions and oxygen, photolysis. Energy is formed and is stored in form of ATP.

9. a) i) Light stage-grana

ii) Dark stage-stroma

b) -Uses the energy formed or produced during light stage.

-Uses the hydrogen ions produced in light stage for carbon dioxide fixation.

10. i) Cuticle -Transparent allowing light to penetrate.

ii) Veins –Xylem vessel transport water to the photosynthesizing cells as it is a raw material

Phloem - Transport manufactured food out of the leaf to create high concentration gradient.

11. a) To hydrolyse/break down the disaccharide (non-reducing sugar).

b) Non-reducing sugar

c) i) Condensation,

ii) Hydrolysis

d) i) Starch,

ii) Glycogen

12. i) Fatty acids and glycerol

ii) Form part of the cell membrane

- Provide insulation of bodies of animals
- A source of metabolic water.
- Provide energy in absence of carbohydrates

13. a) L - Blue-black

M -Yellow

N - Blue Black

b) Absorb carbon (IV) oxide in the jar.

CHAPTER 6

NUTRITION IN ANIMALS

1. a) Rhizobium

   b) Symbiosis

2. a) Activate enzymes

   b) Magnesium/zinc

3. Scuvy

4. - Rickets

   - Goitre

5. Sharp/ hooked/ strong beaks for killing/ripping off flesh from bones, sharp claws for grabbing/holding prey.

6. i) Salting -This removes / absorbs water by osmosis from micro-organism cell. Which then die due to dehydration. Meat also becomes dehydrated and thus unsuitable for microbial growth.

   ii) Refrigeration –Low temperature renders the micro-organism inactive (Enzymes do not work at low temperature).

   iii) Canning –Boiling kills all micro-organism in the food. Sealing under pressure excludes all micro-organisms and ensures that growth takes place.

7. Similarity: Both are heterotrophic.

   Difference: Predators kill to get food while parasites obtain foods without killing the host.

8. Pancreatic juice containing digestive enzymes is prevented from reaching food. Insulin and glycogen hormones which regulate sugar are released directly into the blood stream.

9. Roughage provide grip needed for peristalsis/lack of roughage results in slow/no movement of food leading to constipation. (Accept: add bulk to peristalsis to take place)

10. a) Breakdown of (complex) food substance by enzymes to simpler compounds which can be absorbed.

    b)
    - Small intestine is long/coiled to offer large surface area for digestion and absorption.
    - The walls are muscular for peristalsis.
    - Inner walls posses mucus glands, goblet cell; that secret mucus for lubrication and protection of the walls from digestive enzymes.
    - The inner walls have digestive glands that secrete digestives enzymes.
    - The inner walls has villi to increase surface area for absorption.
    - The villi have numerous blood vessels for transport of the end products of digestion.
    - The villi also have lacteal vessels, for transport of fats/lipids.

11. - Quicken healing of wounds
    - Forms connective tissues of the teeth and jaws.
    - Provides resistance to body infections

12. a) Homodont-Organism has same number of teeth, type of teeth and the same

size.

b) Slice fish and crush bones

c) $I=\frac{0}{3}$, $C=\frac{0}{3}$, $PM=\frac{3}{3}$, $M\frac{3}{3}$

13.

a) -Has alkaline salts that help create alkaline media to neutralize acidic food from stomach.

-Enhance emulsification of fats into droplets

b) As the substance concentration decreases the rate of enzyme action decreases.

14. -Hydrogen ions, ATP molecules

-Oxygen gas

15. a) i) - Premolar tooth

ii) - Presence of two roots

- Presence of cusps on the crown.

b) Has a blood vessel that provides nourishment to the tooth and remove waste products.

16. a) Vitamin D, Vitamin K

b) - Transmission of nerve impulses.

- Ionic balance/osmotic balance

- Contraction of muscles.

17. a) In the stomach there is acid medium and ptyalin only acts at slightly alkaline medium.

b) High temperatures above 40^0C.

c) -Villi

- Microvilli

18. a) Ingestion is the taking of food into the body.

b) Digestion is the breakdown of large and insoluble molecules that can be absorbed.

c) Absorption is the uptake of soluble food materials from lumen of digestive tract across the epithelial lining of the gut into blood stream.

d) Assimilation is the utilization of absorbed food molecules by the body to provide energy or the materials necessary for growth, repair and reproduction.

e) Egestion in the elimination of undigested waste food materials from the body.

19. a) They produce saliva. Saliva contains the enzyme salivary amylase (ptyalin) which begins the digestion of starch breaking it to maltose. It also lubricates food making it suitable for swallowing.

b) It produces pancreatic juice. Contains $NaHCI_3$ which neutralizes the acid of chime and creates a PH of 7-8 which is the Optimum PH for the action of pancreatic juice are;-

- Trypsin which digests protein to peptides.
- Amylase which digests starch to maltose
- Lipase which digests fats to fatty acids and glycerol

c) It produces bile. Bile salts droplets a process called emulsification. This increases the surface area of the fat enhancing the action of pancreatic lipase.

20. i) It lubricates food

ii) It prevents digestion of the gut wall by proteolytic enzymes

iii) It makes food particles to adhere to one another during swallowing and during gestation.

21. Hydrochloric acid in the stomach denatures salivary amylase stopping its activity.

22. A sheep has the following herbivorous adaptations.

- It has a thick horny pad on the upper jaw over which vegetation is pressed by chisel-like incisors and canines on the lower jaw during feedings.
- If has a diastema which provides space for tongue movements that separate grass which is being chewed by check teeth and grass that is newly gathered by front teeth.
- Its premolars and molars have large top surface, which is worn out unevenly forming cusps which help in crushing and grinding of vegetation.
- The joints of the jawbones are loose allowing up and down as well as sideways movement of the lower jaw, which aids in the grinding of vegetation.
- Its rumen contains microorganisms that ferment cellulose releasing simple fatty acids that are absorbed by the animal.

23. i) Pepsin-digests proteins to peptides

ii) Rennin-Coagulation of milk proteins to peptides

iii)
- Hcl-converts pepsinogen to pepsin
- Kills bacteria in food
- Provides an acidic pH (pH 1.5-2.5) which is the optimum pH for action of Pepsin
- Unfolds proteins enabling pepsin to work on them.

24. This leads to lack of bile salts, which emulsify fats.

25.

	Nutrient	Food Source	Role in the body
a)	Vitamin A	Carrots, Liver, Egg yolk	Synthesis of rhodopsin (for proper function of retinal).
b)	Iron	Liver	Manufacture of hemoglobin
c)	Iodine	Iodized salt, sea food	Manufacture thyroxin
d)	Vitamin D	Fish, liver, plant oil, egg yolk	Aids assimilation of calcium phosphate for making teeth and bones.
e)	Protein	Meat, milk seed of legumes, fish	Making new cells/growth and repair of tissues.

FORM 2 WORK

CHAPTER 1

TRANSPORT IN PLANTS

1. a) To investigate the effect of light on the rate of transpiration.

   b) More water was lost in the light than in the dark. Rate of transpiration was greater in light than in the dark. This is because the stomata are fully open in light but less open or closed in the dark. In light, photosynthesis takes place hence no water used.

2. The leaves exposed a smaller surface area to the sun thus reducing transpiration. Excessive water loss.

3.

   - Temperature- high temperature increase transportation. Low temperature lower transpiration.
   - Light intensity-more light increase transpiration, dim light lower transpiration.
   - Wind – strong wind increase transpiration, calm weather lower transpiration.
   - Humidity – High humidity lower transpiration, low humidity increase transpiration.
   - Atmospheric pressure- High atmospheric pressure increase transpiration.
   - Water availability-more water increase transpiration due to opening of stomata while little water lowers transpiration.

4. a) A - Epidermis

   B - Pith

   b) C - Transport manufactured food/translocation

D - Produce new cells/divide giving new cells

E - Transport mineral salts and water.

c) -Xylem in centre/star shaped.

-Phloem in arm of xylem

-No pith in root

-Roots hairs present in root

5 a) K - Root hair

L - Xylem vessel

b) Water moves from the soil into the root hair by osmosis. Because concentration of cell sap is higher than water in the soil; the cell sap of the root hair is diluted thus making it less concentrated than neighbouring cells; therefore water moves into the neighbouring cell. It is then actively secreted into L.

c) Active transport/diffusion.

6.

- Lignified/thickened to prevent collapsing.
- Narrow to facilitate capillarity
- No cross wall for continuous flow of water.
- Side walls pitted to allow lateral movement of water and mineral salts.

7.

- Turgidity
- Presence of xylem vessels
- Presence of collenchyma

8. a) R- sieve pore/plate

   S-Cytoplasm strand

   T- Companion cell

   b) Translocation

   c) Thickened, - Lignified

9. a) Lignin,

   b) Phloem

10. They are strengthened by lignin hence supporting the stem.

11. - Xylem - Transports water and mineral salts to photosynthesizing cells
    - Phloem - Transports manufactured foods from the leaves creating high concentration gradient.
    - Veins - Supports the leaf to be upright for maximum absorption of light for photosynthesis.

12. a) Ovule

    b) Ovary

13 a) -Xylem vessels

   -Sclerenchyma

   b) -Turgidity of parenchyma cells

   -Presence of collenchyma cells

14.

   a) Dicot root

   b) i) Presence of root hairs

   ii) Phloem between rays of xylem (star shaped xylem).

c) J - Epidermis

K - Phloem

L - Xylem

d) Absorbs water and mineral salts from soil.

15.
- Adhesion- force of attraction between unlike molecules
- Due to the force of adhesion water tends to stick to the walls of vessels containing it.
- Cohesion- forces of attraction between like molecules
- Cohesion between water molecules prevents the water column from breaking.
- Root pressure- due to pressure generated by the root's endodermis
- Capillary due to narrowness of xylem.
- Transpiration pull- as water evaporates from the leaf's surface more is absorbed
- After the water reaches the leaves cells, it passes the cells by osmosis from the xylem. Water vapour diffuses out through stomata.

16.
- Absence of cuticle to allow diffusion of water.
- Thin walled to reduce distance of diffusion.
- Elongated to increase surface area for absorption of water and mineral salts.
- Presence of large vacuole to increase concentration gradient between cell sap and soil water.

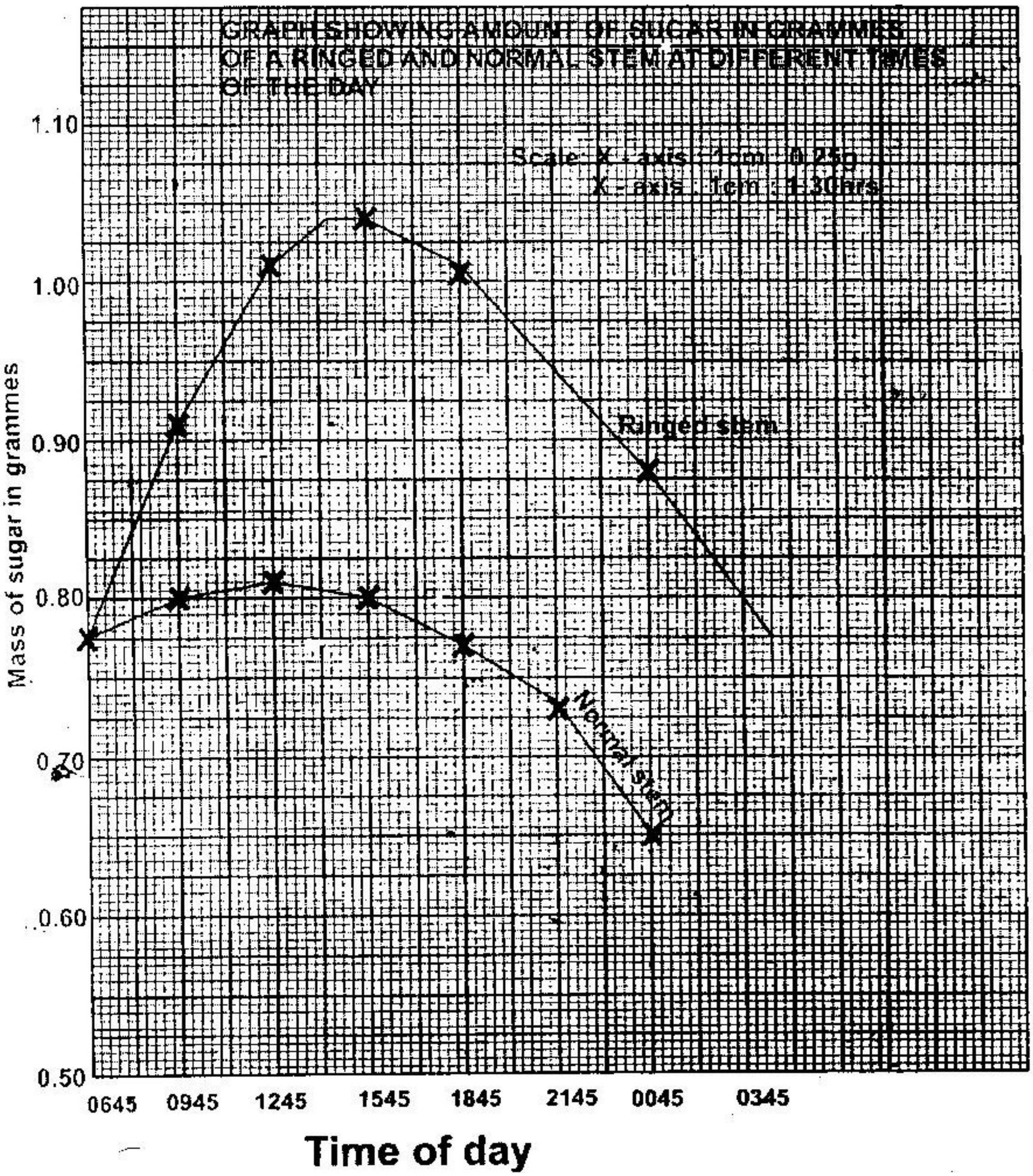
17.

a) Phloem tissues

b) K- Companion cell

c) Supply nutrients and energy to the sieve tubes.

18. a) Graph



b) i) 15:45

ii) 12:45

c) 0.79 ± 0.02 grammes

d) The food that had been manufactured the previous day had been converted to soluble sugars and was being translocated to other parts of the plant.

e) i) 06 45 hours and 15 45 hours.

- There was low concentration of sugar early in the morning as there was little translocation.
- As day progresses the light intensity increases and more food is manufactured thus more translocation increasing concentration of sugars.

ii) 15 45 and 00 45

- The light intensity is decreasing reducing rate of photosynthesis. Less food is manufactured, hence less is translocated.
- As it turns dark there is no photosynthesis reducing concentration of sugar translocated.

iii) Sieve plates

- Cytoplasmic strands

f) - Amino acids

- Soluble fats/lipids.

19. i) Reduce transpiration

ii) Eliminate excretory wastes on the leaf

20. a) - Maintain transpiration stream

- Cool the plant
- Remove excess water
- Enhance absorption and distribution of water and mineral salts.

b)
- Few and small leaves
- Reduced leaf size
- Sunken stomata
- Thick cuticle.

CHAPTER 2

TRANSPORT IN ANIMALS

1. Carboxyhaemoglobin
2. Blood group **A** has antigens **A** on red blood cells and antibodies **b** in plasma. Recipient's blood group **B** has **B** antigens and **a** antibodies. When blood group **A** from donor is transferred antigen **A** will react with antibody **a** in the recipient's blood. Clumping or agglutination of the red blood cells will take place: the clumped red blood cells block capillaries and this hinders the flow of blood and may result in death.
3. In a closed circulatory system, blood flow is confined to enclosed vessels while in open circulation blood is not confined to vessels but flows in cavities (sinuses) and is in direct contact with tissues.
4. a) i) Arthropoda

   ii) Chordata

   b) When blood is confined within vessels, it generates high pressure. This results in a faster rate of circulation, over long distances, ensuring efficient transportation of material e.g nutrient to all parts of the body, which renders the animals more active than those with open circulatory system.
5. i) They contain haemoglobin, a molecule that readily combine with oxygen.

   ii) They are biconcave discs without a nucleus, allowing more haemoglobin to be packed in cells so that each cell can carry more oxygen.
6. a) i) - Capillaries

   ii) - They are thin-walled (one cell thick), thus allowing diffusion of materials.

b) - Have a small diameter to increase pressure thus allow materials to diffuse out.

- They are intimately associated with tissues in order to allow exchange of materials

- They are numerous- to provide a large surface area for exchange of materials.

c) i) Pulmonary arterioles contain more carbon dioxide than pulmonary venules.

ii) Pulmonary arterioles contain less oxygen than pulmonary venules.

7. It does not dissociate easily hence leads to suffocation

8. i) They contain haemoglobin, a molecule that readily combine with oxygen.

ii) They are biconcave discs without a nucleus, allowing more haemoglobin to be packed in cells so that each cell can carry more oxygen.

9. i) Platelets (Thrombocytes)

ii) Calcium, Ca^{2+}

iii) Fibrin.

10. a) Anemia/low blood volume/low haemoglobin leading to low oxygen, loss of nutrients and dehydration.

b) Blood clotting

c) Transfusion, taking fluids/eating iron in foodstuff/taking iron tablets.

11. a) - Thrombosis

- Arteriosclerosis

- Varicose veins

b) - Regulate body temperature

- Regulate pH of fluids

- Regulate osmotic pressure

12. a) Presence of valves

b) - Have biconcave shape to increase surface area for absorption of gases.

- Absence of nucleus and other organelles

- To increase packaging of haemoglobin.

- Presence of red pigment haemoglobin that has high affinity for oxygen.

13. - During birth

- Breast feeding

14. - Red blood cells have a biconcave shape, which increases the surface area for gaseous exchange. They have a thin plasma membrane, which allows rapid diffusion of gases. They contain haemoglobin, which readily combines with oxygen in areas of high oxygen tension (lungs) and releases it readily in areas of low oxygen tension (other body tissues). They have no organelles with whole internal space being filled with haemoglobin. They contain the enzyme carbonic anhydrase which help in the transport of carbon dioxide.

- Some white blood cells are phagocytic which enables them to engulf and destroy invading micro-organisms. They are also capable of amoeboid motion, which enables them to squeeze between cells of the capillary wall and into infected tissues where they proceed to engulf invading micro-organisms other white blood cells called lymphocytes are able to recognize antigens of invading micro-organism and to form antibodies against them.

- Platelets are able to aggregate at the site of a damaged blood vessel forming a temporary platelet plug which stops blood loss. They also produce the substance called thromboplastin which initiates the blood clotting mechanism.
- Plasma is composed mainly of water which is a solvent for a large variety of substance. This enables it to act as a medium for transport of a large number of water soluble substances. It has a high heat capacity that enables it to transport heat from highly active tissues to the rest of the body.

15. Blood: Tissues which consist of a liquid part called plasma in which several types of cells are suspended.

    Plasma: Liquid part of the blood

    Serum: Plasma from which the blood clotting protein called fibrinogen has been removed. It does not clot.

    Tissue fluid: Liquid part of blood without plasma proteins. It is derived from the blood by the process of ultra filtration.

    Lymph: is a tissue fluid, which drains into lymphatic vessels instead of going back into the blood vessels.

16. a) The patient's red blood cells have antigen A on their membrane and his

plasma has anti-b antibodies .

The donor's red blood cells have antigen B on their membrane and his plasma has anti-a antibodies. After transfusion, the anti-b antibodies in the patient's plasma reacted with B antigens on the donor's red blood cell membrane. This led to clumping together of the donor red blood cells a process called haemagglutination. This may have caused blockage of capillaries in a vital organ like the heart or brain leading to death.

b) i) A,B,AB,O

ii) He is universal recipient. His plasma' lacks antibodies.

17. Active immunity-that is produced when an animal's body reacts to an antigen by producing antibodies.

Passive immunity- Immunity that is produced when antibodies are transferred from one individual to another.

18. Antibodies formed against common cold viruses remain in the body and provide immunity for only a few days. Therefore, once a person has recovered from cold, he/she is only protected for a few days. Those antibodies formed against measles virus remain in the body and provide immunity throughout the person's life. Therefore, once a person has recovered from measles, he or she is protected for life.

19. PH of blood plasma is not altered homeostasis is maintained. Within the red blood cells, there is an enzyme (carbonic anhydrate) which help in fast loading/combination and offloading/dissociation of carbon dioxide.

20. Through tissues fluid, Oxygen and other food substance pass from the blood to the cells. Carbon dioxide waste substance passes from the cells to the blood through it.

CHAPTER 3

GASEOUS EXCHANGE

1. a)
   - Air enter into tracheal system through spiracles
   - It moves onto the tracheoles then moves on to the tips of tracheoles.
   - Air rich in oxygen dissolves in a fluid at the tip of the tracheoles. There is low concentration of oxygen in tissues as compared to the fluid.
   - Oxygen diffuses into the tissues due to concentration gradient. It is used in metabolic activities.
   - In tissues there is high carbon dioxide concentration than in the fluid in tracheoles.
   - Carbon dioxide diffuses from tissues into tracheole due to concentration gradient. It moves into trachea then out of the body through spiracles.

   b)
   - Water enters through the mouth when it opens its mouth. When it closes the floor is raised and water flows over the gills.
   - Oxygen diffuses into the gills blood capillaries while carbon dioxide diffuses from the blood capillaries along concentration gradient.
   - Flow of water and blood in gill filaments is by counter current flow.

2. a)
   - Large number of alveoli-increase surface area.

- Alveoli moist-dissolve diffusing gases.
- This walls- allow quick diffusion of gases
- Rich blood supply- transport oxygen and carbon dioxide.

b) i) Carbon dioxide diffuses into the cells. It moves in the plasma or red blood cells.

- Carbonic acid in plasma or carbamino haemoglobin in red blood cells or hydrogen carbonate.
- At the lungs hydrogen carbonate, carbonic acid and carbomino haemoglobin dissociates releasing cavity due to concentration gradient.

ii) Due to metabolic activities carbon dioxide is released from mesophyll cell. It diffuses into the intercellular spaces.

- Due to concentration gradient the gas diffuses into the sub-stomatal air spaces.
- When stomata open carbon dioxide is released into the atmosphere.

3. a) Carbon dioxide diffuses into the tracheoles then into the trachea and out into the atmosphere through spiracles.

b)
- Stomata.
- Lenticels
- Cuticle

4.
- To facilitate transportation of gases/exchange of gases i.e. oxygen and carbon dioxide.
- Create high concentration gradient.

5 a)
- External intercostals muscle contract while internal intercostals muscles relax.
- Diaphragm contract flattening. Volume in thoracic cavity
- Air rushes into the lungs.

b) Opening During the day photosynthesis takes place and sugar is formed in guard cells
- Osmotic pressure increases and water is drawn from neighbouring cells by Osmosis.
- The guard cells become turgid, bulge outward causing opening of stomata.

Closing During the night there is no photosynthesis and sugar is converted to starch.
- Osmotic pressure decrease and water is lost to the neighbouring cell osmosis.
- Guard cells become flaccid, closing the stomata.

6.
- Stomata
- Lenticels
- Cuticle

7.
- High number of stomata on the upper surface of the leaf.
- Absence of cuticle to allow diffusion of carbon dioxide and oxygen.

8. a)
- Pneumatophores
- Aerenchyma tissues
- Cuticle

b) - The diaphragm flattens.

- Volume in thoracic cavity increase.
- Pressure decreases compared to atmospheric pressure. Air rushes into the lungs through the nostrils.

9. a) K- Pleaural membranes

L - Alveolus

M- Intercostals muscles

b) - Has c-shaped cartilage rings that support it, preventing it from collapsing and allow free flow of air.

- Inner lining has mucus secreting cells that trap fine dust particles and micro-organisms.
- Inner lining has hair like structures called cilia that enhance upward movement of the mucus to the larynx.

c) Diffusion

d) Mycobacterium tuberculosis

10 - Highly folded to increase surface area.

- High network of blood capillaries
- Thin walled
- Moist

11. The trachea are strengthened by rings of cartilage which prevent them from collapsing.

12. - The epidermis of the root hair cells do not have cuticle and gaseous exchange takes place.

- When soil is water logged oxygen cannot diffuse into the root tissues hence no respiration. Metabolic activities stop leading to death.

13. - Air is cleaned by the cilia in nostrils
    - Controlled amount of air is taken in through nose
    - Individual is able to detect the smell of air breathed in.
14. - Spongy mesophyll cells are loosely packed allowing diffusion of gases.
    - Spongy mesophyll cells have a film of moisture on the surface to dissolve diffusing gases.
    - Large sub-stomatal air space in order to create high concentration gradient of diffusing gases.
    - Presence of stomata where gases enter or leave the leaf.
15. - Carbon dioxide
    - Water vapour
    - Oxygen
16. - Skin
    - Mouth
17. - Mammals –alveoli
    - Fish – gill filaments
    - Leaves – spongy mesophyll cells
    - Amoeba – cells membrane
18. Diffusion
19. Support the trachea and prevent it from collapsing when there is reduced pressure.

CHAPTER 4.

RESPIRATION

1. a) To derive off air or oxygen

   b) To avoid killing yeast/Denaturing enzymes in yeast

   c) To prevent air from getting into the yeast and glucose mixture.

   d) Lime water turn to white precipitate

   e) Use boiled yeast/glucose without yeast/yeast without glucose

2. - Lactic acid is toxic to tissues and must be removed from muscles to liver.

   - To increase supply of oxygen to tissues

3. a) Anaerobic respiration

   b) Brewing/Beer making

4. - Ethanol

   - Energy (ATP)

5. - Lactic acid

6. a) Adenosine triphosphate (ATP)

   b) i) Beer brewing/wine making

      ii) Baking using yeast.

7. - Have thin epithelium/wall to reduce distance of diffusion of the gases.

   - Moist to dissolve the diffusing gases

   - Highly folded to increase surface area for diffusion of gases.

   - Well supplied with blood or vascularized to help maintain high concentration gradient.

8. a) A mouse has high surface area to volume ratio and tends to lose heat

faster. It required more energy to replace it.

A dog has low surface area to volume ratio and lose less heat. Less energy is required to replace it

b) Lactic acid

9. a) i) Ethanol and carbon (IV) oxide.

ii) Lactic acid

b) It is the state when human body undergoes anaerobic respiration producing lactic acid. Oxygen has to be taken into the body to break the lactic acid.

10. a) Ratio of carbon dioxide produce to oxygen used up during breakdown of a food substrate.

b) $R.Q = \frac{CO_2 \text{ produced}}{O_2 \text{ used up}}$

$R.Q = {}^{102}/_{145}$

$R.Q = 0.7$

c) Fat/ Lipid

11.

Aerobic respiration	Photosynthesis
- Take place in both plants and animals - Takes place in all body cells - Takes place during the day and night - Oxygen is taken up while carbon dioxide is removed.	- Only takes place in plants. - Takes place in cells containing chloroplast - Takes place during the day only. Carbon dioxide used up while oxygen is given off.

12. a) Mitochondrion

b) A - Outer membrane

B - Inner membrane

C - Matrix

D - Cristae

c) Increase surface area over which respiration takes place:

d) ATP

CHAPTER 5

EXCRETION AND HOMEOSTASIS

1. Pancreatic juice containing digestive enzyme is prevented from reaching food. Insulin (and glucagons), which regulates sugar, is released directly into the blood stream.

2. a) Heat from the body metabolism is not lost to the surrounding through sweating because evaporation of sweat will be low; as air is already saturated with moisture.

   b) Hypothalamus

3. a) Sweat produces does not evaporate due to high humidity and the body does not cool, hence more sweat produces leading to accumulation

   b) Hypothalamus

4. - Elimination of uric acid requires less water than ammonia, hence (more) water is conserved.
   - Uric acid is less toxic than ammonia hence safer to excrete where there is less water.

5. a) Regulation of blood sugar lowers blood sugar level/controls the conversion of blood sugar to glycogen/maintain correct blood sugar level (90-100mg/100cc of blood)

   b) Controls the absorption of water in the kidney (tubules) nephron/regulation of water in the body/osmotic pressure in the blood.

6. More water will enter the amoeba (by osmosis) rate of water discharge by contractile vacuole will increase. Contractile vacuoles will be formed to discharge the excess water.

7. i) Proteins/plasma; protein/fibrinogen; albumin, globulin, prothrombin.
   ii) Blood cells, RBC/white blood cells/Platelets.

8. - Tests/React/Boil urine with Benedicts/Fehlings: positive results/Orange red precipitate is an indication of the disease diabetes mellitus.
   - Brick red instead of orange, use of Benedict's solution with boiling/heating.

9. After vigorous activity when blood glucose falls below normal.

10. a) Diabetes insipidus
    b) Anti-diuretic Hormone/ADH/ vasopressin

11. Maintenance of constant level of water, salts, osmotic pressure for optimum conditions for metabolism, suitable condition for cellular functions.

12. Converted into fats and stored as adipose tissue.

13. a) - Most enzymes in the body function with a narrow range of temperature
    - High temperature denatures enzymes
    - Low temperature inactivates/inhibit enzymes
    b) Sugar is a raw material for respiration therefore less sugar leads to low rate of respiration hence less energy available to the body/low rate of metabolism.

14. a) Heat loss by conduction/convection from the blood vessels, the skin enters general circulation cooling the body.

b) Vasoconstriction, thus less blood flowing to the skin surface thus reducing heat loss. Sweating ceases. Heat produced by shivering through metabolism is retained in the body.

15. a) Sebum

b) - Cooling the body when water content evaporates.

- Excrete excess salts, lactic acid and urea.

16. - Regulates the blood sugar level in the body by converting glucose into glycogen.

17. - Adhesion- force of attraction between unlike molecules

- Due to the force of adhesion water tends to stick to the walls of vessels containing it
- Cohesion – forces of attraction between like molecules.
- Cohesion between water molecules prevents the water column from breaking.
- Root pressure-due to pressure generated by the root's endodermis.
- Capillary due to narrowness of xylem
- Transpiration pull-As water evaporates from the leaf's surface, more is absorbed.
- After the water reaches the leaves cells, it passes the cells by osmosis from the xylem.
- Water vapour diffuses out through stomata.

18. a) i) Maintenance of a constant internal environment of cells.

ii) Regulation of the concentration of water and salts in the body fluid.

b) - Insulin - Glucagon

19. a) - The amino acids are broken into amino group (NH_2) and carboxyl group

(COOH). The amino group combines with hydrogen forming highly toxic ammonia. It immediately carbines with carbon (IV) oxide forming urea that is less toxic.

- The carboxyl group are converted to carbohydrates and then oxidized or converted into neutral fats and deposited on parts of the human.

b) - Bowman's capsule
- Proximal convoluted tubule
- Distal convoluted tubule

c) i) Less water reabsorbed in the blood stream and dilute urine is produced.
ii) Diabetes insipidus

20. a) Excretion is the removal of metabolic waste products from the body of an organism.
b) Secretion is the removal of a substance from a cell where it is formed and its transfer to another part of the body where it serves a useful function
c) Egestion is the removal of undigested food material from the body of an organism.

21. Blood cells and plasma proteins

22. a) Ultra filtration
b) Selective reabsorption
c) Because the pores in the glomerular capillaries are too small for plasma protein to pass through.
d) Blood cells

e) Most of the water in the glomerular filtrate is reabsorbed by the urine is formed whereas very little urea is reabsorbed.

23. As moisture from the urine or saliva evaporates from the surface of the skin, it reabsorbs latent heat of vaporization from the body thus cooling it.

24. Being exothermic, fish do not spend any part of their food intake in the maintenance of body temperature. This is unlike the case with mammals which spend a significant part of their food on temperature maintenance. Therefore fish are able to spend more of their food intake on growth.

25. During hot dry weather, the humidity difference between the surface of the skin and atmospheric between air is high. Under such conditions, sweat evaporates easily from the skin surface. This cools the body due to absorption of latent heat of vaporization. When the weather is hot and humid the humidity difference between the surface of the skin and atmospheric air is low. Evaporation of sweat takes place slowly with the result that sweat accumulates on the person's skin. Therefore the cooling effect of sweat on the body is greatly reduced.

26. Negative feedback refers to a regulatory mechanism whereby a deviation of the entity being regulated above or below the normal range triggers a sequence of event to bring it back to normal.

27. a) A - Hepatic artery

B - Hepatic portal vein

C - Hepatic vein

b) i) B

ii) B

iii) C

iv) A

v) C

c) During fasting there is no glucose from the alimentary canal making glucose concentration in vessel B low. Vessel C obtains glucose derived from the hydrolysis of glycogen in the liver.

FORM 3 WORK

CHAPTER 1 – CLASSIFICATION II

1. - Food spoilage
   - Food poisoning
   - Cause disease

2. a) A- Sorus

   B- Rhizomes

   b) Pteridophyta

3. Arthropoda

4. When they interbreed freely giving rise to a viable/fertile offspring.

5. Arachnida

6.

	Organism	Reason
Insecta	-Praying Mantis -Tsetse fly	- 3 body parts - 3 pairs of legs
	-Centipede -Millipede	-Many segments -Many legs
	-Tick -Spider	-2 body parts -4 pairs of legs

7. - Presence of rhizoids
   - Absence of vascular tissues
   - Body parts not differentiated into roots, stem and leaves

8. - Brewing industry

- Baking of bread
- Manufacture of medicine/antibiotics
- Source of food
- Manufacture of vitamin K and B12

9. Interbreed to produce fertile/viable offspring
10. Cephalothorax; prosona.
11. Chordata
12. Class insecta
13. Arachnida
14. a)
    - Fungi
    - Saprophytic bacteria

    b)
    - Refrigeration
    - Very low temperature inactivates the organism and metabolic activities are very low and they do not reproduce
    - Cooking –High temperatures kill the micro-organism and they cannot reproduce
    - Preservatives – create unsuitable acidic media in which micro-organisms cannot grow.
    - Salting – Create high osmotic pressure and micro-organisms become dehydrated.
15. a)
    - They are closed circulatory system
    - They are homoeothermic
    - Both use lungs for gaseous exchange

b) - They have mammary glands

- Skin covered with fur or hair
- They have diaphragm separating thoracic and abdominal cavities

16. - Have notochord in embryonic stage

- Have endoskeleton

17. a) A-Capsule B- Rhizoids

b) Division Bryophyta

c) Gametophyte

d) - Vascular tissues absent

- Body not differentiated into roots, leaves or stem.
- Display alternation of generations.

18. - Number of body parts

- Number of appendages
- Presence of wings

19. a) - Algae have chlorophyll but fungi do not have.

- Algae are single celled while fungi are multicellular.

b) - Source of food for aquatic animals

- Manufacture of gels and paints

20. - Source of agar used in cultivating micro-organism

- Manufacture of gels and paints
- Source of agar used in cultivating micro-organisms.

21. a) The spore producing structure (asexually) gives rise to the gamete producing structure (sexual) and they alternate.

b) Division bryophyta

Division pteridophyta

22. a) - Segmented bodies

- Jointed appendages

- Exoskeleton

- Body divided into parts

b) i) Second name should be in small letter. The names should be underlined.

ii) Tuberrasum

c) Division pteridophyta

23. Class diplopoda

24. - Number of body parts

- Number of legs

- Number of wings

- Number of antennae

25. Binary fission

CHAPTER 2

ECOLOGY

1. - May kill soil micro-organism that decompose humus to release mineral salts
   - Soil structure interfered with encouraging soil erosion.
2. - Drought/food shortage /overgrazing
   - Fire
   - Emigration
3. a) The fish were caught, their age determined and the 2 year olds were retained and their length measured and recorded.
   - This was done repeatedly until a large number were measure; calculation was done by dividing the total length of all fish by the total number of fish.

   b) Lake A has hard water with more calcium while Lake D has soft water with no calcium. Calcium is necessary for bone formation. Fish in Lake A grow faster and greater bone length than fish in Lake D. Lake C has more food which fish eat than Lake D.

   c) Lakes C and D have little or no calcium which is necessary for the formation of the shell in snails.

   d) i) Light temperature, carbon dioxide concentration, oxygen concentration, PH and salinity.

   ii) Light-Affects the rate of photosynthesis.

   Temperature- Affects enzyme activities hence photosynthesis.

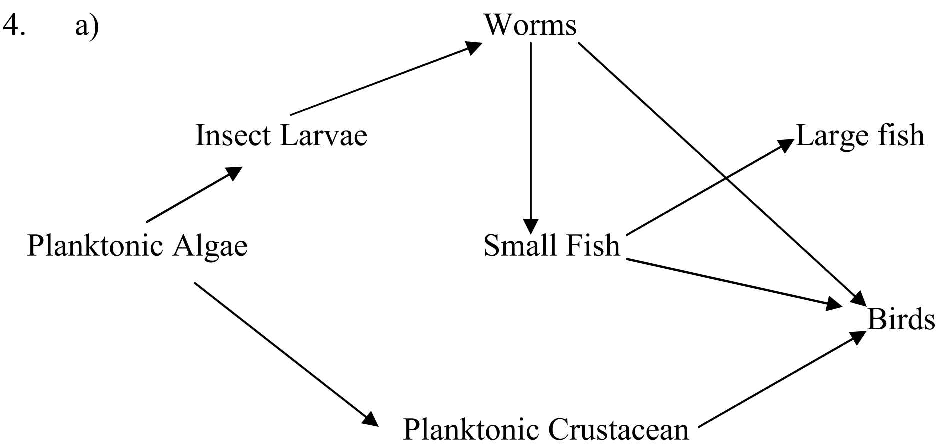
   CO_2 concentration – Determines rate of photosynthesis.

O_2 Concentration- affects rate of respiration.

Salinity – Osmoregulation in plants and animals.

PH – Affect enzyme activities.

4. a)



b) i) Planktonic algae → Planktonic crustacean → Birds

ii) Planktonic algae → Planktonic crustacean → Small fish → Large fish

c) Producers must always have a higher biomass than consumers because they support the consumers which are at higher trophic levels

d) i) Pollution – herbicides and pesticides, over fishing, bird hunting

ii) Herbicides and pesticides – kill insect and planktonic algae reducing their number.

- Fishing increases planktonic crustaceans and insect larvae leading to over consumption of algae depleting them.
- Decrease in number of insect larva.

5. i) A- Desert, arid and semi-arid

B- Aquatic, marshy land

6. a) i) 1968

ii) Hot water, sewage and industrial waste.

iii) High temperature reduces dissolved oxygen causing suffocation of fish.

- Sewage leads to eutrophication reducing oxygen concentration and reduces penetration of light.
- Industrial waste- toxic substance kills the organism.

b) i) A = 7512-20 = $^{7492}/_{5}$

= 1498.4 fish per year

B = 617- 23 = $^{594}/_{5}$

= 118.8 fish per year

Difference = 1379.6 fish per year

ii) - Reproductive rate
- Competition
- Predation
- Sex ratio

c) i) Capture/recapture method

ii) Marks may disappear

During marking fish may be killed

Predation/death may interfere.

7. The pollutants may be absorbed by aquatic plants which in turn may be eaten by fishes. The pollutants therefore get into man through the food chain.

8. - Improving sanitation to prevent infection by parasite.
- Insecticide to kill vector like mosquitoes or tsetse fly.

- Avoid indiscriminate sexual intercourse to prevent spread of parasites.

9. a) Green plants → Grasshoppers → Lizards → Snakes

   Green plants → Grasshoppers → Lizards → Cats

   Green plants → Mice → Snakes → Hawk

   Green plants → Mice → Snakes → Cats

   b) Mice

   c) Lizard, cats, hawk, snakes.

   d) Most plants will die

   Some organisms will starve and die

   Some organisms may migrate.

10. a) Population size

   $$N = \frac{374 \times 400}{80} = 1870$$

   b) There was even distribution of crabs; No movement in and out of lagoon/on migration. There was random distribution of crabs after first capture

   c) Capture, mark, release then recapture/capture/recapture method.

.

11. a) Grass → Grasshopper → Guinea fowls

   Grass → Termite → Guinea fowls

   b) Lions would complete with leopards

   Gazelle number would reduce

   Grass would be increased

c) Grass

12. a) E -Denitrifying bacteria

J - Nitrifying bacteria

b) F- Nitrogen fixation

H-Decomposition

c) G-Plants

13. a) Community – it is the total number of plants and animals living together in an area.

- Population- total number of organisms of a given species occupying an area at a certain trophic level

b) - Use of net to capture the grasshopper

- Which are then counted and marked. They are then released.
- Total number of grasshoppers is determined by multiplying the grasshoppers captured second time by those captured and marked first time.
- The sum is divided by number of grasshoppers marked in second capture.

14 - During manufacture of sulphuric acid and nitric acids oxide of sulphur and nitrogen are released into air causing acid rain.

- Motor vehicle exhaust fumes release carbon monoxide a respiratory poison.
- Combustion of fuels and coal increase concentration of carbon dioxide creating greenhouse effect.

- Aerosols containing CFC in herbicides and perfumes deplete the ozone layer.
- Smoke from factories mix with fog forming smog which reduces visibility.
- Exhaust fumes from vehicles contains lead from leaded petrol that poisons the body.
- Deforestation exposes top soil to air currents encouraging sheet erosion
- Leaded petrol that poisons the body.
- Loud noise from factories, aeroplanes and Jua Kali workshops can lead to poor hearing ability.
- Radio active emissions can lead to mutations.

15. Curved sharp hooked strong beaks for killing or tearing flesh from bones.
    Curved strong sharp claws for holding prey.
16. a) Crop- potato/tomato
    b) Disease- potato blight/tomatoes rot
    c)
    - Use of fungicides
    - Uprooting and burning infected plants
    - Crop rotation
    - Use biological control
    - Use disease resistant varieties
17. Cattle are mainly grazers while most wild animals are browsers.
18. a) i) Study of a single species within a community or ecosystem.

ii) Study of different species of organisms in a natural community in an ecosystem.

b) A-Aquatic/Fresh water

B- Forest

C- Arid- Semi Arid

c) - Sunken stomata,

- Reversed rhythm

- Small stomatal pores

19. Entamoeba historical

20. a) Photosynthesis

b) Heterotrophic

c) Aquatic (pond) and terrestrial (forest)

d) Algae → Zooplankton → Small fish → Bird J → Large bird

e) - Number of snails would increase

- Green plants would decrease.

- Bird M would increase

f) In a lower trophic level energy is lost through respiration, excretion and death of some organisms.

g) i) - Vultures

- Decomposers

ii) Vultures- control population of the large birds.

Decomposers- cause decay of dead organisms recycling the nutrients.

h) i)
- Deforestation
- Bird hunting
- Over fishing

ii) Removal of the trees destroys the habitat for birds, they therefore migrate.
- Bird hunting kills the birds reducing their number and increasing small fish, mussels and snails.
- Over fishing reduces the number of small fish increasing zooplankton and reducing the algae

21. - Cells have large air spaces between them to enhance buoyancy.
- Cells are air filled reducing their density.

22. **Ligh**t- high light intensity increase the rate of photosynthesis

Temperature- low temperatures lower metabolic activities while moderate temperature increase metabolic activities. High temperature increases transpiration.

Wind- Strong air current increase rate of transpiration and deforms the plants according to direction of the wind.

Atmosphere pressure-High pressures decrease the rate of transpiration and also reduces rate of photosynthesis.

Ph value-some plants thrive well in acid soils while others thrive better in alkaline soils.

Radioactive radiations- Cause mutations of the offspring

Oxides of sulphur and nitrogen-cause acid rain that corrodes plant leaves.

23. In different continents, regions with similar climatic conditions and lie in the same latitudes have plants and animals not identical.

24. a) Decomposers – Cause breakdown of organic matter enhancing recycling of nutrients

b) Predation - The organism feeds on whole or part of another organism and therefore control their population.

25. X- Denitrification

Y- Animals

Z- Nitrification

26. Offspring (brown fur) = $^2/_4$ x 100 = 50%

a) Pyramid of number Way numbers of individuals occurring at each trophic levels of a food chain may be diagrammatically represented.

Pyramid of biomass The way the total amount of living matter occurring at each trophic level of food chain may be diagrammatically represented.

b) Loss through excretion e.g. egestion

- Heat
- Respiration

c) Two parallel strings are laid down over a determined length and width within study area.

- No. of organisms in belt transect are counted.
- Area transect is worked out.

- Number of organisms per unit area is worked out.

27. a) Population – It is all members of a given species in a particular habitat at a particular time.

Community – All organisms belonging to different species that interact in the same habitat.

b) i) Capture and recapture method.

ii) Line transect

28. - Produce large number of eggs for increased survival.
- Produce enzymes to digest human skin when penetrating.
- Can withstand low oxygen concentration.
- Have hook-like structures to attach to the intestinal walls.

29. - It is addition of substances into water that may cause harm to organisms and are destructive to the ecosystem.

The causes of water pollution include;

- Industrial effluents that may be toxic chemicals which may kill the aquatic organisms. It can be controlled by treating the affluent before discharging them.
- Hot water that reduces concentration of oxygen, killing the animals. It is controlled by placing high penalties on factories discharging hot water.
- Oil spillage from oil tankers that reduces oxygen in water, penetration of light intensity and clog feathers of marine birds. It can be controlled by regular servicing of oil tankers.
- Domestic effluents that include;

- Untreated sewage that causes water borne diseases. It can be controlled by treating sewage before being discharged.
- Detergents that cause eutrophication causing reduced oxygen concentration. It is controlled by banning phosphate based detergents.

- Agricultural effluents that include;
  - Pesticides and herbicides that have heavy metals that they may gammulates along the food chain killing the higher animals. It is controlled by banning phosphate based detergents.
  - Inorganic fertilizers that have nitrates and sulphates that cause eutrophication. It is controlled by use of organic fertilizers.
- Silting due to soil erosion that reduces penetration of light to the plants and clog respiratory surfaces of animals. It is controlled by proper methods of soil erosion control and proper farming methods.

30. a) It is use of natural predator to kill a prey e.g pest instead of use of pesticide.
    b) The aphids are pest found in plants. The ladybirds can be used to control the aphids as they feed on them but not destroy the plants
    c) Prey is the source of food for predators. If the number of prey is smaller than the predators they would be depleted.

31. a) Antelopes are grazers while giraffes are browsers.
    Antelopes have brown fur being camouflaged by the colour of grass while giraffes are camouflaged by the trees.

b) Trees are camouflage against the herbivores preventing them from being spotted by predators. In open grassland herbivores are easily spotted.

c) The population will first increase leading to competition of resource e.g. food or mates. This causes death of the weak herbivores or migrations to new habitats.

32. - As it gets deeper light penetration decreases reducing rate of photosynthesis hence less productivity.
- As it gets deeper carbon dioxide concentration decrease hence reducing rate of photosynthesis hence less productivity

33. a) Plant protein

b) X-Nitrification

Y-Nitrogen fixation

Z- Dentrification

c) Proteins

34. a) i) 25-10 = 15 birds

ii) 20-5 = 15 birds

b) i) The number of species in forest are more than in the number in the savannah hence higher change.

Fruits more abundant in forest than in savannah

Selectively reduces with forest birds because they are many and competition is stiffer than savannah

ii) Seeds more abundant in savannah than in forest-they they are more exposed but seeds in forest plants are inside the fruit.

Birds in savannah are less selective than forest birds.

c) i) B

ii) - Emigration in big numbers

- High death rate during unsuitable condition and disease.
- Predation increase due to attraction of predator due to their high number.

d) - Bush fire –avoid lighting fires
- Eliminating all predators of one herbivore
- Limited predator to maintain high biological control.
- Felling trees- replanting trees
- Having high concentration of industries that provides that cause acid rain. Use of fuels that do not produce the oxides.

CHAPTER 3

REPRODUCTION IN PLANTS AND ANIMALS

1. Prophase

2. Integuments, triploid nucleus

3. - Blood entering placenta has more oxygen, more food substances, less nitrogenous wastes and less carbon dioxide.
   - Blood leaving placenta has less oxygen, less food substance, more carbon dioxide and nitrogenous wastes.

4. Corpus luteum in the ovary secretes progesterone, which maintains pregnancy/development of uterus after four months pregnancy is maintained by progesterone form placenta.

5 - Protandry / protogyny / male and female parts mature at different times
  - Stigma positioned higher than stamen
  - Incompatibility /sterility.

6 - Presence of special structures that attract agents of pollination.
  - Protandry /protogyny

7. - To increase the chances of fertilization and survival of species.

8. a) Wind
   b) To enable it trap pollen grains in the air.

9. - Blood transfusion
   - Use of unsterilized instruments / sharing (contaminated) instruments.
   - Infected mother to foetus; infected mother to newborn.

10. Bring about change or genetic material, which leads to variation that enables organisms to exploit new environment resistance to disease.

11. - Lack of variations
    - Lack of hybrid vigour
    - Disadvantageous traits are retained within species.

12. a) Meiosis

    b) Ovary

    c) n- gametes

    2n- parents

13. a) i) Conditions where other floral parts arise / positioned above the ovary / inferior ovary.

    ii) Male flower

    b) - Large anthers loosely attached to the filament to be easily shaken in the wind.

    - Small / smooth / light pollen grains – easily carried by wind.

14. a) Ovule

    b) Ovary

15. a) - Sister Chromates separates.

    - Chromatids start moving to opposite poles with centromere first.

    b) - Ensure that gametes formed have half the number of chromosomes found in original cell.

    - Formation of sex cells.

    - Leads to variation of genetic material during crossover.

16. a) i) Protoandry Stamens with pollen grains matures before carpel (stigma)

of the same flower.

ii) Self-sterility Pollen grain of anthers cannot grow into pollen tube on the stigma of the same flower.

b) - Mixing of genetic composition of different plants.
- Offspring produced has high yield.
- Offspring is more resistant to disease and adverse conditions.

17. a) Amnion

b) i) – Umblical vein
- Umblical artery

ii) – Umblical vein – rich in nutrients and oxygen.
- Umblical artery – rich in CO_2 and waste like urea.

c) - Has thin membrane to reduce diffusion distance.
- Has villi which increase surface area for exchange.
- Highly vascularized.

d) - Cushions foetus against shock
- Supports the foetus
- Keeps foetus moist (prevent dehydration)

18 a) i) Anaphase I

ii) - Homologous chromosomes separate at the equator.
- Chromosomes start migrating to opposite poles
- Sister chromatids attached at the centromere.

b) Spindle fibers.

19. - Harmful characteristics from the parents may be passed on to the off springs.

- Takes a longer time
- Few offsprings are produced at a time.

20. a) i) Protandry — Stamens mature and pollen grains are shed off before the stigma matures.

ii) Self sterility — Pollen grains from the anthers cannot grow on the stigma of the same flower or plant.

b) i) Q- Antipodal cells

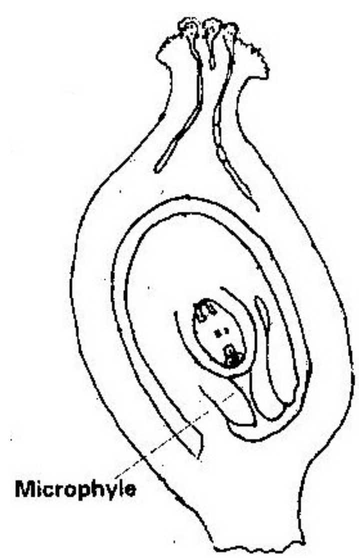
R- Polar body / polar nucleus

S- Egg cell

ii) Path through which the male gametes reach the embryo sac to enhance fertilization.

iii) Prevent other pollen grains from developing into pollen tubes hence no multiple fertilization of embryo sac.

c)



21. a) i) Large brightly coloured corolla / inflorescence / forests / tracts to attract insects.

ii) Scented to attract insects

iii) Have secreted nectar to attract that direct flowers secrete nectar to attract insects.

iv) Pollen grains rough/spiky sticky surface to stick on insect's body.

v) Special shaped corolla tube to enable the insect land.

b) i) Repair / heal endometrium / wall of uterus, which is destroyed in menstruation. Stimulates pituitary gland to produce the luteinising hormone

ii) Stimulates the thickening of the uterus, increases the blood supply to the endometrium. Inhibits the production of follicle stimulating hormone

iii) Responsible for maturation of the graafian follicle / causes ovulation. Stimulates corpus luteum to secrete progesterone.

22.
- Interior lobe of pituitary glands secretes follicle stimulating hormones (FSH). FSH causes Graafian follicle to develop in the ovary. It also stimulates tissues of ovary / all of graafian follicle to secrete oestrogen.
- Oestrogen causes repair /healing of uterine wall; oestrogen stimulates interior lobe of pituitary to produce Luteinsing Hormone which causes ovulation. It also causes graafian follicle to change into corpus luteum and stimulates corpus luteum to secrete progesterone.

- Progesterone causes proliferation of uterine wall in preparation for implementation.

  Oestrogen/progesterone inhibits the production of FSH by interior lobe of pituitary thus no more follicles develops and production reduces.
- In the next two weeks, progesterone level lowers and inhibits production of LH from interior lobe of pituitary.
- The corpus luteum stops secreting progesterone and menstruation occur when the level of progesterone drops. Interior lobe of pituitary start secreting FSH again.

23. i) It forms a large surface area for the diffusion of nutrient from the maternal blood to the foetal blood. Glucose, amino acids and salts are transferred.

    ii) The placenta isolates the foetus from the higher blood pressure of the mother and from direct connection of the two blood systems. Excretion materials can easily pass from foetus to mother.

CHAPTER 4

GROWTH AND DEVELOPMENT

1. IAA /auxins produced by terminal bud; inhibits growth of lateral buds, when cut the suppression cease thus auxiliary buds sprouts.
2. Food stored is used in (mobilized) up for respiration and growth.
3. - They promote cell division
   - Promote fruit formation without fertilization/ parthenocarpy.
4. a) Oxygen is necessary for germination

   b) Germination in B, no germination in A.
5. The adult and larvae exploit different food riches; do not compete for food.
6. Endosperm material was converted into new cytoplasm/ the stored food endosperm is used up to the germination seed while the embryo is growing and adding on more protoplasm.
7. a) Condition necessary for the germination of seed /to show that water, oxygen and warmth are needed for germination.

   b) To absorb all oxygen from the jar

   c) C- to show water is needed for germination of seeds.

   d) Jar A – seeds would not germinate

   Jar B – seeds would has germinated

   e) i) Scarification i.e. scratching to make impermeable seed coat permeable

   ii) Varnilasation – Cold treatment e.g. species of wheat.
8. a) Apical bud produce auxins which inhibits the development of lateral buds.

Removal of terminal buds cause the growth and development and sprouting of lateral buds.

b) The pruning of coffee/tea.

c) More yield /production

9. a) Low oxygen and increase in CO_2

b) Germinating seeds respire using O_2 and release CO_2 only.

c) Absence of light, impermeability of seed coat to water, immature embryo, lack of growth hormones presence of inhibitors.

10.

- Epigeal germination – Epicotyle grows very fast pushing out of soil surface with the cotyleons.

Hypogeal germination – Epicotyle grows very fast and plumule grows out forming first foliage leaves cotyledons remain underground.

11. a) Graph

b) i) 68 ± 1

ii) 130mm

c) Shoot A- Removal of apical bud promotes growth of lateral buds, due to removal of auxins hormones which inhibit lateral bud development.

Shoot B- Gibberellic acid promotes growth of lateral branches

Shoot C- Presence of apical bud inhibit lateral bud development due to reserve of auxins. This is called apical dominance.

d) As a control experiment to show the effect of hormones (auxins) on lateral bud development.

e) - Promotes flowering.
- Promote lateral bud development hence increase yields.
- Break seed dormancy (promote germination)

f) - Germination
- Flowering
- Activate hydrolytic enzymes

12. a) - Absence of water (moisture)
- Unsuitable temperature.
- Lack of oxygen
- Lack of light

b) Hypocotyls

13. a) - Increase in dry mass
- Increase in cell number
- Irreversible increase in volume of cytoplast
- Increase in differentiation.

b) i) Light intensity influence rate of photosynthesis.
ii) Temperature – influence metabolic rate via enzyme action.

c)

Name of hormone	Site of hormone production	Effect
Thyroxin	Thyroid gland	Control basal metabolic rate
Follicle stimulating hormone	Anterior pituitary gland	Maturation of Graafin follicle
Auxins	Stem of apex Root apex	Cell elongation
Gibberellins	All young plant tissues	Stimulates cell growth

14. a) It indicated the amount of organic material present which is a measure of change in mass cytoplasm.
    b) Weigh, reheat at 110^0C for several hours, and cool constant Mass.
    c) Most of mass is starch which is converted to sugars and used up in respiration and other metabolic activities.
    d) Cellulose is synthesized during growth of new cell walls.
    e) Starch → Glucose → Cellulose

FORM 4 WORK

CHAPTER 1 –

GENETICS

1. a) - Ribonucleic acid

   - Has the base 'U' uracil

   b) -G- C –A – G

2. RNA | DNA

	RNA	DNA
a)	Has ribose sugar	Deoxyribose sugar
b)	Has uracil as one	Has thyamine of its bases
c)	Single strand	Double strand

3. Haemophilia (sickle cell anaemia)

4. Controls / regulates enzyme / synthesis for the material for inheritance.

5. Gametes form new offspring

6. Co-dominance / incomplete dominance

7. - High yielding Hybrid vigour
   - Resistance to disease, early maturity.

   Resistance to drought early maturity.

8. Y Chromosome – Hairy pinna, tuft and hair sprouting from the pinna, baldness.

   X Chromosome – Colour blindness; Haemophilia

9. a) Smooth seed coat is dominant to wrinkled seed coat. Let R represent gene for smooth and parental genotype RR x rr

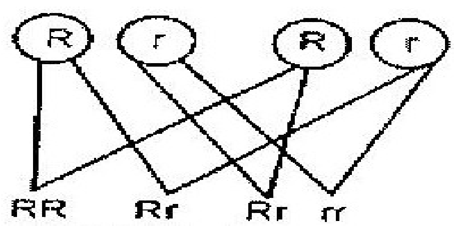
R R r r

	R	R
R	Rr	Rr
R	Rr	Rr

All F1 are ………………….Rr

Parental genotypes Rr X Rr

Gamete



F2 genotypes

i) Genotypic ratio 1RR : 2Rr: 1rr

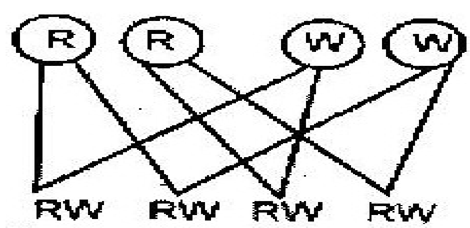
ii) Phenotype ratio 3 smooth: 1 wrinkled

iii) Wrinkled number ¼ x 7324 = 1831

10. a)

Parental genotypes RR x WW

Gametes



F1 genotypes

b) 1:2:1 for ratio, 1 white: 2 pink: 1 red

c) Co-dominance / incomplete dominance / partial dominance / equal dominance.

11. a) White

- Fewer number / lower rates / absence of white in parent and its presence in offspring.

b) Heterozygous/ Rr

c) Homozygous / rr / double recessive.

12 a) i) Haemophilia

ii) Sickle cell anaemia

iii) Colour blindness

iv) Leukemia

v) Albinism

b) i) Inversion- A result of a chromosomal break up and rejoining with the middle piece turned by 180^0

ii) Translocation- A section of chromosome breaking and joining a homologous chromosome.

c) Phenotype: Black mice x Brown mice

Genotype	Bb	x	bb	
Gametes	B	b	b	b
F1 genotype	Bb	Bb	bb	bb
Phenotype	2 black		:	2 brown

13. a) It is alternative form of a chromosome been similar in structure but may have different composition.

b) i) Occurs when the nucleotides of a gene break off and disappear

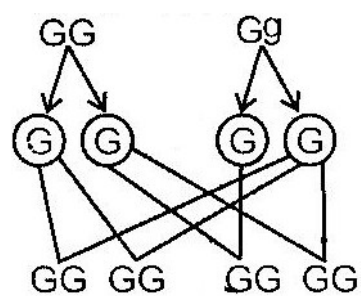
ii) Occurs when the nucleotides of a part of a gene become inverted by taking a 180^0 turn.

c) Testing the genotype of an individual by crossing with the recessive trait.

14. a) i) Parents Homozygous x Heterozygous phenotype purple grains purple grains

Genotype

Gametes



1st filial generation

(Off springs)

- The genotype ratio:
- 2 homozygous purple coloured grains
- 2 heterozygous purple coloured grains

ii) All purple coloured grained maize plants maize plants.

b) Deliberate modification of a characteristics are of an organism by manipulates genes and DNA by transferring genes from one organism to another.

c) It is when best characteristics are developed from both parents and offspring better than either parent.

15. i) Alleles are alternative forms of the same gene which control the inheritance of contrasting features of the same position in homologous chromosomes.

ii) Genotype is the genetic makeup or composition of an organism.

iii) Phenotype is the outward appearance of an organism with reference to a particular trait.

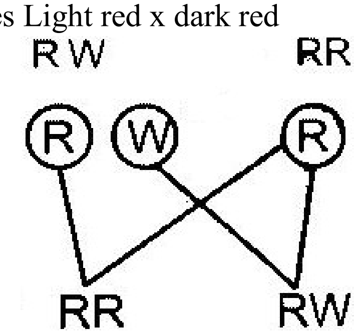
16. a) The genes for dark red colour and white colour are co-dominant. Since the calf is heterozygous it gets a coat colour that is intermediate between dark red and white

b) Let R represent the gene for dark red coat colour and W the gene for white coat colour. The light red bull must be heterozygous (RW) and the dark red cow must be homozygous (RR)

Parental phenotypes Light red x dark red

Parental genotype

Gametes



Offspring genotypes

Offspring phenotypes Dark Red Light red

17. a) Linked genes are those genes that are found in the same chromosomes. They are usually inherited together.

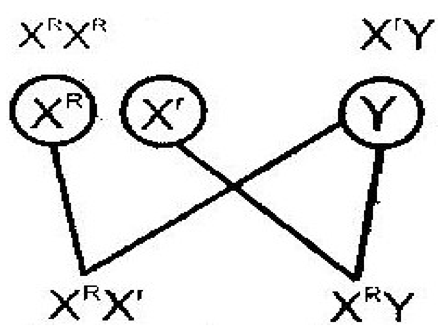
b) These observation show that eye colour in fruit flies is a sex-linked trait. Since it is well known that the Y chromosomes carries very few genes, we can assume that eye colour in fruit flies is an x-linked trait.

- When a true-breeding, red-eyed female is mated with a white-eyed male, all the offspring received an X-chromosome carrying the dominant gene from the mother. Because of this, they all develop red eyed. This is illustrated below.

Parental phenotypes Red eyed female x white eyed m

Parental genotype

Gametes



Offspring genotypes

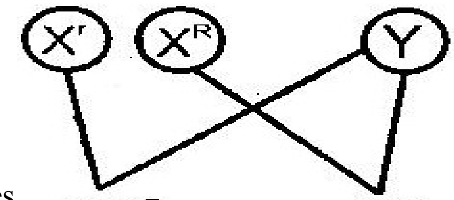
Offspring phenotypes	Red eyed females	Red eyed males

When a true breeding white eyed female is mated with a red-eyed male, the female offspring receive an X-chromosome carrying the dominant gene from the father and develop red eyed. The male offspring received an X-chromosome carrying the recessive gene from the mother. They also receive a Y chromosome contains no genes for eye colour the male offspring develop white eyes.

i) Parental phenotypes White eyed female x Red eyed male

Parental genotype X^rX^r X^RY

Gametes



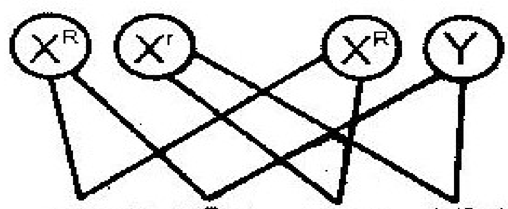
Offspring genotypes X^rX^R X^rY

Offspring phenotypes Red eyed females White eyed males

ii) Parental phenotypes red eyed female x white eyed male

Parental genotype X^RX^r X^RY

Gametes



Offspring genotypes X^RX^R X^RY X^RX^r X^rY

Offspring phenotypes Red eyed Females White eyed Males Red eyed females White eyed males

18. a) Variation refers to the difference in specific characteristics that exist between members of a species e.g. in humans, characteristics that exist between members of a species e.g. in human, characteristics such as height, blood group.

b) Discontinuous variation refers to the existence of two or more distinct forms between them e.g. pink or white flowers in pea plants

- Continuous variation refers to the existence of a characteristic in a continuous gradation between two extremes e.g. height, weight and fruit size in trees.

c) Genetic variation provides the raw materials for evolution by natural selection. It increases the chances of survival in an ever-changing environment. If a particular species is highly adapted to a specific habitat, it might find it difficult to survive in case there is a sudden change in the environmental conditions. In absence of variation, such a change may lead to death of all members have a variation that adapts them to the new conditions they will survive. They will reproduce and multiply rapidly in the absence of competition from other forms. Since the variation is genetic, the adaptation is passed on to subsequent generations.

d) When DDT was first used in the 1950's it was very effective in killing mosquitoes. In areas where mosquito populations were previously large, their numbers were greatly reduced. However, in the 1960's the number of mosquitoes in these areas began to rise again despite the continued application of DDT.

- It then became evident that a few mutant forms had a variation that made them resistant to DDT. In the absence of competition from other forms, the resistant forms reproduced and multiplied very fast. The result was that the number of mosquitoes started to arise again despite the continued application of DDT.

Thus the presence of a variation in the few mutant forms saved mosquitoes from extinction.

19. a) They represent the bases guamine, thymine, cytosine, adenine

    b) It is a DNA strand because it contains the base thymine which is absent in RNA.

    c) C A A T C G A C T

    d) C A A T C G A C U

20. Human females produce one type of egg containing an X chromosome. Males produce two types of sperms; half contains a X-chromosome and the other half contains a Y chromosome.

    Fertilization of the egg by a sperm carrying an X chromosome gives rise to a baby girl.

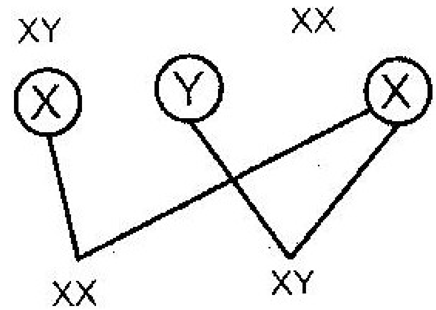
    Fertilization of the egg by a sperm carrying a Y-chromosome gives rise to a baby boy.

Parental phenotypes Man x Woman

Parental genotype

Gametes

Offspring genotypes



Offspring phenotypes Girls Boys

CHAPTER 2

EVOLUTION

1. - Lamarckism- a character acquired in the life of an organism which is favourable in its adaptation to the environment in inherited.
   - Darwinian Theory – As a result of variations, some organisms become better adapted to the environment. They therefore survive better and mature giving rise to more adapted off springs.
   - The less adapted organisms die before maturity hence been eliminated from the environment.
   - The less adapted organisms die before maturity hence are eliminated from the environment.
2. Adaptive radiation / divergent evolution 1995
3. - Evidence does not support Lamarck' theory.
   - Acquired characteristics are not inherited/inherited characteristics are found in reproductive cells only.
4. Fossils/ (records), palaeontology, Geographical distribution, comparative anatomy/taxonomy; cell biology; comparative serology; comparative embryology; comparative immunology.
5. Assists to eliminate disadvantageous characteristics / perpetuate advantageous characteristics.
   - Allow better-adapted organisms to survive (adverse changes) in environment/less adapted organisms are eliminated by adverse changes in the environment.
6. a) Gives evidences of types of plants, animals organism that existed at certain

geological age/ long ago.

b) Gives evidence of relationship among organism / common ancestry of a group of organisms.

7. Nature selects those individuals who are sufficiently well adapted; rejects those that are poorly adapted

8. - For a new species to be formed, a population of organisms must become completely isolated or separated from the others; Over long period of time so that any new variation that rise will not therefore flow to other population.
   - Geographical isolation – this is due to physical barriers e.g. oceans / seas / deserts
   - Ecological Isolation- a barrier resulting form the occupation of different types of habitats from the original type.
   - For reasons of feeding/ predation / breeding as well as <u>environmental changes</u> (e.g. climate and vegetation which may result in population living in different habitats so becoming, ecologically separated from one another)
   - <u>Behavioural isolation</u> alteration in behaviour proceeding mating which include courtship behaviour / lack of attraction between males and females in different chemicals / pheromoes / coloration /songs e.t.c

   <u>Reproduction isolation</u>: a barrier to successful mating between individuals of population; due to structural differences in reproductive organs as well as failure in fertilization/ incompatibility.

   <u>Genetic isolation</u> – Even if fertilization takes place the zygote may be inferior / fails to develop; however if the zygote develops the offspring may be inferior or sterile.

9. a) It is the emergence of present forms of organisms gradually from pre-existing ones some of which no longer exists).

   b) It is the drifting apart of the continents from one land mass (Pangaea).

10.

   a) When organisms of the same origin become adapted (modified) in different ways in order to fit in the environment. The organisms are separated due to natural factors.

   b) When an organism is exposed to drug for sometime it becomes modified (adapted) to living in presence of the drug. The offspring produced therefore survive in presence of the drug. Hence drug resistant.

11. a) <u>Homologous structures</u> – structure / organs that have arisen from a common but they have assumed different functions

   b) <u>Analogous structures</u> – Structures/organs that have originated from different ancestors but they perform the same function.

12. a) <u>Natural selection</u> is a process where nature selects those organisms that are well adapted to the prevailing environmental conditions enabling them to survive to reproductive maturity. Those organisms that are poorly adapted die young leaving no offspring and their characteristics are eventually eliminated from the population.

   b) <u>Mutation</u> brings about new hereditary characteristics (or hereditary variation) in a species. Some of the new characteristics are favourable but others are unfavorable. Favourable characteristics enable the organism possessing them to compete better in the struggle for existence. The result is that most of them survive to adulthood and give rise to offspring of the next generation. Since characteristics resulting from mutation are inheritable they are passed on to new generation. On the other hand, only very few of those organisms with unfavorable characteristics survive to adulthood and give rise to young ones. The final result is that the favourable characteristics are propagated in the population giving rise to organisms that are better adapted to the environment. The unfavorable characteristics are gradually weeded out and may eventually get eliminated from the population.

13.(a) A hybrid is an offspring of across between different varieties or breeds of the same species.

(b) Hybrid vigour refers to the improved qualities, such as increased yields, fertility, resistance to diseases and toughness seen in offspring of different.

15. The peppered moth usually on trunks and branches of tress, industrial cities, tree trunks and branches are normally dark in colour due to deposits of soot and other pollutants. A white moth resting on such a trunk or branches is highly conspicuous and is easily picked and eaten by preying birds. A dark moth resting at the same places is effectively camouflaged by the dark background and is not easily seen by preying birds.

- In rural areas, tree trunks and branches are normally white in colour due to growth of lichens. A white moth resting on such a trunk or branch is effectively camouflaged by the white background and is not easily seen by preying birds. A dark moth resting at the same place is highly conspicuous and is easily picked and eaten by praying birds.
- Therefore, dark moths are adapted for survival in industrial areas. Here most of them reach maturity and reproduce more dark moths. On the other hand, only a few white moths survive to maturity and reproduce in industrial areas. In rural areas, most white moth survives to maturity and reproduces more white moths. Here only a few dark moths survive to maturity and reproduce.

16. (a) Special creation is a concept which proposes that all living things were made by God at a specific time and have remained unchanged since.
 (b) Organic evolution is a concept which proposes that all living things arose from a few ancient simpler forms through gradual modification.

CHAPTER 3

RECEPTION, RESPONSE & CO- ORDINATION

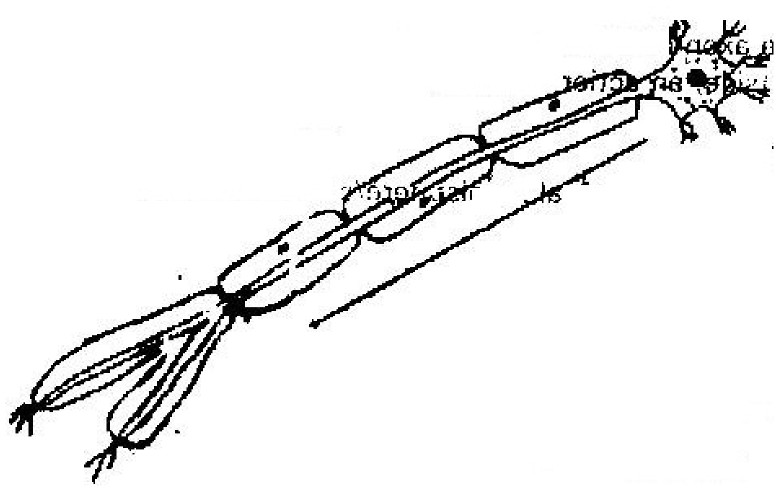
1. **Structural difference**- the cell body in motor neurone is terminal (at the end) and inside the central is nervous system. While the cell body in sensory neuron is not terminal but has axon on both end i.e. bipolar.

   Functional differences – motor neurone carries impulses from CNS to the effectors i.e. muscles, while sensory neurons carry impulse for receptor to CNS.

2.

Hormone	Site of production	Function
Oestrogen	Ovary	Initiate and control development of secondary sexual characteristics
Aldosterone	Aldernal gland	Mineral reabsorption

3. (i)



(ii) P- Protection/ insulation

Q- Impulses transmitted/depolarization is faster.

4. (a)

Adaptation	Function
Conjunctiva - An epithelium colourless	Protects eyeball
Cornea: Transparent/curved	Allow light/ refract light entering the eye
Aqueous/vitreous: clear	Allow light to pass through/ refract light. Maintain the shape of the eye
Iris. Opaque and contractile	Controls light intensity/amount of light entering the eye.
Ciliary muscle/ body contractile	Control curvature of lens, secretes humour
Suspensory ligaments: are fibrous	Hold lens in position
Lens is transparent; lens is biconvex	To allow light to go though/to refract light/ to FREE light.
Retina- contain light sensitive cells	Where an image is formed which perceive light.
Cones: contain pigments	For colour vision/ bright light/ light of high intensity
Rods contain pigments	For dim light vision

Forea centralis: High concentration of cones	For accurate vision
Choroid: Layer has blood vessels	For nutrition and pigments. Reduce light reflection and absorb stray light
Sclera- Tough/ non elastic or fibrous	Gives eye shape and protects
Optic nerve	Contain sensory neurones for transmitting impulses from retina to brain.

(b)

Defect	**Correct**
Myopia/ short sightedness	Biconcave/diverging lenses
Hypermetropia/long sightedness	Biconvex/ converging lenses
Astigmation	Use of cylindrical lenses with combined curvature

5. When an impulse passes along the axon, the membrane of the axon becomes depolarized to sodium ions to diffuse into the axon; the inside of the axon becomes positively charged relative to the outside, an action potential is generated.
6. Cerebrum/ cerebral hemisphere/ cerebral cortex
7. (a) B- Cerebellum

   C- Medulla oblongata

   (b) - Control locomotion

   - Control voluntary movement

- Vision/ hearing/ smell/ taste
- Intelligence/ memory
- Personality speech
- Mediates cranial co-ordination

(c) Loss of muscle coordination/ balance

8. (a) Cones- Discrimination of colour/ sensitive to high light intensity / bright light.

   (b) Rods- Dim light vision/ low light intensity

9. (a) (i) -Receive sound waves

   -Transforms sound waves into vibrations

   -Transmit vibration to the ear ossicles

   (ii) Equalizes the air pressure in the middle ear to that in the outer ear.

   (iii) Amplify/ transmit vibrations from the lympanic membrane in the inner ear.

   (b) There are three semi- circular canals arranged in a plane at right angles to each other. At the end of each canal is a swelling called ampulla which contains receptors.

   Movement of the beat cause movement of the fluid in at least one canal/ the fluid movement deflect the coperta and stimulate the receptors/sensory hairs. Nerve impulses are transmitted to the brain by the auditory nerve.

10. The organisms move towards light so as to absorb it for photosynthesis.

11. (a) Thigmotropsm

   (b) -That part of plants is offered support

   -The leaves become more exposed to sunlight increasing photosynthesis.

-Flowers become exposed to pollinating agents.

12.

(a) X- Motor neurone

Y- Receptor

(b) Acetylcholine

13. (a) Alter the shape of the lens during accommodation

(b) - Rods- sensitive to dim light

- Cannot distinguish colour

- Cones- Sensitive to colour

- Enhance high clarity of vision

14. (a) Ear ossicles Magnify sound wave vibrations from the ear drum

(b) Cochlea Receives sound vibrations from the oval window and transmits into the auditory nerve.

(c) Semi- circular canals Structures that help maintain body balance

(d) Eustachian tube Enhance equalizing of pressure between outer and the middle ear.

15. (a) In the central nervous system (spinal cord)

(b) (i) Motor neurone

(ii) P – Dendrites

Q- Axoplasm (Axon)

(c) Insulates the axon

16. (a) Auxin

(b) Growth response due to touch of a part of a plant e.g. tendrils

17. The ear is an organ involved in perceiving sound and maintaining body balance and posture. It is made of the following sections.

- Pinna- That is funnel shaped structures made of skin and cartilage. It receives sound waves and directs them to the ear tube.
- External /auditory meatus- That is a canal lined with hair and wax. It allows passage of sound waves to the middle ear. The hairs and wax trap dust particles that enter the ear.
- Tympanic membrane that is a thin flexible sheet-like structure receives sound waves and passes the vibration to the ossicles.
- Middle ear that is composed of:

  Tiny bones known as ossicle. They are stapes, anvil and incus. They amplify vibration from the tympanic membrane.
- Eustachian tube that connects the ear to the nasal cavity. It balances pressure on both sides of the tympanic membrane.
- Oral window that is a thin flexible membrane that opens into the inner ear. it receives vibrations from the ossicles and passes them to the inner ear.

Inner ear that is composed of:

- Vestibular apparatus that are the semi circular canals, utricles and the saccules. They help in maintenance of body balance and posture.
- Cochlea that is a coiled structure that has sensory cells for hearing. It is connected to the auditory nerve that is involved in transmission of sounds to the brain.

18.
- Presence of rods having rhodopsin pigment that is sensitive to dim light.
- Rods are more sensitive to motion and easily notes movement from the cornea of the eye.
- More than 120 million rods present on the retina.

19. Perceive sound waves.

Maintain body balance and posture.

20.

Response of human eye	Response of flowering plant
• Quick response	• Slow response
• Does not result to growth	• Results to growth
• Mediated by nerve impulses and brain	• Mediated by growth hormones (auxins)
• The response is not permanent	• Response is more permanent

21. (i) Thigmotropism

(ii) Auxins on the stem are sensitive to touch. They migrate to opposite side. Growth is more on the touched side. This causes bending.

(iii) Have more chlorophyll to trap sunlight
- Have stomata for entry of carbon dioxide.
- Thin and transparent cuticle to allow entry of light into the photosynthetic cells
- Presence of veins for transportation of raw materials to the leaf or food for the leaf.

22. Euglena have chlorophyll and are autotropic. They move towards light source (positive phototactic) to absorb sunlight for photosynthesis.
23. Acetylcholine is a chemical substance present at the synaptic knob. When a nerve impulse reaches the synapsis, acetylcholine forms in vesicles moving to the membrane.
24. In the spinal chord.
25.
    - Tar is deposited on parts of the respiratory tract causing cancer.
    - Hardening the blood vessels and can cause heart attack.
    - Irritation of the respiratory tract resulting to frequent coughing.
    - Smoke can cause air pollution.
26. (i) Cones on retina.

    (ii) Vitreous humour.

    (iii) Suspensory ligaments.

CHAPTER 4

SUPPORT AND MOVEMENT IN PLANTS AND ANIMALS

1. (a) K - Facet for articulation, with the next vertebra

   L - Transverse process for attachment of muscles

   (b) Cervical or neck region

2.
   - Skeletal muscles have actin and myosin which facilitate concentration and relaxation.
   - High density of mitochondria to provide energy for contraction.
   - Elongated fibres to allow change in length

3. (i) Ball and socket joint

   (ii) Biceps (flexor muscles) relax triceps (extensor muscles) contract.

4.

	Biceps	Gut muscles
(i)	Striated	Un- striated
(ii)	Multinucleated	Un- nucleated
(iii)	Long fibre	Short fibred
	cylindrical	Spindle shaped

5. (a) Femur

   (b) Ball and socket joint

6.
   a. Attachment of powerful back muscles
   b. Maintain posture

c. Maintain flexibility of vertebral column

7. (a)

a. Hydrostatic
b. Exoskeleton
c. Endoskeleton

(b) <u>Cervical vertebrae</u>

- Presence of vertebraterial canal for passage of vertebral artery. Atlas had (broad) surfaces, for articulation with condyles of skull to permit nodding
- Axis has adontoid process/ projection Centrum to permit rotary/ turning. Act as a pivot for atlas.
- Branched/ forked/ short and broad transverse processes for attachment of neck muscles
- Presence of zygopophysis for articulation between vertebrae
- Has short reduced neural spine for attachment of neck muscles. Has wide neural canal for passage of spinal cord and protect it.

Lumbar

- Broad / long neural spine for attachment of powerful back muscles.
- Large and well developed transverse processes for attachment of muscles
- Has metamorphosis and hypothesis for muscle attachment. Large thick centrum for support.
- Prezygopophysis and post zygopophysis present for articulation between vertebrae

Sacral vertebrae

- Interior has well developed transverse processes which are fused to the pelvic girdle.
- Vertebrae fused for strength transmit weight of the stationary animal to the rest of the body
- Sacrum has a broad base/ short neural spine for attachment of back muscles

8.

(a) Ulna

(b) Radius

(c) Humerus

9.

(a) Inter- vertebral discs/ Fibro cartilage

(b) Absorb shock and reduce friction between the bones

10. Side walls have deposition of lignin to strengthen them

11.

(a) Y- Femur

Y- Tibia

Z- Fibula

(b) (i) Synovial fluid

(ii) Absorb shock/ reduce friction between joints

(c) Ligament

(d) Ball and socket – allow movement in all direction

Hinge joint- Allow movement in one plane only

(e) Sigmoid notch

12.

(a) Have short neural spines

(b) - Xylem tissues

- Collenchymas tissues

- Sclerenchyma tissues

- Parenchyma tissues

13. (a)

Type of muscle	Where found
(i) Skeletal	Attached bones and skeleton
(ii) Smooth	Walls of tubular structures
(iii) Cardiac	Heart muscles

(b) Ball and socket joint – allows movement in all directions i.e 360^0

Hinge joint- Allows movement only on one plane i.e 180^0

(c) It is a slippery fluid that lubricates the joints reducing friction during movement.

(d) - Prevents drying out of organism

- Controls size of the organism

- Provides protection against microbial infections and mechanical injury.

14.

a. Support and protects inner delicate tissues

b. Prevents excessive loss of water from body tissues

c. Provides surfaces for muscle attachment.

15.

(a) Dorsal fin – Prevented rolling or yawing

(b) Pectoral and pelvic fins- used for steering and prevent pitching

(c) Caudal fin – steering and forward propulsion

16.

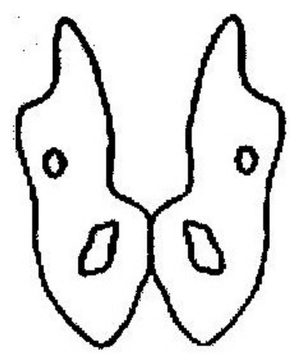
a. Contract spontaneously and do not fatigue.

b. Innervated by the autonomic nervous system

c. Contractions are initiated from within the muscles

d. Thy are myogenic

17. (a)



c) Femur – Articulates with acetabulum

Sacrum – articulates with ilium

18. Tendons – Tissues between muscles and bone in a joint

Ligaments – Tissues between bone and another bone in a joint

19. They are muscles that contract while the others relax e.g triceps and biceps muscles.

20. (a)

- Xylem vessels
- Collenchyma
- Sclerenchyma

(b) Xylem- lignified on the side walls

Collenchyma – thickened by deposition of cellulose and pectic compounds

Sclerenchyma – lignified on the cell walls.

21. (a)

- Immovable joints
- Synovial (movable) joints)
- Glinding/ sliding joints

(b)

- Immovable joint – Cranium / skull
- Synovial joint – between limbs
- Glinding / sliding joint- vertebral column

22.

- Turgidity of the parenchyma cells
- Presence of collenchyma tissues

23. **Skeletal muscle**

a. Attached to the skeleton
b. They are striated/ fibres that allow contractions
c. Presence of mitochondria to provide energy for contractions
d. Have antagonistic contractions to enhance movement

Cardiac muscle

a. They are the heart muscles

b. Highly connective tissues to allow harmonious contraction

c. They do not fatigue

d. Ends are intercalated to transmit impulses throughout the heart

Smooth muscle

a. Walls of tubular organs

b. Capable contracting slowly

c. Innervated by autonomic nervous

d. System/ involuntary movement

PAPER 1

1. (a) Flagellum

   (b) Cilia

2. (a) Genes located on sex chromosomes and are transmitted along with them

   (b) Colour blindness, hairy pinna, haemophilia, baldness

3. (a) Genus - Bidens

   (b) Species- pilosa

4. (a) Carbon (IV) Oxide

   - Uric acid

   (b) Oxygen, gum, carbon (IV) Oxide

   - Tannins, quinine water vapour, latex

5. - Small and light to be carried easily by wind
   - Have hair- like structures or floss to increase buoyancy in the air
   - Develop wing- like structures or floss to increase surface area for increased buoyancy in the air

6. The pollen tube directs the male nuclei or gametes into the ovules in the embryo sac.

7.

DNA	RNA
a. Double stranded	a. Single stranded
b. Deoxyribose sugar	b. Ribose sugar
c. Thymine base	c. Uracil base
d. Less oxygen molecules	d. More oxygen molecules

8. (a)

Meiosis	Mitosis
a. Takes place in reproductive cells	a. takes place in body cells
b. Division are double (I and II)	b. division is only one
c. Daughter cells are identical to prevent cell (Diploid)	c. daughter cells are not identical to prevent cells (Haploid)
d. Homologous chromosomes associate	d. Homologous do not associate
e. Four daughter cells formed	e. Two daughter cells formed
f. Chiasmata formed and crossing over occur	f. Chiamata not formed and there is not crossing over

(b)
- DNA replicates
- Cell organelles replicate
- Cells build up a store of energy to carry out the division process through to the end

9.
- Oviduct
- Trachea or tracheal epithelium

10. (a) A- Epidermal cell

B- Guard cell

(b) Carbon (IV) Oxide

(c) - Thick inner wall and thin outer wall to control the opening and closing of them.

- Presence of chloroplasts to carry out photosynthesis

11. - Human

- Mosquitoes

12. Female – Ovaries

Male - Testes

13. - Are brightly coloured to attract animals

- Have a sweat scent to attract animals

- are succulent, juicy and edible

- Have seeds that resists egestion of enzyme

- Have hooks for attachment on the body of animals

14. (a) Aerobic respiration, mitochondria

(b) R.Q = $\frac{\text{Carbon (IV) Oxide produced}}{\text{Oxygen used}}$

$= \frac{18}{26} = 0.692 = 0.7$

(c) Lipid/ fat

15. - Has villi and micro villi

- Long length

16. (a) A- Synaptic Knob

    B- Synaptic cleft

    (b) C- Transmits impulses

17. It prevents the formation of solutions which would otherwise interfere with osmotic pressure of the tissues

18. (a) Cotyledon pushes above the ground

    (b) Shoot pushes above the ground but cotyledon remains underground.

19. Assimilation is the process by which the body uses up the absorbed products by which the end products of digestion are taken into the epithelial cells of the ileum by diffusion or active transport.

20. - Regulation of blood glucose
    - Regulation of amino acids
    - Excretion of cholesterol and bile
    - Production of heat

21. Lactic acid fermentation is the breakdown of glucose in a limited supply of oxygen in muscle tissues while alcoholic fermentation is the breakdown of glucose in absence of oxygen in plant tissues.

22. Water logging reduces the oxygen concentration in the soil hence plants die due to lack of oxygen for respiration.

23. - Plasmolysis occurs when plant cells are placed in hypertonic solution. They lose water by osmosis and shrink.
    - Haemolysis occurs when red blood cells are placed in hypertonic solution. They absorb water by osmosis swell and burst.

24. Father X^HY

    Mother H^YX^Y

25. A- Collenchyma- provides support for the stem

    B- Sclerenchyma – provide mechanical support for the plant

    C- Parenchyma- for support and storage of food

    D- Xylem- transports water and minerals salts from the roots to other to parts of the plant, also gives support to the plant

26. (a) AIDS- Human immunodeficiency virus

    (b) Bilharzia- Schistoma mansoni or Schistoma spp

    (c) Cholera- Vibrio cholerae

27. Lungs, gills, skin, buccal cavity, book lung or tracheole.

28. - Production of food
    - Production of oxygen
    - Removal of carbon (IV) Oxide from the air.

29. (a) Continental drift- the drifting apart of the continents from one land mass called Pangaea. Pangaea split to form Gondwana and Glaurasia which further split into different continents.
    (b) Fossils are the remains of organism that lived long ago which have been preserved naturally in the earth's crust.

30. Phototropism

PAPER 2

1. Genes being carried together on the same chromosomes and inherited and inherited together.

(b) (i)

Phenotypes	Normal carrier Woman	x	Normal man	
Genotypes	X^HX^h	x	X^HY	
Gametes	X^H	X^h	X^H	Y
Offspring genotype	X^HX^H	X^HY	X^HX^h	X^hY
Offspring phenotype	Normal female	Normal male	Normal female	Haemophiliac male

(ii) Male requires only one recessive allele to be a hemophilic while the female require two recessive alleles to be hemophilic.

(iii) Red- green colour blindness/ hairy nose and ears

2. (a) (i) 4.0 – 0.04 = 3.96%

(ii) Oxygen – exhaled air contains less oxygen because some of the oxygen in inhaled was used up for respiration.

Carbon (IV) oxide: exhales air contains more CO_2 produced during metabolism/ respiration.

Nitrogen – No inhaled nitrogen is used up in the body.

(b) - Have stomata on their surfaces

- Have thin cuticle which allows for diffusion of gases

- Have spongy mesophyl layer with air spaces to increase ventilation.

3. (a) X- Nitrifying bacteria

Y- Denitrifying bacteria

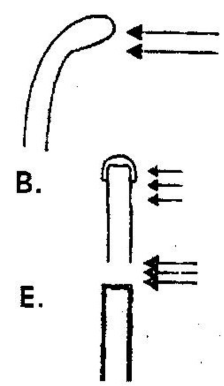
(b) Nitrogen gas

(c) A- Assimilation

B- Consumption

C- Nitrifying

4. (a) kkkk



NB: No change in size E

(d) Auxins produce at the tip, migrate to the shaded side promoting faster growth on that side resulting to a grown curvature towards light.

(e) Set up B and C are control experiments showing it's the short tip that produces auxins that promotes phototropism.

5. (a) Process whereby fertilization takes place inside the body of the female.

(b) - Pose health risk to the pregnant animals e.g. during birth the mother may die due to excessive bleeding.

- Pregnant animals is vulnerable to predators

- Its too demanding to the mother in terms of nutrients.

(c) - Secrete hormones/ endocrine gland

- Medium for gaseous exchange

- Medium through which nutrient are supplied to the foetus

- Medium through which waste products are removed from the foetus.

(d) (i) - Causes Graafian follicle to develop in the ovary

- Stimulate ovary to secrete oestrogen hormone

(ii) - Healing and repair of uterine wall after menstruation

- Stimulates the pituitary gland to secrete oestrogen hormone

(iii) - Cause ovulation

- Stimulates the corpus luteum to secrete progesterone hormone.

6. (a) Graph of the rate of growth of growth of femur and head.

(b) (i) Intermittent growth/ discontinuous growth

(ii) Phylum arthropoda

Reason: Shows continuous growth/ intermittent growth.

(c) (i) Length of femur remains constant/ no change in length; growth has not taken place because of the presence of rigid exoskeleton/ cuticle which limits expansion of tissues.

(ii) Length of femur increased because moulting/ ecdysis/ shedding of exoskeleton has occurred allowing growth/ expansion of tissues.

(d) - Juvenile hormone

- Moulting/ ecdysone hormone.

7. - Soil particles are surrounded by a film of water

- The cell sap of the root hair is more concentrated than soil water.

- Cell membrane of root hair acts as a semi- permeable membrane.

- Due to the concentration difference between cell sap and water in the soil water moves into the root hair by osmosis.
- This reduces the concentration of the cell sap in the root hair hence water moves into neighbouring cells (by osmosis). This continues through cell sap to cell sap; Cytoplasm and through intercellular spaces.
- Minerals are absorbed by either diffusion or active transport.
- Diffusion occurs where a concentration gradient exist/ concentration of mineral salts is more in soil water than in cell sap of root hair.
- Once in the xylem water moves up the plant aided by narrowness of the xylem vessels/ capillary; root pressure; attraction of water molecules to each other/ cohesion attraction of water molecules to the walls adhesion.
- From the stem xylem water enters the xylem of the leaves
- Once in the leaves water enters the mesophyll and by osmosis moves from cell to cell until it reaches the sub-stomatal chamber, where it evaporates into the air creating a transpiration pull.

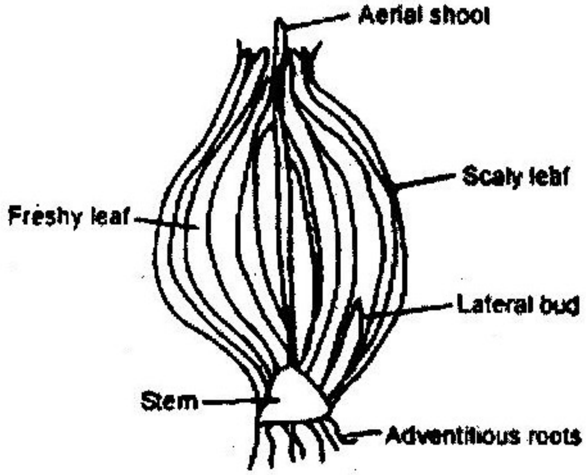
(Total marks 25, max marks 20 marks)

8. (a)
   - Secretion is discharge of useful substances such as hormone and enzymes.
   - Excretion is the elimination of waste products of metabolism
   - Egestion is the removal of indigestible and undigested materials from the alimentary anal.

(b)
- Has renal artery to supply blood rich in metabolic wastes.
- Has renal vein that drains purified blood from the kidney
- Renal artery branches into arterioles that serve individual nephrons
- Efferent arteriole is narrower than the afferent arteriole to create high pressure required for ultra filtration.
- Glomerular capillaries are very narrow to create high pressure to enhance ultra filtration.
- Has a capsule barrier that selectively allows some substance to pass through.
- Has convoluted tubule that is highly coiled to slow down rate of flow of filtrate to allow more time for re-absorption.

PAPER 3

1. (a)



(b) (i)

Food	Procedure	Results	Conclusion
Starch	Place about 3 drops of extract on a white tile. Add a few drops of iodine solution	Yellow colour of iodine remains	Starch absent
Reducing sugars	Place about 2 cm^3 of extract in a test- tube and a few drops	Colour turns from blue to green yellow orange, orange then red- brown.	Reducing sugars present

(iii) When food is manufactured by the green aerial leaves it is then transported into the fresh leaves and stored for future use and development of new lateral buds.

(c) (i) - Distilled water – the stalks were firm and more elongated. The water is Hypotonic and water is absorbed by osmosis into the cells which become turgid.

- 20% salt solution – the stalks were flaccid and shrunk.
- The salt solution is hypertonic and water is absorbed by osmosis and they become flaccid.
- Extract- The stalks remained size as the original. The extract is isotonic to that of the cell sap water moves in and out of the cells freely and turgidity of the cells is not affect.

(ii) Help in support of plant by maintaining turgidity of the cells

(d)
- Have outer scaly that protect inner delicate parts.
- Have fresh leaves that store the manufactured food.
- Stem holds the bulb firmly into soil.
- Adventitious roots absorb water and minerals.

2.

Leaf	Step followed	Identity
P	1a, 2a	Hibiscus
Q	1b, 4a	Nandi Flame
R	1a, 2b, 3a	Mango
S	1b, 4b	Bean
T	1a, 2b, 3b	Morning glory

3. (a) Phylum - Arthropoda

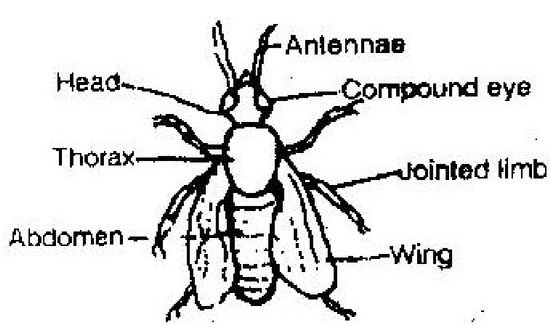
Reasons - Segmented body

- Jointed limbs
- Bilateral symmetry
- Has exoskeleton

(b) Class - Insecta

Reasons - Three body parts

- A part of compound eyes
- Three pairs of walking legs
- Two pairs of wings



(c)

Specimen B	Man
a. Has exoskeleton b. Body has three body parts c. Presence of wing d. Open circulatory system e. Presence of antennae poikilothermic	a. Has endoskeleton b. Body compact having head, trunk and limbs. c. Closed circulatory system d. Absence of antenna e. Homeotherms

(d)
- Dorsal ventral streamlined in order to fit in the small thin cracks
- Exoskeleton is dark in order to be camouflaged against the dark background.
- Wings aligned to the smooth exoskeleton in order to reduce friction when moving in the cracks.

PAPER 231/3

1. (a) J1 and J2 Rosales/ dicotyledonae

Reason Net veined/ net venation/ reticulate venation / two cotyledons

K1 and K2 Graminales/ monocotyledonae

Reason Parallel veined/ parallel venation/ one cotyledon/ fibrous root system.

(b) (i) Hypocotyle

(ii) -Protect the plumule/ shoot tip/ first foliage leaves

-Opens space through the soil for cotyledons out of the soil.

(c) Exposure of curvature to light, Auxins migrate to lower side. Faster growth of cells on that lower side, hence stem straightens.

(d) Plumule sheath/ coleoptile

(e) (i) J1 and J2

(ii) Root nodules

(iii) Rhizobium/ bacteria

(iv) Symbiotic

(f) (i) Cotyledons/ seeds leaves

(ii) -Photosynthesis

-Stores food/ food reserve

(g) (i) Hypogeal

(ii) Remains of grains/ cotyledons remain underground.

(h) -Tap root (system)

-Fibrous toot (system)

2. (a) M - Trachea/ windpipe/ part of piece of trachea.

N - Lung/ piece or part of lungs

(b) Specimen M - Neck/ neck. Region/ throat/ cervical region

Specimen N - Thoracic region/ chest cavity/ thorax/ chest/ Rib cage

(c) M is a passage/ leads air into N Both M and N are part breathing structures/ breathing system/ respiratory system.

Specimen	**Features**	**How features adapts specimen to its functions**
M	(i) Rigid/firm cartilage (ii) Hollow/ tabular trachea/ lumen (iii) Elastic muscles (iv) Mucus lining surface (v) Moist surface	- Prevents collapsing keeps it open - Allow passage of air - Cause movement / allow for compression and flexibility - Top trap foreign bodies/ filter the air (Bacteria, dust etc) - To moisten the air.
N	(i) Spongy/ porous/ soft/ air sacs/ air spaces (ii) Elastic tissue (iii) Vascularized (iv) Moistened surface (v) Bronchioles (vi) Pleural membrane	- Increase surface area for gaseous exchange/ store air. - Allow for stretching / expansion - Facilitate transport of gases. - Air passage into and out of the lungs - protection/ reduces friction

3. (a) (i) Specimen Q- Lumbar verterbrae

Reasons

1. Large / broad Centrum
2. Long/ broad transverses processes
3. Presence of metamorphosis
4. Presence of anapophysis
5. Broad/ wide neurospine

(ii) Specimen R- Cervical vertebrae

Reasons

1. Pointed/ short/ small neural spine
2. presence of verbraterial/ canals
3. Winged/ forked/ branched/ divided transverse processes
4. Presence of cervical ribs

(b) - Presence of neural canal for passage of spinal cord

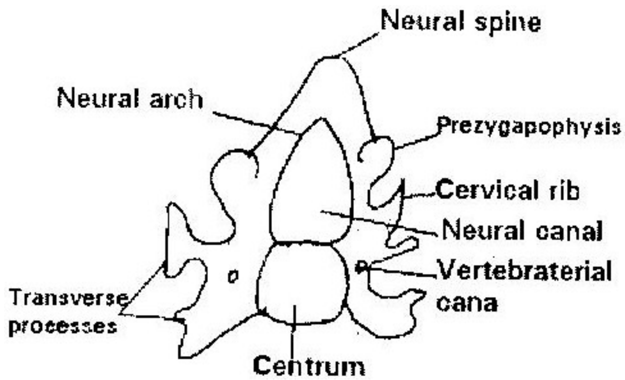
- Neural spine for attachment of muscles
- Has facets for articulations with other vertebrae
- Vertebraterial canals for passage of blood vessels
- Has neural arch and Centrum for protection of the spinal cord.

(c)

Q	R
a. Vertebraterial canal absent	a. Vertebraterial canal presence.
b. Large, unbranched transverse processes	b. Small, forked/ branched transverse processes.
c. Broad/ large neural spine	c. Neural spine small/ narrow
d. Narrow neural canal	d. Wide neural canal

e. Presence of matapophyses or anapophyses	e. Absence of metapophyses or anapophyses

(d)



Drawing marks

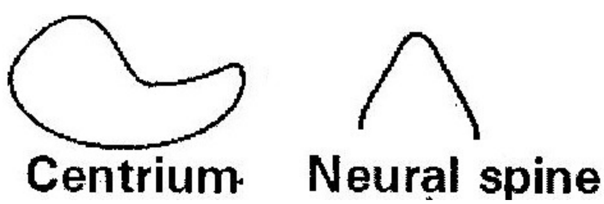
D1 Complete outline and proportionality

Proportionality = size of Centrum to be smaller than neural canal

D2 Branched Transverse processes. Vertebraterial canals properly drawn-

near the point where neural; arch and Centrum comes into contact.

D3 Centrum and neural spine properly drawn.



BIOLOGY PAPER 3

1. (a) Vertebral column

(b) K - Atlas

M - Axis

N - Cervical vertebrae

(c) - Wing – like transverse processes/ broad transverse process

- No centrum

- Presence of verbraterial canal

- Reduced neural canal

- Wide neural canal

(d) Odontoid process

Spinal cord

(e) S (facet) - Articulate with axis

T - Passage of vertebral blood vessels and vertebral nerves

(f) Condyles of the skull

(g) U - Neural pine

Y - Odontoid process

R - Centrum

L_1 - distilled water

L_2 - Salty water

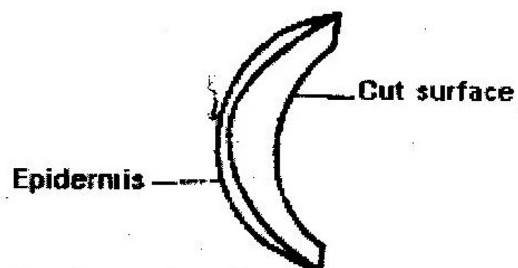
2. (a) (i) Become firm/ turgid – L_1

L_1 become flabby/ flaccid

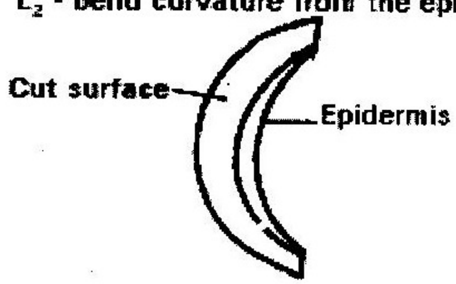
(ii) L_1 – Hypotonic solution as compared to cell sap of cells. Gains water by osmosis and cell becomes turgid.

L_2- Solution hypertonic as compared to cell sap of cells/ Loses water by Osmosis become flaccid/ plasmolysed.

(b) (i) L_1 Bend curvature from the cut surface



L_2 - bend curvature from the epidermis



(ii) The cuticle prevented water from being absorbed by cells (it is water proof)

3. (a) Set A - normal condition

Set B - in the dark

Set C - in the dark with unilateral / source of light

(b)

A	B
- Green leaves	- Yellow leaves

- Short internodes	- Long internodes
- Strong stem	- Weak stem
- Broad leaves	- Small leaves

(c) (i) Etiolation

(ii) Grows tall/ long in an attempt to reach light.

(d) Positive phototropism

(e) Auxins diffuse on the side away from light result to more cell division (more growth on that side resulting to bending towards light.

BIOLOGY

Paper 1

1. Name the tissues in plants responsible for:

   (a) Transport of water and mineral salts

   (b) Transport of carbohydrates

   (c) Primary growth (3 marks)

2. State the importance of the following processes that take place in the nephrons of a human kidney:

   (a) Ultra filtration (1 mark)

   (b) Selective reabsoption (1 mark)

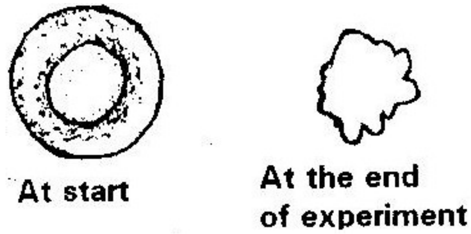
3. (a) Name a disease of the liver whose symptom is jaundice (1 mark)

   (b) State the causative agent of:

   (i) Cholera (1 mark)

   (ii) Candidiasis (1 mark)

4. The diagram below show a red blood cell that was subjected to a certain treatment



(a) Account for shape of the cell at the end of the experiment (2 marks)

(b) Draw a diagram to illustrate how a plant cell would appear if subjected to the same treatment. (1 mark)

5. (a) State two factors that effect enzymatic activities. (2 marks)

   (b) Explain how one of the factors stated in (a) above affects enzymatic activities (2 marks)

6. (a) What is meant by non- disjunction? (1 mark)

   (b) Give two examples of continuous variations in humans (2 marks)

7. (a) What is a fossil? (1 mark)

   (b) How does convergent evolution occur? (3 marks)

8. The diagram below shows a stage in mitosis in a plant cell

   (a) Name the stage of mitosis (1 mark)

   (b) Give two reasons for your answer in (a) above (2 marks)

   (c) Name the part of the plant from which the cell used in the preparation was obtained (1 mark)

9. Give three factors that determine the amount of energy a human requires in a day (3 marks)

10. (a) Name the antigen that determines human blood groups (2 marks)

 (b) State the adaptation that enables the red blood cells to move in blood capillaries (1 mark)

11. (a) What is homeostasis? (1 mark)

 (b) Name three processes in the human body in which homeostasis is involved (3 marks)

12. State two functions of the endoplasmic reticulum (2 marks)

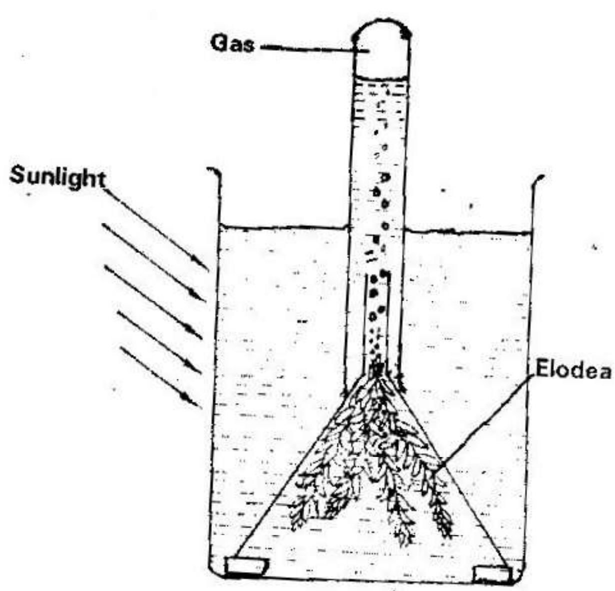
13. (a) Name the part of the retina where image formed on the retina (1 mark)

 (b) State two characteristics of the image formed on the retina. (2 marks)

14. Describe the three characteristics of a population (3 marks)

15. Explain what happens when there is oxygen debt in human muscles (2 marks)

16. The diagram below represents a set up that was used to investigate a certain process in a plant



(a) State the process that was being investigated (1 mark)

(b) State a factor that would affect the process (1 mark)

17. Account for the following phases of a sigmoid curve of growth of an organism:

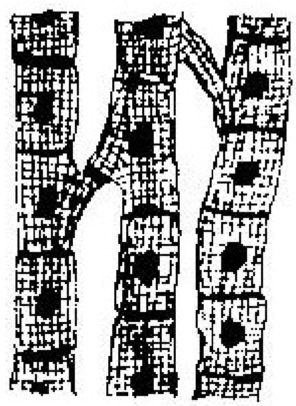
(a) Lag phase (1 mark)

(b) Plateau (1 mark)

18. How is the epidermis of a leaf of a green plant adapted to its functions?

(2 marks)

19. The diagram below represents a tissue obtained from an animal



(a) Identify the tissue (1 mark)

(b) State the function of the tissue named in (a) above (1 mark)

20. (a) What is single circulatory system?

(b) Name an organism which has single circulatory system (1 mark)

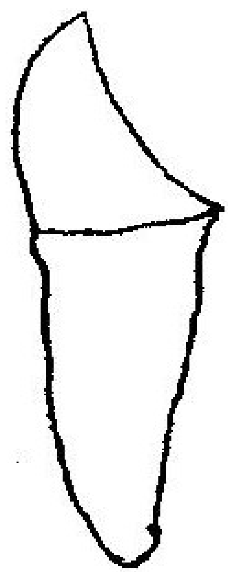
(c) Name the opening to the chamber of the heart of an insect (1 mark)

21. (a) What is seed dormancy? (1 mark)

(b) Name a growth inhibitor in seeds (1 mark)

22. State two characteristics of aerenchyma tissue (2 marks)

23. The diagram below shows a human tooth



(a) Identify the tooth (1 mark)

(b) How is the tooth adapted to its function (1 mark)

(c) State the role of the following vitamins in human body

(i) C (1 mark)

(ii) K (1 mark)

24. Name the sites where light and dark reactions of photosynthesis take place

Light reaction

Dark reaction

25. Giving a reason in each case, name the class to which each of the following organisms belong (4 marks)

Bean plant

Reason

Bat

Reason

26. State one use of the following excretory products of plants:

    a. Colchicines (1 mark)

    b. Papain (1 mark)

27. Explain how anaerobic respiration is applied in sewage treatment

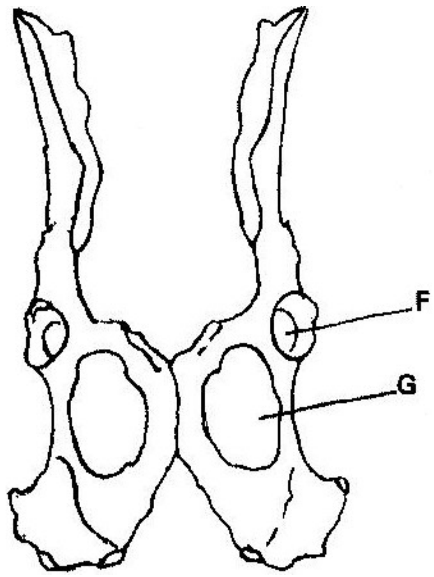
(1 mark)

28. (a) State the mode of asexual reproduction in yeast (1 mark)

    (b) Distinguish between protandry and protogyny (2 marks)

29. State a function of amniotic fluid (1 mark)

30. The diagram below shows two fused bones of a mammal.



(a) Identify the fused bones (1 mark)

(b) Name the

(i) Bone that articulates at the point labeled F (1 mark)

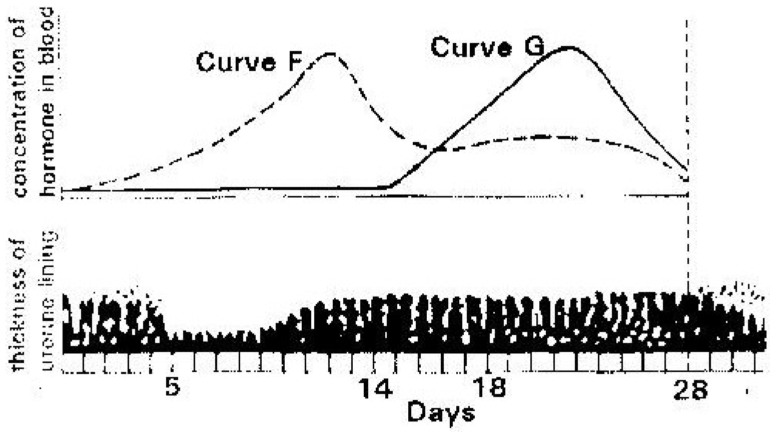
(ii) The hole labeled G (1 mark)

BIOLOGY PAPER 2

SECTION A (40 Marks)

Answer all the questions in this section

1. The figure below shows changes that take place during menstrual cycle in human



(a) Name the hormones whose concentrations are represented by curves F and G

(2 marks)

(b) State the effects of the hormones named in (a) above on the lining of the uterus (2 marks)

(c) (i) Name the hormone which is released by the pituitary gland in high concentration on the 14^{th} day of the menstrual cycle.(1 mark)

(ii) State two functions of the hormone named in (c) (i) above.

(2 marks)

(d) State the fertile period during menstrual cycle. (1 mark)

2. A pea plant with round seeds was crossed with a pea plant that had wrinkled seeds. The gene for round seeds is dominant gene state

(a) The genotype of parents if plant with round seeds was heterozygous

(2 marks)

(b) The gametes produced by the round and wrinkled seed parents

Round seed parent

Wrinkled seed parent

(c) The genotype and phenotype of F1 generation. Show your working

(3 marks)

(d) What is a test- cross? (1 mark)

3. The equation below represents a process that takes place in plants

$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$

(a) Name the process (1 mark)

(b) State two conditions necessary for the process to take place (2 marks)

(c) State what happens to the end- products of the process (5 marks)

4. (a) Give three reasons in each case why support is necessary in:

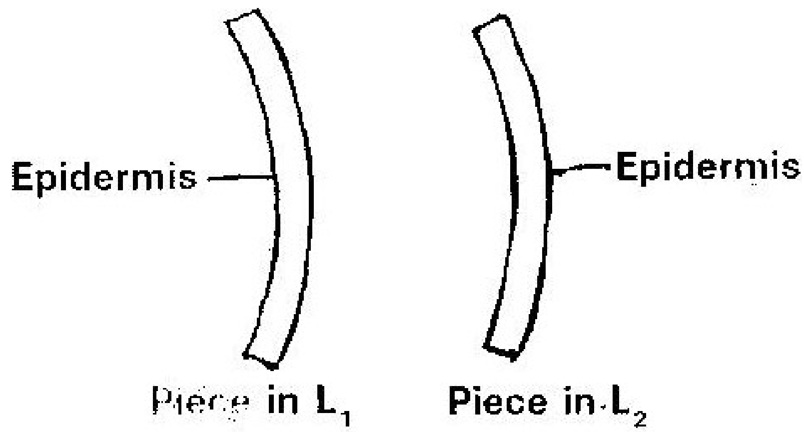
(i) Plants (3 marks)

(ii) Animals (3 marks)

5. A freely obtained dandelion stem measuring 5 cm long was split lengthwise to obtain two similar pieces.

The pieces were placed in solutions of different concentrations in Petri dishes for 20 minutes

The appearance after 20 minutes is as shown



(a) Account for the appearance of the pieces in solutions L_1 and L_2 (6 marks)

L_1 ________________________________

L_2 ________________________________

(b) State the significance of the biological process involved in the experiment

(2 marks)

SECTION B (40 Marks)

Answer questions 6 (compulsory) and either questions 7 or 8

6. An experiment was carried out to investigate transpiration and absorption of water in sunflower plants in their natural environment with adequate supply of water. The amount of water was determined in two hour intervals. The results are shown in the table below.

Time of day	Amounts of water in grammes	
	Transpiration	Absorption
11 00 - 13 00	33	20

13 00 - 15 00	45	30
15 00 - 17 00	52	42
17 00 - 19 00	46	46
19 00 - 2100	25	32
21 00 - 23 00	16	20
23 00 - 01 00	08	15
01 00- 03 00	04	11

(a) Using the same axes, plot graphs to show transpiration and absorption of water in grammes against time of the day (7 marks)

(b) At what time of the day was the amount of water the same for transpiration and absorption? (1 mark)

(c) Account for the shape of the graphs of:

(i) Transpiration (3 marks)

(ii) Absorption (3 marks)

(d) What would happen to transpiration and absorption of water if the experiment was continued till 05 00 hours? (2 marks)

(e) Name two factors that may affect transpiration and absorption at any given time (2 marks)

(f) Explain how the factors you named in (e) above affect transpiration.

(2 marks)

7. Describe the nitrogen cycle (20 marks)

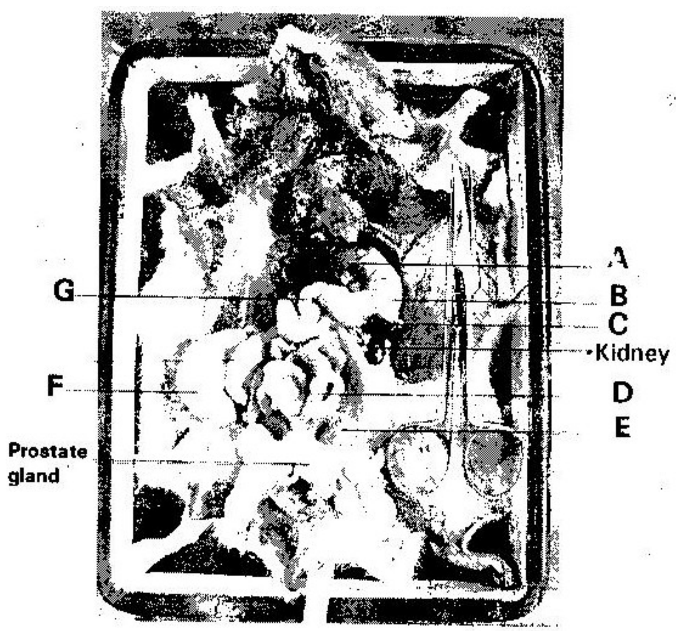
8. (a) State four characteristics of gaseous exchange surfaces (4 marks)

   (b) Describe the mechanism of gaseous exchange in a mammal (16 marks)

BIOLOGY PAPER 3

PRACTICAL

1. Below is a photograph of a dissected mammal. Examine the photograph.



(a) Name the parts labeled A, B, C, D and G (5 marks)

A ___________________

B ___________________

C ___________________

D ___________________

G ___________________

(b) State the function of the structures labeled E and F

E ___________________ (1 mark)

F ___________________ (1 mark)

(c) In the photograph label the structure where vitamin K is produced

(1 mark)

(d) (i) Name the sex of the mamma; in the photograph (1 mark)

(ii) Give a reason for your answer in (d) (i) above (1 mark)

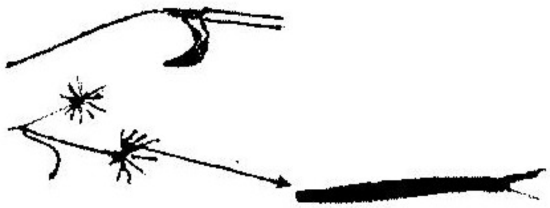
(e) (i) The actual length of the dissecting scissors in the photograph is 5 cm. Calculate the magnification of the photograph (2 marks)

(ii) Calculate the actual length of the mammal from the tip of the nose to point X on the tail (2 marks)

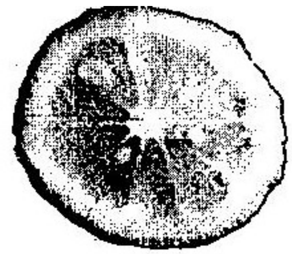
2. You are provided with substances labeled S, T,U X, and Y. S. T and U are food substances, while X is 10% sodium hydroxide solution and Y is 1% copper sulphate solution. Carry out tests to determine the food substance (s) in S, T and U. (9 marks)

Substance	Food substance being tested for	Procedure	Observations	Conclusion
S				
T				
U				

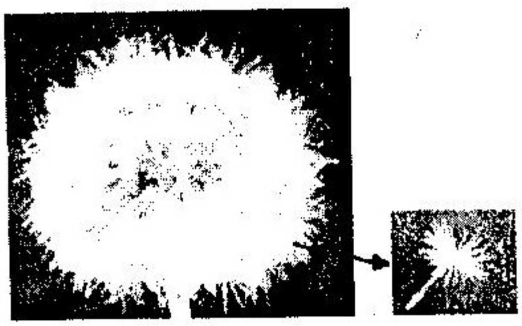
3. Below are photographs of specimens obtained from plants. Examine the photographs.



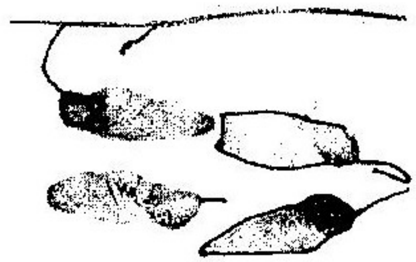
SPECIMEN K



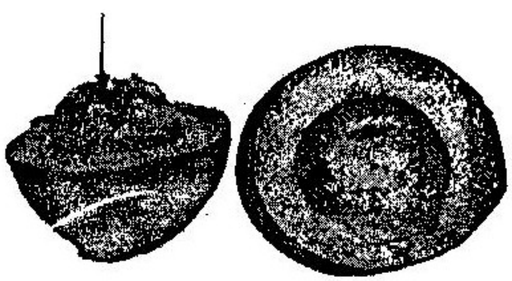
SPECIMEN L



SPECIMEN M



SPECIMEN N



SPECIMEN P

SPECIMEN Q

(a) In the table below name the mode of dispersal and the features that adapt the specimen (s) to that mode of dispersal (12 marks)

Specimen	Mode of dispersal	Adaptive
K		
L		
M		
N		
P		
Q		

(b) (i) Label any two parts on specimen L (2 marks)

(ii) State the type of placentaion in specimen L (1 mark)

(c) Name the structure labeled W on specimen P (1 mark)

MARKING SCHEME

PAPER 1

1. (a) Xylem

   (b) Phloem tissues

   (c) Apical meristem

2. (a) Remove the soluble substances; both useful and waste products from the blood stream

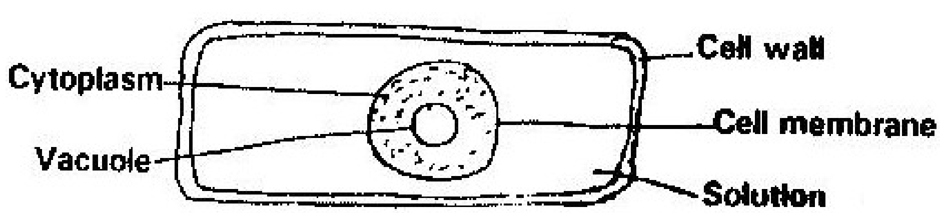
   (b) The reabsorb the useful substances that had been ultra filtrated and leave the waste products.

3. (a) Liver cirrhosis

   (b) (i) Vibrio cholerae

   (ii) Candida vaginitis

4. (a) The blood cell was placed in a hypertonic solution/ highly concentrated solution. It lost water to the surrounding by osmosis until it lost its shape and became crenated.



5. (a)

- Temperature
- pH value

- Enzyme concentration
- Substrate concentration
- Enzyme co- factors and co- enzymes
- Enzyme inhibitors

(b)

- Temperature – very low temperatures inactivate the enzymes. Very high temperatures denature the enzymes. Therefore the optimum temperature should be maintained for maximum enzyme activity.
- pH value- Some enzymes act best in acidic or basic medium. Therefore optimum pH value should be maintained for maximum enzyme activity.
- Enzyme concentration – when the enzyme concentration is increased the rate of enzyme activity also increases as long as there are enough molecules of the substrate.
- Substrate concentration – when the substrate concentration increases the rate of enzyme activity also increases as long as there are enough molecules of the enzymes. However further increase of the substrate may not increase the rate as all active sites of the enzymes will be occupied.
- Enzyme co- factors are the non- proteinous substances that activate the enzymes. Most enzymes will not work without the co-factors. They are metallic ions.
- The co- enzymes are organic non- protein molecule that work in association with enzymes. They are derived from vitamins.

- Enzyme inhibitors – they are substances that slow down or stop the enzyme activity if present. They fit into the active site of the substrate has no chance to fit into the active site.

6. (a)

- It is when the homologous chromosomes fail to segregate/ separate in anaphase resulting to one gamete or cell having more chromosomes and the other having less chromosomes

(b)

- Body height
- Skin colour
- Body weight
- Finger print types

7. (a)

- It is a remain of an ancestral form of an organism that has been accidentally preserved in a naturally occurring materials e.g. in sedimentary rocks.

(b)

- It is when structures from different origins become modified to perform similar functions.
- As a result of competition of resources and in order to exploit new habitats/ environments, the structures become modified to look similar and have similar function e.g. wings of insects, birds and bat.

8. (a)

- Early telophase

(b)

- The chromatids have moved and are very close to the poles
- Spindles fibres have disappeared

(c)

- Tip of root or shoot
- Flower when in form of bud
- A fertilized ovule developing into seed

9.

- Type of occupation
- Gender
- Age
- Body size
- Basal metabolic rate

10. (a)

- Antigen type A and antigen type B.

(b)

- They are small in size and can change their shape in order to fit in the pores of capillary walls.

11. (a)

- It is the maintenance of a constant internal environment of cells:

(b)

- Osmotic pressure regulation
- Temperature sugar regulation
- Ionic balance
- Blood sugar regulation

12. Rough endoplasmic reticulum – Transport substances (proteins) within the cell. Smooth endoplastic reticulum – Site of synthesis of lipids and destruction of foreign chemicals e.g. drugs.

13. (a) Forea centralis or yellow spot
    (b) - It is virtually inverted
    - It is not a real image

14.

- Population density – The number of individual per unit area.
- Sex – Ratio – the number of males and females in an ecosystem
- Age structure – number of individuals in the reproductive age
- Birth and mortality rate- The number of births comparing to the number of deaths in the population
- Population growth – the rate of increase in the number of individuals of a species.

15.

- As a result of a vigorous exercise, anaerobic respiration takes place resulting to formation of lactic acid. It is toxic and causes muscle cramps.
- It therefore has to be broken down into carbon (IV) oxide and therefore extra oxygen has to be taken up for this process. The extra oxygen is oxygen debt.

16. (a) Photosynthesis

(b)

- Sunlight
- Temperature change
- Carbon (IV) oxide concentration

17. (a) The cells are few and immature and have not yet started dividing rapidly.

(b) There are very many cells present and the rate of death/ destruction of old cells is equal to the rate of formation of new cells.

18.

- It is transparent to allow light to penetrate through to the photosynthesis cells
- The cells are packed from end to end preventing entry of micro- organisms and offering mechanical protection.
- The cells secrete cuticle that is waxy and water proof preventing excessive water loss from the leaf.
- Lower epidermis has specialized cells known as guard cells that control opening and closing of stomata.

19. (a) Cardiac muscle tissue

(b) Contract and relax rhythmatically enhancing the heart beat

20. (a) It is when the transport fluid flows from the heart once before passing through the oxygenating sites and body tissues then back to the heart.

(b) Fish

(c) Ostium

21. (a) It is a state when there is inactivity in a seed. Growth slows down or stops

completely.

(b) Abscisic acid

22.

- The cells are vertically arranged from end to end, from the upper epidermis to the lower epidermis.
- Large intercellular air spaces present

23. (a) Canine

(b)

- Long root to support it into the jaw bone
- Curved and pointed to tear the flesh/ meat

(c) (i) Vitamin C- Protection against infections

(ii) Vitamin K- Prevention of excessive bleeding

24. Light reaction – Granum/ Grana

Dark reaction – Stroma

25.

Bean plant – class dicotyledonae

Reason

- Two cotyledons in the seed
- Net work veined leaves and are broad
- Tap root system
- Experience secondary growth

Bat- Class mammalian

Reasons

- Has mammary glands
- Body covered with fur
- Gives birth to its young

26.

(a) Enhance non- disjunction resulting to polyploidy

(b) It is used as a meat tenderizer

27. Anaerobes are introduced into a septic tank and they breakdown the human refuse anaerobically

28. (a) Budding

(b)

- Protandry – stamens manure earlier and shed their pollen grains before the stigma matures to receive them.
- Protogyny- Stigma matures earlier before the stamens ripen to release the pollen grains.

29.

- Create aquatic environment for growth of the foetus
- Offer mechanical protection of the foetus from external pressure

30. (a) Pelvic girdle

(b) (i) Femur

(ii) Obsturator foramen

BIOLOGY PAPER 2

1. (a) F - Oestrogen

   G - Progesterone

   (b) F - Enhance healing of uterine wall after menstruation

   G - Increasing supply of blood to the endometriom/ uterine wall.

   (c) (i) Lutenising hormone

   (ii) Release of the ovum from the ovary, ovulation

   Reorganization of the remains of Graafian follicle into corpus

   Luteum

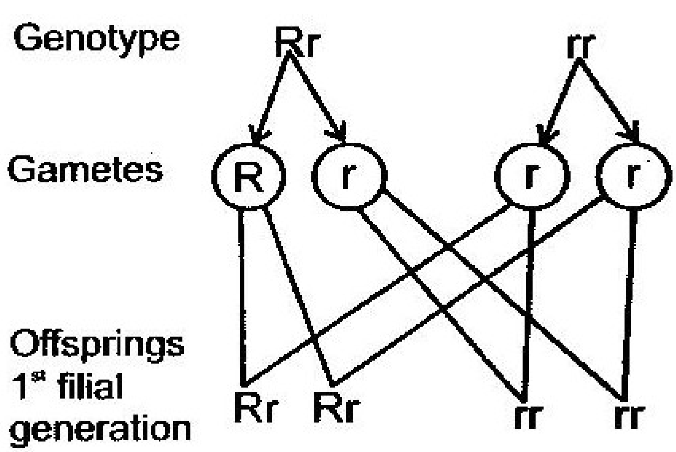
2. (a) - Male or 1st parent Rr
   - Female or 2nd parent rr

   (b) - Round seed parent R r

   - Wrinkled seed parent r r

   (c) Parents male/1st X Female/2nd

Phenotype Round seeded Wrinkled Seeded



- Genotype 2 – heterozygous round seeded plants

  2 – Homozygous wrinkled seeded plants
- Phenotype 2 – Round seeded plants

  2 – Wrinkled seeded plants

(d) It is the crossing of an organism with the recessive trait in order to determine the genotype of the organism

3. (a) Photosynthesis

   (b)

- Presence of sunlight
- Presence of chlorophyll

(c)

- Oxygen gas diffuse to spongy mesophyll cells and used in respiration to produce energy.
- Excess oxygen gas diffuses into the atmosphere through the stomata.
- Glucose is stored in parts of the plant as starch
- Glucose is converted into starch for temporary storage then converted into sucrose for translocation to other parts of the plant
- Glucose is broken down in the process of respiration to produce energy.

4. (a) (i)

- To expose the leaves for maximum absorption of sunlight hence maximizing photosynthesis
- To expose the flowers to enhance pollination hence fertilization.

- To enhance parts of the plant to withstand forces of the environment e.g. air current

(ii)

- To give the body its posture and shape
- Enhance the movement of the animal by the articulation of bones and muscles
- Offering of mechanical protection of delicate organs by the skeleton

(b)

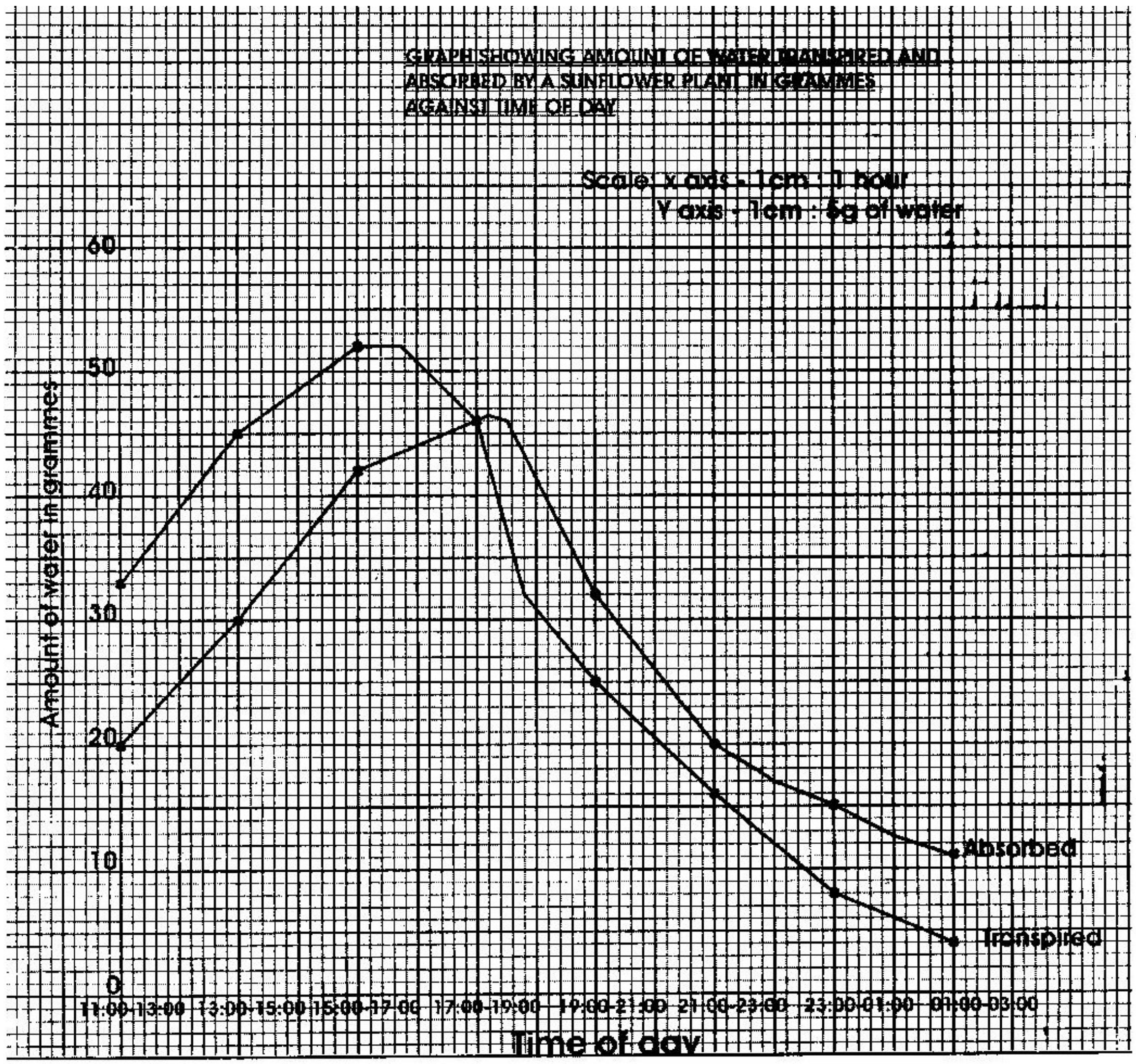
- To avoid predation
- In search of food, mate and shelter
- To colonize new areas
- To move away from unfavourable conditions e.g. foods, fires and earthquakes

5. (a)

- L_1 the piece was placed in a hypotonic solution. The cells absorbed water by Osmosis and became turgid. However the epidermis is water proof and the cells could not absorb water. The piece therefore bulged/ swelled outward on the cut part.
- L_2 the piece was placed in a hypertonic solution. The cells lost water to the surrounding by osmosis and became flaccid. However the epidermis is waterproof and the cells did not lose water. The piece therefore shrunk on the part not affecting the epidermis.

(b) Maintaining the osmotic pressure helps to maintain the turgidity of plant cells hence supporting the plant.

6. (a) Graph



(b) Between 17: 00 and 19.00

(c) (i) **Transpiration**

- Between 11: 00 to 17: 00 the rate of transpiration increased.
- Between these hours, the light intensity, temperature and air currents also increase. These conditions favour high rate of transpiration.
- Between 17:00 to 03:00 the rate of transpiration decreases steadily. The temperatures and light intensity decrease as night falls. Therefore conditions favouring transpiration are absent hence decrease in transpiration.

(ii) **Absorption**

- Between 11:00 to 19:00 the absorption of water increases steadily. Between these hours conditions favouring transpiration are present hence high loss of water from the plant.
- A suction force is created and more water is absorbed.
- Between 19:00 to 00:300 the absorption of water decreases steadily. Between these hours the rate of transpiration is decreasing and more water is retained in the plant.
- Less water is absorbed

(d) Both transpiration and absorption would be lower. At 05:00 the temperatures are very low and there is no light. Hence there is very little loss of water.

(e)

- Temperature

- Light intensity
- Air currents
- Atmospheric pressure
- Humidity

(f)

- **Temperature**- As temperature increases the rate of evaporation also increases. More water is lost from the surface of the leaf hence increasing rate of transpiration. As temperature decreases less water evaporates from the surface of the leaf hence decreasing rate of transpiration.
- **Light intensity** – As light intensity increases, it enhances opening of stomata. More water therefore is lost in form of water vapour.
  As light intensity decreases (1 dim or dark) the stomata close and little or no water is lost from the leaf hence lower rate of transpiration.
- **Air currents** – Strong air currents drives away the diffuse water vapour from the surface of the leaf creating high diffusion gradient. This increases rate of transpiration. Calm condition allows water vapour to accumulate on the surface of the leaf reducing the diffusion gradient hence lower rate of transpiration.
- **Humidity**- when the amount of water vapour in the air is high, there is low diffusion gradient and less water is lost. When the air is dry, there is high diffusion gradient and more water is lost, hence high rate transpiration.

7.

- It is the cycling of nitrogen compounds in nature. Free nitrogen of the air cannot be used by the plants but has to be converted into ammonium compounds and nitrates in order to be absorbed. It is done by the following ways:
- Fixation by lighting and thunderstorms
- The atmospheric oxygen and nitrogen combine and nitric and nitrous acids are formed. They combine with minerals in the soil forming nitrates that are then absorbed

- Fixation by nitrogen- Fixing bacteria
- The bacteria can be symbiotic that live in root nodules of leguminous plants e.g. Rhizobium
- They absorb the nitrogen and convert it into nitrates that are then used by the plant.
- Other bacteria are free living in the soil e.g. Azotobacter. They absorb the nitrogen and then absorbed by the plants
- Nitrifying bacteria e.g Nitrosomonas and Nitrococcus
- When plants and animals die they decompose releasing the proteins in form of ammonium compounds. The nitrifying bacteria oxidize the ammonium compounds into nitrates then to nitrates. They are then absorbed by plants.
- This process is referred to as ammonification.

- However there some bacteria in the cycle that convert the nitrates to nitrites, ammonia and nitrogen gas that cannot be used by the plant. They are known as the denitrifying bacteria. They reduce the nitrates to obtain the oxygen for their respiration e.g. psocoudomonas denitrificans.

8. (a)

- They are moist to dissolve the diffusing gases
- They are highly folded to increase surface area over which gaseous exchange takes place.
- They are thin walled to reduce distance over which diffusion of gases takes place.
- They are well supplied with blood capillaries to carry the diffusing gases hence creating high diffusion gradient.

(b)

- Gaseous exchange takes place in the alveolus
- The inhaled air has high concentration of oxygen compared to that of blood capillaries. The oxygen first dissolves in the moisture on the surface of the alveolus. It then diffuses across the alveolar wall then through the capillary wall into the red blood cells.
- The hemoglobin in the red blood cells combines with the oxygen forming oxy- hemoglobin.
- The blood is said to be oxygenated and is transported to the heart via the pulmonary vein.

- The carbon (IV) oxide is more concentrated in the blood capillaries than in the alveolar cavity.
- It therefore diffuse across the capillary wall and then through the alveolar wall into the alveolar spaces.
- It is then expelled during exhalation.

PAPER 3 (PRACTICAL)

1. (a) A - liver

B - Stomach

C - Pancreas

D - Ileum

G - Duodenum

(b) E - Caecum/ appendix – has enzyme cellulose that digests cellulose into maltose.

F - Rectum- absorption of water and being of faecus before being released through the anus.

(c) On the diagram

(d) (i) Male

(ii) Presence of prostrate gland

(e) (i) Magnification = $\frac{\text{Length of drawing}}{\text{Actual length}}$

Magnification = $\frac{9 \text{ cm}}{15 \text{ cm}}$

Magnification = X 0.6

(ii) Magnification = $\frac{\text{Length of drawing}}{\text{Actual length}}$

$$X\ 0.6 = \frac{14.5 \text{ cm}}{X}$$

$$X = \frac{14.5}{0.6}$$

Actual length = 24.17 cm

2.

Substance	Food substance being tested for	Procedure	Observations	Conclusion
S	Protein	Place a portion of S in a test tube. Add an equal portion of sodium Hydroxide. Shake well. Add a few drops of copper (II) sulphate solution.	Colour turns to purple/ violet	Proteins present
T	Protein	Place a portion of T in a test tube. Add equal portion of sodium hydroxide. Shake well. Add a few drops of copper (II) sulphate solution	Colour turns to pale purple/ pink	Traces of proteins present
U	Protein	Place a portion of U in a test tube. Add an equal portion of sodium hydroxide solution. Shake well. Add a few drops of copper(II) sulphate solution	Colour remains yellow/ orange	Proteins absent

3.

Specimen	Mode of dispersal	Adaptive features
K	Animal dispersal	Hooks have developed on the calyx for attachment to fur or clothes of passing animal.
L	Animal dispersal	Brightly coloured and edible pericap. Seed coats tough and not digested.
M	Wind dispersal	Seeds have hairy and feather- like projections to increase surfaces so that they are blown by the wind
N	Animal dispersal	Succulent mesocarp and tough indigestible endocarp. Seed not destroyed.
Q	Explosive mechanism	Dry pod with structure/ line of weakness that breaks releasing seeds

(b) (i) On the diagram

(ii) Free central placentation

(c) Endocarp

WORKED OUT PAST KCSE PRACTICAL QUESTIONS

Example 1

1. You are provided with a specimen labeled K and solution labeled P and Q. Cut the specimen into two halve. From one half remove the outer and an inner leaf of the specimen. Examine them.

    K- Onion bulb

    P- Distilled water

    Q- Saturated sodium chloride solution

    (a) State the observable features of the outer and inner leaves of the specimen.

    Answer

    Outer leaf

    - Thin/ membranous papery (1 mark)
    - Scaly/ dry/ dehydrated (1 mark)
    - Pigmented / coloured/ brown/ purple/ red (1 mark)

    Inner leaf

    - Fleshy /succulent/ juicy (1 mark)
    - Thick swollen (1 mark)
    - White/ cream yellow/ purple/ red/ green (1 mark)

    (b) State the functions of the inner and outer leaves of the specimen (4 marks)

Answer

Outer leaf- protection (1) protection against transpiration/ drying/ evaporation/ loss of water/ injection/ desiccation/ mechanical damage/ injury

(1) Inner leaf - storage (1); storage of food / water

(c) (i) - Name the type of reproduction exhibited by specimen K. (1 mark)

- Asexual / vegetative propagation/ vegetative reproduction/ natural vegetative reproduction.

(ii) Give a reason for your answer in c (i) above. (1 mark)

Answer

- Presence of bud(s)

Using the outer half of specimen K, remove some of the inner leaves. Cut the leaves along their lengths into nine strips. Each strip should be about 2 mm wide. Place three strips into the solution labeled P. Place another three strips into the solution labeled Q and leave the last three strips in a Petri dish labeled R. Allow the experimental set ups to stand for 10 minutes.

(d) Use your finger to feel the texture of the strips

Record your observations (2 marks)

Strip in solution P

Answer

Firm/turgid/hard/stiff

Strips in solution AQ

Answer

Soft/ flabby/ flaccid/ flexible/ tender/ limp

(e) Account for the texture of the strip in solution Q. (4 marks)

Answer

- Q is more concentrated / hypertonic than the cell cell sap.
- Cells lost water by osmosis
- Hence the cells become flabby/ flaccid/ plasmolysed / shrank

(f) Suggest the concentration of solution P in relation to the cell sap in the strips of the specimen.

Answer

P is dilute/ less concentrated/ hypotonic

Give a reason for your answer (1 mark)

Answer

Strips remain firm/ turgid/ hard/ stiff/ remain the same/ absorbs some water/ did not become flaccid

(g) State the aim in the set up R (1 mark)

Answer

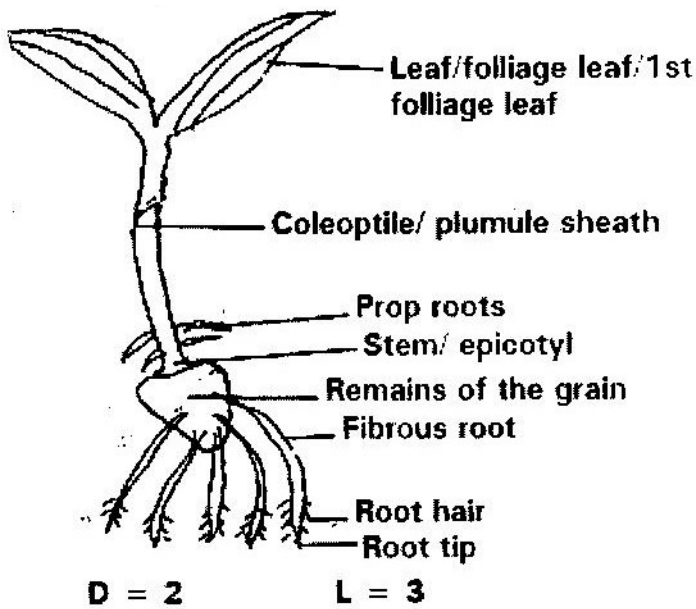
Control experiment

Example 2

2. You are provided with a specimen labeled W which was grown in the dark examine the specimen

W- Young Maize seedlings

(a) Draw and label all the observable parts of the specimen (5 marks)



Note:

- Roots (fibrous root) should be single or double lines
- Roots should average from remains
- 2 foliage leaves must be shown
- Outline must be correct.

(b) State the function of any three parts you have labeled (3 marks)

Name of the part	**Function**
(i) Leaf (ii) Coleoptile/plumule sheath (iii) Remains of grain (iv) Fibrous roots (v) Root roots (vi) Prop roots (vii) Testa (vii) Root tip (ix) Stem	- For photosynthesis/ manufacture of food/ gaseous - Protest the 1st foliage leaf - Stores food for the young germinating seedling - Anchoring the seedling to the ground / support. - Transport or conduction of water - Absorption of water and minerals salts from the soil. - For anchorage / support - Protection against desiccation/ damage - Region of cell division/ growth - Supports other parts of plant/ helps in conduction of water.

(c) Cut off the shoot and keep the rest of the specimen to be used in question. Crush the shoot on a white tile using a glass rod. Carry out the following food tests. Record your observations and conclusion in the table below.

(4 marks)

Test	**Observation**	**Conclusion**
(i) Add a drop of iodine solution to a portion of the crushed shoot on the white tile	Colour of iodine remains/ no colour change/ remain yellow/ brown/ red/ orange	Starch absent
(ii) Place another portion	Turn yellow/ orange/ brown /	Reducing sugar present

of the crushed shoot in a test tube. Add 1 cm^3 of Benedict's solution. Shake the mixture and heat.	red Greenish - yellow	Traces of reducing sugar

(d) Remove the grain from the remaining part of the specimen. Crush it and carry out the following tests. Record your observations and conclusions in the table below.

Test	**Observation**	**Conclusion**
(i) Add a drop of iodine solution to a portion of the crushed shoot on the white tile	Blue black/ Blue/ black	Starch present
(ii) Place another portion of the crushed shoot in a test tube. Add 1 cm^3 of Benedict's solution. Shake the mixture and heat.	Colour changes from blue to yellowish green, orange/ brown/ brick red/ red	Simple sugar/ reducing sugar present glucose present

(e) Account for your results in (c) and (d) above (7 marks)

Answer

- The grain stores starch during germination. Some of the starch in the grain is converted/ broken down/ hydrolysed by enzyme/ amylase/diastase to reducing/simple sugars. The sugars are translocated/ moved to the shoot. The shoot did not photosynthesize.